



ECONOMICS

Student's Book

Senior Four



Experimental version

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FOREWORD

Dear Student,

Rwanda Basic Education Board is honoured to present to you Economics book for Senior 4 which serves as a guide to competence-based teaching and learning to ensure consistency and coherence in the learning of economics subject. The Rwandan educational philosophy is to ensure that you achieve full potential at every level of education which will prepare you to be well integrated in society and exploit employment opportunities.

The government of Rwanda emphasizes the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn and the competence you acquire. Those factors include quality instructional materials available, assessment strategies for the students among others. Special attention was paid to activities that facilitate learning process, develop your ideas and make new discoveries during concrete activities carried out individually or with peers. In Competence-Based Curriculum (CBC), learning is considered as a process of active building and developing knowledge and meanings by the student where concepts are mainly introduced by an activity, a situation or a scenario that helps the student to construct knowledge, develop skills and acquire positive attitudes and values. For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of skills;
- Share relevant information with other students through presentations, discussions, group work and other active learning techniques such as role play, case studies, investigation and research in the library, from the internet or from your community;
- Participate and take responsibility for your own learning; and
- Draw conclusions based on the findings from the learning activities.

To facilitate you in doing activities, the content of this book is designed to be self-explanatory so that you can use it independently, acquire knowledge, and assess your competences. The book is organized into units. Each unit includes the key unit competence, followed by an introductory activity before the development of Economic concepts that are connected to real world situations. The number of lessons in each unit depends on its length. Each lesson begins with a learning activity, and each unit ends with an end-of-unit assessment.

I wish to sincerely extend my appreciation to REB staff who organized the editing process of this textbook. Special gratitude also goes to lecturers, teachers, illustrators and designers who supported the exercise throughout. Any comment or contribution would be welcome to the improvement of this textbook for the next edition.

Dr. MBARUSHIMANA Nelson
Director General, REB



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UNIT 1

BASIC ECONOMIC CONCEPTS AND PRINCIPLES



Key Unit Competence:

Analyze real-life situations using basic economic principles to make rational economic decisions at individual, community, and national levels.

Learning objectives

Knowledge and understanding

- Define economics and explain its meaning, origin, and development.
- Explain the importance of economics to individuals, society, and national development.
- Identify and explain basic economic terms, including needs, wants, resources, goods, and services.
- Distinguish and correctly use key economic terms such as price, production, distribution, exchange, economic agents, externalities, ceteris paribus, and economic variables.
- Describe the nature of economics as an art, a science, and a social science.
- Differentiate between positive and normative economics using relevant examples.
- Explain the scope of economics and identify its main branches, namely microeconomics and macroeconomics.
- Compare microeconomics and macroeconomics by stating their meanings, examples, relationships, and differences.
- Explain the fundamental principles of economics, including scarcity, choice, and opportunity cost, and show how they are related.
- Describe the Production Possibilities Frontier (PPF), its meaning, assumptions, and importance.
- Interpret points under, on, and beyond the PPF using diagrams and examples.

1

- Construct and explain a hypothetical PPF schedule showing alternative production possibilities.
- Explain shifts in the PPF and identify different types of PPFs.
- Identify and explain basic economic questions and economic activities.
- Describe different types of economic systems and analyze their advantages and disadvantages.

Skills

2

- Apply basic economic concepts to analyze real-life situations at individual, community, and national levels.
- Use economic terms correctly in oral and written explanations.
- Analyze situations of scarcity and make rational choices by identifying opportunity costs.
- Construct, label, and interpret PPF diagrams to demonstrate efficient, inefficient, and unattainable production outcomes.
- Analyze real or hypothetical economic problems using fundamental economic principles.
- Compare different economic systems and assess their suitability for different societies.
- Make reasoned economic decisions related to consumption, production, and resource allocation.
- Work collaboratively to discuss, explain, and solve simple economic problems.

Attitudes and values

3

- Appreciate the need for wise and responsible use of limited resources.
- Demonstrate rational decision-making in managing personal and shared resources.
- Show awareness of fairness and efficiency in economic activities and resource distribution.
- Develop a positive attitude toward saving, budgeting, and planning.
- Recognize the role of economics in promoting sustainable development.
- Respect diverse economic systems and viewpoints while engaging in discussions.
- Develop responsibility and accountability in economic choices affecting individuals, communities, and the nation

Have you ever wondered why your pocket money cannot buy everything you want, why market prices keep changing, or why families and governments must decide which needs to meet first? For example, you may have to choose between buying data

bundles and saving for school materials; farmers must decide which crops to grow on limited land; and the government cannot build all roads, hospitals, and schools at the same time. These familiar situations reflect a fundamental economic reality: although human needs and wants are unlimited, the resources available to satisfy them are limited.

It is this challenge of making choices under limited resources that economics seeks to explain. In this unit, you will study key economic concepts such as needs and wants, goods and services, scarcity, resources, economic agents, economic activities, and economic systems. You will also learn how economics developed as a subject and how its ideas apply to everyday situations like budgeting, production, market decisions, and government planning. By the end of this unit, you will be able to apply economic thinking in daily life, make informed decisions, and understand how the efficient and fair use of resources supports sustainable development.



Introductory activity

Read each scenario and complete the questions that follow.

- A. A Senior Four student receives 2,000 FRW for the weekend but wants to buy a snack, a notebook, and a pen. The money is not enough for everything.
- B. A farmer in Gicumbi district must decide whether to plant potatoes, beans, or vegetables due to limited land.
- C. A shop owner in Huye district notices increased demand for phone chargers but has a limited budget for restocking.
- D. The Government of Rwanda must choose whether to first build a school, repair a road, or expand electricity in a rural sector.

Questions

- 1. What do the four situations have in common? Identify two economic problems shown in these scenarios.
- 2. Choose one scenario and identify the main needs, main wants, and the limited resources involved.
- 3. Explain the economic choice in your selected scenario and identify the opportunity cost.
- 4. Think about a recent decision you made involving limited resources. What options did you have, what influenced your choice, and what was the opportunity cost?
- 5. Why is it important for individuals, families, and governments to make fair and responsible economic decisions? Give an example from Rwanda.

1.1 Meaning, origin and development of economics

Learning Activity 1.1



"Read the scenario below carefully and answer the questions that follow."

Way back in ancient Rwanda, people continually faced choices in their daily lives, such as deciding whether to use clay pots for storing food or for exchange, choosing between growing sorghum or beans, or determining how to share land among families. These early choices involved managing limited resources, much like the decisions individuals and governments make today in response to scarcity.

Questions

- a) Based on the scenario, in your own words, what do you think economics is about?
- b) Why do you think people needed economic thinking long before economics was taught in school?
- c) How are the choices made by people in ancient Rwanda similar to the choices people and governments make today?
- d) How are the choices made by people in ancient Rwanda similar to the choices people and governments make today?

Have you ever wondered what economics is really about and how it came to be studied as an important subject in schools and society? This lesson introduces students to the background of economics by exploring how the subject emerged and gradually evolved over time. It highlights key stages in the growth of economic thought and the contributions of early scholars who shaped the field. Students will examine how economic ideas have changed in response to real-world challenges and how the subject has expanded to address modern issues. By the end of the lesson, students will gain a clear understanding of how economics developed into a structured discipline and why it remains relevant in today's society.

1.1.1 Meaning of economics

Every day, individuals, families, businesses, and governments must make choices because they cannot satisfy all their needs and wants at once. This happens because resources such as money, land, time, and labour are limited. Economics is the subject that helps us understand how these choices are made.

Economics is **the social science that studies how people use limited resources to satisfy unlimited needs and wants**. It explains how goods and services are produced, distributed, and consumed. It explores why individuals must choose between alternatives and how societies decide what to produce, how to produce it, and for whom the goods and services will be produced.

Definitions of economics by key economists

Different economists have explained economics in different ways:

- **Adam Smith (1723–1790) - wealth and markets:** Adam Smith viewed economics as the study of how a nation creates wealth through production and trade. He argued that when individuals act in their own economic interest within a free market, their actions naturally support overall national growth, a principle known as the “invisible hand. invisible hand is a metaphor introduced by Adam smith that describes how self interested individuals in free markets can unintentionally benefit society by responding to supply and demand signals. Exemple, when farmers in Nyabihu district specialise in potatoes production while traders transport the harvest to Kigali, efficiency increases reflecting Smith's ideas.
- **Alfred Marshall (1842–1924) – Human Welfare:** Marshall considered economics as *a study of people in their daily economic activities*, such as earning and spending income. Example: Government subsidies on fertilizer improve farmers' incomes showing Marshall's welfare focus.
- **Lionel Robbins (1898–1984) – Scarcity and choice:** Robbins defined economics as *“the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses.”* He explained that economics is about how people make choices because their needs and wants are unlimited, but the resources available to satisfy them are limited.

Since these scarce resources have alternative uses,choosing one use means forgoing another,which creates opportunity cost. Exemple: a farmer in Musanze district choosing between beans and maize faces opportunity cost.

- **Paul Samuelson (1915–2009) – Production and distribution:** Samuelson described economics as how societies allocate scarce resources to produce and distribute goods over time. He emphasised efficiency and long-term planning. Example: budget allocation for health, education, and infrastructure reflects Samuelson's view of resource distribution.
- **Dr. Donald Kaberuka- Development and What Works :** According to Donald Kaberuka, the Rwandan economist and former President of the African Development Bank, economics is the practical management of resources and policies to achieve sustainable development and improve people's welfare, with emphasis on solutions that work in real-world conditions rather than on ideology alone.

Hence, economics is about making choices because resources are limited. It helps to understand everyday issues like prices, saving, spending, production, jobs, and development.

1.1.2 Origin of economics.

Economics comes from the Greek word **“Oikonomos,”** meaning household management, derived from *oikos* (household) and *nomos* (management). At its core, economics is concerned with how people manage limited resources to meet unlimited needs (scarcity). From early times, families and communities had to decide how best to use land, food, tools, and time, which marked the beginning of economic thinking. In Rwanda, traditional households practiced this by organizing labor, sharing harvests, and cooperating through community activities such as **Umuganda**. Economics

today applies to many areas of life, including education, employment, housing, and transport, and examines how individuals, firms, and governments make choices under conditions of scarcity.

Historically, economic ideas have evolved alongside society. Ancient Greek thinkers like **Aristotle** reflected on wealth and trade, distinguishing between meeting basic needs and profitseeking. During the Middle Ages, scholars such as **Thomas Aquinas** emphasized fairness in economic dealings through the idea of a “just price,” a concept also reflected in traditional societies where trade was guided by customs and mutual trust. Later, between the 16th and 18th centuries, **mercantilism** shaped European economic thought by focusing on trade, exports, and wealth accumulation. Although Rwanda was not part of this system, its communities engaged in regional trade by exchanging surplus goods based on experience and cultural values. These historical developments show how economics grew from simple household management into a broad field explaining how societies organize production, exchange, and consumption.

1.1.3 Development of Economics

By the 18th century, economics became a formal discipline through the work of **Adam Smith**, who explained in *The Wealth of Nations* (1776) that individuals pursuing their own self-interest can unintentionally benefit society, a process he called the “invisible hand.” This idea is seen when farmers or businesses produce goods to earn income while also meeting community needs, as happens in Rwanda through household production and cooperatives. Other classical economists added important ideas, including **David Ricardo’s** theory of comparative advantage, which explains why countries like Rwanda specialize in exports such as coffee and tea, and **Thomas Malthus’s** warning that rapid population growth can strain limited resources.

Over time, economics expanded as thinkers focused on supply, demand, prices, and the role of government. **John Maynard Keynes** later argued that governments should increase spending during economic downturns to stimulate growth, an idea reflected in Rwanda’s investment in infrastructure and education. Economics has thus evolved from moral and philosophical thinking into a systematic study of how scarce resources are produced, distributed, and consumed. In Rwanda, this evolution is seen in the shift from traditional practices such as barter and communal farming to modern economic policies and national development strategies like **Vision 2050**, which aim to build a sustainable, knowledgebased economy.

Application activity 1.1



1. Why do people and families have to make choices when using money, land, or time? Give one exemple from your daily life.
2. Economics comes from a word that means managing a household. How did families in traditional Rwanda manage their resources? Give one exemple.
3. What does Lionel Robbins mean when he says economics is about scarce resources and choice? Explain using a simple exemple.
4. How has economics evolved from early practices like barter and communal farming to modern planning in Rwanda today? Why is this change important?

1.2 Importance of economics

Learning Activity 1.2



1. When you receive pocket money, what do you usually consider before spending it?
2. Have you ever saved money for a school need or future expense? Explain why.
3. Why do you think planning how to use money is important?
4. Which public services provided by the government benefit your community?
5. Why is it important for citizens to contribute taxes?
6. How do government programs like Girinka or Ejo Heza help families?

Studying economics is important because it helps us understand how individuals, businesses, and governments make choices when resources are limited. It equips students with knowledge and skills to make informed decisions about money, resources, and development, and to better understand real-life issues such as prices, employment, poverty, and national growth. Thus, economics helps people make wise choices, manage scarce resources, and promote development at both individual and national levels, as explained below.

- Studying economics is important because it helps us understand how individuals, businesses, and governments make choices when resources are limited. It equips students with knowledge and skills to make informed decisions about money, resources, and development, and to better understand real-life issues such as prices, employment, poverty, and national growth. Thus, economics helps people make wise choices, manage scarce resources, and promote development at both individual and national levels, as explained below.
- Economics is an important subject because it helps individuals, families, businesses, and governments make informed decisions when resources are limited. Studying economics provides practical knowledge and skills that support personal development and national growth.
- Better financial decision making: Studying economics helps students understand budgeting, saving, and choice under scarcity. Students and families learn how to allocate limited income among competing needs, such as transport, food, school fees, and savings, leading to wiser financial decisions.
- Improved problem solving and critical thinking: Economics trains students to analyse real-life problems such as rising prices, unemployment, and shortages. By understanding causes like seasonal changes or low productivity, students can think critically about practical solutions that improve livelihoods.
- Understanding how markets work: Economics explains how prices are determined by demand and supply and how producers and traders respond to market changes. This knowledge helps students understand everyday market behaviour, such as price increases when supply falls or production expansion when demand rises.

- Preparation for national development: Through economics, students understand government roles in budgeting, taxation, investment, and development planning. This helps explain how national programs and public spending contribute to economic growth and improved living standards.
- Efficient and sustainable use of resources: Economics teaches the importance of using scarce resources wisely and avoiding waste. Students understand sustainable practices such as proper land use, water conservation, and reducing post-harvest losses, which support long-term development.
- Support for entrepreneurship and job creation: By studying economics, students gain skills in cost management, profit calculation, and identifying business opportunities. This knowledge encourages self-employment, innovation, and the creation of small businesses and cooperatives.
- Promotion of responsible and informed citizenship: Economics helps students understand why taxes are necessary and how public funds are used to provide essential services. This knowledge promotes responsible citizenship and support for national development efforts.
- Increased awareness of global economic connections: Economics helps students understand international trade, investment, and global markets. This awareness shows how Rwanda connects with other countries through exports, imports, tourism, and foreign investment.
- Studying economics helps students apply economic reasoning in real-life situations, make informed decisions, support household welfare, and contribute to Rwanda's sustainable development.

Application activity 1.2



1. Why is it important for individuals and families to study economics in their daily lives?
2. Explain how the study of economics helps people make better choices when resources are limited.
3. Give two examples of real-life issues (at home, school, or in Rwanda) that economics helps us understand.
4. Why is studying economics important for government decision-making and national development?
5. Which different fields can someone who has studied economics work in, and which one would you choose?

1.3 Basic and key economic terms used in economics

Learning Activity 1.3



Read the scenario below and answer the questions that follow.

Mr. Gato has several needs and wants that he hopes to satisfy with his monthly income of RWF 100,000. He allocates this income to essential expenditures such as rent, food, his children's school fees, medication, and loan repayments. He also hopes to purchase a car and travel abroad with his family during the school holidays.

- a) Based on this scenario, identify Mr. Gato's needs and wants.
- b) Which ones (from (a) above) should he satisfy first? Support your answer.

In economics, *basic terms* are simple words or definitions used to describe fundamental ideas (such as needs, wants, resources, wealth, goods, and services), mainly helping understand the subject's vocabulary, while *key terms* are broader, more analytical ideas (such as price, externalities, ceteris paribus, production consumption, exchange etc.) that explain how economic systems work and are used to analyze real-world situations.

1.3.1 Basic economic terms

i. Needs:

Needs are essential requirements for survival and healthy living, such as food, water, shelter, clothing, healthcare, and education. Examples: a family in buying food stuff; a student in needing notebooks and school shoes; a household in needing clean water.

ii. Wants:

Wants are human desires that are not essential for survival but help to make life more comfortable and enjoyable. They may be *material* (tangible items such as cars, TVs, phones, and fridges) or *immaterial* (intangible items and qualities such as help, prestige, or power). Wants can also be *private*, belonging to individuals (for example, a personal car, radio, or bicycle), or *public*, shared by society (such as roads, bridges, playgrounds, and churches). Generally, wants are unlimited, insatiable, and dynamic, may be recurring, complementary, competitive, or alternative, and they vary in urgency, person, place, and time, making prioritization necessary due to limited resources.

Table1.1. Difference between needs and wants

Needs	Wants
Basic necessities required for survival	Desires that improve comfort or enjoyment
Essential for life and basic functioning	Not essential for survival
Limited and universal for all people	Unlimited and vary from person to person
Without them, life becomes difficult or even impossible	Without them, life can still continue
Exemples: food, water, shelter, clothing, healthcare	Exemples: smartphones, luxury clothes, holidays, cars
Must be satisfied first	Satisfied after needs are met

iii. Resources (factors of production):

Resources are all the productive inputs used to create goods and services. Resources can be classified into three main categories: natural resources, which are resources provided by nature, such as land, water, air, minerals, and sunlight; human-made resources, which are resources created by people, including buildings, machinery, tools, and equipment; and human resources, which consist of the labour, skills, knowledge, and abilities that people contribute to production. Resources are also classified as **renewable**, meaning they can be used repeatedly and replaced quickly, such as solar, wind, and water energy, and **non-renewable**, which are limited and get exhausted when used, such as coal, petroleum, natural gas, and minerals like gold and diamonds.

iv. Wealth:

Wealth refers to the stock of valuable assets owned by an individual, organization, firm, or country at a particular point in time. It has monetary value, is relatively scarce, can be transferred from one owner to another, and provides utility or satisfaction to its owner. Wealth exists in different forms, including social wealth, which consists of public assets such as forests, roads, libraries, and hospitals; *business wealth*, which belongs to business organizations; and *individual wealth*, which refers to privately owned assets and may be marketable (buyable and sellable) or non-marketable, such as pension contributions that cannot be transferred.

v. Services:

Services are intangible activities that satisfy non-material wants. They include direct services, such as education and healthcare, which benefit individuals personally, and indirect services, such as banking, mobile money transfers, transport, and communication, which support individuals and businesses. Because services are intangible, they cannot be seen or touched before they are delivered. They are usually consumed at the time they are provided, cannot be stored for future use, and depend on service providers and supporting facilities for their delivery.

vi. Goods:

Goods are tangible products produced to satisfy human wants. They can be seen, touched, and stored for future use. Examples of goods include food, clothes, books, furniture, and vehicles.

Types of Goods

- a) Free goods:** These are goods that are freely available in nature and do not require human effort or payment to obtain because they are abundant. Examples include air, sunshine, rainwater, and sea water. Free goods have no cost, unlimited supply, no opportunity cost, and cannot be owned or exchanged, since they are available to everyone without scarcity.
- b) Economic goods/commercial goods:** Economic goods are goods that satisfy human wants and are scarce relative to demand. Because they are limited in supply, they have value, require resources to produce, and can be bought and sold in the market. Their use involves an opportunity cost, as choosing one good may require giving up another. Examples of economic goods in Rwanda include maize flour, beans, milk, school textbooks, uniforms, motorcycles, mobile phones, and cement used in construction.
- c) Public goods:** Public goods are goods that are collectively owned and enjoyed by society and are non-exclusive (no one can be prevented from benefiting). Examples include national defense, public roads, street lighting, public parks, and public clocks. They are not consumed individually but enjoyed collectively, and one person's use does not reduce availability to others. Public goods are characterized by non-excludability (users cannot be excluded), non-rivalry (one person's use does not reduce others' use), indivisibility (cannot be divided into individual units), and collective consumption (used jointly by society), often generate positive externalities (benefits spill over to others), and have unpredictable demand (usage cannot be accurately measured or forecast).

Because individuals can benefit from public goods without paying, they tend to free-ride (enjoy benefits without contributing). This free-rider problem makes public goods difficult to price and unprofitable for private firms, since there is no clear market price (no direct payment by users). Consequently, public goods are typically provided and financed by the government through taxation.
- d) Private goods:** Private goods are goods owned and used by individuals. One person's use of these goods prevents others from using them at the same time. They are obtained through direct payment. Examples include a mobile phone, a bicycle, a school uniform, a textbook, a personal car, and Inyange products such as milk, yoghurt, and juice.
- e) Merit goods:** These are goods and services that are considered beneficial to individuals and society. Examples include education, healthcare, immunization, public libraries, and clean water. These goods improve individual welfare and generate positive benefits for society as a whole, as a result, governments usually provide, subsidize, or encourage their consumption.
- f) Economic bads (demerit goods):** These are goods or services that are harmful to individuals and society and tend to be over-consumed if left to the

free market. Examples of demerit goods include alcohol, tobacco, illicit drugs, gambling, and pollution-causing products. Because people may ignore or underestimate their negative effects, they can lead to health problems, social issues, and other negative effects on society, as a result, governments often discourage their consumption through taxes, regulations, or bans.

- g) Intermediate goods:** These are goods used up or transformed in the production of other goods or services and do not remain in their original form in the final product. They are not for direct consumption but are processed to make final goods, such as flour for bread, cotton for clothes, steel for vehicles, and fuel for production.
- h) Final goods:** These are goods that are ready for final use and not used as inputs to make other goods within the time period. They include all goods used by consumers, businesses, or government for final use. Examples include Radio or television bought for home use, Bus bought by a transport company to carry passengers, medical equipment bought for a public hospital etc.
- i) Producer goods:** These are goods that are used to produce other goods and services rather than to satisfy human wants directly. They are durable items that assist in the production process over a long period and are not used up after a single use. Examples include machines, tools, factory buildings, equipment, and vehicles used for business purposes.
- j) Consumer goods:** These are goods that are produced for direct use and satisfaction of human wants. They are bought by final consumers and are not used to produce other goods. Examples include food, clothing, shoes, books, furniture, and household appliances.
- k) Complimentary goods:** These are the type of goods that are used together. To get the full utility of one good, the other complementary good must be used along with it. For example, car and petrol are two complementary goods.
- l) Substitute goods:** These are goods that can be used in place of one another to satisfy the same want because they serve a similar purpose. When the price of one good rises, consumers may switch to its alternative. Examples include tea and coffee, butter and margarine, rice and maize, and gas and charcoal, LG and Samsung TVs, beans and peas, soft drinks and natural juice etc.
- m) Inferior goods:** These are goods whose demand falls with an increase in the consumer's level of income. They are usually cheaper and bought mainly because of low income, not preference. A good example is a second-hand cloth. When the income of people increases, people avoid buying second-hand clothes and shift to more expensive clothes.
- n) Veblen (Luxury) goods:** These are luxury goods whose demand rises as price and consumer income increase because they are valued as symbols of status, prestige, and comfort. They are conspicuously consumed to display wealth, such as expensive cars, luxury watches and jewelry, high-end smartphones, and luxury holidays.
- o) Giffen goods:** These are goods for which people buy more when the price rises and less when the price falls, which is the opposite of normal goods. This usually happens with cheap basic foods that poor consumers depend on. When the price of such a good rises, people cannot afford better alternatives,

so they are forced to buy more of the cheaper staple. Examples may include maize, rice, or bread in low-income households under special conditions.

- p) Normal goods:** A normal good is a good whose demand increases when consumer's income rises and decreases when their income falls, assuming other factors constant. Examples include Inyange milk and yoghurt, smart-phones, better-quality clothing, furniture, motorcycles, etc.

1.3.2 Key economic terms

- a) Price:** This refers to the rate at which a commodity is exchanged in the market, or the value of a good expressed in monetary terms. Prices take different forms: **the market price**, which is the price prevailing at a given time; **the equilibrium price**, which is set when quantity demanded equals quantity supplied; **the normal price**, which is the long-run price after short-term fluctuations; and **the reserve price**, which is the minimum price a seller is willing to accept for a commodity.
- b) Economic Agents:** These are the main decision-making units in an economy that influence production, consumption, and resource allocation. They include **households**, which are groups of individuals making joint economic decisions; **firms**, which use factors of production to produce goods and services; **the state/government**, which has political and economic power to regulate and influence the activities of households and firms; and **the foreign sector**, which consists of exports and imports and represents all economic activities with the rest of the world outside the country's boundaries.
- c) Production:** This refers to the process of using resources (factors of production) such as land, labour, capital, and entrepreneurship to create goods and services that satisfy human needs and wants. It involves changing raw materials or providing services to make them useful to consumers. Examples: Farmers producing bananas, factories producing cement, tailors making school uniforms.
- d) Distribution:** This refers to the way goods, services, and income are shared among individuals or groups in a society. It explains how resources and wealth such as wages, profits, and rent are allocated to different people after production has taken place. Example, Transporting Irish potatoes from Rubavu to Kigali; distributing school materials to districts.
- e) Consumption:** This refers to the use of goods and services by individuals or households to satisfy their needs and wants. It is the final stage of economic activity, where people buy and use products such as food, clothing, or services for personal benefit rather than for further production. Examples: Families eating food; students using textbooks; households using electricity and water
- f) Exchange:** This refers to the process of trading goods and services between individuals, businesses, or countries, usually using money as a medium. Examples: Mobile money payments in rural markets; tourists exchanging currency; farmers selling produce to cooperatives.
- g) Externalities:** These are the side effects of an economic activity that affect people who are not directly involved in that activity, and these effects are not

included in the market price of the goods or services produced. When the side effects are harmful, they are called **negative externalities** because they impose costs on others, such as factory pollution, noise from bars or construction, and vehicle emissions, which can harm health and the environment. When the side effects are beneficial, they are known as **positive externalities** because they create advantages for others, such as education, vaccination, and tree planting, which improve society by raising skills, protecting public health, and conserving the environment.

- h) Ceteris paribus** is a Latin term meaning “**all other things remaining constant.**” In economics, it is used to study the effect of one factor at a time on an outcome while keeping other factors unchanged. This is important because it helps economists simplify analysis and clearly understand cause-and-effect relationships in complex economic situations.
- i) Economic variables:** These are measurable economic quantities that change over time or vary across situations, such as prices, income, population, or employment, and are used to analyze economic behavior and outcomes. Economic variables are commonly classified as follows:
- i. Dependent variables:** Outcomes that change in response to other variables (e.g., demand depending on price).
 - ii. Independent variables:** Variables that cause changes in other variables (e.g., price influencing demand).
 - iii. Exogenous variables:** Factors that originate outside the economic system and are not easily controlled (e.g., weather, foreign demand).
 - iv. Endogenous variables:** Factors determined within the economic system (e.g., prices, wages, quantities).

Application activity 1.3



Carefully analyse the case study below and use it to answer the questions that follow.

Nyabihu District in Rwanda is well known for Irish potato farming. Marie is a household head living in Nyabihu with her family. Her family grows Irish potatoes using hoes and pangas. Some potatoes are consumed at home, while the rest are transported to different markets in Rwanda.

With the income earned, Marie pays school fees, buys food, and uses MTN Mobile Money to pay for electricity and water. She also bought a smartphone recently. The government has constructed a public road in the area, making transport easier.

Nearby, a small factory processes potatoes into crisps. However, some residents complain that waste from the factory pollutes a nearby stream used by villagers. At the same time, a tree planting program started in the area to protect the environment.

From the case study,

- a) Identify two needs and two wants in Marie's household.
- b) List two goods and two services and support your answer.

- c) Identify and classify the following:
 - i. public goods
 - ii. private goods
 - iii. producer goods
 - iv. consumer goods
 - v. merit goods
- d) Identify one negative externality caused by the potato factory.
- e) Suggest one government measure to reduce this problem.
- f) Explain one positive externality from the tree planting program.

1.4 Nature of economics

Learning Activity 1.4



Carefully read each situation below. Then answer the questions that follow.

Situations

- A: Fuel prices increased by 15% last year, leading to higher transport costs.
- B: The government should reduce fuel taxes to lower the cost of living.
- C: When prices rise, households reduce their quantity demanded of goods.
- D: The government decides to invest more money in education to improve future economic growth.
- E: It is better to use renewable energy to protect the environment and future generations.
- F: Healthcare ought to be free for all citizens.
- G: As interest rates increase, borrowing by firms decreases.
- H: The government should increase spending on agriculture to improve food security.

Questions

1. For each situation (A–H), state whether the statement:
 - a) describes a fact, or
 - b) gives an opinion or advice
2. Identify which statements show economics as:
 - a) factbased explanations using data or causeandeffect,
 - b) policy recommendations or guidance for action.
3. Briefly explain why both types of statements are important in an economy.

The nature of economics refers to the fundamental characteristics of economics as a discipline, showing that it is concerned with how people make choices in allocating scarce resources to satisfy their unlimited wants. Owing to differing perspectives, economists have debated its nature, leading to three broad interpretations: **economics as a science**, which uses systematic methods and laws to explain economic relationships; **economics as an art**, which applies economic principles to solve practical problems and guide policy decisions; and economics as **a social science**, since it studies human behavior and interactions within society in relation to scarce resources.

1.4.1 Economics as a Science, an art and social science

a) Economics as a science

Economics is regarded as a science because it applies systematic and logical methods to study economic problems and the decision-making behavior of individuals, groups, and organizations within an economy. It involves the collection of facts, the formulation of theories, the testing of ideas, and the drawing of conclusions based on empirical evidence. By using both inductive reasoning (from observations to general principles) and deductive reasoning (from theory to specific outcomes), economists analyze and predict economic phenomena. For example, the law of demand demonstrates scientifically that, other factors remaining constant, an increase in the price of a good generally leads to a decrease in the quantity demanded.

b) Economics as an art:

Economics is also described as an art because it involves the practical application of economic knowledge, principles, and theories to solve real-life economic problems and guide decision-making. It helps individuals, firms, and governments make informed choices about budgeting, pricing, production planning, resource allocation, and policy formulation. Unlike economics as a science, which focuses on explaining economic facts and relationships, economics as an art emphasizes what should be done to achieve desired goals such as economic growth, employment, stability, and improved social welfare. In this way, economics provides practical guidance for action and helps society use scarce resources efficiently to meet its needs.

c) Economics as a social science:

Economics is described as a **social science** because it studies **human behavior and relationships in society** as people, firms, and governments make decisions about how to use **scarce resources** to satisfy **unlimited needs and wants**. It examines how individuals, households, firms, and governments choose, interact, and influence one another in activities such as production, distribution, exchange, and consumption. Unlike natural sciences, economics focuses on people and social institutions, recognizing that economic decisions are shaped by **social, cultural, political, and ethical factors**.

In studying these interactions, economics uses **positive economics** to explain objective facts and describe "what is," and **normative economics** to express value judgments about "what ought to be" by recommending policies and improvements, thereby helping society understand economic problems and make informed decisions.

1.4.2 Positive and Normative Economics

i. Positive economics

Positive economics is the branch of economics concerned with **objective facts and cause-and-effect relationships** that can be tested and verified using evidence and logical analysis. It focuses on **what is, what was, or what will be** in the economy, rather than on opinions or value judgments, and seeks to explain how the economy actually functions. Because positive economic statements are **descriptive and measurable**, they can be proven true or false using data. Examples include statements such as *"Fuel prices are increasing this year," "Unemployment was higher in 2020 than in 2019," "An increase in fuel prices leads to higher transport costs," "Higher taxes reduce consumers' disposable income," "Rwanda's coffee exports increased by 10% last year,"* and *"If interest rates rise, inflation will decrease in the future", "Higher interest rates may reduce borrowing by households and businesses", "An increase in fertilizer use can increase agricultural productivity, other things being equal."* All these statements are positive economic statements because they are based on observable facts or predictable outcomes and can be empirically tested.

ii. Normative economics

Normative economics is the branch of economics that deals with **value judgments, opinions, and recommendations** about **what ought to** be done in the economy. It focuses on **what should be**, rather than what is, and is influenced by *personal beliefs, ethics, and societal goals* such as fairness, equity, and improved social welfare. Normative economics is **prescriptive** rather than **descriptive**, as it evaluates economic outcomes and proposes policies based on values instead of objective facts. Its statements cannot be tested or proven true or false and commonly use words like **should, ought to, better, must or desirable**. For example, statements such as *"The government should reduce taxes on basic goods," "Income ought to be distributed more equally to reduce poverty," "Rwanda should invest more in education to reduce poverty,"* or *"It would be better for the government to increase spending on healthcare"* are normative because they express opinions and societal preferences rather than measurable facts. The main purpose of normative economics is to guide policy and decision-making in line with a society's values and priorities.

Table 1.2 Difference between Normative and positive economics

Aspect	Positive Economics	Normative Economics
Meaning	Studies economic facts and relationships	Deals with opinions and value judgments
Focus	What <i>is</i> , what <i>was</i> , or what <i>will be</i>	What <i>ought to be</i>
Nature	Descriptive and objective	Prescriptive and subjective
Basis	Facts, data, and evidence	Personal beliefs, ethics, and social values
Testability	Can be tested and proven true or false	Cannot be tested or proven true or false
Type of Questions	How does the economy work?	What should be done?
Language Used	Uses statements like ' <i>is</i> ', ' <i>was</i> ', ' <i>will be</i> ', ' <i>may</i> ', ' <i>can</i> '	Uses words like ' <i>should</i> ', ' <i>ought to</i> ', ' <i>better</i> ', ' <i>must</i> ', ' <i>desirable</i> '
Role in Economics	Explains economic behavior	Guides policy and decisionmaking
Category in Nature of Economics	Economics as a science	Economics as an art
Exemple	"Higher taxes reduce consumers' disposable income" "Many households in Rwanda are using mobile money services" "Access to credit can help businesses expand" "Rising fuel prices may increase the cost of living."	"Taxes should be reduced to improve living standards" "Government should subside school fees" "More resources ought to be allocated to education" "The government must create more employment opportunities for young people" "It would be better to improve rural roads to support farmers" "It is desirable to reduce income inequality."

Application activity 1.4



Read the situations below carefully.

Scenario 1: The price of maize flour increased by 20% last year, causing many households to reduce their consumption.

Scenario 2: The government should subsidize basic food items to improve the welfare of low-income households.

Scenario 3: A business studies past sales data to decide how much output to produce next year.

Scenario 4: The government decides to increase spending on healthcare and education to promote long-term economic development.

Scenario 5: Families choose how to spend their limited income on food, housing, education, and transport.

Questions

1: Identify the Nature of Economics

For each scenario (1–5), state whether it best shows:

- Economics as a **science**
- Economics as an **art**
- Economics as a **social science**

Give a brief reason for your choice.

2: From the scenarios above:

1. Identify **two statements** that represent **positive economics**.
2. Identify **two statements** that represent **normative economics**.
3. Explain your answers using the definitions you have learned.

3: Answer the following in **4–5 sentences**:

Explain how economics helps governments make decisions about the use of scarce resources by combining positive and normative economics

1.5 Scope and Branches of economics

Learning Activity 1.5



Read the two situations carefully and answer the questions that follow.

Situation 1: Sellers in **Nyaminyonga market** decide whether to raise the price of bread after the cost of flour increases.

Situation 2: The **government** introduces a nationwide program to reduce unemployment by investing in roads and schools, affecting jobs, income, and economic growth across the country.

1. In each situation, who is making the decision?
2. Does each decision affect a single market or the entire country?
3. Which situation focuses on the behaviour of individual buyers and sellers?
4. Which situation deals with national issues?
5. Based on scale and impact, which situation best describes small scale issue and which a large - scale issue and why?

Have you ever thought about how the many choices people make every day, at home, in businesses, and within governments, are studied in an organized way? Economics focuses on how these decisions are made in using limited resources to satisfy needs and wants, leading to different economic activities. To better understand them, the subject is divided into areas that examine both individual units and the economy as a whole, providing a clear foundation for studying the scope and branches of economics.

1.5.1 Scope of Economics

Economics, as explained in the study of its nature, focuses on how people make choices in using scarce resources to satisfy their needs, which forms the basis of its scope. The scope of economics therefore covers a wide range of activities such as production, distribution, consumption, and decision-making at different levels of society. Because these activities occur at both individual and national levels, economics is divided into branches to facilitate their study and analysis. These divisions allow economists to examine individual economic units and the economy as a whole in an organized and systematic manner.

1.5.2 Branches of economics

Branches of economics are the divisions of the subject that organize the study of economic activities into specific areas based on the level or aspect being examined. By dividing the wide scope of economics into branches, such as the study of individual decision-making and the functioning of the entire economy, complex economic issues can be analyzed more clearly and systematically. There are two main branches of economics; Microeconomics and Macroeconomics.

a) Microeconomics

Microeconomics is the branch of economics that studies the economic behaviour and decisions of individual units in the economy, such as households, consumers, workers, and firms. It focuses on how these individuals and businesses make choices about producing, buying, selling, and using resources, and how prices of specific goods and services are determined in particular markets. For example, it examines how a family decides how to spend its income, how a farmer decides what crop to grow, or how a shop sets the price of its products. It actually looks at smallscale economic activities and individual decisionmaking rather than the economy as a whole.

b) Macroeconomics

Macroeconomics is the branch of economics that studies the economy as a whole. It focuses on broad economic issues such as economic growth, inflation, unemployment, government spending, national income, and international trade. Unlike microeconomics, which examines individual economic units, macroeconomics analyses the performance and behaviour of the entire economy.

Examples of macroeconomic issues include government policies aimed at reducing unemployment, efforts to increase economic growth and GDP, measures taken by the National Bank of Rwanda to control inflation, Rwanda's trade with other countries, the balance of payments, exchange rate management, public debt, and national budget planning..

Relationship between micro and macro levels

Microeconomic and macroeconomic issues are closely linked because individual decisions add up to affect the whole economy, and national conditions influence individual choices. For example, when many farmers increase maize production (micro), national agricultural output rises (macro). And if the government increases investment in roads (macro), farmers can transport goods more easily, which affects their production and sales decisions (micro). This shows that what happens at the small scale shapes the big picture, and the big picture, in turn, guides small-scale decisions.

Table 1.3: Difference between Microeconomics and Macroeconomics.

Aspect	Microeconomics	Macroeconomics
Area of study	Studies individual economic units such as consumers, firms, and markets	Studies the economy as a whole
Main focus	Individual decisionmaking and price determination	National income, overall economic performance
Key issues	Demand and supply, pricing, production costs, consumer behavior	Inflation, unemployment, economic growth, national income
Level of analysis	Small or individual level	Large or national level
Exemples	Price of a commodity, wages of workers, output of a firm	GDP, inflation rate, unemployment rate
Policy concern	Efficient allocation of resources	Economic stability and growth

Application activity 1.5



Read each statement carefully and base your analysis on daily life experiences in Rwanda and what you have learned about the scope and branches of economics to answer the questions thereof.

- a) Transport fares often increase when fuel prices rise. Explain how this situation shows the relationship between production, exchange, and consumption.
- b) A shopkeeper decides to reduce prices to attract more customers.
 - i. Is this a microeconomic or macroeconomic issue?
 - ii. Give a reason for your answer.
- c) The Government of Rwanda introduces a national program to reduce youth unemployment. Explain why this is a macroeconomic issue and mention one way it affects households.
- d) Many farmers increase tomatoes production across different districts in Rwanda. Explain how these microeconomic decisions can lead to a macroeconomic outcome.

1.6 Fundamental Principles of Economics

Learning Activity 1.6



You have 10,000 FRW to spend in one day, but your needs are many. You must decide how to use this money among different options such as food, school shoes, clothes, transport, phone airtime or data, and entertainment. Because the money is limited, choosing to spend on some items means giving up others, forcing you to make careful decisions about what is most important.

Questions:

1. Why couldn't you choose everything you wanted?
2. How did you decide what was most important?
3. What did you have to give up when you made your choice?
4. Do families, schools, and governments face similar problems?

The fundamental principles of economics provide a foundation for understanding how economic decisions are made. They help explain how scarce resources are allocated and guide individuals and societies in making informed choices in everyday economic situations.

1.6.1 Meaning of the fundamental principles of economics:

Fundamental principles of economics are the basic ideas that explain how individuals, businesses, and governments make decisions in the face of limited resources. They show how scarcity forces people to make choices, compare alternatives, and give up the next best option when a decision is made, helping us understand how trade-offs

are handled and how an economy functions. These principles are **Scarcity, Choice and Opportunity Cost**.

a) Scarcity

Scarcity refers to the limited availability of resources in relation to the unlimited desires of individuals and society. In economics, it means that resources such as time, money, and natural resources are not sufficient to satisfy all human needs and wants. Because of scarcity, people must economize by making decisions and setting priorities in the use of available resources. If resources were abundant, there would be no need for choice or careful allocation. Therefore, scarcity is the fundamental economic problem that forces individuals and society to choose among competing needs and wants. Examples: A farmer in Nyagatare district has only 2 hectares of land and must decide which crops to grow because land is scarce or a student with 5,000 FRW must decide whether to buy airtime or a snack because money is limited.

b) Choice

Choice refers to the act of selecting the most appropriate option from available alternatives. It arises because of scarcity, which forces people to decide which wants to satisfy immediately, which to postpone, and which to give up. In making these decisions, **time preference** plays an important role, as individuals differ in how they value present benefits compared to future ones.

Some people place greater value on immediate consumption, preferring smaller rewards now rather than larger rewards later, hence **positive time preference**, while others are willing to wait for greater future benefits, influencing spending, saving, and investment behavior, hence **negative time preference**.

To make rational decisions about present and future needs, individuals arrange their wants according to importance by creating **a scale of preference**, which ranks needs from the most pressing to the least urgent. Similarly, individuals, businesses, and governments evaluate available alternatives by considering trade-offs and **opportunity costs**. Through this process, **economic decisions** are made to allocate scarce resources efficiently for production, consumption, and investment, with the aim of maximizing benefits and minimizing costs.

c) Opportunity cost:

Opportunity cost is the value of the next best alternative that is given up whenever a choice is made. It does not refer to all the options forgone, but only the one that ranks immediately after the chosen option on the scale of preference. Opportunity cost therefore represents what could have been achieved with the same resources if a different decision had been taken. For example, if a government with limited funds chooses to build a school instead of a hospital, the hospital becomes the opportunity cost; a person may decide to go to the university for three years, the opportunity cost of that decision is the income that could have been gained by working for those three years. In general, for every choice made, the item that comes next on the scale of preference is regarded as the opportunity cost.

1.6.2 Relationship among the fundamental principles of economics

Scarcity, choice, and opportunity cost are closely related ideas in economics and together explain how individuals and societies make decisions.

Because scarcity exists, meaning that resources such as land, labor, capital, and time are limited while human wants are unlimited, people and societies are forced to make choices about how to use those limited resources. Every choice involves selecting one alternative over others, and when a choice is made, the next best alternative that is given up is known as the opportunity cost. Therefore, scarcity leads to choice, and choice results in opportunity cost. In simple terms, because resources are scarce, we cannot have everything we want, so we must choose, and the cost of that choice is the benefit we sacrifice by not choosing the next best option. The relationship between scarcity, choice, and opportunity cost is best shown by the production possibilities frontier (PPF) diagram.

Application activity 1.6



1. You have 10,000 FRW to spend this weekend, and you must decide how to use it among three options: buying notebooks for your lessons, buying clothes to go out with friends, or saving the money to buy a chicken for your home project.
 - a) Arrange these options in a scale of preference according to your personal priorities (from the most preferred to the least preferred).
 - b) From your scale of preference:
 - i. Identify the option you choose.
 - ii. State the opportunity cost of your choice.
 - iii. Explain why this option is the opportunity cost.
2. Explain, using an example, how opportunity cost arises when a person makes a choice.

1.7 The Production Possibility Frontier (PPF)

Learning Activity 1.7



Imagine your school has limited resources such as land, labor, time, and tools, which can be used for only two purposes: **growing crops** like maize or beans to support the school feeding program, or **constructing a playground** for co-curricular activities. Because these resources are limited, allocating more land, labor, and time to crop production reduces the resources available for developing the playground, and allocating more resources to the playground similarly limits the amount of crops the school can grow.

Questions:

Assume the school has 10 equal units of resources (land, labor, or time).

- i. What happens to playground development when more resources are used for growing crops?
- ii. Why can't the school grow the maximum amount of crops and build a full playground at the same time?
- iii. When the school chooses to allocate more resources to crops, what does it give up?
- iv. Which combinations show the best use of the school's limited resources?
- v. How does this activity help you understand that choices must be made because resources are scarce?

Having previously examined the fundamental principles of economics, we now turn to how these principles are applied in analyzing economic decisions concerning the use of resources. This lesson focuses on illustrating how limited resources are allocated and how choices necessarily involve trade-offs within an economy. In doing so, it highlights the close relationship between economic theory and its practical application in real-life situations.

1.7.1 Meaning and assumptions of the production Possibilities Frontier (PPF):

A production possibilities frontier (PPF) is a diagram that shows the maximum possible combinations of two goods or services that an economy can produce using its available resources and existing technology, assuming all resources are used efficiently; it illustrates scarcity, trade-offs, and opportunity cost, with points on the curve showing efficient production, points inside indicating inefficient use of resources, and points outside representing unattainable levels of production at that time. It is also called the transformation curve or the opportunity cost curve.

Opportunity cost can be calculated for decision-making between alternatives thus the formula;

a) Opportunity Cost = Value of next best alternative forgone.

Or can be calculated for production possibility frontier (PPF) and comparative advantage (per unit), thus the formula;

$$\text{b) Opportunity cost of Good X} = \frac{\text{Alternative forgone}}{\text{Actual production}} = \frac{\text{Quantity of good Y forgone}}{\text{Quantity of good X gained}} = \frac{\Delta Y}{\Delta X}$$

Exemple 1: Production Possibility

A farmer has land that can produce either 100 bags of maize, or 50 bags of beans. If the farmer decides to produce 100 bags of maize, the opportunity cost is the beans not produced.

Calculation:

Opportunity Cost of 100 bags of maize = 50 bags of beans

To find the opportunity cost per bag of maize:

$$\begin{aligned}\text{Opportunity cost} &= \frac{\text{Alternative forgone}}{\text{Actual production}} \\ &= \frac{50 \text{ bags of beans}}{100 \text{ bags of maize}} = 0.5 \text{ bag of beans}\end{aligned}$$

Opportunity cost of 1 bag of maize = 0.5 bags of beans

While to find the opportunity cost per bag of beans:

$$\begin{aligned}\text{Opportunity cost} &= \frac{\text{Alternative forgone}}{\text{Actual production}} \\ &= \frac{100 \text{ bags of maize}}{50 \text{ bags of beans}} = 2 \text{ bags of maize}\end{aligned}$$

Opportunity cost of 1 bag of beans = 2 bags of maize.

Exemple 2: Choosing a Business

You have two business options: Poultry farming: Profit = RWF 800,000 Pig farming: Profit = RWF 1000,000

a) If you choose poultry farming.

Calculation

Opportunity Cost = Value of next best alternative forgone = RWF1000,000

Opportunity cost of choosing poultry farming = RWF 1000,000 because by choosing poultry farming, you gave up the chance to earn RWF 800,000 from pig farming.

b) If you choose pig farming.

Calculation

Opportunity Cost = Value of next best alternative forgone = RWF 800,000

Opportunity cost of choosing Pig farming = RWF 800,000 because by choosing Pig farming, you gave up the chance to earn RWF 1000,000 from poultry farming.

Assumptions of the PPF

The production possibilities frontier (PPF) is based on several key assumptions that simplify economic analysis:

- **Fixed resources:** The quantity and quality of factors of production (land, labor, capital, and entrepreneurship) are given and do not change.
- **Fixed technology:** The level of technology remains constant over the period considered.

- Only two goods are produced: The economy produces only two types of goods or services for simplicity.
- **Full employment:** All available resources are fully and efficiently employed.
- **Efficient production:** Resources are used in the best possible way, with no waste.
- **Resources are not equally adaptable:** Factors of production cannot be easily shifted from producing one good to another, leading to increasing opportunity costs.
- **Constant time period:** The analysis applies to a specific moment in time.

1.7.2 A hypothetical example of the opportunity cost schedule showing alternative production possibilities of commodities X and Y

Table 1.4 Opportunity cost schedule

Combination	Quantity of X	Quantity of Y
A	0	200
B	40	170
C	100	100
D	140	50
E	160	0

This schedule shows the different combinations of Quantity X and Quantity Y that can be produced with limited resources. At point A, all resources are used to produce Y, while at point E, all resources are devoted to producing X. As production moves from A to B, 40 units of X are gained at the cost of 30 units of Y; from B to C, 60 more units of X require giving up 70 units of Y; from C to D, an additional 40 units of X cost 50 units of Y; and from D to E, just 20 more units of X require sacrificing the remaining 50 units of Y. This pattern **shows increasing opportunity cost**, because producing extra units of X requires giving up increasingly larger amounts of Y, resulting in a **concave (bowed-out)** Production Possibility Frontier (PPF).

1.7.3 Illustration and interpretation of the PPF:

The production possibilities frontier (PPF) is typically downward sloping and **concave** (bowed outward) to the origin. This shape reflects the existence of scarcity, since resources are limited, and shows that producing more of one good requires sacrificing some quantity of another.

The concave shape of the PPF is explained by the increasing **Marginal Opportunity Cost (MOC)**. Resources are not equally suited for the production of all goods, so as production shifts from one good to another, less efficient resources are used. As a

result, increasing amounts of one good must be given up to produce additional units of the other, leading to increasing opportunity cost.

Points on the PPF represent efficient use of resources, where all resources are fully employed. Points inside the curve indicate inefficiency or unemployment of resources, while points outside the curve are unattainable with the current level of resources and technology. In theory, the PPF may be a **straight line** if the Marginal Rate of Transformation (MRT) is **constant**, or **convex** if the MRT is **decreasing**. However, such cases are rare. In most real-world situations, the MOC increases, which makes the PPF concave in shape.

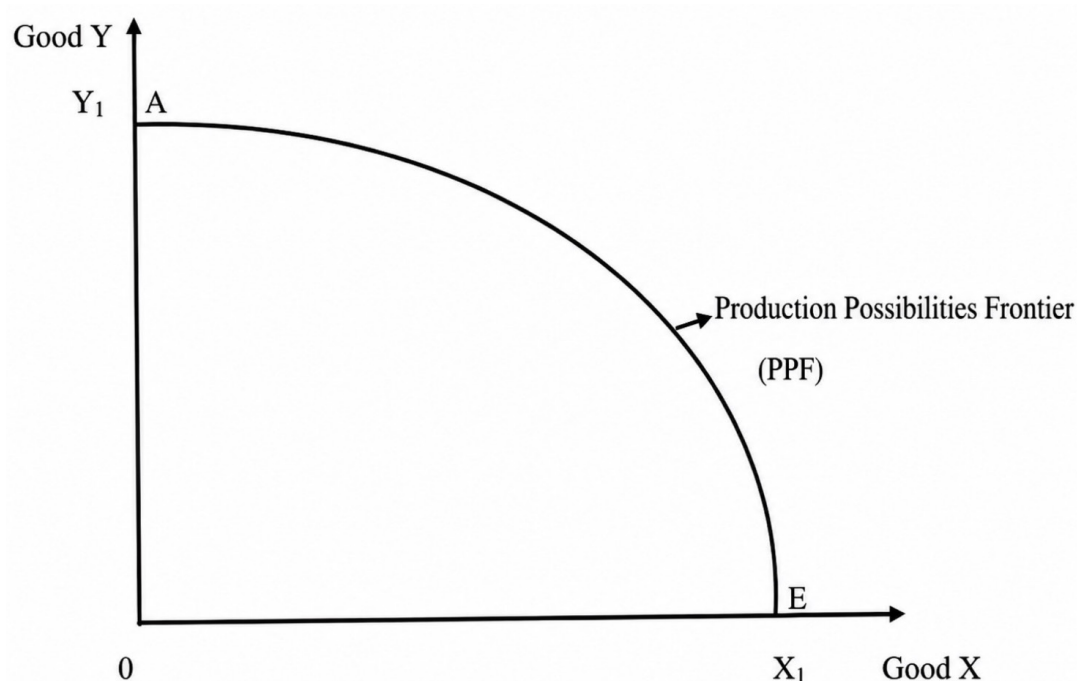


Figure. 1.1: The Production Possibilities Curve.

The diagram shows a Production Possibilities Frontier (PPF) illustrating the trade-off between producing Good X (on the horizontal axis) and Good Y (on the vertical axis) using limited resources. Point A represents maximum production of Good Y with no Good X, while point Z represents maximum production of Good X with no Good Y. The downward-sloping, concave curve indicates that as production of Good X increases, increasing amounts of Good Y must be sacrificed, reflecting increasing opportunity cost due to resource specialization.

Points under, on and beyond production possibility curve.

We have seen that a Production Possibility Frontier (PPF) shows the maximum combinations of goods an economy can produce efficiently. However, production may occur below, on, or beyond the PPF, raising questions about efficiency and resource use. This creates the need to examine these points and understand what they reveal about economic performance and decision-making.

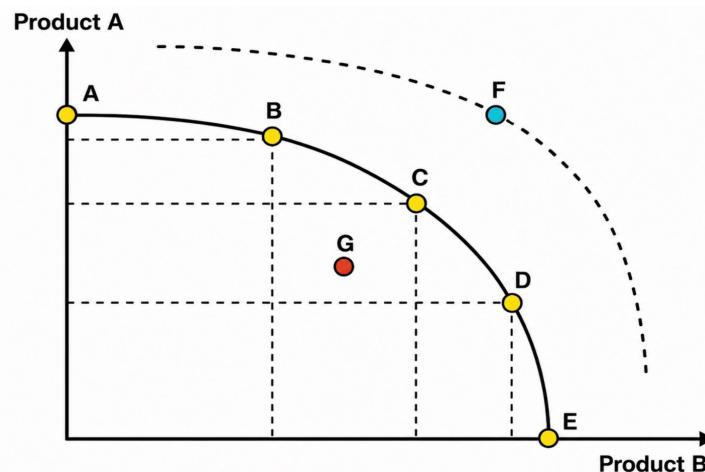


Fig 1.2 Points under, on and beyond production possibility curve

The diagram shows a production possibilities frontier (PPF) for Product A and Product B. Points on the PPF (such as A, B, C, D, and E) represent efficient production, where all available resources are fully and efficiently used. A point inside the PPF (such as G) **indicates inefficiency or underutilization of resources**, meaning the economy could produce more of one or both goods without sacrificing the other. A point above (or outside) the PPF (such as F) is **unattainable** with current resources and technology though **desirable**, showing combinations of output that cannot be achieved unless resources increase or technology improves.

1.7.4 Importance, shifts and types of the PPF.

i. Importance of the PPF

The Production Possibility Frontier (PPF) is important because it helps explain key economic ideas in a simple, visual way. Its importance includes:

- **Shows scarcity:** The PPF demonstrates that resources are limited, so an economy cannot produce unlimited amounts of goods.
- **Explains choice and trade-offs:** It shows that producing more of one good requires giving up some of another, helping students understand decision-making.
- **Illustrates opportunity cost:** Movement along the curve shows how much of one product must be sacrificed to gain more of another.
- **Indicates efficiency:** Points on the curve represent efficient use of resources, while points inside the curve show underutilization or inefficiency.
- **Helps understand economic growth:** An outward shift of the PPF shows growth (more resources or better technology), while an inward shift shows a decline.
- **Guides resource allocation:** It helps governments and firms decide how best to allocate scarce resources between different uses.

Overall, the PPF is a powerful tool for understanding how economies operate under conditions of scarcity and how choices affect production.

Exercise 1.1

Analyse the table below and answer the questions that follow

1. Utexrwa Industries produce two products: Shirts (Y) and Trousers (X). It can produce the following quantities using the same level of resources.

Combination	Quantity of trousers (X)	Quantity of shirts (Y)
A	0	150
B	20	100
C	30	80
D	40	60
E	50	40
F	60	0

Refer to the table above.

- a) Derive the production possibility frontier (PPF) by plotting the quantities of Y on the y-axis and the quantities of X on the x-axis.
- b) Identify the maximum production possibilities for commodity X and Y respectively.

ii. Shift in the Production Possibilities Frontier (PPF)

The PPF may shift inwards or outwards. A shift of the PPF inwards indicates **economic decline** while a shift outward indicates **economic growth**. The Production Possibility Frontier (PPF) is not fixed. It can change over time depending on what happens in an economy. These changes are shown by shifts of the entire curve. When the PPF shifts outwards, it means the economy can produce more goods and services than before.

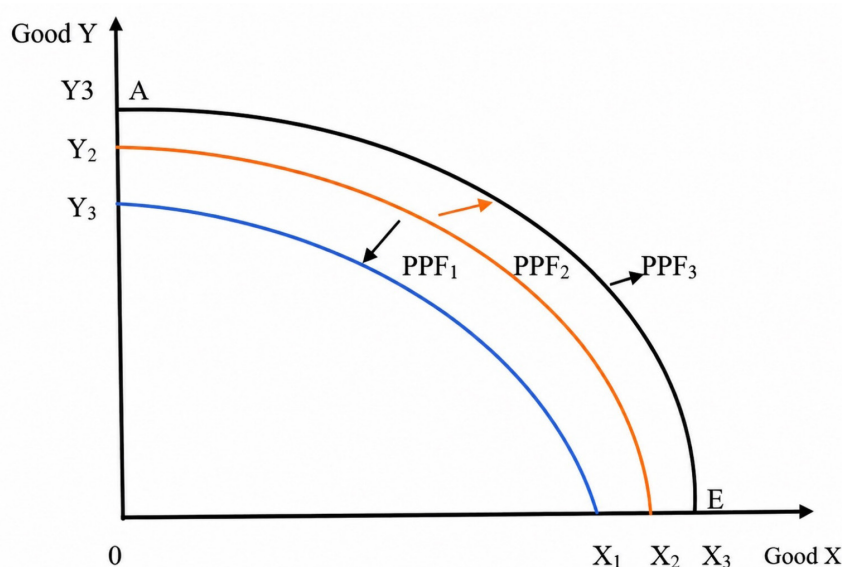


Fig 1.3 Shift of the PPF

The graph shows shifts in the PPF (PPF₁ to PPF₃) Each curve represents the maximum possible combinations of the two goods that can be produced with the avail-

able resources and technology at a given time. PPF_1 is the innermost curve and represents the lowest level of production, while PPF_2 and PPF_3 represent progressively higher production levels.

The outward shift from e.g., from **PPF_2 to PPF_3** indicates **economic growth** which may be due to;

- Technological advancement that improves efficiency
- Increase in the labour force due to population growth or better education
- Capital accumulation such as new machines, tools, and infrastructure
- Discovery of new natural resources
- Improvement in skills, training, and productivity
- Better organization and management of resources

This shift shows that the economy's productive potential has expanded beyond the level of PPF_2 .

While the inward shift from **PPF_2 to PPF_3** indicates **economic decline**

- Natural disasters such as droughts, floods, or earthquakes destroying resources
- Loss of labour due to disease, war, or migration
- Destruction of capital like factories, machines, or infrastructure
- Decline in technology or failure to maintain equipment
- Poor governance, political instability, or conflict
- Economic recession leading to underutilization of resources

This shift indicates that existing resources are fewer or less productive than when the economy operated at PPF_2 .

Exercise 1.2

Analyse the table below and answer the questions that follow

1. Utexrwa Industries produce two products: Shirts (Y) and Trousers (X). It can produce the following quantities using the same level of resources.

Combination	Quantity of Good X	Quantity of Good Y
A	0	200
B	40	170
C	80	120
D	120	60
E	160	0

Refer to the table above.

- a) Derive the production possibility frontier (PPF) by plotting the quantities of Y on the y-axis and the quantities of X on the x- axis.

- b) Suppose a new technology improves production efficiency in the manufacture of the two goods in a PPF model. Draw and describe how the PPF would change and explain whether this indicates economic growth or decline.
- c) Suppose a new technology improves production efficiency in the manufacture of good X in a PPF model. Draw and describe how the PPF would change and explain what this implies for both commodities.

1.7.5 Types of PPF

The shape of a production possibility frontier (PPF) varies depending on the nature of resources and technology. These different types show how opportunity cost changes as production shifts, helping students understand how economies allocate resources under different conditions. These types include:

- a) **Bowed-out (Concave) PPF:** A bowed-out PPF represents increasing opportunity cost, where each additional unit of one good requires sacrificing larger amounts of the other good. This happens because resources are specialized and not equally suited for all uses; as production shifts, less suitable resources are employed, making production progressively less efficient. This is the most realistic and commonly observed shape of the PPF.

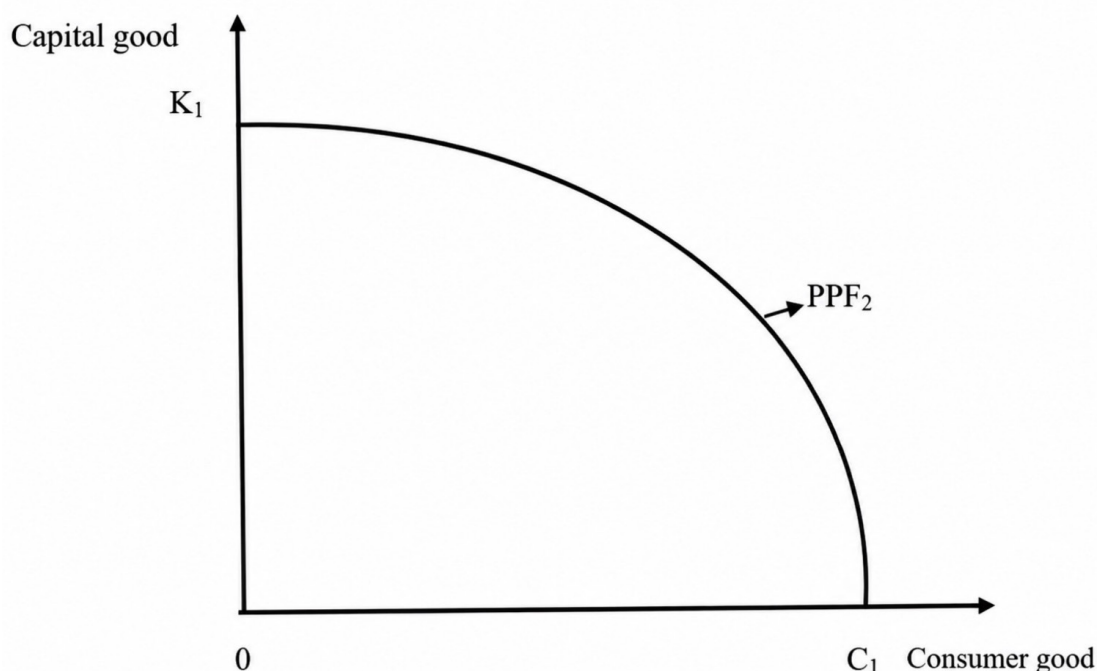


Fig 1.4 Bowed-out (Concave) PPF

This bowed-out PPF shows the trade-off between capital and consumer goods, indicating increasing opportunity cost as producing more consumer goods requires sacrificing progressively more capital goods due to resource specialization. A concave PPF is also known as **a common/Normal PPF**.

b) Inverted/ Convex PPF: A convex Production Possibility Frontier (PPF) is a curve that bends inward toward the origin. It shows **decreasing opportunity cost**, meaning that as more of one good is produced, less of the other good is given up. This happens when resources can easily adjust and be used efficiently in producing different goods. However, this type of PPF is rare in real life because most resources are not easily adaptable to all types of production.

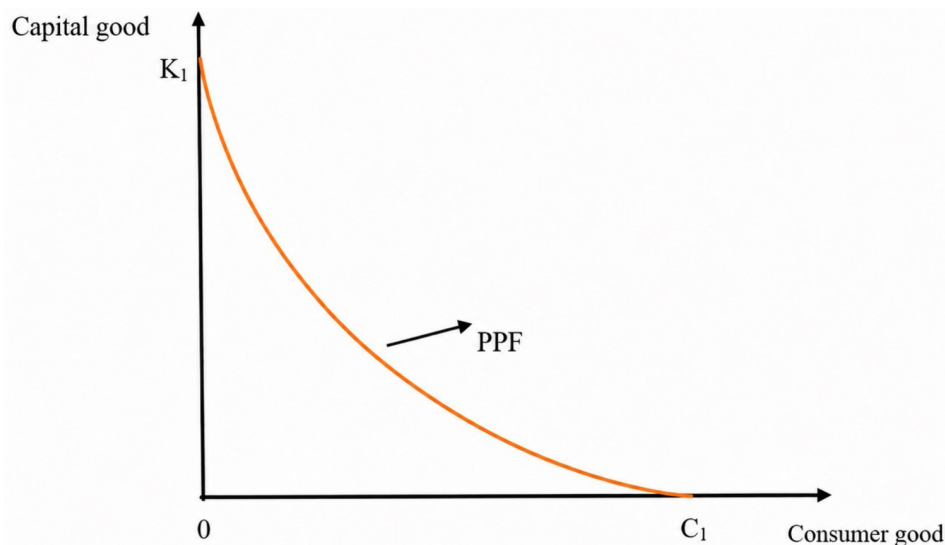


Fig 1.5 Inverted/ Convex PPF

The curve above is a **convex PPF** because it bends inward toward the origin. It shows **decreasing opportunity cost**, meaning that producing more consumer goods requires giving up smaller amounts of capital goods and vice versa. This happens when resources can easily adjust to different uses, though this is rare in real life.

c) Straight-line / Linear PPF: A straight-line production possibilities frontier shows a constant opportunity cost, meaning that producing more of one good always requires giving up the same fixed amount of the other good. This occurs when resources are perfectly adaptable and equally efficient in producing both goods, so shifting resources from one product to another does not change productivity or costs.

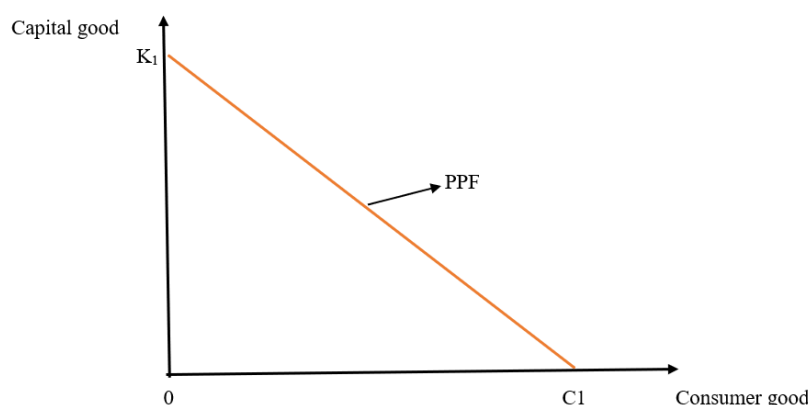


Fig 1.6 Straight-line / Linear PPF

The curve is a constant or linear Production Possibility Frontier (PPF) because it is a straight line sloping downward from left to right. It shows **constant opportunity cost**, meaning that producing more consumer goods always requires giving up the same amount of capital goods. This happens when resources are equally efficient in producing both goods

d) Biased PPF: This is a PPF that shifts outwards in favor of one commodity only. In this case, the PPF shifts outwards more in one direction than the other. This is caused by improvement in technology and increased factors of production that are in favor of one commodity.

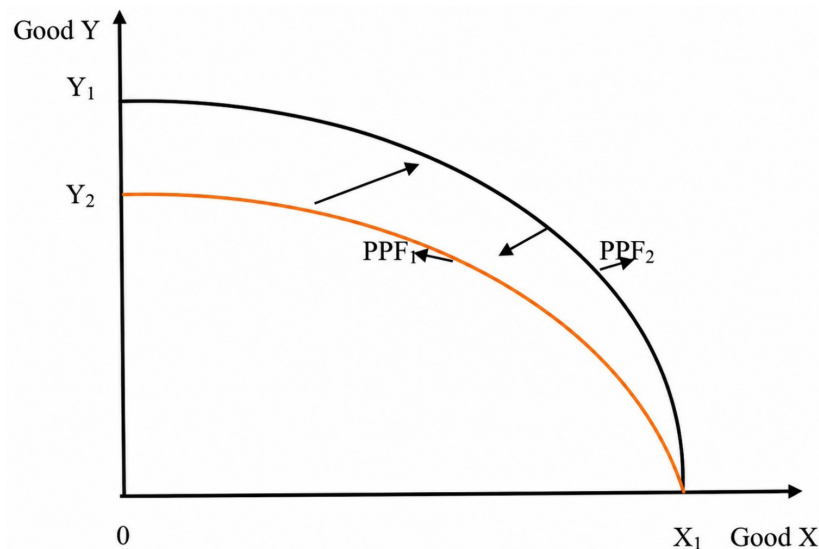


Fig 1.7 Biased PPF of Good X

The above illustration shows an increase in production of Good Y while there is no increase in production of Good X. This implies that the improvement in technology and increase in resources caused increase in production of Good Y only.

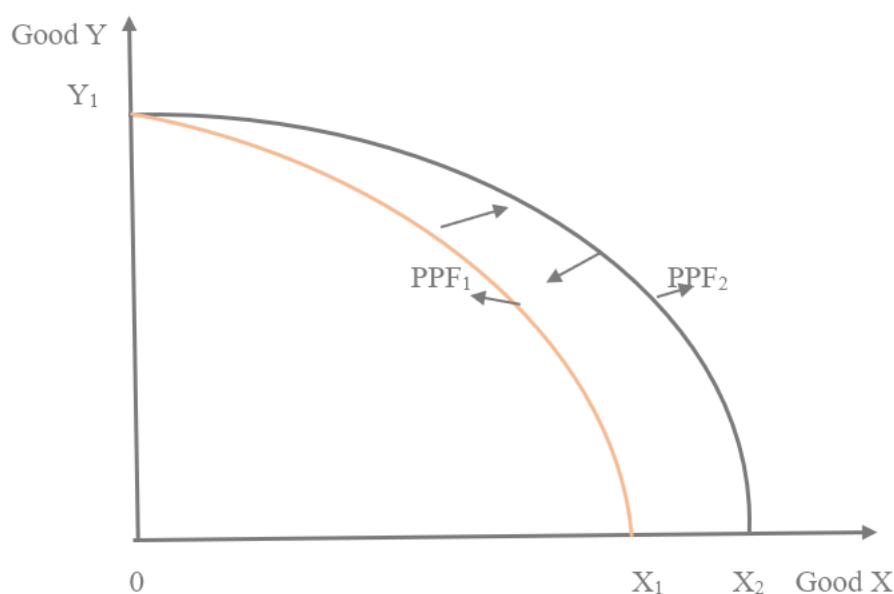


Fig 1.8 Biased PPF of Good Y

The above illustration shows an increase in production of Good X with no increase in production of Good Y. This implies that the improvement in technology and increase in resources was in favor of production of Good X only, causing an increase in production of Good X.

e) Unbiased PPF: An unbiased shift occurs when productive capacity for both Good X and Good Y increases or decreases equally by the same proportion, causing the PPF to shift parallel outward (growth) or inward (decline).

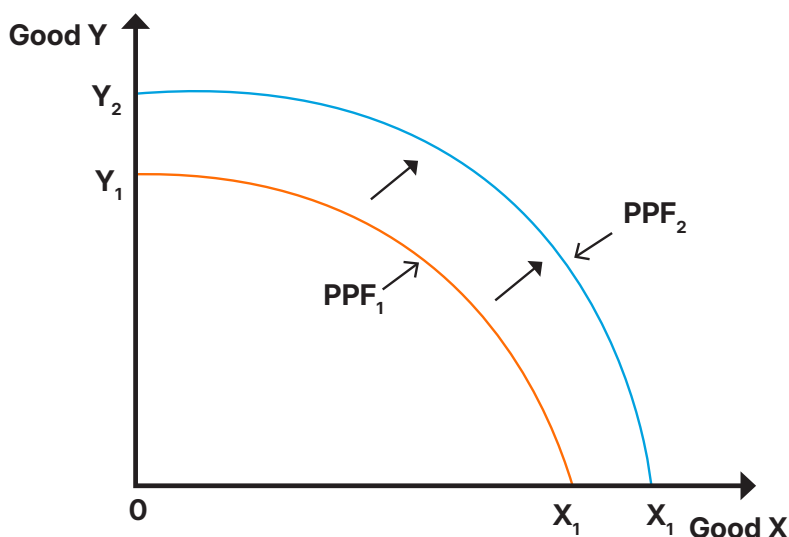


Fig.1.9 Unbiased PPF

The figure above shows an **unbiased shift of the PPF** by the same proportion, causing the PPF to shift **parallel outward** from PPF₁ to PPF₂ representing **economic growth** and may result from factors such as overall technological improvement across all sectors, an increase in the labour force, better education and training, discovery of resources that can be used to produce both goods, or improved infrastructure like roads, electricity, and communication or **parallel inward shift** from PPF₂ to PPF₁ representing **economic decline** and may be caused by natural disasters, war or political instability, epidemics that reduce labour supply, or an economywide recession leading to destruction or underuse of capital.

Application activity 1.7



A school agriculture club has limited land, labor, and farming tools to produce **maize** and **beans** during one season. The club cannot produce unlimited amounts of both crops at the same time because resources are scarce.

The following production possibilities are available:

The club is currently producing at point **C**, but it wants to increase production.

Using the information above:

- Draw the PPF curve using the schedule provided.
- State the assumptions of the PPF shown in this scenario.
- Explain the opportunity cost of moving from point B to point C.

- iv. Describe the type of PPF formed and give a reason.
- v. Explain two importances of the PPF to the school agriculture club.
- vi. Suggest two factors that may shift the PPF outward.
- vii. Explain whether the outward shift would be biased or unbiased.
- viii. Why can the club not produce maximum maize and beans at the same time?
- ix. What happens when more maize is produced?
- x. What does a point beyond the curve indicate?
- xi. How can the club increase its production capacity?

1.8 Economic questions and economic activities.

Learning Activity 1.8



Imagine your school is a small village with limited land, workers, and simple tools. Some people grow food, others make clothes, and some build houses. Because not everything can be done at the same time, the villagers must choose which needs are most important, decide how and where work should be done, agree on the right time to produce certain goods, and determine how what is produced will be shared among the people. These choices help the village use its resources carefully so everyone can live better.

From the scenario, identify:

- i. The activities that are taking place in the village.
- ii. The decisions being made by the village people for their activities.
- iii. Why can the village not produce all goods in large amounts at the same time?
- iv. What do all these choices tell us about living in a community with limited resources?
- v. Why are these decisions important for the survival and well-being of the village?

Every society is faced with the challenge of using limited resources to satisfy unlimited wants. As a result, decisions must be made about how resources are allocated and how goods and services are produced and distributed. These decisions are carried out through various economic activities, which shape how an economy functions and meets the needs of its people.

1.8.1 Economic Questions.

Resources such as land, labour, capital, and skills are limited, but people have many needs and wants. Because of this, societies cannot produce everything they want and must make choices. These choices are called **economic questions**, and they help decide what to produce, how to produce it, who will receive it, and when and where production should take place. These decisions guide how scarce resources are used to meet people's needs. They include the following;

- a) What to produce:** Societies must decide which goods and services to produce because resources are limited and cannot satisfy all needs at once. Producing more of one good means sacrificing another, so choices must reflect the most urgent needs of the people. By prioritizing essential goods such as food, shelter, and healthcare, societies ensure that scarce resources are used to improve welfare rather than being wasted on less important items.
- b) How to produce:** Deciding how goods and services should be produced involves choosing the most suitable methods and resources, such as whether to rely more on labour or machines. The right production method helps lower costs, reduce waste, and make better use of scarce resources. Efficient methods make it possible to produce more goods using the same limited inputs, improving overall productivity.
- c) For whom to produce:** Since goods and services are limited, societies must decide how they will be shared among people. These decisions determine who benefits from production and who may go without. Fair and well-planned distribution of goods helps promote social harmony, reduce inequality, and ensure that scarce resources contribute to the well-being of the whole community.
- d) When to produce:** Choosing when to produce means deciding the most appropriate time for production. Producing goods at the right time such as crops in the rainy season or goods when demand is high helps prevent shortages and waste. Proper timing ensures that scarce resources like labour and raw materials are not misused or lost.
- e) Where to produce:** Deciding where to produce involves selecting the most suitable location for production activities. Producing goods in places where resources, skills, and markets are available reduces transport costs and saves time. This helps avoid unnecessary expenses and ensures that scarce resources are used efficiently and effectively.

Economic questions are important because they guide societies on how best to use limited resources to meet unlimited needs. By making informed decisions about production, methods, timing, location, and distribution, societies can minimize waste, improve efficiency, and increase overall welfare. Without answering these questions carefully, scarce resources would be misallocated, leading to shortages, inefficiency, and lower standards of living.

1.8.2 Economic activities.

Economic activities are the actions people and organizations carry out to produce, distribute, exchange, and consume goods and services in order to satisfy human needs and wants. These activities take place in **four main sectors**: the **primary sector**, which involves extracting raw materials such as farming, fishing, and mining; the **secondary sector**, which focuses on processing and manufacturing these raw materials into finished goods; the **tertiary sector**, which provides services like transport, trade, health care, and banking; and the **quaternary sector**, which involves knowledgebased activities such as education, research, information services, and the use of technology. Together, these sectors support the functioning and growth of an economy.

Application activity 1.8



Analyse the scenario below and answer the questions that follow;

Rutare is a village with fertile land, a river, skilled workers, and simple machines. The people grow crops and rear animals, make furniture and clothes, and sell these goods in the village market. Some villagers work as teachers, health workers, and agricultural advisers who share knowledge and information to improve production, while others use phones and computers to communicate and keep records. Because resources are limited, the village cannot support all these activities equally at the same time.

From the scenario;

- a) Identify different activities carried out by the people of Rutare.
- b) Classify each activity you identified into its correct group of economic activities.
- c) Explain why each activity is important to the village.
- d) State any three decisions the people of Rutare must make because their resources are limited.
- e) Explain how the village might decide:
 - i. Which goods to produce
 - ii. How the goods should be produced
 - iii. Who should benefit from the goods
- f) Give one exemple showing when and where one good should be produced in the village, and explain why.

1.9. Economic systems

Learning Activity 1.9



1. Answer the Questions

1. Who decides what your family buys for daily use at home?
2. Who decides which goods a business in your community will sell?
3. Who decides how much tax people should pay in Rwanda?

2. Examine the following situations and then answer the question that follows:

- i. A shopkeeper sets a price for bread.
- ii. Government sets national exam dates.
- iii. A farmer chooses which crops to grow.
- iv. Government regulates moto taxi fares.

Questions

- a) Which of these decisions are made by individuals?
- b) Which are made by the government?
- c) Why do you think some decisions are made by government and others by individuals?

After examining how societies make choices and organize economic activities, it becomes clear that these decisions do not occur in isolation. Instead, they are guided by the way an economy is organized to allocate resources and manage production and distribution. These organized approaches to decision-making within an economy are known as economic systems.

1.9.1 Meaning of Economic systems

An economic system is the structure a society uses to organize the production, distribution, and consumption of goods and services to meet people's needs and wants. It sets the rules that guide key economic agents like households, firms, government, and the foreign sector, by determining how resources are controlled, decisions are made, prices are set, and income is distributed, while also answering fundamental economic questions such as *what, how, for whom, where, and when to produce*.

Economic systems vary in approach, including the traditional system, where customs and traditions guide decisions; the hands-off (market) system, driven by demand and supply with minimal government intervention; and the hands-on system, where the government plays an active role, thus, leading to systems such as traditional, capitalism, socialism, and mixed economies.

1.9.2 Types of Economic Systems

This lesson introduces students to the different types of economic systems and explains how each system organizes the use of resources in a society. It highlights how decisions on what to produce, how to produce, and for whom to produce are made under different systems, helping students understand how these systems influence economic activities and outcomes in real life. The main types of economic systems to be studied include the traditional, command, market, and mixed economic systems.

a) Traditional economic system:

A traditional economic system is one in which economic activities are guided by customs, traditions, and long-established practices. Production is mainly based on subsistence activities such as farming, hunting, and gathering, with methods passed down through generations. Decisions about what, how, and for whom to produce are determined by cultural norms and community needs rather than market forces or government control. This system is characterized by limited use of technology, low specialization, reliance on barter instead of money, strong community cooperation, and a high level of stability, though it tends to resist change and innovation.

Advantages of a Traditional Economic System.

A traditional economic system is rooted in customs and long-established practices that guide how resources are used and how economic activities are carried out. This system offers several benefits, particularly in promoting stability, cooperation, and the preservation of cultural values within a society. These advantages/ benefits are explained below.

- **Stability and security:** Customs and traditions guide economic activities, creating predictability and reducing uncertainty.
- **Low unemployment:** Clearly defined roles based on community needs ensure that most people are engaged in productive work.
- **Self-sufficiency:** Individuals and families produce most of their basic needs, reducing reliance on external markets.
- **Preservation of culture:** The system maintains cultural heritage, traditions, and customs, strengthening community identity.
- **Environmental sustainability:** Use of simple, local methods promotes conservation and harmony with nature.
- **Strong social bonds:** Emphasis on cooperation and shared responsibilities fosters unity and mutual support within the community.

Disadvantages of a Traditional Economic System

While a traditional economic system has its strengths, it also presents certain challenges that may limit growth and flexibility within an economy. These disadvantages are explained below.

- The reliance on customs and traditions can hinder rapid economic growth and development, as necessary changes are often resisted or slow to implement.
- There is usually little encouragement for scientific and technological advancements, which restricts progress and innovation.

- The system often provides little incentive for entrepreneurship, resulting in fewer choices for consumers and potentially lowering the overall standard of living.
- With minimal surplus production and little access to diverse goods and services, the overall standard of living can be lower compared to more modern economies.
- The focus on subsistence production means fewer choices for consumers, as goods are produced primarily for local needs rather than market demand.
- Communities may resist new ideas and practices, making it difficult to adapt to changing circumstances or global economic trends
- Decisions based on tradition rather than market forces can lead to inefficiencies in resource use and distribution, potentially resulting in shortages or surpluses.
- With individuals performing multiple roles, there is often less specialization in labor, which can limit productivity and efficiency.
- Traditional economies may be more susceptible to fluctuations in weather or natural resources, as their reliance on local production makes them less resilient to external changes.

b) A free enterprise (free market or laissez-faire or capitalistic or unplanned) economy.

A free-market economy is an economic system in which productive resources are privately owned and economic decisions are largely determined by the forces of demand and supply with minimal government intervention. Prices are set through open market competition, allowing businesses to respond to consumer preferences, while consumers influence production through their purchasing choices (*consumer sovereignty*). The system is characterized by profit motive, strong competition, entrepreneurial freedom, and individualism, with resource allocation guided by the “invisible hand” of the market. Although it promotes efficiency, innovation, and economic dynamism, it can also lead to income inequality, as individual welfare largely depends on one’s ability and effort. It should be noted that no country today practices a pure laissez-faire economy; most operate mixed economies with varying degrees of government involvement.

Advantages of a Free Enterprise Economic System

A free enterprise economic system offers several benefits that arise from allowing individuals and businesses to make independent economic decisions. These advantages reflect how the system promotes efficiency, innovation, and the effective use of resources within the economy, as explained below.

- **Efficient allocation of resources:** Competition among firms ensures that resources are used efficiently, reducing costs and improving productivity.
- **Encourages innovation:** The desire to earn profit motivates businesses to develop new products and improve production methods.
- **Wide consumer choice:** Consumers have access to a variety of goods and services, allowing them to choose according to their preferences.
- **Improved living standards:** Profit incentives encourage productivity, which can lead to higher incomes and better living conditions.

- **Rapid market adjustment:** Prices and production adjust quickly in response to changes in consumer demand and market conditions.
- **Consumer sovereignty:** Consumers influence production decisions through their purchasing choices, guiding what is produced in the economy.
- **Entrepreneurial freedom:** Individuals are free to start and operate businesses in any sector, promoting economic diversity and specialization.
- **Wealth creation and investment:** The system encourages wealth accumulation, which promotes investment and contributes to economic growth and job creation.
- **Limited government intervention:** The government plays a minimal role, allowing businesses to operate freely while focusing on essential public functions such as security and law enforcement.
- **Global competitiveness:** Competition improves efficiency and quality, making the economy more attractive to investment and international trade.

Disadvantages of a Free Enterprise Economic System

Despite its benefits, a free enterprise economic system also has certain limitations that may affect fairness and stability within an economy. These disadvantages highlight some of the challenges that arise from relying largely on individual decision-making and market forces, as explained below.

- **Income inequality:** Wealth and income are unevenly distributed, as success depends largely on individual ability and opportunity.
- **Market failures:** Essential goods and services, such as healthcare and education, may be underprovided by the market.
- **Monopolies and oligopolies:** Lack of regulation may allow large firms to dominate markets, reducing competition and consumer choice.
- **Worker exploitation:** Firms may minimize costs by offering low wages or poor working conditions in pursuit of profit.
- **Environmental degradation:** Profit motives may encourage overuse of natural resources and pollution due to weak regulation.
- **Economic instability:** Free markets are prone to fluctuations, leading to periods of inflation, recession, and unemployment.
- **Limited access to basic services:** Individuals with low incomes may struggle to afford essential goods and services.
- **Negative externalities:** Some production activities create social costs, such as pollution, that are not reflected in market prices.
- **Weak social safety nets:** Individuals who are unemployed or unable to work may receive limited support, increasing vulnerability to poverty.
- **Delayed adjustments:** Changes in demand and supply may take time to stabilize, causing short-term economic imbalances.

c) A centralized (planned, command, or socialist) economy:

A command is an economic system in which the government owns and controls the major resources and makes key decisions about what to produce, how to produce, and for whom to produce, rather than relying on market forces. Its main features include public ownership of land, factories, and enterprises, central planning by the government, prices and wages set by the state, limited or no role of private profit, reduced competition, and production aimed at meeting social needs rather than maximizing profit, with the government also controlling distribution to promote equality and reduce income differences.

Advantages of a Planned Economy

A planned economy involves a high level of organisation in economic activities, where decisions about resource use and production are centrally coordinated. This approach can lead to several benefits, particularly in promoting stability, equitable distribution of resources, and the efficient achievement of national economic goals, as explained below:

- **Easy implementation of government policies:** Since the government controls production, policies are implemented without resistance.
- **Reduced income inequality:** Private ownership and competition are limited, helping to narrow the gap between the rich and the poor.
- **Efficiency in achieving social objectives:** Production is planned by the state to meet national and social needs rather than individual profit.
- **Promotion of social welfare:** The main aim of production is to improve community wellbeing, not profit maximization.
- **Economic stability:** Major economic decisions are made by a central planning authority, reducing economic uncertainty.
- **Protection of consumers:** The government provides essential goods and services at low prices, reducing exploitation.

Disadvantages of a Planned Economy

Although a planned economy has certain advantages, it also has limitations that may reduce efficiency and limit individual freedom. These challenges arise from the centralized nature of decision-making and the reduced role of market forces, as explained below.

- **Poor quality of goods:** Lack of competition and weak response to consumer preferences often lead to low-quality products.
- **Lack of motivation and innovation:** Absence of private rewards reduces individual initiative and creativity.
- **Bureaucracy and delays:** Central planning involves too much administration, leading to slow decision-making and inefficiency.
- **Limited consumer sovereignty:** Consumers have little freedom of choice, as production decisions are made by the state.
- **High administrative costs:** Managing production and distribution increases government expenditure.
- **Mismanagement of resources:** Corruption, nepotism, and poor supervision may lead to inefficient use of resources.

d) Mixed economic system:

Mixed economic system is an economic system in which both the government and the private sector participate in making economic decisions and owning resources, combining features of the market and command systems. Its main features include private ownership of businesses alongside government ownership of key industries, Public–Private Partnerships (PPPs), prices are mainly determined by demand and supply but influenced by government regulation, freedom of enterprise with state control to protect public interest, government provision of public and merit goods, and use of taxes, subsidies, and laws to reduce inequalities and correct market failures. Examples of countries operating mixed economies include Rwanda, Kenya, South Africa, the United Kingdom, the United States, India, and China, where market forces operate but government intervention remains significant.

Rwanda operates under a **mixed economic system**, where both the government and the private sector play important roles in economic activities. The government is actively involved in planning, regulation, and provision of key services such as electricity, water, health, education, and public infrastructure for example, through state institutions like REG (energy) and large public road projects. At the same time, private individuals and firms freely engage in agriculture, trade, manufacturing, transport, banking, and telecommunications, such as private farmers selling produce in markets, private companies operating factories, and telecom firms like MTN and Airtel providing services. This combination of government control in strategic areas and market freedom for private enterprise clearly shows that Rwanda practices a mixed economy.

Advantages of a Mixed Economy.

A mixed economy combines elements of both market and planned systems, allowing for cooperation between the government and the private sector in economic activities. This arrangement can lead to several benefits, particularly in balancing efficiency with social welfare and promoting overall economic stability, as outlined below.

- A mixed economy creates more employment opportunities from both the government sector (such as teachers, health workers, and civil servants) and the private sector (such as factory workers, traders, and service providers).
- Government intervention helps ensure that resources are not allocated only to people who can pay more. Through subsidies, taxation, and social programmes, the state supports vulnerable groups, leading to a fairer distribution of income and resources.
- In a mixed economy, the government deliberately invests in less developed or rural regions, promoting balanced regional development and reducing inequality.
- The government uses anti-monopoly laws and competition policies to prevent large firms from dominating markets as monopolies. This protects consumers from high prices and poor-quality goods and services.
- Governments use price controls, subsidies, and buffer stock policies to reduce extreme price fluctuations, especially for essential goods such as food, fuel, and transport, thereby protecting consumers.

- Private ownership and profit incentives encourage domestic and foreign investment, while government support such as infrastructure, incentives, and stable policies further boosts investor confidence and investment levels
- A central planning authority guides national priorities (such as health, education, and infrastructure), while the market mechanism handles consumer goods. This combination promotes efficient and purposeful resource allocation.
- The government provides public goods like roads, security, street lighting, education, and healthcare, which the private sector would not supply adequately because they are not profit oriented.

Disadvantages of a Mixed Economy

Although a mixed economy offers many benefits, it also faces certain challenges that may affect efficiency and decision-making. These drawbacks arise from the involvement of both the government and the private sector, which can sometimes lead to conflicts and inefficiencies, as explained below.

- Excessive regulation, taxation, or bureaucracy can discourage private businesses, reduce efficiency, and limit innovation.
- When the government dominates certain industries (such as utilities or transport), it may create state monopolies, which can lead to inefficiency, poor service delivery, and lack of competition.
- Both the government and private sector may provide similar services (such as health or transport), leading to wastage of resources and inefficiency.
- With both public and private producers operating simultaneously, there may be excessive use of natural resources if regulations are weak, leading to environmental degradation.

Application activity 1.9



Carefully study each of the scenarios reflecting decisions and interactions that people and institutions make in their daily lives concerning the use of resources presented below and respond to the questions that follow.

Scenario 1: In a remote rural community, families grow sorghum and beans mainly for their own consumption using simple tools such as hoes and machetes. Farming methods and crop choices are based on longstanding customs passed down from elders. Any small surplus is exchanged within the village through barter, such as trading beans for milk or firewood. Skills, roles, and economic activities are learned through family and community traditions rather than formal training.

Scenario 2: The government owns and operates the national electricity company, which controls the generation, transmission, and distribution of electricity across the country. It sets electricity prices and decides how power is supplied to different areas, and private companies are not allowed to compete in this sector. The aim is to ensure affordable and reliable electricity for all citizens, especially low-income households.

Scenario 3: A privately owned shoe factory decides what type of shoes to produce based on consumer demand and available resources. Prices are determined by demand and supply, and customers are free to choose where to buy shoes. The factory competes with other businesses to attract buyers.

Scenario 4 The government provides public schools to ensure all children can access basic education, while private individuals and organizations run private schools offering different facilities and learning approaches. The government sets rules and standards for education, but private schools are allowed to operate alongside public ones, giving families choice while protecting public interest.

Questions

For each scenario, identify.

- i. The main economic activities carried out.
- ii. The type of economic system illustrated and its characteristics
- iii. One advantage of the system identified to consumers, producers and government.

End of unit assessment



Knowledge and understanding

1. Explain four ways in which Economics is important to individuals or society.
2. (a) Differentiate between wealth and resources, giving examples.
(b) What are the characteristics of wealth?
3. Classify each statement as positive or normative economics and give a reason.
 - a) "Coffee exports in Rwanda increased last year."
 - b) "The government should reduce taxes on basic foods."
4. What are public goods? Give examples of public goods.
5. Explain why most modern economies are mixed economies.

Skills questions

6. Analyse why Economics is considered a social science rather than a natural science.
7. Using examples from Rwanda, explain how scarcity influences what individuals, households, and the government can produce or consume.
8. Differentiate between microeconomics and macroeconomics and analyse how the two levels interact within Rwanda's economy.

9. Why are public goods always provided by the state?
10. (a) Analyse the significance of points under, on, and beyond the PPF.
(b) How do shifts in the PPF affect a country's production capacity?
(c) Analyse the importance of a PPF.

Values and Attitudes question.

11. a) How does knowledge of scarcity encourage people to plan better for the future?
b) How does learning economics help people become responsible members of society?
12. Why should society care about externalities, such as pollution, when making economic decisions?
13. Why is it important for governments to aim for production points on or near the PPF?

UNIT 2

DEMAND AND SUPPLY ANALYSIS



Key Unit Competence:

Apply demand and supply interactions to explain price determination, market equilibrium, and economic changes in everyday decisions.

Learning objectives

Knowledge and understanding

- Define price and explain the different types of prices found in markets.
- Explain the meaning of a market and identify different types of markets.
- Describe how prices are determined in the market through demand and supply.
- Identify and explain the factors that determine prices in the market.
- Define demand and distinguish between different types of demand.
- State and explain the law of demand, including demand schedules and demand curves for individuals and the market.
- Explain the slope of the demand curve and the factors that support its shape.
- Describe abnormal demand curves and situations that cause them.
- Explain the factors that determine demand in the market.
- Distinguish between a change in demand and a change in quantity demanded.
- Define supply and explain the different types of supply.
- State and explain the law of supply, including individual and market supply schedules and curves.
- Describe abnormal supply curves and their causes.
- Explain the factors that determine supply in the market.
- Distinguish between a change in supply and a change in quantity supplied.
- Define equilibrium, equilibrium price, and equilibrium quantity.
- Explain how market equilibrium is determined using joint demand and supply schedules, curves, and simple equations.
- Explain the effects of changes in demand and supply on market equilibrium.

1

Skills

2

- Apply demand and supply concepts to explain how prices are determined in real markets.
- Construct and interpret demand and supply schedules and curves for individual markets and the market as a whole.
- Analyze changes in demand and supply and explain their effects on prices and quantities.
- Use diagrams and simple equations to determine equilibrium price and quantity.
- Explain real-life price changes (e.g., food prices, transport fares, data bundles) using demand and supply interactions.
- Analyze everyday economic situations to make informed and rational consumer or producer decisions.
- Interpret abnormal demand and supply situations and relate them to actual market behavior.

Attitudes and values

3

- Appreciate the role of demand and supply in determining fair market prices.
- Demonstrate rational decision-making as consumers and producers based on price signals.
- Develop awareness of market forces and how economic changes affect individuals and communities.
- Show responsibility and fairness in market participation, such as buying, selling, and pricing decisions.
- Value informed decision-making when responding to changes in prices, shortages, or surpluses.
- Recognize the importance of market equilibrium in promoting economic stability and wellbeing.

Economics examines how scarce resources are used to satisfy unlimited human wants, with the price system playing a central role in this process. Prices guide consumers on what to buy and producers on what to produce. This unit introduces price theory by explaining how prices are determined in different markets through the interaction of demand and supply. It defines prices and markets, uses demand and supply schedules and curves to explain price formation, and shows how equilibrium price and quantity are established when demand equals supply.



Introductory activity

Analyse the scenario below and answer the questions that follow:

Imagine it is a very hot day at school and, during break time, a seller brings bottles of water to sell to students. When the price of water is low, many students are willing to buy it, but as the price increases, fewer students are willing to purchase the water. On the other hand, the seller is willing to sell more bottles when the price is high and is less willing to sell many bottles when the price is low. At a particular price, the number of bottles of water that students want to buy becomes equal to the number of bottles the seller is willing to sell.

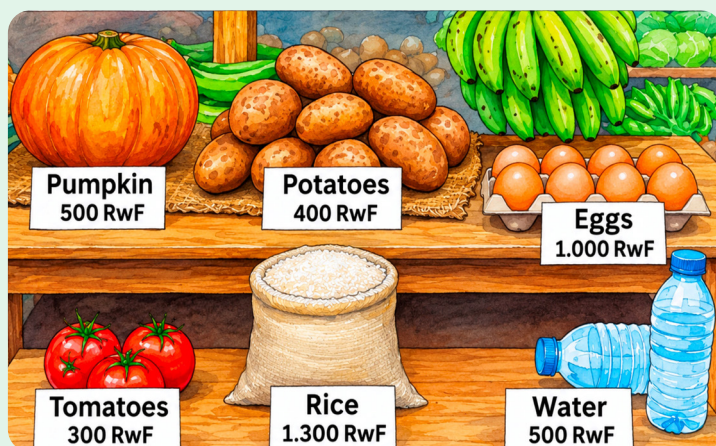
Questions:

Based on the scenario:

- When the price of bottled water is low, do many or few students want to buy it? Why?
- What happens to the number of buyers when the price of bottled water increases?
- At which price would the seller prefer to sell more bottles: a high price or a low price? Why?
- What happens when students want more bottles than the seller has available?
- What happens when the seller has more bottles than students want to buy?
- What do you call the price at which the number of bottles wanted by buyers is equal to the number supplied by the seller?

2.1 Introduction to price theory

Learning Activity 2.1



Observe the photo and answer the following questions:

- a) Which product in the image has the highest price, and how much does it cost?

- Compare the price of rice (1,300 RWF) and eggs (1,000 RWF)? which one is more expensive and by how much?
 - Where can these commodities be from?
2. You normally buy commodities from different places; how do you think such prices are reached at?

In any economy, prices play a vital role in guiding the buying and selling of goods and services. Price theory helps us understand how these prices are determined through the interaction of demand and supply in the market. This lesson introduces the basic ideas of price and markets, their different types, and explains how prices are set and influenced by various factors in the economy.

2.1.1 Meaning and types of prices and markets.

a) Meaning of prices.

Price is the amount of money paid in exchange for a good or service, agreed upon by both the buyer and the seller. It represents the value of a product in monetary terms and plays a key role in economic decisions. Building on this idea, price theory is a branch of microeconomics that explains how prices are determined in the market and how they influence consumer choices and producer decisions. It shows how consumers seek to maximize satisfaction while producers aim to maximize profit through the interaction of demand and supply. There are different types of prices and these include;

Types of prices

- **Reserve price:** A reserve price is the minimum price a seller is willing to accept for a good. For example, If a farmer sets a reserve price of RWF 400,000 for a cow, any bid below RWF 400,000 will be rejected. It is common in Auctions, Government sales, Public tenders, Online marketplaces etc..
- **Equilibrium price:** This is the price ruling in the market at which the buyers wholly clear what has been brought to the market by suppliers. At this price, quantity demanded is equal to quantity supplied. At this price, there is neither shortage nor surplus. Buyers and sellers are satisfied. For example, if at RWF 1000 per kilogram of tomatoes, consumers demand 70 kg and sellers supply 70 kg, then RWF 1000 is the equilibrium price and 70 kg is the equilibrium quantity.
- **Market price:** A market price is the actual price at which a good or service is being bought and sold in the market at a given time. It is the price consumers pay and sellers receive, regardless of whether it is the equilibrium price or not. For example, suppose in a Kamonyi market a kilogram of beans is currently selling for RWF 1,000, then market Price = RWF 1,000 per kg. This is simply the prevailing or actual price in the market. It may be equal to, above or below the equilibrium price.
- **Normal price:** It is the long-run price of a good to which market prices tend to return over time. For example, If the normal price of a kilogram of maize flour is RWF 1,000: A shortage may push the market price up to RWF 1,300, an oversupply may reduce the market price to RWF 900. Over time, producers adjust production, and the price tends to return to around RWF 1,000. Thus, RWF 1,000 is the normal price..
- **Relative price:** This is the price of a good or service expressed in terms of another good or service that must be sacrificed to obtain it. It shows how much of one good must be given up to obtain another. For example;

Suppose 1 kg of rice = RWF 2,000 and 1 litre of milk = RWF 800. The relative price of rice

$$= 2,000 \div 800$$

= 2.5 litres of milk

1 kg of rice is worth 2.5 litres of milk.

Suppose moto fare = RWF 1,200 and a loaf of bread = RWF 600, the relative price of a moto ride

$$= 1,200 \div 600$$

= 2 loaves of bread

One moto ride cost as much as 2 loaves of bread.

- **Absolute price (also called nominal price):** This is the price of a good or service measured in monetary terms, that is, the actual amount of money paid to purchase it. For example, A loaf of bread costs RWF 1,000, A kilogram of rice costs RWF 2,000, A litre of milk costs RWF 700 etc.

b) Meaning of market

A market is a place or system where people buy and sell goods and services. It brings together buyers who want to purchase and sellers who want to sell. A market does not have to be a physical location. It can be a physical market, such as a vegetable market, supermarket, or shopping mall; an online market, such as Amazon, Jumia, or eBay; or a financial market, where stocks, bonds, and currencies are traded. Regardless of its form, every market involves buyers who want goods or services, sellers who provide them, an exchange of money or other value, and prices that are determined by the forces of supply and demand, although governments may sometimes influence or regulate prices. The main purpose of a market is to make trading easy and fair.

Types of markets

- **Commodity market:** This is a market that is involved in the exchange of consumer goods. Producers of such goods meet with consumers and they negotiate the exchange. Examples of goods include food, clothes, shoes, books, stationery, furniture, and kitchen items.
- **Capital market:** A capital market is where financial securities like shares and bonds are bought and sold. It helps firms and governments raise long-term funds for investment and expansion. Transactions are carried out through brokers and agents. An example is the Rwanda Stock Exchange.
- **Factor market:** A factor market is where factors of production (land, labour, capital, and entrepreneurship) are bought and sold. Firms pay for these factors through wages, rent, interest, and profit.
- **Foreign exchange market (forex market):** A foreign exchange market is where different currencies are bought and sold. It is needed because of international trade and globalisation, which require currency exchange between countries. The main participants are banks, importers, exporters, and forex bureaus.

- **Free market:** A free market is a market where prices are determined by demand and supply with no government control. Buyers and sellers freely agree on prices, and resources are allocated through market forces
- **Open Market:** An open market is an economic system where buyers and sellers trade freely with minimal government interference, and prices and activities are mainly determined by competition and market forces of demand and supply. Entry and exit are easy, with few or no restrictions such as licensing requirements, taxes, subsidies, or trade barriers. Although it promotes free trade, in reality governments still apply some basic laws to regulate contracts, financial obligations, and ensure fair transactions in all markets.

2.1.2 Price determination and factors affecting prices in the market.

Prices in a market are formed through the interaction of buyers and sellers and are influenced by various conditions within the market. This lesson focuses on how prices are determined and the factors that cause them to change, such as shifts in demand and supply and other market forces. Understanding this forms the basis for studying **price determination and factors affecting prices in the market**.

i. Price determination

Price determination refers to the process through which the price of goods and services is established in a market. In a market economy, the price of goods and services is not fixed in one way, but is determined through different mechanisms depending on the situation. Prices may be set by producers, agreed upon through negotiations between buyers and sellers, or fixed by the government through regulation. In other cases, prices are shaped by competition, market forces, or agreements among firms. These different **methods** explain how prices are formed in real markets and lead to various systems of price determination, which are discussed below:

- **Demand and supply:** Prices are mainly determined by the interaction of demand and supply. When demand is higher than supply, prices rise; when supply is higher than demand, prices fall. When both are equal, the market reaches equilibrium price, where what is bought equals what is sold.
- **Sales auction (bidding):** This is a system where many buyers compete for one seller's goods. Each buyer offers a price, and the highest bidder wins the commodity. It is commonly used in areas like contract awarding and auction sales.
- **Price legislation:** Here, the government sets the price of goods and services. It may fix a maximum price (ceiling) or a minimum price (floor). It is illegal to sell above or below the set price, depending on the law.
- **Haggling (bargaining):** This involves direct negotiation between buyer and seller. The seller starts with a higher price, the buyer offers a lower one, and both adjust until they agree on a final selling price.
- **Resale price maintenance:** In this case, the producer fixes the selling price of a product, and retailers must sell at that price. Examples include newspapers, magazines, and airtime cards, where the price is usually printed on the item.
- **Treaties (agreements):** These are price agreements between firms or sellers to avoid unfair competition, especially price undercutting. They agree on a common price level for a commodity to maintain stability in the market.

- **Price leadership (imperfect collusion):** This occurs when the largest or most efficient firm sets the price, and other firms follow. The leading firm influences prices because of its low costs, strong position, and market experience.
- **Cartel arrangements (perfect collusion):** Cartel arrangements (perfect collusion) occur when firms in the same industry agree to work together to fix prices, control output, and reduce competition, mainly to avoid price wars and high costs, with an example being the OPEC which coordinates oil production and pricing among member countries.

ii. Factors determining prices in the market

Factors that determine prices in the market are the conditions or influences that cause the price of goods and services to be set at a certain level and to change over time. They explain why prices rise, fall, or remain stable, mainly through their effect on demand and supply by influencing how much buyers are willing to pay and how much sellers are willing to accept. Understanding these factors helps explain how market prices are formed and why they change in real-life situations. These factors include the following;

- **Cost of Production:** The cost of production includes all expenses incurred in making a good or service, such as wages, raw materials, power, transport, rent, and other overhead costs. Producers must cover these costs and earn a profit, so higher production costs usually lead to higher prices. If costs fall due to efficiency or cheaper inputs, producers may lower prices while still remaining profitable.
- **Level of Competition in the Market:** Competition affects how much control producers have over prices. In highly competitive markets with many sellers offering similar products, producers often lower prices to attract customers. In markets with little or no competition, sellers may charge higher prices because consumers have few alternatives. Competition can also shift from price to quality, branding, and customer service.
- **Government Policy:** Government policies influence prices through measures such as taxes, subsidies, and price controls. Taxes increase the cost of goods and raise prices, while subsidies reduce costs and make goods more affordable. Price controls are often applied to essential commodities like fuel or electricity to protect consumers from excessive price increases.
- **Demand and Supply of the Commodity:** Prices are largely determined by the interaction of demand and supply. When demand is high and supply is limited, prices increase as consumers compete for scarce goods. When supply exceeds demand, sellers reduce prices to attract buyers and clear excess stock. Changes in income, tastes, population, or production levels can shift demand and supply, causing prices to change.
- **Type of Product:** The nature of a product influences its price. Processed and manufactured goods are usually more expensive than raw products because they involve added costs such as processing, packaging, and marketing. Branded and specialized products also tend to have higher prices due to perceived quality, reputation, and customer loyalty.

Application activity 2.1



1. A farmer wants to sell 100 kg of maize at the market. He sets a minimum price of 300 FRW per kg, but buyers are only willing to pay 250 FRW. Later, after negotiation, the market clears at 310 FRW per kg.

Question:

- i. Identify the reserve price and the market-clearing price in this situation,
 - ii. Explain why the market cleared at a price higher than the minimum price the seller was willing to accept.
 - iii. What do you understand by the term market?
 - iv. Explain any 2 advantages and disadvantages of free market to consumers, producer and the entire economy.
2. How are prices of prices in different commodities determined in markets in the country?
 3. A local bakery in Rusizi produces bread. The bakery's costs include flour, sugar, electricity, wages for workers, and rent for the shop. There are also several other bakeries in the area producing similar bread.

Questions:

- i. Explain how the cost of production and level of competition influence the price of bread in this market.
- ii. Suggest a strategy the bakery could use to remain competitive without reducing quality.

2.2 Demand analysis.

Learning Activity 2.2



Read the market situation carefully and answer the questions that follow:

Imagine you are planning to buy sugar from a local market in Rwanda. One shop sells sugar at 1,000 FRW per kilogram, while another shop sells the same sugar at 1,200 FRW per kilogram. You also notice that some people in the market can afford to buy more sugar than others.

Questions:

- a) Which shop would you choose to buy sugar from? Why?
- b) If the price of sugar increases further, what would happen to the quantity of sugar you buy?

- c) Why do some people buy more sugar than others even when the price is the same?
- d) Are there people who want (desire) sugar but cannot buy it? What could be the reason?
- e) From this situation, what do you think must be present for someone to buy sugar?

In everyday life, people constantly make choices about what to buy and how much to buy depending on their needs, preferences, and available income. These choices influence what is available in markets and affect the decisions made by producers and sellers. Studying this lesson helps you understand how consumers make purchasing decisions and why buying patterns change under different economic conditions. This knowledge will enable you to interpret market behavior, make wise personal economic decisions, and better understand how resources are allocated in society.

2.2.1 Meaning and types of Demand

Demand refers to the desire backed by the willingness and ability of consumers to buy a particular good or service at various prices over a given period of time. This means that for demand to exist, consumers must not only want (desire) the product, but must also have the income and readiness to pay for it at the stated price. Quantity demanded, on the other hand, refers to the specific amount of a good or service that consumers are willing and able to buy at a particular price over a given period of time. It varies with price because consumers tend to buy more at lower prices and less at higher prices, holding other factors constant. Demand can be of different types which include the following:

Types of Demand (Interrelated Demand)

Types of demand focus on interrelated demand and explain how the demand for one good or service is influenced by changes in the demand for another. This helps in understanding the relationships between goods in real-life situations and how these relationships affect consumer choices and market behaviour.

- 1. Joint / Complementary / Twin Demand:** Joint or complementary demand occurs when two or more goods are used together to satisfy a single want. The demand for one good is directly linked to the demand for the other. When demand for one increase, demand for the other also increases, and when demand for one fall, demand for the other falls. For example, cars and petrol or phones and SIM cards, guns and bullets, camera and camera films, Bicycles and bicycle tyres, Pen and ink, Bulbs and electricity/sockets etc are complements because one cannot be effectively used without the other. A rise in the price of one usually reduces demand for both.
- 2. Competitive (Substitute) Demand;** Competitive demand refers to demand for goods that serve the same purpose and can replace each other. These goods compete for the same consumers. When demand for one good increases, demand for its substitute decreases. For example, tea and coffee or butter and margarine, beef and chicken, wood and charcoal, Coca-cola and Pepsi, moto travel and bus travel, satisfy similar needs. If the price of one rise, consumers

may switch to the cheaper alternative, increasing its demand and reducing the demand for the more expensive one.

- 3. Composite Demand:** Composite demand occurs when one commodity has several uses, and its total demand is the sum of demand for all those uses. For example, electricity is used for lighting, cooking, manufacturing, and heating; water for cooking, drinking, washing, irrigation; timber for construction, furniture; land for settlement, farming, livestock, industrial development etc. An increase in demand for one use (such as industrial use) can reduce availability for other uses. Each use is independent, but together they determine the overall demand for the commodity.
- 4. Derived Demand:** Derived demand is demand for a good not for its own sake, but because it is used to produce another good or service. The demand for factors of production like labour, land, and capital depends on the demand for the final products they help produce. For instance, increased demand for bread leads to higher demand for flour, labour, and baking equipment.
- 5. Independent Demand:** Independent demand refers to demand for goods whose consumption does not depend on the demand for other goods. These goods satisfy separate needs, and a change in demand for one does not affect the demand for another. Examples include clothing and furniture. Consumers buy such goods based on personal needs, preferences, and income.
- 6. Dependent Demand:** Dependent demand is demand for components or intermediate goods used in producing final products. It depends directly on the demand for the finished goods. For example, the demand for car engines depends on the demand for cars. If car sales increase, demand for engines will also increase.
- 7. Direct Demand:** Direct demand refers to demand for goods and services that directly satisfy consumers' wants. It is also called consumer or final demand. Examples include food, clothing, smartphones, or transport services bought for personal use. This demand comes directly from consumers, not from producers.
- 8. Latent Demand:** Latent demand exists when consumers desire and are willing to buy a good, but lack the purchasing power to do so. For example, a student may want a laptop but lacks the money to buy it. This demand does not appear in the market until income or purchasing power increases.
- 9. Effective Demand:** Effective demand is demand that is backed by both willingness and ability to pay at a given price. It is the demand that actually shows up in the market and leads to purchases. Effective demand depends on income, prices, tastes, and overall economic conditions. Only effective demand influences market prices and production decisions.
- 10. Individual Demand:** Individual demand refers to the quantity of a good or service that a single consumer is willing and able to buy at different prices over a given period of time, holding other factors constant. It shows how one person's quantity demanded changes as the price changes. For example, at a lower price of sugar, one consumer may buy more kilograms, while at a higher price, they buy less.

11. Market Demand: Market demand is the total quantity of a good or service that all consumers in a market are willing and able to buy at different prices over a given period of time. It is obtained by adding together the individual demands of all consumers in the market at each price. Market demand therefore reflects the combined buying behavior of all consumers and determines the overall demand faced by sellers.

2.2.2 The law of demand, demand schedule and curve:

The law of demand states that, **all other factors remaining constant (ceteris paribus), when the price of a good or service rises, the quantity demanded falls, and when the price falls, the quantity demanded rises.** This shows an inverse (opposite) relationship between price and quantity demanded.

The law of demand shows how consumers normally react to price changes. When prices are low, consumers can afford to buy more and are encouraged to increase their purchases. When prices rise, consumers buy less or switch to cheaper alternatives because their purchasing power is reduced. As a result, higher prices lead to lower quantity demanded, while lower prices lead to higher quantity demanded.

The law of demand is commonly demonstrated using the following illustrations: *demand schedule and demand curve.*

a) The demand schedule

A demand schedule is a table that shows different prices of a commodity and the corresponding quantities demanded. It clearly illustrates that when price falls, quantity demanded increases, and when price rises, quantity demanded decreases. A demand schedule can be derived for an individual consumer or whole market.

An individual demand schedule is a table showing the quantities of a commodity that a single consumer is willing and able to buy at different prices over a given period of time, assuming other factors remain constant. It shows how one buyer's quantity demanded changes as the price changes.

Table 2.1 A hypothetical example of an Individual demand schedule for Sugar

Price (FRW per kg)	Quantity Demanded (kg)
1,500	1
1,200	2
1,000	3
800	4
600	5

The schedule above shows that as the price of sugar falls, the consumer buys more. This illustrates the **law of demand** at the level of a single consumer.

A market demand schedule is a table that shows the total quantities of a commodity that all consumers in a market are willing and able to buy at different prices over a given period of time. It is obtained by adding together the quantities demanded by all individual consumers at each price.

Table 2.2 A hypothetical example of a market demand schedule for Sugar

Price (FRW per kg)	Consumer A (kg)	Consumer B (kg)	Consumer C (kg)	Market Demand (kg)
1,500	1	2	1	4
1,200	2	3	2	7
1,000	3	4	3	10
800	4	5	4	13
600	5	6	5	16

From the table above, at each price, the market demand is the **sum of individual demands**. Like individual demand, market demand also increases as price falls.

b) The demand curve

A demand curve is a graphical representation of the relationship between the price of a good or service and the quantity demanded by consumers over a given period of time, assuming other factors remain constant. It is drawn with price on the vertical axis (Y-axis) and quantity demanded on the horizontal axis (X-axis). The demand curve normally slopes downward from left to right, showing that as the price of a commodity falls, the quantity demanded increases, and as the price rises, the quantity demanded decreases. This downward slope illustrates the law of demand. It can be drawn for an individual consumer or the whole market for the same commodity.

An **individual demand curve** shows the quantities of a commodity that one consumer is willing and able to buy at different prices. It reflects the buying behavior of a single buyer.

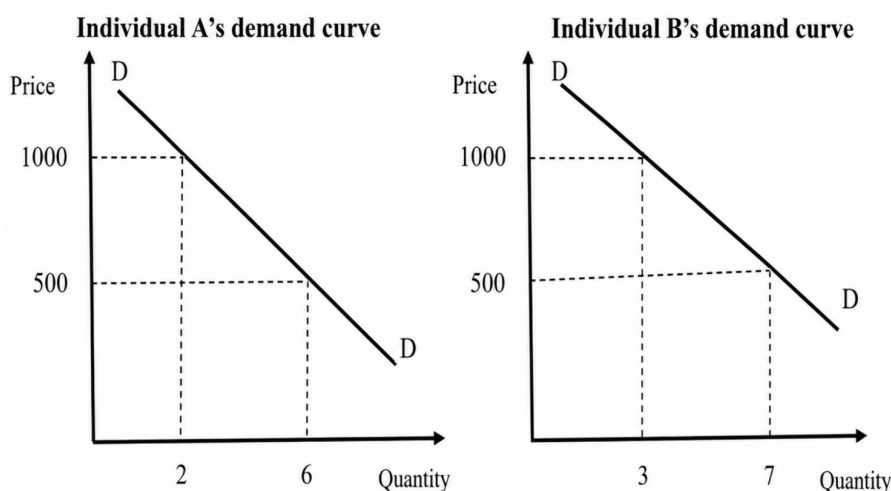


Fig 2.1 individual demand curves

A market demand curve shows the total quantity of a commodity that all consumers in a market together are willing and able to buy at different prices. It is formed by adding up (horizontally summing) all the individual demand curves for that commodity at each price level. This means that at every price, the market demand reflects the combined purchases of all buyers.

Because it combines the demands of many consumers, the market demand curve is usually flatter than an individual demand curve, as changes in price affect different consumers in different ways. Like individual demand curves, the market demand curve slopes downward from left to right, showing the law of demand. This downward slope indicates that as the price of a good increases, the quantity demanded decreases, since fewer consumers are willing or able to buy the good at higher prices.

Market demand curve

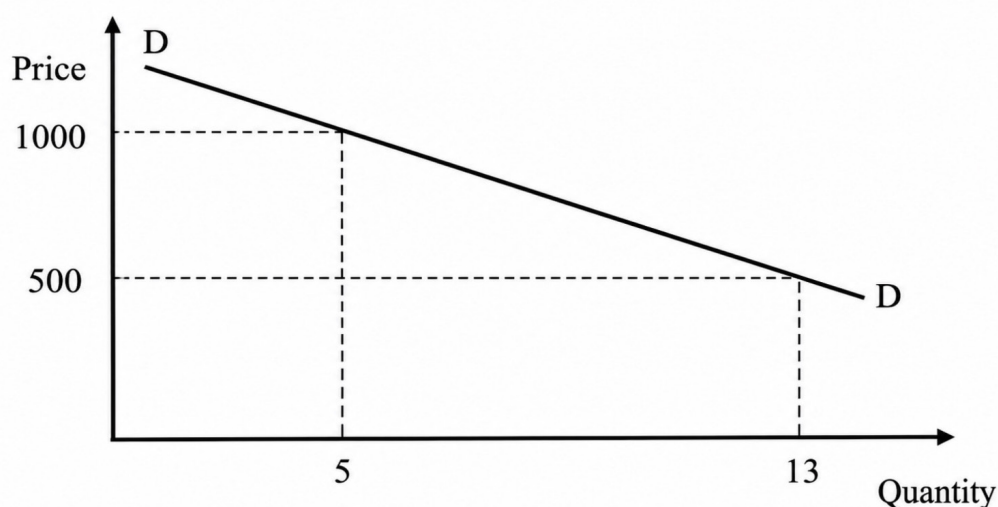


Figure 2.2 A market demand curve:

Exercise: 2.1

Analyse the demand schedule below and answer the questions that follow.

A hypothetical Individual Demand Schedule for meat

Price per unit (FRW)	Quantity Demanded (Units per week)
12,000	1
10,000	3
8,000	7
6,000	12
4,000	18

Questions

- Plot the data on a graph.
- Is the schedule or curve for and individual or market? Explain

2.2.3 Slope of the demand curve

The demand curve has a negative slope, meaning it slopes downward from left to right. This shows an inverse relationship between price and quantity demanded: as price falls, quantity demanded rises, and vice versa. The negative slope is explained by diminishing marginal utility, the substitution effect, and the income effect. The curve may be linear, showing a constant rate of change in demand, or non-linear (curved), showing varying consumer responsiveness at different price levels; however, in all cases, the slope remains negative.

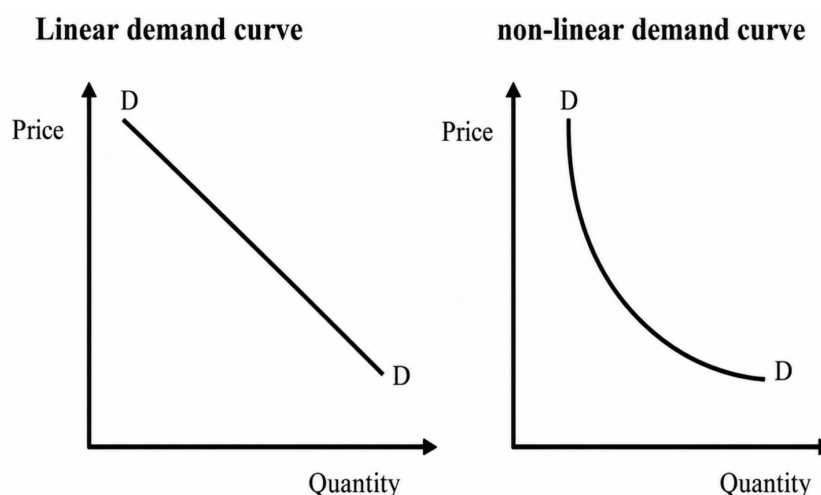


Fig 2.3 The slope of the demand curve

The demand curve, whether linear (a straight line) or non-linear (curved) as illustrated above, generally slopes downward from left to right under normal conditions, indicating an inverse relationship between price and quantity demanded.

Although real-world demand is often curved because consumers respond differently to price changes at various price levels, the direction of the curve remains downward in all cases, as supported by the factors explained below.;

- **Law of Diminishing Marginal Utility:** The law states that as a consumer consumes more units of a commodity, the additional satisfaction (marginal utility) gained from each extra unit decreases. Because each successive unit gives less satisfaction, consumers are only willing to buy more units if the price falls. Thus, high prices correspond to high marginal utility and low quantities demanded, while low prices correspond to low marginal utility and high quantities demanded. This inverse relationship between price and quantity demanded leads to a downward-sloping demand curve.
- **Real Income Effect:** Real income is the purchasing power of income. It shows how much a person's money can actually buy. When the price of a good falls, the purchasing power of consumers' income increases, even if their money income remains unchanged. Consumers can now afford to buy more of the good with the same income. As a result, a fall in price leads to an increase in quantity demanded, reinforcing the downward slope of the demand curve.

- **Substitution effect:** When the price of a good decreases, it becomes relatively cheaper compared to substitute goods. Consumers respond by substituting the cheaper good for more expensive alternatives. This shift toward the lower-priced good increases its quantity demanded as price falls, contributing to the negative slope of the demand curve.
- **Psychological factors (price effect):** Consumers often perceive lower prices as bargains or better value for money, which psychologically encourages higher purchases. Conversely, higher prices may discourage buying because consumers perceive the good as too expensive or poor value. These perceptions influence buying behavior and help explain why demand increases when prices fall.
- **Presence of low-income earners:** Low-income consumers are more sensitive to price changes. When prices fall, they significantly increase their quantity demanded; when prices rise, they sharply reduce consumption. Since low-income earners form a large part of the market, their behavior strongly contributes to the inverse relationship between price and quantity demanded.
- **Many uses of the commodity:** Goods with multiple uses tend to be demanded more when prices are low because consumers can afford to use them for both essential and non-essential purposes. When prices rise, consumers restrict usage to only essential purposes, reducing quantity demanded. This variation in usage at different prices creates a downward-sloping demand curve.

Application activity 2.2



Below is a hypothetical individual demand schedule, use it to answer the questions that follow;.

A hypothetical Individual Demand Schedule for meat

Price per unit (FRW)	Quantity Demanded (units per week)
10,000	2
8,000	4
6,000	6
4,000	8
2,000	10

Questions:

- Using the information in the table above, draw a demand curve on a graph.
- Describe the shape and direction of the demand curve you have drawn.
- Explain the relationship between price and quantity demanded as shown by the curve.

- d) State whether the demand curve has a positive or negative slope and give a reason for your answer.
- e) Identify the law of demand and explain how the curve drawn supports it.

2.2.4 Abnormal demand curves

Learning Activity 2.2.4



Read the scenario provided and discuss the questions that follow. Use the situation to explain how consumer behavior in this case differs from the normal situation.

One evening, people hear on the news that heavy rains may damage roads, making it difficult for shops to restock basic goods like salt. The next morning, shops increase the price of salt. Normally, people expect demand to fall when prices rise. However, instead of buying less salt, many households rush to the shops and buy more salt than usual, worried that prices may increase further or that salt might completely run out. As a result, even though the price of salt has increased, the quantity demanded has also increased.

Questions

- a) Under normal conditions, what usually happens to quantity demanded when the price of a commodity rises?
- b) Why did people decide to buy more salt despite the higher price?
- c) What happened to the relationship between price and quantity demanded in this situation?
- d) Does this behavior follow or go against the normal law of demand?
- e) What does this tell us about consumer behavior in special or unusual situations?

In economics, demand usually falls when price rises, following the law of demand. However, in some special or unusual situations, consumers behave differently. In these cases, a rise in price may lead to an increase in quantity demanded. Such situations give rise to what are known as **abnormal or regressive demand curves**.

An abnormal demand curve is a demand curve that slopes upward from left to right, showing that price and quantity demanded move in the same direction, unlike the normal demand curve. This abnormal behavior occurs when consumer decisions are influenced by factors such as fear of shortages, expectations of future price changes, status or prestige motives, or extreme poverty. In such situations, demand may increase when prices rise (for example, during panic buying or for luxury goods) or decrease when prices fall (for instance, when consumers expect prices to fall further), thereby violating the normal law of demand. The abnormal demand curves

are otherwise known as the exceptions of the law of demand because they violate it. This can be explained such factors as below;

- **Giffen Goods (Extreme Poverty Case):** Giffen goods are inferior goods mainly consumed by very poor households, such that when their price rises, consumers are forced to buy more of them because they can no longer afford better substitutes. Named after Sir Robert Giffen, these goods are typically essential, low-income items with few substitutes. True Giffen goods are very rare in real life so, some examples are usually theoretical or potential examples rather than proven cases. The upward-sloping demand curve of Giffen goods is driven by a significant income effect that outweighs the substitution effect, causing quantity demanded to rise with price. This directly violates the law of demand.

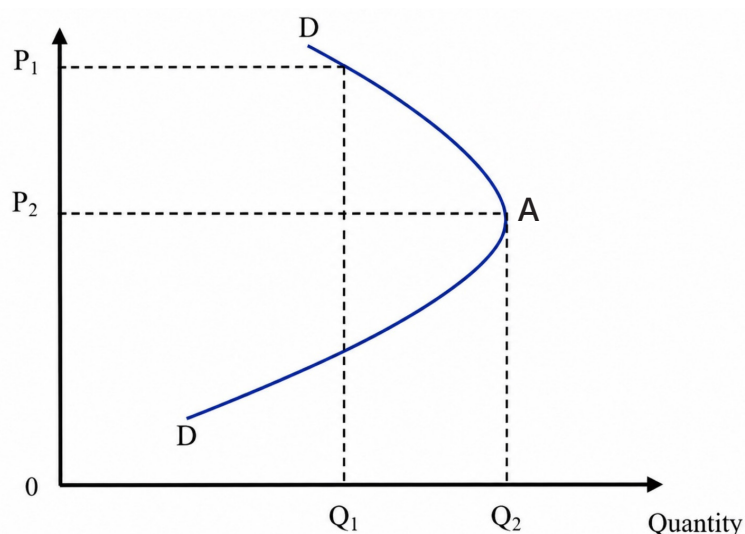


Fig 2.4: Giffen good demand curve

The Giffen demand curve has an unusual shape because, for very poor consumers, the income effect outweighs the substitution effect. In the figure, below point A, as the price rises to P_2 , the quantity demanded increases to Q_2 . This occurs because the good is a basic necessity, and the higher price reduces consumers' real income, forcing them to buy more of the inferior good and fewer alternative goods. Above point A, demand behaves normally: as the price increases from P_2 to P_1 , the quantity demanded falls from Q_2 to Q_1 because consumers can afford less of the good. Therefore, a Giffen good is an exception to the law of demand below point A, but follows the law of demand above point A.

- **Prestige or Veblen goods (status effect):** Prestige goods are luxury goods whose high prices give them social status and exclusiveness. For such goods, a rise in price can increase demand because consumers associate higher prices with higher prestige. Conversely, a fall in price may reduce demand as the good loses its status value, making demand move abnormally.

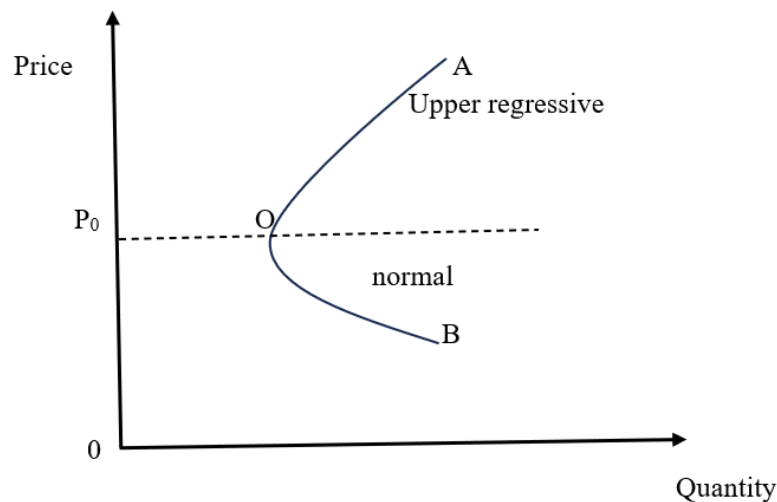


Fig 2.5 Veblen good demand curve

This curve shows that at price P_0 (point O) the good behaves like a normal good, where a lower price would increase quantity demanded, but beyond this point (curve OA), it becomes a Veblen good where the demand curve slopes upward, meaning that as price increases, quantity demanded also increases due to the desire for status and prestige; however, below P_0 (curve OB), it still follows the normal downward-sloping demand behavior.

- **Speculative Demand (Future Price Expectations):** Speculative demand arises when consumers expect prices to rise further in the future. Even if prices are already high, consumers may increase current purchases to avoid paying still higher prices later. As a result, an increase in price leads to an increase in quantity demanded due to expectations rather than current utility.
- **Panic buying and fear of shortages:** During emergencies, crises, or rumors of shortages, consumers may buy unusually large quantities of goods even when prices rise. Fear of future unavailability dominates price considerations, leading consumers to stock up. In such situations, higher prices do not reduce demand, creating an exception to the law of demand.
- **Ignorance, habits, and perceived Quality:** Sometimes consumers believe that higher prices indicate better quality, especially when they lack adequate information. Due to habit or ignorance, they may prefer and demand higher-priced goods, assuming they are superior. This mistaken belief can cause demand to rise with rising prices.
- **Economic depression:** During an economic depression, incomes fall, unemployment rises, and consumer confidence is low. As a result, people may reduce their purchases even when prices fall, choosing to save or because they lack purchasing power. This causes demand to fall despite falling prices, making it an exception to the law of demand.

Application activity 2.2.4



Analyze the real-life situations provided below. For each situation, identify whether it shows normal demand or an exception to the law of demand. Where demand is abnormal, explain why consumers behave differently and describe how the demand curve would look.

Situations for Analysis

A: The price of maize flour rises sharply after poor harvests. Despite the price increase, very poor households buy more maize flour and reduce their consumption of meat and other foods.

B: A rumour spreads that fuel prices will increase heavily next week. Even though fuel prices are already high, many people rush to buy and store fuel.

C: The price of a famous designer handbag increases, and more wealthy consumers begin buying it because it makes them feel prestigious.

D: During an economic recession, the prices of household items such as furniture fall, but consumers continue to buy less than before.

Questions

- Does demand increase or decrease as price changes in each situation?
- Which situations show an exception to the law of demand?
- What factor causes the abnormal behavior in each case?
- Would the demand curve in each case slope downward or upward? Explain.
- How does each situation differ from normal demand behavior?

2.2.5 Factors determining demand

Learning Activity 2.2.5



Carefully read and answer the questions below by using examples from your own daily experiences.

- When the price of a product you like increases, do you usually buy more of it or less of it? Why?
- If your pocket money increases, how does this affect the amount of goods you buy? Give an example
- Why do people buy more cold drinks on hot days and fewer on cold days?
- Suppose there is change in taste or fashion of a shoe you have been using, how do you think it will affect your desire for it?
- If the price of tea increases, what do you think might happen to the demand for coffee? Explain your answer.
- Why do people buy more school materials at the beginning of a term than during the holiday?
- From these examples, identify and list the factors you think influence the demand for goods and services.

In daily life, people make decisions about what to buy and how much to buy depending on different situations and circumstances. At times, more of a product is purchased, while at other times less is bought, even when it is available in the market. This shows that demand is not determined by price alone, but by several other factors that influence people's willingness and ability to buy goods and services. These factors can be expressed in a **demand function**, written as $Q_d = f(P, Y, T, P_r, \dots)$, which shows the relationship between quantity demanded and its determinants. A demand function is a mathematical expression that shows how the quantity demanded of a good or service is related to its determinants. The effect of any one determinant is analyzed while the other determinants are held constant (*ceteris paribus*).

Where:

Qd = Quantity demanded	Pr = Prices of related goods (substitutes and complements)
P = Price of the good	T = Tastes and preferences
Y = Income of consumers	... = Other factors (e.g., population, expectations)

Among these, several key factors play an important role in determining demand, as discussed below.

- **Price of the Commodity:** When the price of a commodity falls, consumers are willing and able to buy more, so demand increases. When the price rises, consumers buy less, so demand decreases, assuming other factors remain constant.
- **Income of the Consumer:** When consumers' income increases, demand for normal goods increases because they can afford to buy more. When income decreases, demand for these goods falls. (For inferior goods, demand may increase when income falls.)
- **Price of Other Commodities:** If the price of a substitute good (e.g., tea and coffee) rises, demand for the other good increases. If the price of a complementary good (e.g., bread and butter) rises, demand for the related good decreases.
- **Weather:** Weather conditions affect demand for some goods. For example, hot weather increases demand for cold drinks and ice cream, while cold weather increases demand for warm clothes and heaters.
- **Government Policy:** Government actions such as taxes, subsidies, bans, or price controls influence demand. Higher taxes increase prices and reduce demand, while subsidies lower prices and increase demand.
- **Tastes and Preferences:** When consumers develop a strong liking for a product due to fashion, advertising, or culture, its demand increases. If a product becomes unpopular, demand decreases.
- **Population:** An increase in population leads to more consumers, thus increasing demand for goods and services. A decrease in population reduces demand.
- **Past Levels of Income:** If consumers earned high income in the past, they may

maintain high levels of consumption even if current income falls, thus keeping demand high. Low past income may limit current demand.

- **Future Price Expectations:** If consumers expect prices to rise in the future, they buy more now, so demand increases. If they expect prices to fall, they postpone buying, and demand decreases.
- **Bandwagon Effect:** The bandwagon effect is when people increase their demand for a product simply because many other people are buying or using it. Demand rises due to popularity or the desire to fit in, and it falls when the product loses popularity.

Application activity 2.2.5



Analyze the scenario and explain how each of the given factors affects the quantity demanded of bread. For each factor, clearly identify it and state whether it causes an increase or decrease in quantity demanded, giving a brief reason.

Scenario:

"The price of bread remains unchanged, but the price of butter increases, the price of rice decreases, a popular health campaign discourages people from eating bread, and the population in the area rises."

Questions:

- a) How is bread related to butter?
- b) How is rice related to rice?
- c) What happens to the quantity demanded of bread when the price of butter increases? Why?
- d) How does a decrease in the price of rice affect the demand for bread? Explain.
- e) In what way does the health campaign influence consumers' demand for bread?
- f) How does an increase in population affect the quantity demanded of bread?
- g) What is the overall effect of these factors on the quantity demanded of bread?
- h) What other factors may affect the quantity demanded of bread in the market other than those mentioned in the scenario?

2.2.6 Change in demand and change in quantity demanded

i. Change in demand

A change in demand refers to an increase or decrease in the quantity of a good or service that consumers are willing and able to buy at the same price, caused by factors other than the good's own price, such as changes in income, tastes and preferences, population size, expectations, or the prices of related goods, and it is shown by **a shift of the demand curve** either to the right (increase) or to the left (decrease).

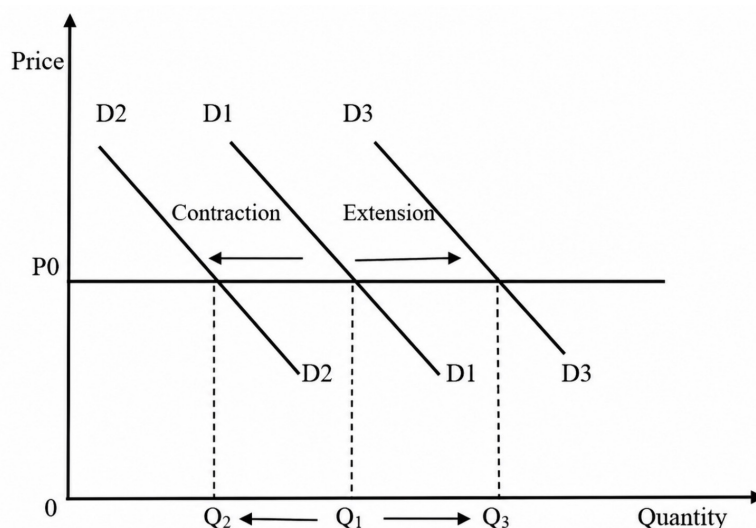


Fig 2.6 Change in demand curve

The curve shows a change in demand, where a decrease in demand shifts the demand curve left from D_1 to D_2 , reducing quantity from Q_1 to Q_2 , and an increase in demand shifts it right from D_1 to D_3 , increasing quantity from Q_1 to Q_3 , all at the same price P_0 .

ii. Change in quantity demanded

A change in quantity demanded refers to an increase or decrease in the amount of a good or service that consumers are willing and able to buy **as a result of a change in its own price**, while all other factors remain constant, and it is shown by a movement along the same demand curve rather than a shift of the curve. A movement along the demand curve is caused **only by a change in the price of the good itself**, with all other factors such as income, tastes, and population remaining constant.

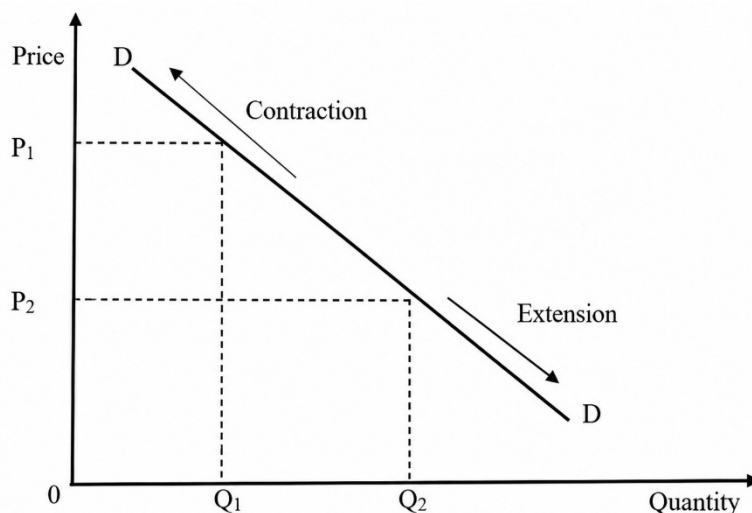


Fig 2.7 Change in quantity demanded curve

The curve shows a movement along the same demand curve (DD) caused by a change in price. When the price falls from P_1 to P_2 , quantity demanded increases from Q_1 to Q_2 , which is called an extension of demand. When the price rises from P_2 to P_1 , quantity demanded decreases from Q_2 to Q_1 , which is called a contraction of demand. This movement occurs because only the price changes, while other factors remain constant.

Application activity 2.2.6



Carefully read and answer the following questions

1. a) Distinguish between latent demand and effective demand.
b) With examples, explain what you understand by derived demand.
2. For each scenario below, state whether it represents a change in demand or a change in quantity demanded. Give a reason for your answer.
 - i. The price of sugar falls and consumers buy more sugar.
 - ii. Consumers' income increases and they buy more cars at the same price.
 - iii. The price of bread rises and people buy less bread.
3. Explain the factors that support the law of demand.

2.3 Supply analysis

Learning Activity 2.3



Analyse the scenario below and answer the questions that follow:

In a local market, tomato sellers bring different quantities of tomatoes each day. On some days, the market is full of tomatoes, while on other days, only a few are available. Prices also keep changing.

Imagine you are tomato sellers, discuss how you would respond to the following situations.

Questions

- i. When the price of tomatoes increases, would you bring more or fewer tomatoes to the market? Why?
- ii. If the cost of production (e.g., transport, fertilizers) increases, how would this affect the quantity you would take to the market?
- iii. How does a good harvest or favorable weather influence the amount of tomatoes you bring to the market?
- iv. What happens to the amount taken to the market when the number of sellers increases?
- v. How would a government tax or subsidy affect your willingness to take tomatoes to the market?
- vi. From your discussion, what do you think "supply" means?
- vii. Apart from price, what other factors influence the quantity supplied?

In the previous lessons, we examined demand and focused on the behavior of consumers, why they are willing and able to buy different quantities of goods and services at various prices. However, a market does not involve buyers alone. For goods and services to be available, there must be producers willing to offer them for sale. In this lesson, we shift our attention from consumers to producers by studying

supply analysis, where we explore the factors that influence sellers' willingness and ability to produce and offer goods and services in the market.

2.3.1 Meaning and types of supply

In everyday life, markets are filled with goods such as food, clothes, and other products brought by sellers. The quantity of these goods varies from time to time depending on different conditions. This leads to the concept of supply, which explains how producers decide the amount of goods and services to offer for sale.

a) Supply: Supply is defined as *the quantity of a good or service that producers are willing and able to offer for sale at various prices over a given period of time.* Closely related is the concept of quantity supplied, which refers to the *specific amount of a good that producers are willing and able to offer for sale at a particular price at a given time.* The key difference is that supply shows the relationship between price and quantities offered at various prices, while quantity supplied refers to one exact quantity at a specific price.

b) Types of supply

Supply can take different forms depending on how producers respond to changes in market conditions. This section introduces the various types of supply, highlighting how supply behavior differs under different situations and helping to explain real-life patterns in the availability of goods and services. These include the following:

- i. Individual Supply:** This refers to the quantity of a good or service that one producer is willing and able to sell at different prices over a given period. For example, a single poultry farmer deciding how many trays of eggs to sell per week depending on market prices is exercising individual supply.
- ii. Market Supply:** This is the total quantity supplied by all producers of a good in a market at various prices during a specific period. For instance, the combined number of egg trays supplied by all poultry farmers in a town each week represents the market supply of eggs.
- iii. Joint Supply:** This occurs when two or more products are produced together using the same resources, so increasing the production of one automatically increases the other. For example, cattle farming, where producing beef also results in hides and bones being supplied at the same time, Sugar and molasses (from sugarcane), Timber and sawdust (from trees), Mutton and wool (from sheep), Timber and firewood, Coffee Beans and Coffee Husks, Bananas and Banana Fibres. You can't really produce one without also getting the other as a byproduct.
- iv. Composite (Competitive) Supply:** This exists when a single resource can be used to produce different goods, and producing more of one means producing less of another. Both goods use the same limited resources. For example, the same cattle herd could be raised primarily for milk production or for beef, a farmer using land to grow either maize or wheat faces composite supply because the land cannot be used for both at the same time.

2.3.2 The law of supply, supply schedule and curve

The law of supply states that, **all other factors remaining constant (ceteris paribus), when the price of a good or service rises, the quantity supplied also rises, and when the price falls, the quantity supplied falls.** This shows a direct (positive) relationship between price and quantity supplied.

The law of supply explains how producers normally react to price changes. Producers aim to maximize profit. When prices are high, producing and selling more goods becomes more profitable, so producers are encouraged to increase output. When prices fall, profits decline, making production less attractive, and producers reduce the quantity supplied. As a result, higher prices lead to higher quantity supplied, while lower prices lead to lower quantity supplied.

The law of supply is commonly demonstrated using the following **illustrations: supply schedule and supply curve.**

a) The Supply Schedule

A **supply schedule** is a table that shows different prices of a commodity and the corresponding quantities supplied. It clearly illustrates that when price rises, quantity supplied increases, and when price falls, quantity supplied decreases. A supply schedule can be prepared for an individual producer or for the whole market.

An **individual supply schedule** is a table showing the quantities of a commodity that a single producer is willing and able to supply at different prices over a given period of time, assuming other factors remain constant. It shows how one producer's quantity supplied changes as the price changes.

Table 2.6 A hypothetical example of an individual supply schedule for Sugar

Price (FRW per kg)	Quantity Supplied (kg)
600	1
800	2
1,000	3
1,200	4
1,500	5

Explanation:

As the price of sugar increases, the producer supplies more sugar. This illustrates the law of supply at the level of a single producer.

A **market supply schedule** is a table that shows the total quantities of a commodity that all producers in a market are willing and able to supply at different prices over a given period of time. It is obtained by adding together the quantities supplied by all individual producers at each price.

Table 2.7 A hypothetical example of a market supply schedule for Sugar.

Price (FRW per kg)	Producer A (kg)	Producer B (kg)	Producer C (kg)	Market Supply (kg)
600	1	2	1	4
800	2	3	2	7
1,000	3	4	3	10
1,200	4	5	4	13
1,500	5	6	5	16

Explanation:

At each price, the market supply is the sum of all individual supplies. Like individual supply, market supply increases as price rises. b) The Supply Curve

A **supply curve** is a graphical representation of the relationship between the price of a good or service and the quantity supplied by producers over a given period of time, assuming other factors remain constant. It is drawn with price on the vertical axis (Yaxis) and quantity supplied on the horizontal axis (Xaxis).

The supply curve normally **slopes upward from left to right**, meaning that as price rises, quantity supplied increases, and as price falls, quantity supplied decreases, which shows the law of supply. The slope can be **linear or non-linear**: it is linear when supply increases at a constant rate with price, and non-linear when supply changes at different rates due to factors like rising costs or limited production capacity.

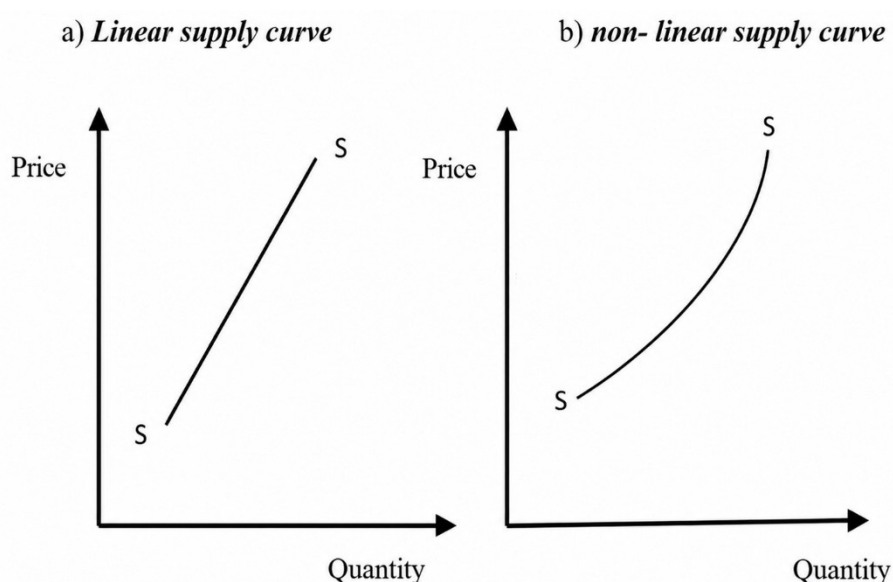
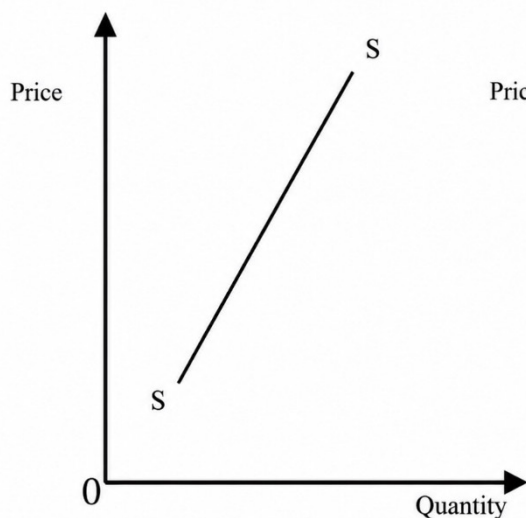


Fig 2.8 Shape of the supply curve

A supply curve can be drawn for **an individual producer or for the entire market**.

An individual supply curve shows the quantities of a commodity that one producer is willing and able to supply at different prices. It reflects the production behavior of a single seller.

a) Individual A's supply curve



b) Individual B's supply curve

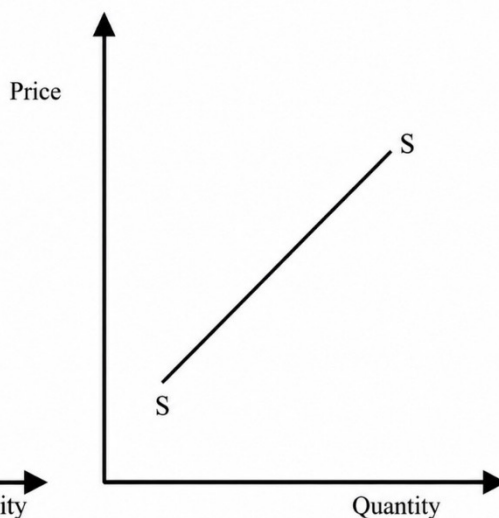


Fig 2.9 Individual supply curves

A market supply curve shows the total quantity of a commodity that all producers in a market together are willing and able to supply at different prices. It is formed by adding up (horizontally summing) all the individual supply curves at each price level. Because it combines the supply of many producers, the market supply curve is usually flatter than an individual supply curve. Like individual supply curves, the market supply curve slopes upward from left to right, showing that as price increases, the quantity supplied by producers also increases.

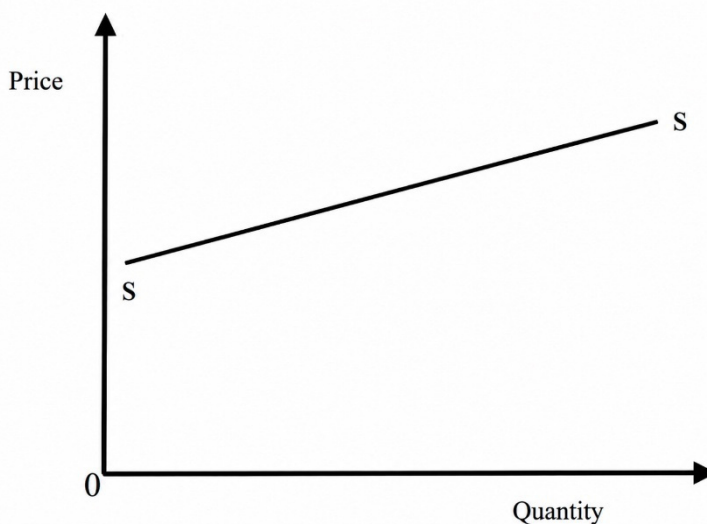


Fig. 2.10 A market supply curve

Exercise. 2.2

Use the table below to answer the questions that follow.

Hypothetical Supply Schedule (For beans)

Price per Unit	Quantity Supplied (kg)
1,000	10
2,000	20
3,000	30
4,000	40
5,000	50

- Plot the information on a graph.
- What shape is the curve obtained? Explain.

2.3.3 Abnormal supply curves.

Learning Activity 2.3.3



Read the scenario below and discuss the questions that follow.

During a harvesting season, traders expect the price of beans to rise sharply in the coming weeks because of shortages in nearby regions. When the price of beans begins to rise, instead of bringing more beans to the market as expected, many traders decide to store their beans, hoping to sell them later at even higher prices. As a result, less beans are supplied to the market when prices are high. Later, when the price of beans starts to fall due to imports from other areas, traders panic and rush to sell their stored beans to avoid losses. This causes more beans to be supplied to the market when prices are falling.

Questions

- Under normal conditions, what usually happens to the quantity supplied when the price of a commodity rises?
- In the scenario, why did traders decide to withhold beans from the market when prices were rising?
- What happened to the quantity of beans supplied when prices started to fall?
- How does the relationship between price and quantity supplied in this situation differ from the normal situation?
- Does this behavior follow or go against the normal law of supply? Explain briefly.
- What does this situation tell us about producer behavior in special or unusual circumstances?

In previous lessons, it was observed that producers normally supply more of a good when its price rises and less when the price falls. However, this is not always the case in real-life market situations. In some unusual circumstances, producers may behave differently, supplying less of a commodity when prices are high and more when prices are low. This lesson introduces abnormal supply curves, which explain these exceptional situations and show why producers sometimes respond to price changes in an unexpected way.

a) Meaning of abnormal supply curves:

An abnormal (or exceptional) supply curve is a supply curve that does not follow the normal law of supply. Normally, higher prices lead producers to supply more, and lower prices lead them to supply less, so the supply curve slopes upward. However, in abnormal situations, the quantity supplied behaves differently due to special conditions which include the following.

Fixed supply in the short run: Production and supply take time because producers must first organize resources. When the available time is too short to increase production, the quantity supplied remains constant even if the price rises, resulting in a fixed supply situation. In this case, the curve may appear as shown below.

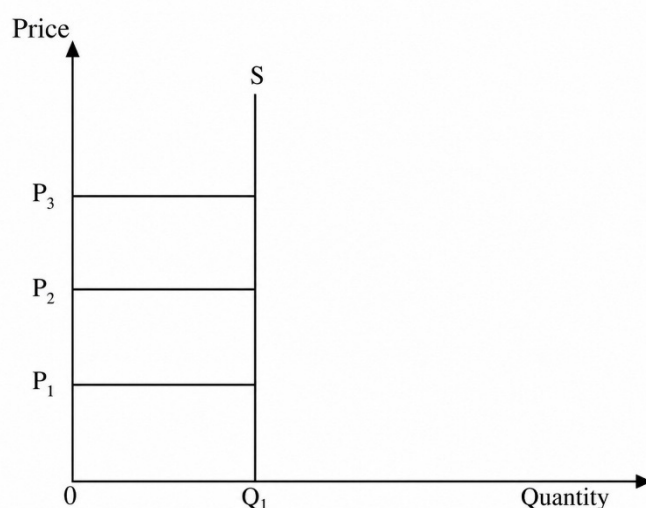


Fig 2.11 Fixed supply curve

As the price increases from OP_1 to OP_2 , and eventually to OP_3 , quantity supplied of the commodity remains constant because time is so short for the supplier to increase supply.

The supply of land in the short run: In the short run, land as a factor of production has a fixed supply. Land includes all natural resources such as soil, minerals, and rainfall. Because the amount of land cannot be increased, its supply remains constant regardless of changes in price, and therefore its supply curve is **perfectly inelastic**.

The backward-bending supply curve of labour: This occurs when workers supply fewer working hours as wages rise beyond a certain level. At high wages, workers prefer more leisure since they can earn enough income by working fewer hours, some stop working more once they reach their income targets, and others reduce work due to insecurity or unsafe conditions. As a result, higher wages can lead to a decrease in the quantity of labour supplied, causing the supply curve of labour to bend backward as illustrated as below;

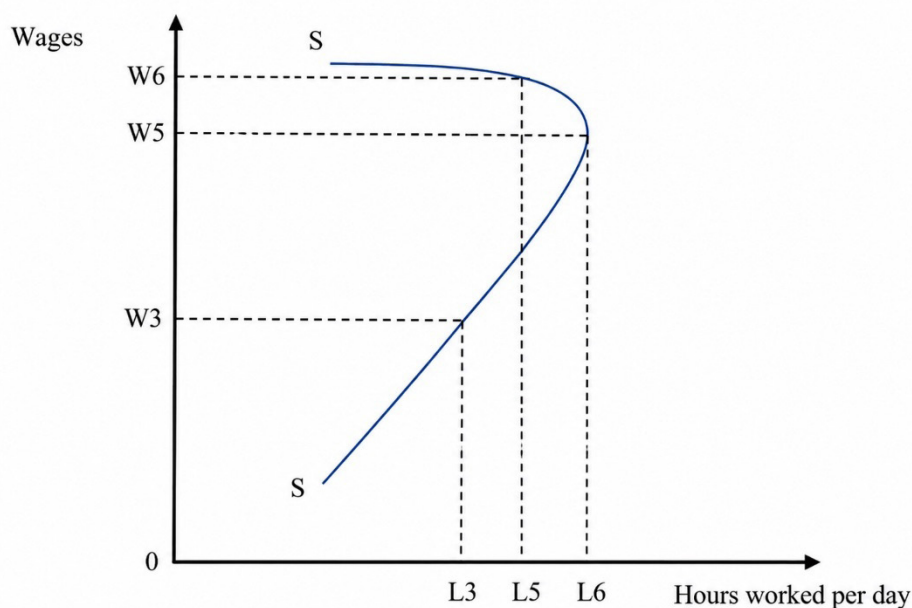


Fig 2.12 The backward bending supply curve of labour

From 0 to W3, an increase in wages to W5 leads workers to increase their working hours from L3 to L6, showing a normal response to higher wages. However, when wages rise further from W5 to W6, the hours worked fall from L6 to L5, meaning workers reduce labour despite higher wages.

Application activity 2.3.3



1. a) Distinguish between supply and quantity supplied.
b) Under what circumstances may the law of supply be violated?
2. Why do you think workers may withdraw from work even when their wages increase?

2.3.4 Factors that determine supply:



Introductory activity 2.3.4

Analyse the scenario below and answer the questions that follow:

In a small town, the quantity of maize, vegetables, milk, and fruits in the local market increased over time. Farmers began using better tools and improved methods, some expanded their farms due to easier access to land and labour, and more people joined farming after encouragement from local leaders. At the same time, good weather led to better harvests. Although a few farmers switched to producing other crops, the overall amount of goods supplied continued to rise even though prices remained unchanged.

- What things in the story helped farmers produce more goods for the market?
- How did farmers' methods or tools affect the amount they produced?
- What role did the weather play in the amount supplied?
- Why did more people decide to join farming?

The supply of a commodity is influenced not only by its price but also by several other factors. These factors are represented in the **supply function**, which is a mathematical expression that shows the relationship between quantity supplied and its determinants. The supply function can be expressed as:

$$Q_s = f(P, P_f, T, C, \dots)$$

Where:

Q_s = Quantity supplied	T = Technology
P = Price of the good	C = Cost of production
P_f = Prices of related goods	$\dots$ = Other factors affecting supply

The quantity of a commodity supplied depends on several factors as shown in the supply function. The major determinants of supply include the following:

- Price of the Commodity (P):** The price of the good itself is the most important determinant of supply. Generally, as the price of a commodity increases, producers are willing to supply more because production becomes more profitable. Conversely, when the price falls, the quantity supplied decreases.
- Cost of Production (C):** The cost of inputs such as raw materials, labour, and energy affects supply. An increase in production costs reduces profit, leading to a decrease in supply. On the other hand, lower costs encourage producers to supply more.
- Technology (T):** Improvements in technology make production more efficient and reduce costs. This increases output and encourages producers to supply more goods. Poor or outdated technology limits production and reduces supply.
- Prices of Related Goods (P_f):** The supply of a good may be affected by the prices of other goods that producers can produce. If the price of an alternative product increases, producers may switch to producing that good, reducing the supply of the original product.
- Number of Sellers:** An increase in the number of producers in the market leads to an increase in total supply. Conversely, when firms exit the market, supply decreases.
- Government Policy:** Government actions such as taxes, subsidies, and regulations affect supply. Taxes increase production costs and reduce supply, while subsidies lower costs and encourage more production.
- Expectations of Future Prices:** If producers expect prices to rise in the future, they may reduce current supply to sell later at higher prices. If they expect prices to fall, they may increase current supply.

- **Natural Factors:** Natural conditions such as weather, climate, and natural disasters strongly affect the supply of agricultural products. Favorable conditions increase supply, while unfavorable conditions reduce it.
- **Availability of Factors of Production:** The availability of resources such as labour, capital, and raw materials influences supply. When these factors are readily available, supply increases; when they are scarce, supply decreases.
- **Freedom of Entry into the Industry:** When it is easy for new firms to enter an industry, supply increases as more producers join. Restrictions or barriers to entry reduce supply.
- **Seasonal Changes:** Some goods, especially agricultural products, depend on seasons. Supply is higher during the appropriate season and lower during the off-season.
- **Gestation Period:** Goods that take a long time to produce (such as crops or large projects) cannot be supplied quickly, even when prices rise. This delays the response of supply.
- **Demand for the Commodity:** High demand for a commodity encourages producers to increase supply, while low demand discourages production and reduces supply.

2.3.5 Change in supply and change in quantity supplied.

Changes in supply and changes in quantity supplied explain how producers respond to different economic conditions. While a change in quantity supplied occurs due to a change in the price of a commodity, a change in supply results from other factors that affect producers' ability or willingness to supply goods.

a) Change in supply

A change in supply refers to an increase or decrease in the quantity of a good or service that producers are willing and able to offer for sale at the same price, **caused by factors other than the price of the good itself**. These factors include changes in the cost of production, technology, weather or seasons, number of producers, government policies, or prices of related goods. A change in supply is **shown by a shift of the supply curve**: a shift to the right shows an increase in supply, while a shift to the left shows a decrease in supply.

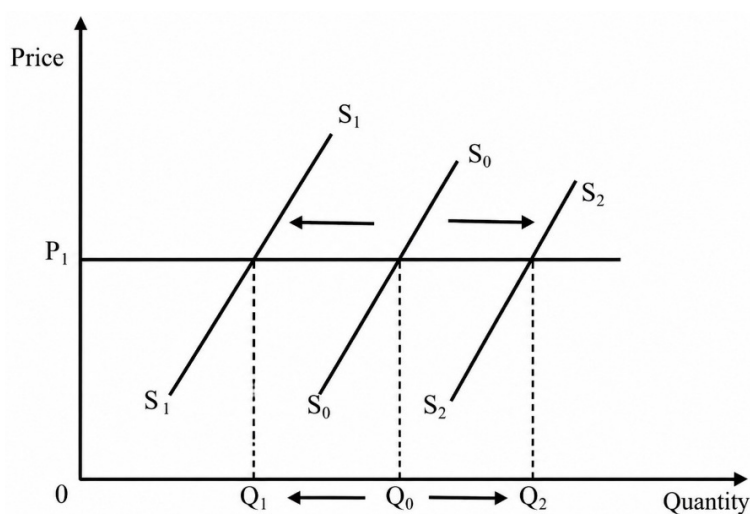


Fig 2.13 A change in supply curve

At price OP_1 , quantity supplied is OQ_0 . The price remains constant, but quantity supplied either decreases to OQ_1 and the supply curve shifts to S_1 or quantity supplied increases to OQ_2 and the supply curve shifts to S_2 . A change in supply may be an increase in supply and the supply curve shifts to the right or a decrease in supply and the supply curve shifts to the left.

b) Change in quantity supplied

A change in quantity supplied refers to an increase or decrease in the quantity of a good or service that producers are willing and able to sell **due to a change in its own price**, while all other factors remain constant. It is shown by a movement along the same supply curve, not by a shift of the curve. When the price of a good rises, producers are willing to supply more, leading to an extension of supply. When the price falls, producers supply less, leading to a contraction of supply. All other factors such as technology, costs of production, taxes, and number of producers remain unchanged.

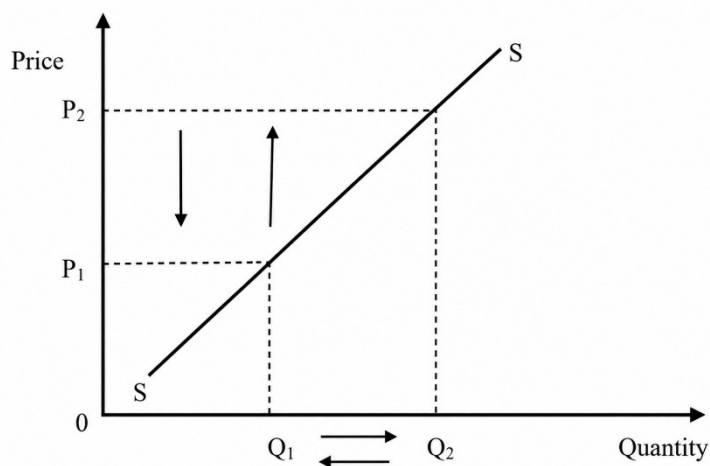


Fig 2.14 A change in quantity supplied curve

As shown, when the price rises from P_1 to P_2 , producers increase the quantity supplied from Q_1 to Q_2 because higher prices provide greater incentive and potential profit, making producers more willing to offer more goods for sale. Conversely, when the price falls from P_2 to P_1 , the quantity supplied decreases from Q_2 to Q_1 because lower prices reduce profit, discouraging producers from supplying large quantities. This increase or decrease in quantity supplied is represented by a movement along the same upward-sloping supply curve, not a shift of the supply curve itself, and it clearly demonstrates the law of supply.

Application activity 2.3.4



1. a) What is meant by supply function?
- b) Explain the factors that affect quantity supplied of a commodity.

2. In December last year, the price of milk in Kayonza remained constant at 400 FRW per liter. However, the amount of milk brought to the dairy rose from 100,000 liters to 180,000 liters per day.

- a) Discuss the factors that may have brought about this increase in the amount of milk supplied to the dairy.
- b) Illustrate the above phenomenon on a curve.

2.4 Equilibrium determination

Learning Activity 2.4



Analyse the table below and answer the questions that follow.

A hypothetical joint demand and supply schedule.

Price per kg of meat (FRW)	Quantity demanded of meat (kgs)	Quantity supplied meat (kgs)
5000	75	40
6000	70	46
7000	66	53
8000	55	55
9000	50	72
10000	44	85
11000	30	100

Questions;

1. At which price do buyers want to buy more than sellers want to sell?
2. At which price do sellers want to sell more than buyers want to buy?
3. Is there any price where buyers and sellers want to exchange the same quantity?
4. What do you think happens to the price when:
 - i. There are too many buyers but few goods?
 - ii. There are too many goods but few buyers?

In everyday markets, buyers want to pay low prices while sellers want to sell at high prices. This lesson will help you understand how a market reaches a price and quantity that both buyers and sellers accept. We will explore how prices change when there is too much or too little of a product, and how the market finally settles at a balance point.

2.4.1 Meaning of equilibrium, equilibrium price and quantity

Equilibrium is a market situation where the quantity demanded by consumers is equal to the quantity supplied by producers. At this point, there is no shortage and no surplus in the market.

The equilibrium price is the price that prevails in the market at which the quantity supplied by producers is exactly equal to the quantity demanded by consumers. At this price, all the goods brought to the market by sellers are fully purchased by buyers, so there are **no excess supply (surplus)** and **no excess demand (shortage)**. The market is therefore in a state of balance and stability. While the **equilibrium quantity** is the amount of a good or service bought and sold at the equilibrium price. It represents the actual quantity exchanged in the market when equilibrium is achieved.

The equilibrium price and equilibrium quantity occur only at the point where quantity demanded equals quantity supplied. However, when this balance does not exist, the market is said to be in **disequilibrium**. Disequilibrium can take two forms: **shortage or surplus**. A shortage (excess demand) occurs when the quantity demanded is greater than the quantity supplied, usually at a price lower than the equilibrium price, causing buyers to compete for the limited goods available. On the other hand, a **surplus (excess supply)** occurs when quantity supplied exceeds quantity demanded, often at a price higher than equilibrium, leading to unsold goods in the market. These situations create pressure for prices to change until equilibrium is restored.

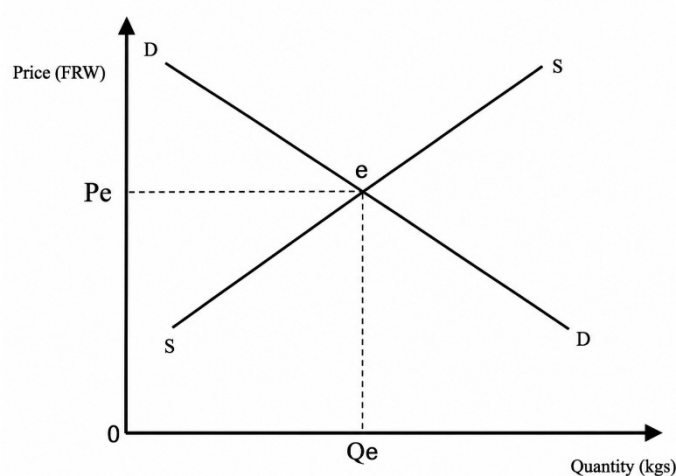


Fig 2.15 Equilibrium situation curve

The curve illustrates market equilibrium, where the downward-sloping demand curve (D) intersects the upward-sloping supply curve (S) at point **e**. This intersection determines the equilibrium price (**Pe**) and equilibrium quantity (**Qe**). At this point, quantity demanded equals quantity supplied, so the market is balanced with no surplus and no shortage.

2.4.2 Determining market equilibrium.

This lesson introduces the concept of market equilibrium, which occurs when demand and supply interact in the market. It explains how equilibrium price and quantity are determined through the forces of demand and supply, and how this balance reflects a situation where there is no tendency for change.

Understanding market equilibrium provides a foundation for analyzing how prices are set and how markets function in real-life situations. The determination of market equilibrium is carried out using two main methods: the use of joint demand and supply schedules, and the use of demand and supply equations, as shown below.

a) Using joint demand and supply schedule and curves.

Market equilibrium can be determined using a demand and supply schedule or graphically by the intersection of the demand and supply curves. This section explains how equilibrium price and equilibrium quantity are derived using demand and supply schedules and curves.

Deriving market equilibrium using joint demand and supply schedule

Table 2.8 Hypothetical Joint Demand and Supply Schedule (Exemple: Beans)

Price (FRW)	Quantity Demanded(kg)	Quantity Supplied (kg)
100	100	20
200	80	40
300	60	60
400	40	80
500	20	100

In this schedule, equilibrium occurs at a price of **300 FRW**, where quantity demanded equal quantity supplied (**60 kg**).

The above information can be illustrated on graphs as below;

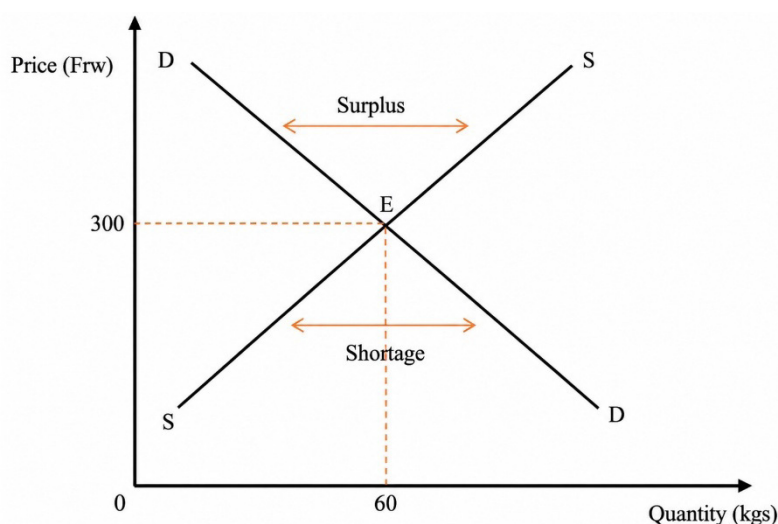


Fig 2.16 Equilibrium curve

The curve shows market equilibrium at point E, where demand equals supply at a price of 300 FRW and quantity of 60 kgs. Prices above equilibrium create a surplus (excess supply), while prices below equilibrium lead to a shortage (excess demand).

b) Determining market equilibrium using equations:

The market is in equilibrium when the quantity demanded equals the quantity supplied ($Q_d = Q_s$). We set the two equations equal to each other to solve for Price (P) to get equilibrium price (P_e). To find the equilibrium quantity, plug the equilibrium price (P_e value) back into either the demand or supply equation. Both should yield the same result.

For example, suppose we have the following market equations:

‣ **Demand:** $Q_d = 100 - 2P$

‣ **Supply:** $Q_s = 10 + 3P$

Step 1: Set the equations equal to each other

$$100 - 2P = 10 + 3P$$

Step 2: Solve for the Equilibrium Price (P_e)

Move the variables to one side and constants to the other:

$$100 - 10 = 3P + 2P$$

$$90 = 5P$$

$$P = 18$$

Thus, equilibrium price (P_e) = 18

Step 3: Solve for the Equilibrium Quantity (Q_e)

Plug $P_e = 18$ back into either original equation (let's use Demand):

$$Q_d = 100 - 2(18)$$

$$Q_d = 100 - 36$$

$$Q_d = 64$$

or

$$Q_s = 10 + 3(18)$$

$$Q_s = 10 + 54$$

$$Q_s = 64$$

Thus, equilibrium quantity (Q_e) = 64 since $Q_d = Q_s$

The market clears at a price of **18** with **64** units exchanged.

Exercise 2.3

Use the following equations to determine equilibrium price and quantity

i. $Q_d = 100 - 2P$

$Q_s = 20 + 3P$

ii. $Q_d = 120 - 2P$

$Q_s = 20 + 2P$

iii. In each situation plot your information on graphs to illustrate the equilibrium situation.

2.4.3 Effects of changes in demand and supply on equilibrium

Market equilibrium changes when either demand or supply shifts in the market. It can be an increase or decrease in demand and supply respectively.

a) Increase and decrease in demand

An increase in demand

Increase in demand shifts the demand curve to the right. It may be due to increase in the size of the population, increase in consumer's income, government subsidizing consumers, reduction in both direct and indirect taxes, reduction in the price of a complementary good, increase in the price of a substitute good, a favorable change in the buyers' tastes and preferences, or a favorable change in seasons. When the demand curve shifts to the right, the equilibrium is extended. This leads to an increase in the equilibrium price and equilibrium quantity.

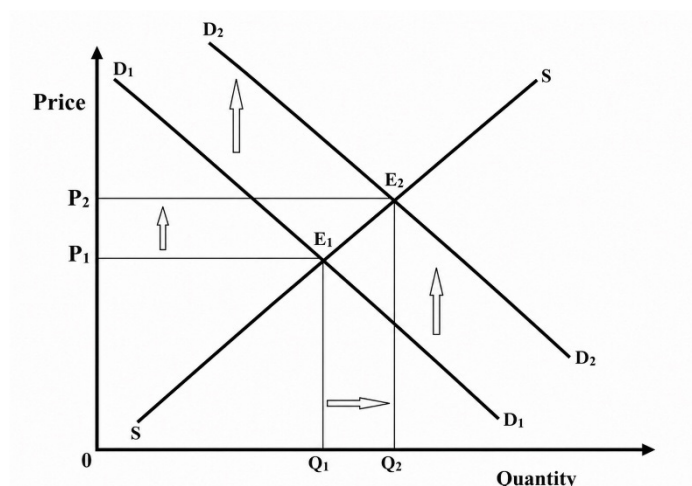


Fig 2.17 Effect of an increase in demand on equilibrium

When demand increases, the demand curve shifts to the right from demand curve D_1 to demand curve D_2 . The equilibrium point is extended from equilibrium point E_1 to E_2 . This leads to an increase in equilibrium price from OP_1 to OP_2 , and equilibrium quantity from OQ_1 to OQ_2 .

A decrease in demand.

A decrease in demand makes the demand curve to shift to the left. It may be due to decrease in the size of the population, decrease in consumer's income, reduction in government subsidies to consumers, increase in both direct and indirect taxes, increase in the price of a complementary good, decrease in the price of a substitute good, unfavorable changes in the buyers' tastes and preferences, or unfavorable change in seasons.

When the demand curve shifts to the left, the equilibrium is contracted. This leads to a decrease in the equilibrium price and the equilibrium quantity.

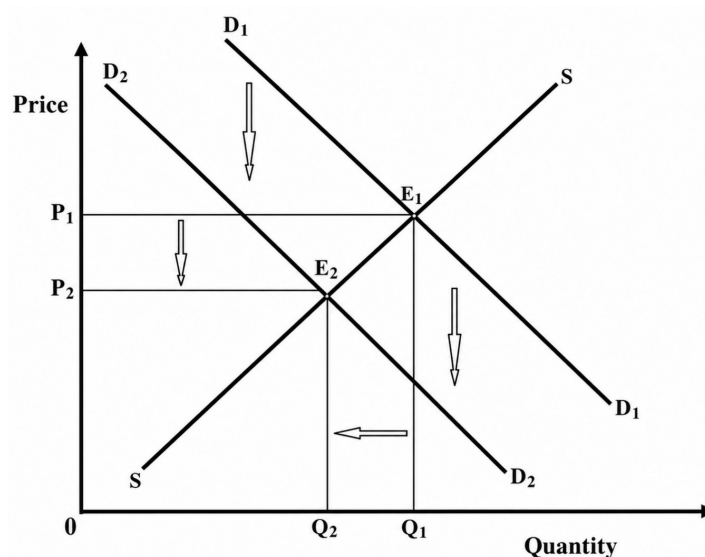


Figure 2.18 Effect of a decrease in demand on equilibrium.

When demand decreases, the demand curve shifts to the left from demand curve D_1 to demand curve D_2 . The equilibrium point is contracted from equilibrium point E_1 to E_2 . This leads to decrease in equilibrium price from OP_1 to OP_2 , and equilibrium quantity from OQ_1 to OQ_2 .

b) Increase and decrease in supply

An increase in supply

Increase in supply shifts the supply curve to the right. This will lead to increase in the equilibrium quantity and decrease in the equilibrium price. An increase in supply of a commodity may be brought about by an increase in the number of producers, improvement in technology, entry of new firms into the industry, decrease in taxes on producers which reduces their production costs, favorable natural factors, increase in demand for the commodity, change in goals of the firm, decrease in prices of factors of production, or increase in price of jointly supplied commodities.

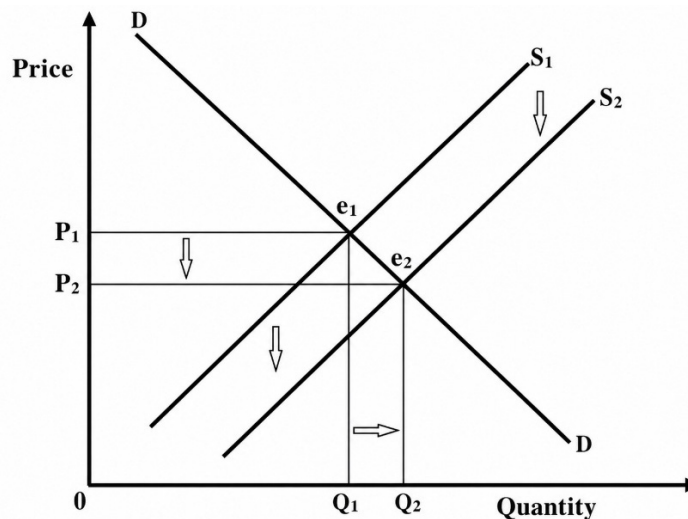


Figure 2.19 Effect of an increase in supply on equilibrium

When supply increases, the supply curve shifts to the right from S_1 to S_2 . Similarly, the equilibrium point is extended from e_1 to e_2 . This leads to increase in equilibrium quantity from OQ_1 to OQ_2 and a decrease in equilibrium price from OP_1 to OP_2 .

A decrease in supply

Decrease in supply shifts the supply curve to the left. The equilibrium point contracts to the left. This will lead to decrease in the equilibrium quantity and increase in the equilibrium price. A decrease in supply of a commodity may be brought about by a decrease in the number of producers, decline in technology, exit of new firms from the industry, increase in taxes on producers which increases their production costs, unfavorable natural factors, decrease in demand for the commodity, change in goals of the firm, increase in prices of factors of production, or decrease in price of jointly supplied commodities.

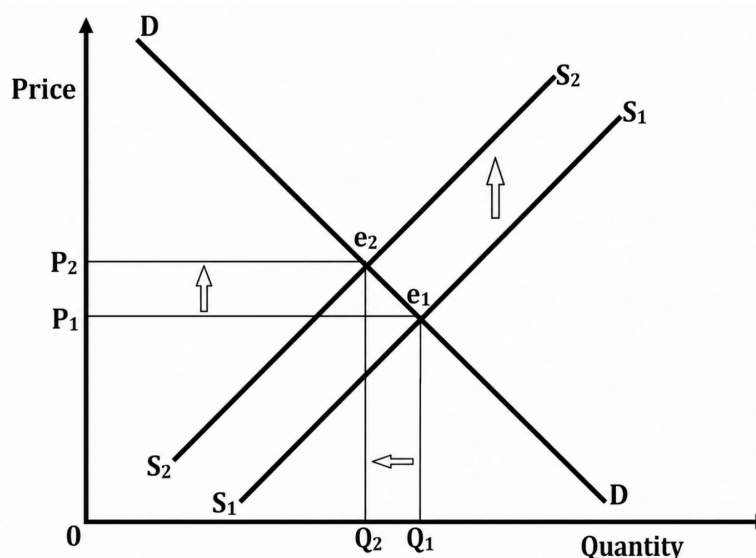


Figure 2.20 Effect of a decrease in supply on equilibrium.

When supply decreases, the supply curve shifts to the left from S_1 to S_2 . Similarly, the equilibrium point contracts from e_1 to e_2 . This leads to a decrease in equilibrium quantity from OQ_1 to OQ_2 and an increase in equilibrium price from OP_1 to OP_2 .

c) Change in demand and supply by the same margins

When both demand and supply change by the same margins and in the same direction, the equilibrium point changes. However, the equilibrium price remains almost the same. The equilibrium quantity either increases or decreases.

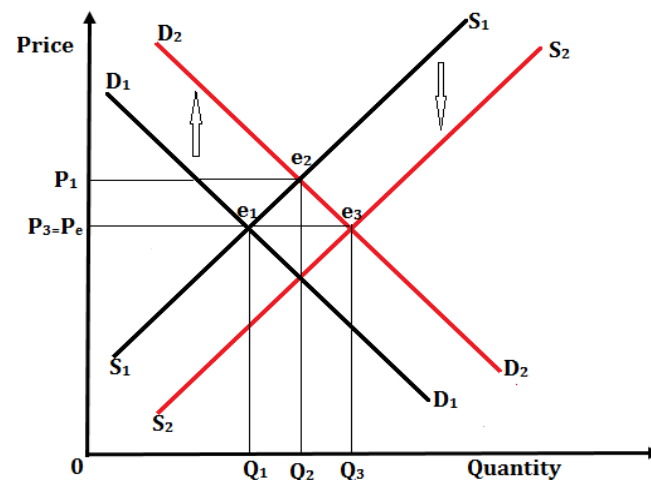


Figure 2.21 Effect of an increase in demand and supply on equilibrium.

As illustrated, when demand increases from D_1 to D_2 while supply decreases from S_2 to S_1 , equilibrium shifts from e_1 to e_2 , raising the equilibrium price from P_e to P_1 and the equilibrium quantity from Q_1 to Q_2 . When demand and supply both increase by the same magnitude ($D_1 \rightarrow D_2$ and $S_1 \rightarrow S_2$), equilibrium shifts from e_1 to e_3 , increasing quantity from Q_1 to Q_3 while leaving the equilibrium price unchanged at P_e . ($P_e = P_3$).

Application activity 2.4



1. a) Given the equations: $Q_d = 80 - 4P$ and $Q_s = 20 + 2P$. Solve for the equilibrium price and quantity.
b) Draw and illustrate a demand and supply curve showing the equilibrium point clearly labeled.
c) Suppose the price is 10, explain what would happen in the market.
2. In Rwanda, a popular new soft drink becomes trendy among young people. At the same time, the government reduces taxes on soft drinks, making them cheaper for consumers.
 - a) What do you think will happen to the demand for the soft drink when it becomes more popular and cheaper? Explain your reasoning.
 - b) If people suddenly lose interest in the soft drink or their incomes decrease, how would this affect its demand in the market? Explain your answer.
3. Farmers in Rwanda adopt modern farming technology that increases crop production.
 - a) Explain how this affects the supply curve.
 - b) Describe the change in equilibrium price and quantity.

**Knowledge and understanding questions**

1. a) Define price and explain its importance in a market economy
b) Explain the meaning of a market and describe any two types of markets you know.
2. a) Distinguish between a change in quantity demanded and a change in demand.
b) Explain the factors that may lead to a change in demand
3. a) Define supply and explain the factors that influence the supply of a commodity.
b) Differentiate between a change in quantity supplied and a change in supply.

Skills questions

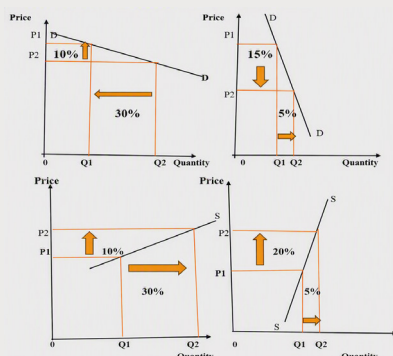
4. Explain the meaning of market equilibrium and describe how it is determined through the interaction of demand and supply.
5. Imagine a demand schedule on your own, draw a demand curve and explain its slope.
6. Using a supply curve, illustrate the effect of improved technology on supply and explain your diagram.
7. Draw a simple demand and supply diagram to illustrate how price is determined in a market. Explain your diagram.

Values and attitudes questions:

8. Explain why consumers should compare prices before buying goods in a market.
9. Explain why the government may control prices of essential goods like fuel or food.
10. Do you think bargaining (haggling) is a good practice in markets? Give reasons for your answer.

UNIT 3

ELASTICITY



Key Unit Competence:

Apply demand and supply responsiveness to make informed everyday economic decisions.

Learning objectives

Knowledge and understanding

1

- Define elasticity and distinguish between its main types, namely elasticity of demand and elasticity of supply.
- Describe elasticity of demand and identify its types, including price elasticity of demand, income elasticity of demand, and cross elasticity of demand.
- Describe price elasticity of demand (PED) in terms of its meaning, formula, categories, determinants, and uses.
- Describe income elasticity of demand (YED) and cross elasticity of demand (CED) with reference to their meanings, formulas, and interpretation of results.
- Identify the determinants of elasticity of demand.
- Define elasticity of supply and outline the factors that influence elasticity of supply.
- Identify and describe the categories of price elasticity of supply.

Skills

2

- Calculate elasticity values of demand and supply accurately using given data.
- Interpret elasticity coefficients to explain consumer and producer responsiveness to changes in price and income.
- Apply price elasticity of demand to explain real-life situations such as price increases, discounts, and taxation.
- Use income and cross elasticity concepts to analyze changes in consumption patterns and relationships between goods.
- Analyze supply responsiveness to price changes using elasticity of supply concepts.

- Make informed everyday economic decisions as consumers or producers based on elasticity information.
- Explain market reactions to price and income changes using elasticity concepts.

Attitudes and values

3

- Appreciate the importance of elasticity in understanding market behavior and price changes.
- Demonstrate rational decision-making when responding to changes in prices and income.
- Value informed consumer and producer choices based on demand and supply responsiveness.
- Develop awareness of market sensitivity to economic changes and shocks.
- Show responsibility and fairness in pricing, consumption, and production decisions.
- Recognize the role of elasticity analysis in supporting efficient resource allocation and economic stability.

Introduction

In the study of demand and supply, it has been observed that changes in price, income, and other factors lead to adjustments in the quantities bought and sold. However, these adjustments are not always the same; some goods respond strongly to such changes, while others change very little. This variation in responsiveness is an important feature of real markets and helps in understanding everyday economic behaviour. In this unit, this idea is explored further by examining how responsive buyers and sellers are under different conditions, providing deeper insight into market dynamics, price determination, and how individuals and firms adjust to changing economic circumstances.



Introductory activity

Analyse the schedules below and answer the questions that follow:

Demand Schedule A

Price of commodity V in FRW	Quantity demanded of commodity V in kgs
1000	200
5000	198

Demand schedule B

Price of commodity X in FRW	Quantity demanded of commodity Y in kgs
1000	250
1050	200

1. Compare how quantity demanded relates to change in price in each of the above demand schedules. What do you derive from your comparison?
2. Illustrate an appropriate demand curve for each schedule. Compare the slopes of your demand curves and make a clear comment on each.
3. Identify and discuss the market conditions that can explain why quantity demanded may respond to changes in price as shown in each of your curves.

3.1 Meaning and types of elasticity

Learning Activity 3.1



Analyse the scenario below and answer the questions that follow:

In a local market, the price of bottled juice increased from 1,000Frw to 1,500Frw. Many customers stopped buying juice and chose water instead. At the same time, juice producers increased the amount of juice supplied to shops because they wanted to earn more profit.

Questions

1. When the price of juice increased, did customers reduce their purchases a little or a lot? What does this tell you about how sensitive they are to price changes?
2. Why did many customers quickly switch from juice to water when the price went up? What does this suggest about the availability of alternatives?
3. If a small change in price leads to a big change in buying behavior, what might this tell you about how people respond to price?
4. When producers increased supply after the price rose, were they reacting strongly or slightly to the price change? What does this tell you about their response to price of juice?

3.1.1 Meaning of elasticity

Elasticity refers to the degree of responsiveness of one variable to changes in another variable. In economics, it specifically measures how a dependent variable responds when an independent variable changes. An independent variable is a factor that influences another variable for example, price of a commodity, consumer's income, price of related commodities etc. while a dependent variable is the outcome that changes as a result of this influence for example demand, supply etc. It quantifies the degree of "stretch" or sensitivity in the market, moving beyond the mere direction of market shifts to measure the exact magnitude of the reaction.

3.1.2 Types of elasticity

Economic responsiveness varies depending on what causes the change in the market. Buyers and sellers may react differently to changes in price, income, or the prices of related goods. To better understand these varied reactions, economists group them into different types of elasticity based on the specific factor involved. Studying these types helps explain how market behavior changes under different economic conditions. There are basically two types of elasticity:

- i. **Elasticity of demand:** measures how responsive the quantity demanded of a good or service is to changes in factors that determine demand such as its price, income, or related goods.
- ii. **Elasticity of supply:** measures how responsive the quantity supplied of a good or service is to changes in factors that affect supply such as its price or other production factors.

3.2 Elasticity of demand.

Learning Activity 3.2



The price of a bottle of cooking oil rises from 1000 RWF to 1200RWF, and as a result, the quantity demanded decreases from 50 bottles to 40 bottles. Calculate:

1. The change in price and quantity demanded.
2. Percentage change in price and quantity demanded.

Elasticity of demand focuses on understanding how strongly consumers react to changes in market conditions. It helps us examine whether buyers respond slightly or significantly when factors like price or income change, giving deeper insight into consumer behaviour in different situations.

3.2.1 Meaning and types of elasticity of demand

i. Meaning of elasticity of demand.

Elasticity of demand is a measure of the degree of responsiveness of quantity demanded of a commodity to changes in factors that affect demand. It is an economic measure that shows how sensitive consumers are to changes in market conditions. It explains how much the quantity demanded of a good or service increases or decreases when factors such as its price, consumer income, or the price of related goods change. While the law of demand shows the direction of change, elasticity of demand measures the **extent of that change**.

Elasticity of demand measures how responsive quantity demanded is to changes in its determining factors, expressed in percentage terms. It shows the percentage change in quantity demanded resulting from a percentage change in factors such as price, income, or the prices of related goods.

$$\text{Elasticity of demand} = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in any of its determining factors}}$$

ii. Types of elasticity of demand

The main types of elasticity of demand are classified according to the factor that causes change in demand:

- Price elasticity of demand (P.E.D): Measures how quantity demanded responds to changes in the **price of the good itself**.
- Income elasticity of demand (Y.E.D): Measures how demand changes when **consumer income changes**.
- Cross elasticity of demand. (C.E.D): Measures how demand for one good change due to a change in **the price of a related good** (substitutes or complements).

3.2.2 Price elasticity of demand (PED)

i. Meaning of price elasticity of demand

Price elasticity of demand refers to a measure of the degree of the responsiveness of quantity demanded of a commodity to changes in its own price. When the price of a commodity increases or decreases, its quantity demanded reacts to a certain extent. Quantity demanded may reduce, increase or even remain constant. Price elasticity of demand therefore measures the extent of this reaction. It shows the responsiveness of consumers to price changes.

It is given by

$$PED = \frac{\text{Percentage change in Quantity demanded}}{\text{Percentage change in price}} = \frac{\Delta Q}{Q} \times \frac{\Delta P}{P} \times \frac{100}{100} =$$
$$= (-) \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Where;

PED price elasticity of demand,

ΔQ Change in quantity.

ΔP Change in Price.

Q is the initial quantity.

P is the initial price.

Because of the law of demand whenever price increases quantity demanded reduces and whenever price reduces quantity demand increases. This means that there is always an inverse relationship between demand and price. This would make the coefficient of elasticity always negative. It is also observed by the slope of the demand curve.

ii. Calculating price elasticity of demand

In this lesson, we move beyond understanding how demand responds to price changes and focus on measuring that response mathematically. By learning how to calculate price elasticity of demand, we are able to determine the exact degree to which quantity demanded changes when price changes, allowing for more precise

analysis of consumer behavior and market reactions.

Elasticity has no units because it is expressed as a ratio of percentages, and its value ranges from zero to infinity. Although it may be calculated as a negative number due to the inverse relationship between price and quantity demanded, the negative sign is ignored and elasticity is expressed in absolute terms. For example, if **PED = -2**, it is written as $-2 = 2$, focusing only on the magnitude of responsiveness rather than its direction.

Worked out example 1:

The price of a good increases from 100 FRW to 120 FRW, and the quantity demanded falls from 50 units to 30 units. Calculate the price elasticity of demand.

Solution:

$$\text{PED} = (-) \frac{\Delta Q}{\Delta P} \times \frac{Q}{P}$$

Calculate the change in quantity demanded: new quantity – old quantity

$$\Delta Q = 30 - 50 = -20 \text{ units}$$

Initial Quantity (Q) = 50 units.

Change in price: new price – old price

$$\Delta P = 120 - 100 = 20 \text{ Frw}$$

Initial Price (P) = 100

Substitute the values in the formula.

$$\text{PED} = (-) \frac{\Delta Q}{\Delta P} \times \frac{Q}{P}$$

$$\text{PED} = (-) \frac{-20}{20} \times \frac{100}{50}$$

$$\text{PED} = (-) \frac{\Delta Q}{\Delta P} \times \frac{Q}{P}$$

$$\text{PED} = (-) -2$$

$$\text{PED} = 2$$

This implies when price increased, demand fell sharply, showing that the good is sensitive to price changes.

Worked out example 2:

Suppose a bakery in Rwamagana sells custom packages of local biscuits. When the price increases from 2000Frw to 2500Frw per package, the customers demand falls for the biscuits, falls by 10%. Calculate the price elasticity of demand.

Solution:**Percentage change in Quantity (% ΔQ) = -10**

We need to calculate percentage change in price.

$$\% \text{ change in price} = \frac{\text{New price} - \text{initial price}}{\text{Initial price}} \times 100$$

$$\% \text{ change in price} = \frac{2500 - 2000}{2000} \times 100$$

% change in price = 25%

$$\text{PED} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$\text{PED} = \frac{-10}{25}$$

$$\text{PED} = -0.4$$

Express in absolute terms

$$\text{PED} = |-0.4| = 0.4$$

Final Answer

$$\text{PED} = 0.4$$

This means customers are not very responsive to price changes, so the increase in price causes only a small decrease in demand.

Exercise 3.1:**Analyse and work out the following:**

- i. The price of a product increases by 20%, and as a result, the quantity demanded falls from 50 units to 40 units. Calculate the price elasticity of demand.
- ii. The price of a life-saving drug increases from 500 FRW to 1000 FRW, but the quantity demanded remains 20 units. Calculate the price elasticity of demand.
- iii. The price of a product remains constant at 2000 FRW, but the quantity demanded changes from 30 units to 60 units. Calculate the price elasticity of demand.

iii. Categories of Price elasticity of Demand

In the previous lesson, we learned how to calculate price elasticity of demand and measure the responsiveness of quantity demanded to changes in price. Building on this, we now examine the different forms of price elasticity of demand based on the values obtained, which helps us interpret these results more clearly. These forms show how sensitive consumers are to price changes whether demand is highly responsive, less responsive, or completely unresponsive, thereby enhancing our understanding of consumer behavior in the market. There are five main forms of price elasticity of demand which include:

a) Perfectly inelastic demand ($PED = 0$):

This is a rare situation where a change in price has absolutely no effect on the quantity demanded; consumers continue to buy exactly the same amount regardless of how high or low the price becomes. This typically applies to goods that are essential for survival and have no substitutes. A real-world example is a unique life-saving drug or an emergency medical treatment that patients must obtain regardless of its cost. This makes the demand curve a perfectly vertical straight line parallel to the Y-axis.

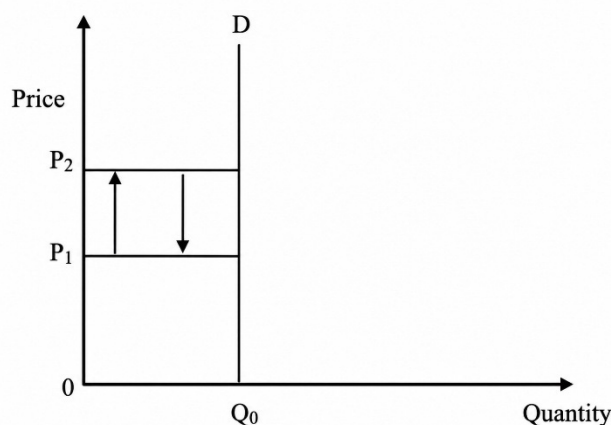


Illustration 3.1 perfectly inelastic demand curve

As price increases from OP_1 to OP_2 (a bigger change), quantity demanded remains constant at OQ_1 .

b) Inelastic demand ($0 < PED < 1$):

This is a situation where consumers do not significantly change the amount they buy when prices change; the percentage change in quantity demanded is smaller than the percentage change in price. This usually occurs with essential goods, habit-forming products, or items with few or no close substitutes, such as basic food staples (like maize flour or rice), fuel, electricity, and water. Even if prices rise sharply, people reduce their consumption only slightly because these goods are necessary for daily life. On a graph, this behavior is represented by a steeply sloping demand curve.

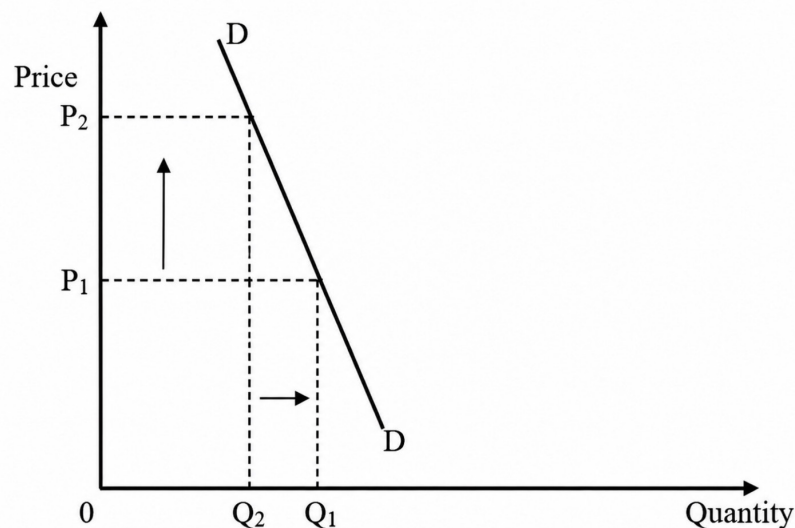


Illustration 3.2 inelastic demand curve

The graph shows a steep downward-sloping demand curve (D), indicating inelastic demand. When the price increases from P_1 to P_2 , the quantity demanded falls slightly from Q_1 to Q_2 . When the price decreases from P_2 to P_1 , the quantity demanded rises only slightly from Q_2 to Q_1 . This small change in quantity compared to the larger change in price shows that demand is relatively unresponsive to price changes.

c) Unitary elastic demand (PED= 1):

This occurs when the percentage change in quantity demanded is exactly equal to the percentage change in price. For example, a 10% increase in price leads to a 10% decrease in quantity demanded, and a 10% decrease in price leads to a 10% increase in quantity demanded. As a result, total consumer expenditure remains constant, and this relationship is shown by a downward-sloping curved line called a rectangular hyperbola.

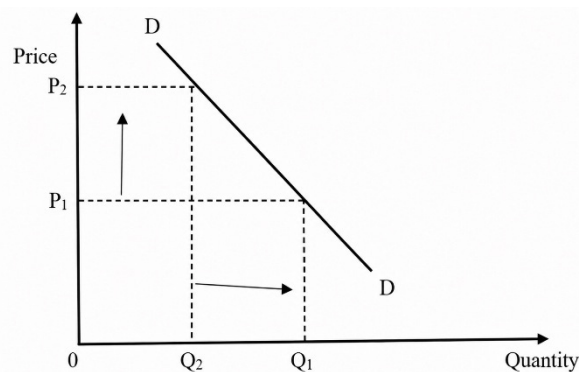


Illustration 3.3 unitary elastic demand curve

When the price increases from P_1 to P_2 , the quantity demanded decreases by an equal proportion from Q_1 to Q_2 . Similarly, when the price decreases from P_2 to P_1 , the quantity demanded increases by the same proportion from Q_2 to Q_1 . This equal percentage change in both price and quantity demanded means that consumers adjust their buying exactly in response to price changes, and total expenditure remains constant along the curve.

d) Elastic demand ($1 < PED < \infty$):

This occurs when consumers are very responsive to price changes, meaning the percentage change in quantity demanded is greater than the percentage change in price. For example, a small increase in price, such as 10%, can lead to a much larger decrease in quantity demanded, while a price decrease causes a more than proportional increase in demand. This is common for luxury goods, non-essential items, or products with many close substitutes, as consumers can easily switch to alternatives. Because of this high responsiveness, the demand curve appears very flat, showing that quantity demanded changes greatly with small price movements.

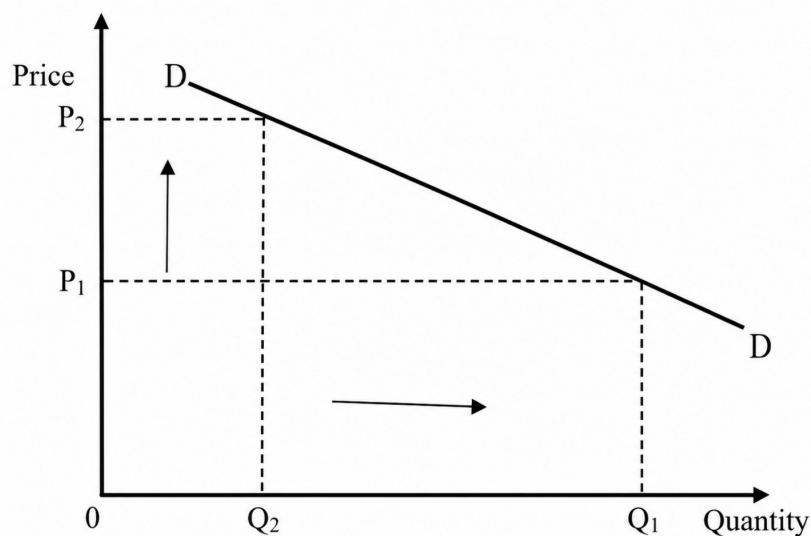


Illustration 3.4 elastic demand curve

The graph shows a flat, gently downward-sloping demand curve (D), representing elastic demand. When the price increases from P_1 to P_2 , the quantity demanded falls sharply from Q_1 to Q_2 . Conversely, when the price decreases from P_2 to P_1 , the quantity demanded rises significantly from Q_2 to Q_1 . This large change in quantity compared to the smaller change in price shows that consumers are highly responsive to price changes, which is the key feature of elastic demand.

e) Perfectly elastic demand ($PED = \infty$):

This describes an extreme situation where consumers are only willing to buy a good at one specific price, and any change in price causes a drastic reaction. If the price increases even slightly above the market price, the quantity demanded immediately drops to zero because consumers can easily switch to identical alternatives. On the other hand, if the price remains at that exact level, consumers are willing to buy any amount (theoretically unlimited). This typically occurs in perfectly competitive markets with identical goods, such as farmers selling the same crop in one market. On a graph, this behavior is shown as a perfectly horizontal demand curve, indicating that quantity demanded is extremely sensitive to even the smallest price change.

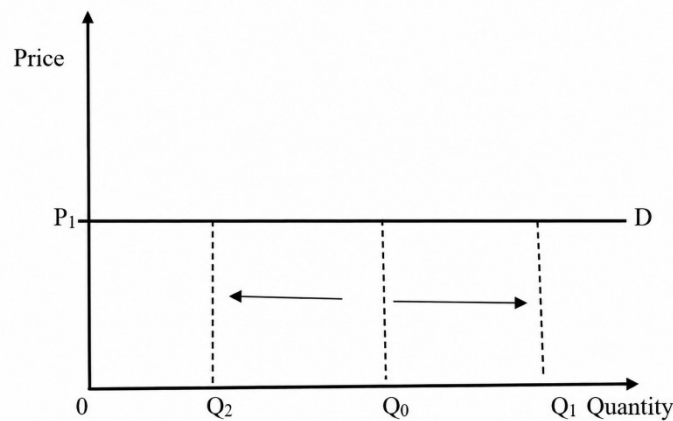


Illustration 3.5 perfectly elastic demand curve

The graph shows a perfectly horizontal demand curve (D) at price P_1 . Quantity demanded can change from Q_2 to Q_0 to Q_1 without any change in price, meaning consumers will buy any quantity at P_1 , but if the price rises even slightly above P_1 , demand drops to zero. <https://courses.lumenlearning.com/wm-microeconomics/chapter/categories-of-elasticity/>

Summary of the forms of PED

Form / Degree of Elasticity	Coefficient Value	Consumer Responsiveness	Shape of Demand Curve
Perfectly Inelastic	$PED = 0$	Completely Unresponsive	Vertical Line
Inelastic	$(0 < PED < 1)$	Relatively Unresponsive	Steep Slope
Unitary Elastic	$PED = 1$	Perfectly Proportional	Rectangular Hyperbola
Elastic	$(1 < PED < \infty)$	Highly Responsive	Flat Slope
Perfectly Elastic	$PED = \infty$	Infinitely Responsive	Horizontal Line

Exercise 3.2

Analyse and calculate the answers to the following questions:

1. A farmer sells identical potatoes at 30 francs per unit, the same as all other sellers. Despite the price remaining at 30 francs, the quantity demanded falls from 50 units to 30 units. Calculate the PED and identify the type of elasticity. Calculate the PED and identify the type of elasticity.
2. The price of a branded shirt increases by 25% increase, and the quantity demanded drops from 120 units to 60 units. Calculate the PED and classify the demand.

3. The price of a life-saving drug increases from 10 to 15 francs, but the quantity demanded remains constant at 100 units. Calculate the PED and identify the type of elasticity.
4. The price of maize flour rises from 20 to 25 francs, and the quantity demanded falls by 10%. Calculate the PED and state the type of demand.
5. The price of a product increases from 50 to 55 francs, and the quantity demanded decreases from 100 units to 90 units. Calculate the PED and identify the elasticity.

iv. Methods of measuring price elasticity of demand (PED).

While classifying demand as elastic or inelastic gives a general idea, economists need exact values to measure it properly. They use methods of calculating Price Elasticity of Demand (PED) based on changes in price and quantity demanded. These calculations show how responsive consumers are and help businesses and governments make better decisions. The two main methods used to measure Price Elasticity of Demand (PED) are point elasticity and arc elasticity.

a) Point elasticity of demand: refers to the measurement of price elasticity at a specific point on the demand curve, providing a very precise and accurate estimate of how quantity demanded responds to a small change in price at that exact point. It is mainly used when changes in price and quantity are minimal, allowing economists to determine elasticity with a high level of accuracy at that particular position on the curve. It can be calculated using the formula below:

$$\text{Point of Elasticity} = \frac{\% \Delta Q}{\% \Delta P} = \frac{\frac{Q_2 - Q_1}{Q_1} \times 100}{\frac{P_2 - P_1}{P_1} \times 100} = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

Where;

P1, Q1 = Initial Price and Quantity

P2, Q2 = New Price and Quantity

ΔP, ΔQ = Change in Price (P2 - P1 and Change in Quantity (Q2 - Q1)

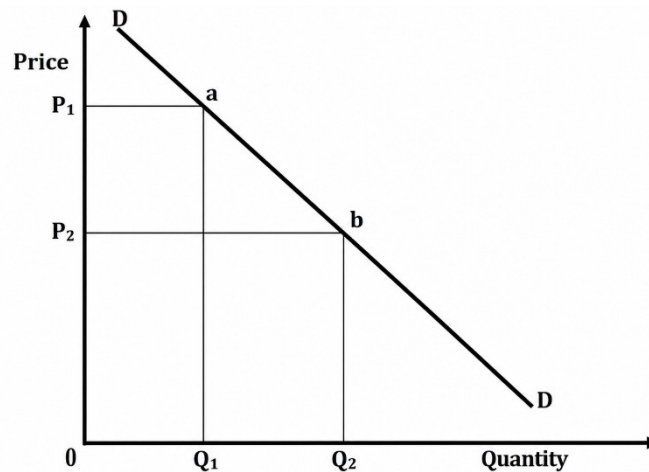


Fig 3.6 Point elasticity of demand:

If prices reduce from p_1 to P_2 , we measure elasticity of demand at Point "a" i.e. the starting point and if prices change from P_2 to P_1 , we measure elasticity of demand at point "b" i.e. point elasticity is measured for the point that represents the initial price before the change.

Example 1:

Price falls from 10 to 9 francs, and quantity demanded rises from 50 to 55 units. Calculate point elasticity of demand.

Solution

$$\Delta Q = 55 - 50 = 5$$

$$Q_1 = 50$$

$$\Delta P = 9 - 10 = -1$$

$$P_1 = 10$$

Substitute in the formula

$$\text{Point elasticity of demand} = \frac{\Delta Q}{\Delta P} \times \frac{P_1}{Q_1}$$

$$PED = -1$$

$$PED = | -1 |$$

PED = 1 (unitary elastic demand)

b) Arc elasticity of demand: This refers to the measurement of price elasticity over a segment (range) of the demand curve rather than a single point, by using the average (midpoint) of the initial and final prices and quantities. It is particularly useful when there are large changes in price and quantity demanded, as it reduces bias and provides a more accurate and reliable measure of responsiveness between two different price–quantity combinations. It can be calculated using the formula below:

$$\text{Arc elasticity of demand} = \frac{\frac{Q_2 - Q_1}{(Q_1 + Q_2)/2}}{\frac{P_2 - P_1}{(P_1 + P_2)/2}} = \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_1 + P_2}{Q_1 + Q_2}$$

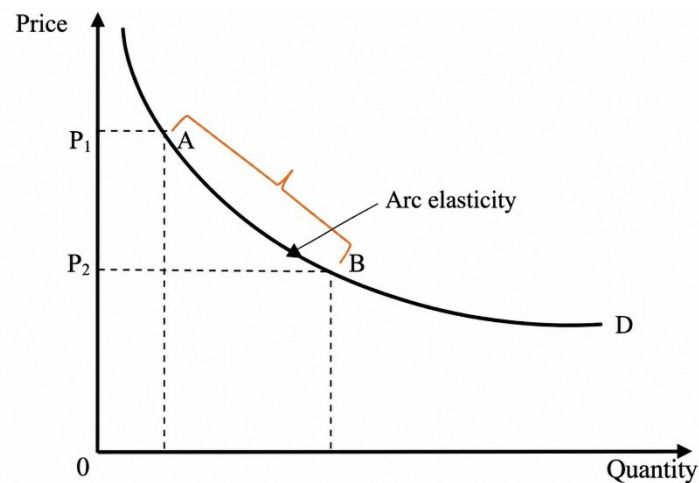


Fig 3.7 Arc elasticity of demand

The curve illustrates arc elasticity of demand between points A and B on the demand curve. As price falls from P_1 to P_2 , quantity demanded increases from A to B, and elasticity is measured using the average of price and quantity, making it more accurate. Because of this averaging method, the elasticity is the same from A to B as from B to A, ensuring a consistent measure regardless of direction.

Example 2

A diagnostic medical clinic in Kigali increases the price of a voluntary wellness checkup scan from 40,000Frw to 60,000Frw, and as a result, the number of patients falls from 200 to 180 patients. Calculate Arc elasticity of demand.

Solution:

$$\Delta Q = 180 - 200 = -20$$

$$\Delta P = 20000$$

$$P_1 = 40,000 \quad P_2 = 60,000$$

$$Q_1 = 200 \quad Q_2 = 180$$

$$\text{Arc elasticity of demand} = \frac{Q_2 - Q_1}{P_2 - P_1} \times \frac{P_1 + P_2}{Q_1 + Q_2}$$

$$\text{Arc elasticity of demand} = \frac{-20}{20,000} \times \frac{40,000 + 60,000}{200 + 180}$$

$$\text{Arc elasticity of demand} = \frac{-20}{20,000} \times \frac{100,000}{380}$$

$$\text{PED}_{\text{Arc}} = -0.001 \times 263.15 = -0.26315$$

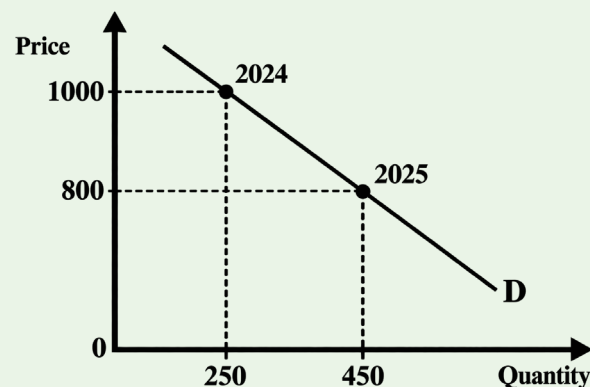
$$\text{PED}_{\text{Arc}} = \approx -0.26$$

$$\text{PED} = |0.26|$$

PED_{Arc} = 0.26. Demand is inelastic, because PED is less than 1. This means even with a large price increase, the clinic retained most of its patients because the scan has few close alternatives.

Exercise 3.3

1. use the curve below to answer the questions that follow:



- Calculate the arc elasticity of demand between the points for 2024 and 2025.
- Calculate the point elasticity of demand at the 2025 point.
- Based on your answers, state the form of price elasticity of demand at the calculated point and over the arc

v. Determinants of price elasticity of demand

Before, you have learned how to measure Price Elasticity of Demand (PED) and classify it into categories such as elastic, inelastic, and unitary. However, an important question remains: what actually causes demand to be more or less responsive to

price changes? To answer this, economists study the determinants of PED the key factors that influence how consumers react when prices change. Understanding these determinants helps explain why some goods have elastic demand while others are inelastic, making it easier for businesses and policymakers to make informed decisions. These factors include;

- **Availability of close substitutes:** The more substitutes a good has, the more elastic its demand because consumers can easily switch when the price rises. For example, if the price of Inyange juice increases, many people may switch to Sina (Urwibutso) juice. However, goods with no substitutes, such as tap water supplied by WASAC, have inelastic demand because people have no alternative.
- **Nature of the good (Necessity vs Luxury):** Necessities have inelastic demand because people must buy them regardless of price. Examples include maize flour (posho), rice, and basic medicines. In contrast, luxury goods like latest smartphones, expensive clothes, or international travel have elastic demand since consumers can do without them if prices rise.
- **Proportion of income spent:** If a good takes a small portion of income, demand is inelastic. For example, a rise in the price of salt or soap will not significantly affect how much people buy. But goods that take a large part of income, such as paying house rent in town or buying a motorcycle, have elastic demand because price changes strongly affect consumers' budgets.
- **Time Period:** In the short run, demand is usually inelastic because consumers need time to adjust. For instance, if fuel prices increase today, people still need to travel to work. However, in the long run, demand becomes more elastic as people adjust, such as using public transport (buses, motos) or moving closer to work.
- **Habit-forming goods:** Goods that people are used to or addicted to tend to have inelastic demand. For example, consumers of cigarettes or alcohol may continue buying them even when prices increase because of habit.
- **Number of alternative uses:** Goods with many uses tend to have elastic demand because consumers can reduce some uses when prices rise. For example, if electricity becomes expensive, households may reduce using it for ironing or outdoor lighting but keep it for essential needs. Goods with only one use, like a pen refill, tend to have inelastic demand since there is little room to adjust usage.
- **Consumer's Income:** Consumers with **high incomes** tend to have **inelastic demand** because they can afford to buy goods even when prices rise. In contrast, **low-income earners** have **elastic demand** since price changes greatly affect their ability to purchase.
- **Price Expectations:** If consumers **expect prices to rise in the future**, demand becomes **inelastic** because they continue buying even when prices increase today. However, if prices are expected to fall, consumers may reduce current purchases, making demand more elastic.
- **Consumer's Ignorance:** When consumers are **not aware of cheaper alternatives**, demand becomes **inelastic** because they continue buying even at higher prices. If consumers are well-informed, demand becomes **more elastic** as they can switch to better options.
- **The durability of a commodity:** The durability of a commodity affects how

consumers respond to price changes because it determines whether purchases can be delayed. Durable goods like cars or furniture can be used for a long time, so consumers can postpone buying them when prices rise, making demand more elastic. In contrast, non-durable goods like milk or bread must be consumed quickly, so people cannot easily delay buying them even if prices increase, resulting in inelastic demand. <https://corporatefinanceinstitute.com/resources/economics/price-elasticity-of-demand/>

vi. Practical applications of price elasticity of demand

Price elasticity of demand is not just a theoretical concept but a practical tool used to guide real-world decisions. By showing how consumers respond to price changes, it helps businesses, governments, and policymakers make informed choices about pricing, taxation, and economic planning. To understand how price elasticity of demand is applied in real life, we now examine its key practical uses in decision-making by businesses and governments as explained below;

a) Consumers:

Consumers also benefit from understanding price elasticity of demand, as it helps them make informed choices about spending and consumption. By knowing how price changes affect demand, they can adjust their buying habits to use their income more efficiently and maximize satisfaction as explained below.

- **Better purchasing decisions:** Consumers use PED to understand how price changes affect their demand, helping them decide when to buy or not buy a product.
- **Budget management:** By knowing which goods are elastic or inelastic, consumers can allocate their income more efficiently, prioritizing essential goods.
- **Substitution decisions:** PED helps consumers switch to cheaper alternatives when prices of elastic goods rise, allowing them to save money.
- **Timing of purchases:** For durable (elastic) goods, consumers may delay buying when prices are high and wait for price reductions.
- **Maximizing satisfaction:** Understanding PED enables consumers to adjust their spending patterns to get the most satisfaction (utility) from their limited income.

b) Firms/ businesses

Firms use price elasticity of demand as a key tool in decision-making, as it helps them understand how consumers will react to price changes and enables them to plan pricing, production, and marketing strategies effectively as explained below;

- **Pricing decisions:** Firms use PED to determine the best price to charge. If demand is inelastic, they can raise prices without losing many customers; if elastic, they may lower prices to increase sales.
- **Revenue prediction:** PED helps firms estimate how changes in price will affect total revenue. For example, a price increase raises revenue when demand is inelastic but reduces revenue when demand is elastic.
- **Production planning:** Businesses adjust their level of output based on how demand is expected to respond to price changes, helping to avoid overproduction or

shortages.

- **Marketing and advertising strategies:** Firms may increase advertising to make demand less elastic by building brand loyalty, so consumers become less sensitive to price changes.
- **Product differentiation:** By making products unique (through branding or quality), firms reduce elasticity, giving them more control over pricing
- **Price discrimination:** PED allows firms to charge different prices to different groups of consumers (e.g., students, adults) based on their sensitivity to price, thereby increasing overall profits.

c) Government:

Governments rely on price elasticity of demand to make effective economic decisions, particularly in areas such as taxation, pricing, and regulation, by predicting how consumers will respond to changes in prices as explained below;

- **Taxation decisions:** Governments use PED to decide which goods to tax. Goods with inelastic demand (such as fuel, alcohol, and cigarettes) are heavily taxed because consumers will continue buying them despite higher prices, ensuring a steady flow of revenue.
- **Revenue prediction:** PED helps governments estimate how much tax revenue they can collect. If demand is elastic, a high tax may significantly reduce consumption and lower revenue, but if demand is inelastic, revenue will increase even after a tax rise.
- **Subsidy policies:** Governments provide subsidies on goods with relatively elastic demand (such as agricultural inputs or education) to encourage higher consumption, since a lower price will lead to a significant increase in demand.
- **Price control policies:** When setting price ceilings (maximum prices) or price floors (minimum prices), PED helps governments predict the likely effects, such as shortages or surpluses, and plan appropriate interventions.
- **Regulation of harmful goods:** PED is used in controlling the consumption of demerit goods like tobacco and alcohol. By raising prices through taxation, governments aim to reduce consumption, especially when demand becomes more elastic over time.
- **Public services and utilities pricing:** Governments use PED when pricing essential services like water, electricity, and public transport to balance affordability for citizens with the need to cover costs and maintain services.
- **Trade and tariff policies:** PED helps governments decide on import and export taxes (tariffs). If demand for imports is inelastic, tariffs can generate revenue with little reduction in imports; if elastic, tariffs may significantly reduce trade.
- **Nationalization of Industries:** Governments may take control of industries producing essential goods with inelastic demand to prevent high prices and ensure fair access.
- **Dévaluation Policy:** PED helps determine whether dévaluation will improve trade; it is effective when demand for exports and imports is elastic.

- **Incidence of Taxation:** PED shows who bears the tax burden. With inelastic demand, consumers bear most of the tax, while with elastic demand, producers bear more.

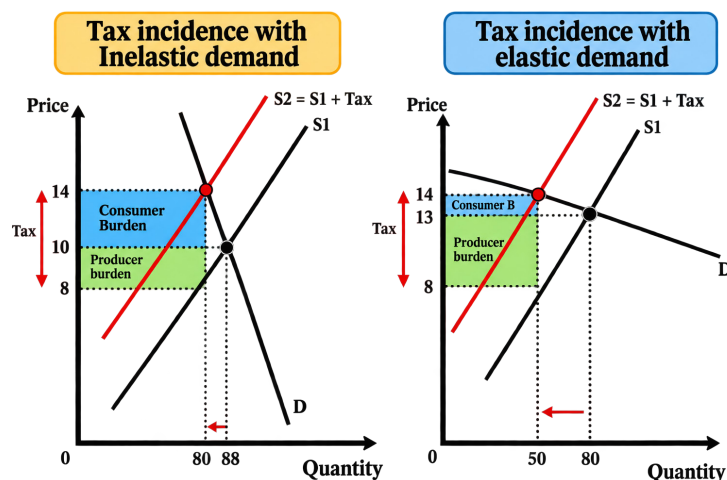


Fig 3.8 Tax incidence under different demand elasticities.

- **With Inelastic demand:** Demand is steep, so consumers are not very responsive to price changes. After the tax, price rises significantly while quantity falls slightly, meaning consumers bear most of the tax burden.
- **With Elastic demand:** Demand is flat, so consumers are very responsive to price changes. After the tax, quantity falls sharply and price rises only slightly, so producers bear most of the tax burden. <https://www.economicshelp.org/microessays/elasticity/price-elasticity-demand/>

d) Trade unions

Trade unions use price elasticity of demand to guide their wage negotiations and decisions, as it helps them understand how changes in labour costs may affect demand, employment, and the overall performance of firms.

- **Wage negotiations:** Trade unions use PED to assess how much wage increase a firm can afford. If higher wages lead to only small changes in demand, unions can confidently demand better pay.
- **Bargaining power:** When demand for the firm's product is inelastic, unions have stronger bargaining power because firms can pass higher wage costs to consumers without losing many sales.
- **Risk of unemployment:** If demand is elastic, wage increases may force firms to raise prices, leading to a large drop in demand. This can reduce production and result in layoffs, weakening the union's position.
- **Balancing wages and employment:** PED helps unions decide whether to demand higher wages or accept moderate increases in order to protect jobs, especially in

highly competitive markets

- **Strategic decision-making:** By understanding PED, trade unions can make informed decisions that balance workers' interests (higher wages) with the firm's ability to remain profitable and maintain employment levels.

Application activity 3.2



Carefully analyze and answer the questions below:

1. What is meant by price elasticity of demand?
2. Explain the different forms of price elasticity of demand.
3. How is knowledge of price elasticity of demand important to consumers, producers, and the government?
4. Should governments impose high taxes on goods with inelastic demand even if it heavily affects low-income consumers? Explain your answer.

3.2.3 Income elasticity of demand (YED)

Learning Activity 3.2.3



Answer the following questions based on your own experience.

- i. What do you usually spend your pocket money on?
- ii. If your pocket money increases, how would your spending on those items change?
- iii. Are there some items you would buy much more of, and others you would buy less of or stop buying as your pocket money increases? Why?
- iv. What does this show about how changes in pocket money affect the quantity of goods you demand?

"While price changes heavily influence consumer behavior, shifts in consumer income represent an equally powerful driver of market demand. Income Elasticity of Demand (YED) provides the logical tool necessary to measure this relationship, quantifying the responsiveness of the quantity demanded for a good relative to a change in the real income of the consumers buying it.

i. Meaning of Income Elasticity of demand

Income elasticity of demand refers to a measure of the degree of responsiveness of quantity demanded of a commodity to change in consumer's income. When a consumer's income rises or falls, the quantity demanded of a good may increase, decrease, or remain unchanged. Income elasticity of demand measures the extent of this response. It is calculated as the percentage change in quantity demanded divided by the percentage change in income.

$$YED = \frac{\% \text{change in Quantity demanded}}{\% \text{change in income}}$$

$$YED = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

ii. Interpretation of Income elasticity of demand

Income elasticity of demand may be:

- **Positive**, which implies that the commodity is a **normal** good/service.
- **Negative**, which shows that the commodity is an **inferior** good/ service.
- **Zero**, which implies that the commodity is a **necessity**.

<https://www.economicshelp.org/microessays/equilibrium/income-elasticity-demand/>

iii. Calculating Income Elasticity of Demand (YED):

Calculating income elasticity of demand is important because it shows how demand for a product change as consumers' income changes, helping to classify goods as normal or inferior and improving understanding of consumer behavior and demand patterns.

Example 1

If the consumer's income increased 10% and demand for beans tea fell 15%. Calculate the cross elasticity of demand.

Solution:

$$YED = \frac{\% \text{ change in Quantity demanded}}{\% \text{ change in income}}$$

$$YED = \frac{-15}{10}$$

$$YED = -1.5$$

The **negative** value shows that beans are an **inferior good**, meaning demand falls as income rises. The magnitude (1.5) indicates the demand is elastic, so demand decreases more than proportionately when income increases.

Example 2:

If demand for apples rose from 40 to 50 apples after a 10% rise in income. Calculate YED

Solution:

$$\text{Percentage change in quantity demanded} = \frac{50-40}{40} \times 100 = 25\%$$

$$YED = \frac{25}{10} = 2.5$$

Since the value is **positive** and greater than 1, apples are **a normal good** with **elastic income demand**, meaning demand increases more than proportionately when income rises.

Example 3:

When income increases from 1000Frw to 1200Frw and the quantity demanded remains constant at 50 units before and after the income change. Calculate Income elasticity of demand

Solution:

Change in income = 1200 - 1000 = 200

Change in quantity = 50 - 50 = 0

Initial Income = 1000

Initial quantity = 50

$$YED = \frac{0}{200} \times \frac{100}{50}$$

YED = 0

This shows that demand does not change when income changes, meaning the good is **perfectly income inelastic (a necessity)**.

Application activity 3.2.3



Read carefully and answer the following questions.

1. Income increases from 30,000Frw to 33,000Frw per year. Quantity demanded for bread increases from 500 to 520 loaves per year. Calculate YED and state the type of commodity.
2. Income rises from 1,500Frw to 1,650Frw per month. Quantity demanded for a budget product falls from 200 to 180. Calculate YED and state the type of commodity.
3. Income rises from 2,000Frw to 2,200Frw per month. Quantity demanded rises by 10%. Calculate income elasticity of demand (YED) and state the type of commodity.

3.2.4 Cross elasticity of demand (CED)

Learning Activity 3.2.4



Analyse the scenario below and answer the questions that follow:

Imagine a community where many people usually take tea with sugar every morning. One day, the price of sugar increases sharply. As a result, some people decide to reduce their use of sugar and instead start using honey in their tea. After some time, shopkeepers notice that while sugar sales have gone down, the demand for honey has increased.

- i. When the price of sugar increased, what change was observed in the demand for honey?
- ii. Why did people choose to buy more honey instead of sugar?
- iii. What does this suggest about the relationship between sugar and honey?
- iv. From this example, how does a change in the price of one good affect the demand for another good?

Every day, people make choices without realizing how connected they are. A change in the price of one product can influence what we buy, avoid, or choose as alternatives. Markets are shaped by these hidden relationships, where goods interact to influence demand and consumer behavior. As you explore this lesson, you will discover how products are linked, sometimes competing, sometimes supporting each other revealing the market as a network where one change can affect many others.

i. Meaning of cross elasticity of demand:

Cross elasticity of demand (CED) is a measure of the degree of responsiveness of quantity demanded of one commodity X to changes in price of another commodity Y. When the price of one commodity increases or decreases, quantity demanded of other related products reacts to a certain extent. Quantity demanded may reduce, increase or even remain constant. Cross elasticity of demand is measured as a percentage change in quantity demanded of one commodity to the percentage change in price of another.

$$CED = \frac{\text{Percentage change in quantity demanded of } x}{\text{Percentage change in price } y}$$

$$CED = \frac{\Delta Q_x}{\Delta P_y} \times \frac{P_y}{Q_x}$$

ii. Interpretation of cross elasticity of demand

Value of CED	Interpretation	Type of Goods
Positive (+)	Demand for one good rise when price of another rises	Substitutes
Negative (-)	Demand for one good fall when price of another rises	Complements
Zero (0)	No relationship between the goods	Unrelated goods

iii. Calculating Cross Elasticity of Demand

Calculating cross elasticity of demand is important because it helps show how goods are related and how a change in the price of one affects the demand for another, improving decision-making and understanding of consumer behavior.

Example 1;

If the price of butter increases by 10% and the demand for margarine rises by 5%, calculate the cross elasticity of demand between the two commodities and state the relationship between them.

Solution:

$$CED = \frac{\text{Percentage change in quantity demanded of } x}{\text{Percentage change in price } y}$$

$$CED = \frac{5}{10}$$

$$CED = 0.2$$

This shows butter and margarine are substitutes since CED is positive.

Example 2:

The price of A increases from 4000Frw to 5000Frw per unit. As a result, the quantity demanded of commodity B falls from 130 units to 100 units. Calculate XED

Solution:

Change in quantity demanded of commodity B = $100 - 130 = -30$ units

Change in price of commodity A = $5000 - 4000 = 1000$ frw

Initial quantity of commodity B = 130

Initial price of commodity A = 4000Frw

$$CED = \frac{\Delta QB}{\Delta PA} \times \frac{PA}{QB}$$

$$CED = \frac{-30}{1000} \times \frac{4000}{130}$$

$$CED = -0.92$$

CED is negative, so commodity A and B are **complementary goods**: The value close to -1 shows they are fairly strong complements.

Application activity 3.2.4



Carefully read and answer the questions below.

1. The price of rice increased from 2,000 FRW to 2,400 FRW per kilogram, and the quantity demanded decreased from 100 kg to 70 kg.

Questions

- a) Calculate the percentage change in price.
 - b) Calculate the percentage change in quantity demanded.
 - c) Determine the price elasticity of demand.
 - d) Interpret the answer.
2. The quantity demanded of commodity X increased from 500 units to 800 units after the price of commodity Y increased by 25%.

Questions

- a) Calculate the cross elasticity of demand (CED).
 - b) State the relationship between commodity X and Y.
3. Explain how the government can use price elasticity of demand when imposing taxes on goods and services.
4. A company producing bottled water discovers that demand for its product is elastic.

Questions

- a) What does elastic demand mean?
- b) Explain how the company can use this information to increase total revenue.

3.3 Elasticity of supply

Learning Activity 3.3



Analyse the scenario below and answer the questions that follow

In a local Rwandan market, the price of coffee increases. Some farmers are able to quickly produce and supply more coffee to the market, while others are unable to increase their production in the short term and continue producing the same amount or switch to other crops like tea or maize.

Questions:

- i. What happens to the quantity of coffee supplied when the price increases?
- ii. Why are some farmers able to increase their coffee production quickly while others are not?
- iii. What factors might limit a farmer from increasing production immediately?
- iv. Why do some farmers choose to grow other crops instead of more coffee?
- v. What does this suggest about how different farmers respond to price changes?
- vi. How does the time needed to produce coffee affect farmers' ability to respond to price increases?

We already know that when prices go up, sellers want to sell more to make higher profits. But wanting to sell more and actually being able to do it are two very different things. Price Elasticity of Supply measures how flexible a business is when the market screams for more of their product, exploring the boundaries between instant production and structural bottlenecks.

3.3.1 Meaning and types of elasticity of supply

i. Meaning of elasticity of supply

Elasticity of supply refers to the measure of degree of responsiveness of the quantity supplied of a good or service to a change in factors that affect supply. It measures how much the quantity supplied will change when any of its determinants change, for example, the product's price, the cost of raw materials, wages, or new technology.

ii. Types of elasticity of supply.

The types of elasticity of supply are classified according to the factors that cause changes in supply. However, since price is the most direct, frequent, and measurable factor influencing producers' decisions, one form stands out as the most commonly used. This is why elasticity of supply is often expressed as **price elasticity of supply**.

3.3.2 Price elasticity of supply (PES):

Supply is mainly influenced by price, so when economists talk about elasticity of supply, they usually refer to how much quantity supplied responds to a change in price. This is what price elasticity of supply (PES) measures. Since producers aim to make profit, and profit largely depends on price, price becomes the key factor in supply decisions. As a result, elasticity of supply is mainly considered in terms of price elasticity of supply.

i. Meaning of price elasticity of supply

Price elasticity of supply refers to the measure of degree of responsiveness of the quantity supplied of a good or service to a change in its own market price. It calculates how much more or less a business will produce when the selling price of their product goes up or down. Price elasticity of supply is expressed as the percentage change in quantity supplied as a result of a percentage change in its own price.

$$PES = \frac{\% \text{ change in quantity supplied}}{\% \text{ change in price}}$$

$$PES = \frac{\frac{\text{Change in quantity supplied}}{\text{Original supply}} \times 100}{\frac{\text{Change in price}}{\text{Original price}} \times 100}$$

$$PES = \frac{\text{Change in quantity supplied}}{\text{Change in price}} \times \frac{\text{Price}}{\text{Quantity}} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Where;

ΔQ = New quantity – initial quantity,

ΔP = New price – initial price,

Q = initial quantity,

P = initial price.

ii. Calculating Price Elasticity Supply. (PES)

It is not enough to know that supply changes when prices change; we also need to know by how much. Price elasticity of supply (PES) helps us measure this by showing how strongly producers respond to price changes.

Example 1;

The price of a good increased from 200 to 260. As a result, the quantity supplied increased from 100 units to 140 units. Calculate the price elasticity of supply (PES).

Solution:

Change in quantity supplied = $140 - 100 = 40$ units

Change in price = $260 - 200 = 60$

Original Quantity supplied = 100

Original price = 200

$$PES = \frac{\text{Change in quantity supplied}}{\text{Change in price}} \times \frac{\text{Price}}{\text{Quantity}}$$

$$PES = \frac{40}{60} \times \frac{200}{100}$$

$$PES = 1.33$$

Example 2:

The quantity supplied of a product increased to 240 units from an original level of 200 units when the price increases by 20% from its original price. Calculate the price elasticity of supply (PES).

$$\% \text{ Change in quantity supplied} = \frac{240-200}{200} \times 100 = 20\%$$

% change in price = 20%

$$PES = \frac{\% \text{ Change in quantity supplied}}{\% \text{ Change in price}}$$

$$PES = \frac{20\%}{20\%}$$

$$PES = 1$$

Exercise 3.4

Analyse and answer the following questions.

1. Distinguish between elasticity of supply and price elasticity of supply.
2. A company that makes wireless headphones notices that when the price of their headphones increases by 15%, the quantity they supply increases by 30%. Calculate the company's price elasticity of supply (PES).
3. The price of wheat increases by 25% due to shortages. However, farmers can only increase the quantity supplied by 5%. Calculate the price elasticity of supply (PES).
4. Use the data in the table below to calculate the price elasticity of supply for premium coffee beans.

Price per kg (Frw)	Quantity Supplied per month (tons)
4000	500
5000	650

5. A manufacturing firm for plastic storage bins operates with a highly elastic supply curve where $PES = 2.0$. Initially, they sell 5000 units per month at a price of 1000 per unit. If they increase their monthly production to 6500 units, calculate the new price required to prompt this supply shift.

3.3.3 Categories of price elasticity of supply

Price elasticity of supply explains how much producers change the quantity supplied in response to price changes, and it can be grouped into different categories based on the level of responsiveness as explained below:

1. Perfectly inelastic supply ($PES = 0$):

A change in price leads to no change in the quantity supplied because supply is completely fixed. Producers do not respond to price changes and continue to supply the same amount as before. Since supply cannot be increased or decreased, it is represented by a vertical curve, with examples such as stadium seats or original artwork that cannot be expanded.

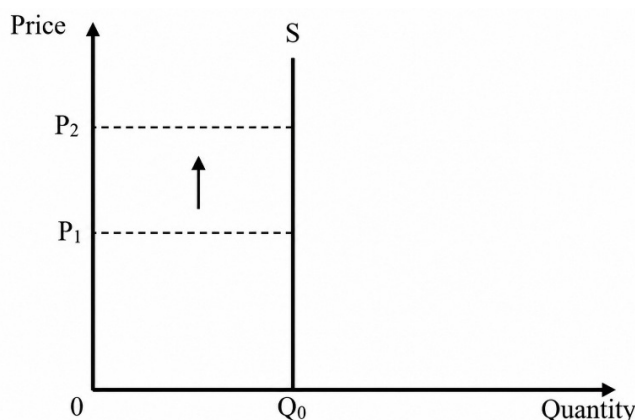


Fig 3.9 Perfectly inelastic supply curve

The curve shows a perfectly inelastic supply curve as a vertical line (S) at a fixed quantity level (Q). This means that even when the price rises from P_1 to P_2 or falls from P_2 to P_1 , the quantity supplied remains completely unchanged. The vertical shape of the curve clearly indicates that supply is fixed, and producers cannot increase or reduce output regardless of changes in price.

2. Inelastic supply ($0 < PES < 1$):

Inelastic supply occurs when the percentage change in quantity supplied is smaller than the percentage change in price, meaning producers do not respond strongly to price changes. This is usually because production takes time or resources cannot be quickly adjusted. The supply curve is a steep upward-sloping line, showing limited responsiveness. A common example is agricultural products like wheat or coffee, where farmers cannot immediately increase output even if prices rise, since crops need time to grow.

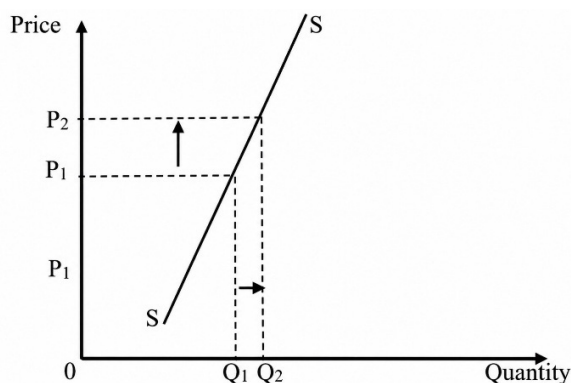


Fig 3.10 Inelastic supply curve

The figure shows an inelastic supply curve (S) where the price rises from P_1 to P_2 , but the quantity supplied increases only slightly from Q_1 to Q_2 , or when the price falls from P_2 to P_1 , the quantity supplied decreases only slightly from Q_2 to Q_1 . This small change in quantity compared to the larger change in price shows that producers respond weakly to price changes, indicating inelastic supply.

3. Unitary elastic supply ($PES = 1$):

Unit elastic supply occurs when the percentage change in quantity supplied is exactly equal to the percentage change in price, meaning a given percentage increase in price leads to an equal percentage increase in output. The supply curve is a straight upward-sloping line that passes through the origin, showing perfect proportional responsiveness, and it is mainly used as a theoretical example in economics to illustrate balanced market adjustments.

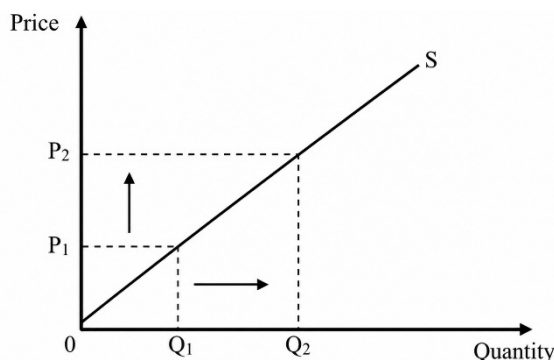


Fig 3.11 Unitary elastic supply curve

The diagram shows a unit elastic supply curve (S) where a rise in price from P_1 to P_2 leads to an equal proportional increase in quantity supplied from Q_1 to Q_2 , and a fall in price from P_2 to P_1 leads to an equal proportional decrease from Q_2 to Q_1 , showing that supply changes at the same rate as price.

4. Elastic supply ($1 < PES < \infty$):

Elastic supply occurs when the percentage change in quantity supplied is greater than the percentage change in price, meaning producers respond strongly to price changes. A small increase in price leads to a large increase in output, and a decrease in price causes a large reduction in supply. The curve is relatively flat (shallow), showing high responsiveness, as seen in manufactured goods like plastic bottles or smartphone cases, where production can be increased or reduced quickly.

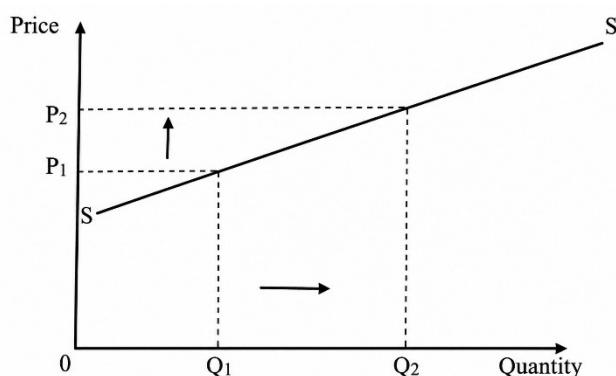


Fig 3.12 Elastic supply curve

The curve shows an elastic supply curve (S) where a small rise in price from P_1 to P_2 leads to a large increase in quantity supplied from Q_1 to Q_2 , and a fall in price from P_2 to P_1 causes a large decrease from Q_2 to Q_1 , showing that producers respond strongly to price changes.

5. Perfectly elastic supply ($PES = \infty$):

Perfectly elastic supply occurs when producers are willing to supply any quantity of a good at one fixed price, but if the price falls even slightly below this level, the quantity supplied drops to zero. This means producers are extremely responsive to price changes. The supply curve is a horizontal straight line, showing that price remains constant regardless of quantity. A typical example is a small farmer in a perfectly competitive market who can sell all output at the market price but cannot sell anything if they try to charge a higher price.

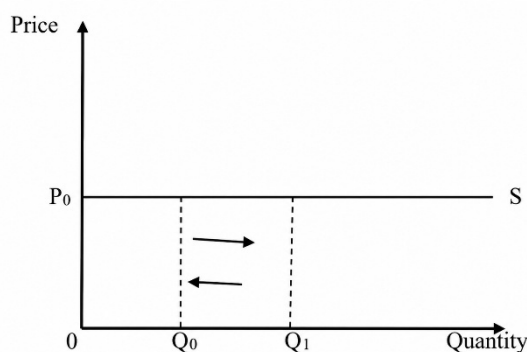


Fig 3.13 Elastic supply curve

The diagram shows a perfectly elastic supply curve (S) as a horizontal line at price P_0 , meaning producers can increase or reduce the quantity supplied (from Q_0 to Q_1 or vice versa) at this fixed price without affecting it. However, if the price falls even slightly below P_0 , the quantity supplied drops to zero, showing extreme sensitivity to price changes.

3.3.4 Determinants of price elasticity of supply

The price elasticity of supply depends on various determinants, and each factor influences whether supply is elastic or inelastic; when a determinant allows producers to quickly and easily adjust production, supply tends to be elastic, while if it limits or slows down the ability to change output, supply becomes inelastic. The following are the key factors that determine this responsiveness.

- **Time period of production:** The length of time available greatly affects the elasticity of supply. For example, if the price of Irish potatoes rises suddenly in Rwanda due to increased demand, farmers cannot immediately increase output because the crop takes several months to grow. Therefore, in the short run, the supply of potatoes is relatively inelastic. However, in the long run, farmers can devote more land to potato cultivation, use improved seeds and farming techniques, or new farmers may enter potato production. As a result, the supply of potatoes becomes more elastic over time.
- **Availability of spare capacity:** Supply tends to be elastic when firms have unused resources. For example, if demand for cement rises in Rwanda and manufacturers such as CIMERWA have spare production capacity, they can increase output by running their plants for longer hours or using additional workers. However, if the plants are already operating at full capacity, production cannot be increased easily without expanding facilities or installing new equipment. In this case, supply becomes inelastic.
- **Ease of storage:** Supply is more elastic for goods that can be stored. For example, Rwanda's coffee cooperatives can store processed coffee and sell it when prices improve. In contrast, fresh tomatoes are highly perishable and must be sold quickly, making their supply relatively inelastic.
- **Mobility of factors of production:** Supply is more elastic when resources can be easily shifted between activities. For example, Rwandan farmers can switch land and labour between growing maize and beans when prices change. However, specialized equipment used in tea processing cannot easily be used for other activities, making supply less elastic.
- **Level of technology:** Advanced technology makes supply more elastic by enabling firms to increase production quickly. For example, modern milk processing plants in Rwanda can respond faster to rising demand than small processors using basic equipment.
- **Cost and availability of factors of production:** Supply is more elastic when production resources are readily available and affordable. For example, construction firms in Rwanda can increase output more easily when skilled labour, cement, and machinery are available at reasonable costs. If these resources become scarce or expensive, supply becomes inelastic.

- **Nature of the commodity:** Durable goods tend to have elastic supply because they can be stored. For example, cement can be stored and sold when market conditions are favourable. In contrast, fresh milk is perishable and must be sold quickly, making its supply more inelastic.
- **Gestation period (maturity period):** Goods with a short production period have more elastic supply. For example, vegetables such as spinach can be produced relatively quickly in Rwanda when prices rise. However, coffee has a long maturity period, making its supply inelastic in the short run.
- **Method of production used:** Modern production methods make supply more elastic. For example, mechanized rice farming in Rwanda can increase output more quickly than traditional farming methods that rely mainly on manual labour.
- **Government policy:** Government policies can affect supply elasticity. For example, subsidies on fertilizers and improved seeds encourage Rwandan farmers to increase production, making supply more elastic. In contrast, high taxes or lengthy licensing procedures may discourage production and make supply less elastic.
- **Freedom of entry of new firms into the industry:** Supply is more elastic when new firms can easily enter the market. For example, if demand for poultry products rises in Rwanda, new poultry farmers can enter the industry and increase total output. If entry is restricted by high capital requirements or regulations, supply becomes less elastic.

3.3.5 Importance of Price Elasticity of Supply

Price elasticity of supply is an important concept that helps explain how producers respond to changes in price. Understanding its importance enables consumers, producers, and the government to make better decisions regarding production, pricing, and resource allocation in the economy, as explained below:

i. To consumers,

- **Anticipating availability of goods:** Consumers can use PES to understand how quickly goods will become available when prices change; if supply is inelastic, they can expect delays or shortages.
- **Understanding low PES and shortages:** When supply is inelastic, consumers know producers cannot easily increase output, so shortages may occur, encouraging them to buy early before goods run out.
- **Making informed purchasing decisions:** PES helps consumers decide when to buy; they may buy quickly if supply is limited or wait if supply is elastic and prices may stabilize.
- **Exploring alternative products:** If a product has low PES and is scarce, consumers can switch to substitutes with more elastic supply to ensure availability.
- **Assessing long-term reliability of products:** Consumers can judge which products will be stable over time; goods with elastic supply are more reliable in price and availability.
- **Guiding purchasing choices:** Overall, PES helps consumers match their buying

decisions with market conditions, avoiding shortages, high prices, and unnecessary spending.

ii. To producers/ firms

- **Planning production and inventory levels:** Producers use PES to decide how much to produce and store; if supply is elastic, they can easily adjust output to meet demand and avoid shortages or excess stock.
- **Reacting to price increases:** When supply is elastic, producers can quickly increase production when prices rise, allowing them to take advantage of higher profits and market opportunities.
- **Anticipating challenges with low PES:** If supply is inelastic, producers know it is hard to increase output quickly, so they plan carefully to avoid losses when prices change.
- **Informing resource allocation:** PES helps producers decide where to use their resources by focusing on products that can respond quickly to price changes and bring better returns.
- **Investing in production capacity:** Understanding PES guides producers on whether to expand production; they invest more in goods with elastic supply and avoid large investments where supply is inelastic.
- **Long-term production planning:** PES helps producers plan for the future by understanding how supply will respond over time, allowing them to grow steadily and avoid sudden problems.

iii. To the government

- **Government decisions on currency dévaluation:** PES helps the government predict whether dévaluation will increase exports; if supply is elastic, producers can quickly expand output and benefit from higher demand, but if it is inelastic, exports will respond little, making the policy less effective.
- **Addressing structural rigidities:** By analyzing PES, the government can identify barriers such as poor infrastructure or lack of inputs that make supply inelastic, and by removing these obstacles, production becomes more responsive and elastic.
- **Policy formulation:** PES guides policy effectiveness; when supply is elastic, incentives like subsidies or tax reductions can significantly boost output, whereas with inelastic supply, such policies may have limited impact.
- **Regional comparisons and tailored policies:** Since PES differs across regions, the government uses it to design targeted interventions; areas with inelastic supply may require improved access to resources, while areas with elastic supply may need less support.
- **Commodity-specific strategies:** Different goods have different elasticity levels, so the government supports commodities with inelastic supply (e.g., agriculture) to improve output, while goods with elastic supply can adjust more easily with minimal intervention.

- **Tax incidence determination:** PES helps determine who bears the tax burden; if supply is more elastic, producers can shift taxes to consumers, but if it is inelastic, producers bear a larger share since they cannot easily reduce output.

Application activity 3.3



1. a) Distinguish between Unitary elastic supply and Inelastic supply
2. Explain why it is important for the government to have knowledge of price elasticity of supply.

End of unit assessment



Knowledge and understanding

1. Explain the concept of elasticity and describe its different types using real-life examples.
2. Explain the determinants of elasticity of demand.

Skills

3. a) Calculate the price elasticity of demand for a product whose price rises from 1000 FRW to 1200 FRW and quantity demanded falls from 50 to 40 units.
b) Interpret the results of elasticity calculations to explain how changes in price affect total revenue for a producer.
c) Describe real market data to determine whether demand or supply is elastic or inelastic and justify your reasoning.
4. Under what circumstances may elasticity of supply be inelastic?
5. Use elasticity concepts to propose a pricing or production strategy for a product in a competitive market.
6. Analyse how elasticity of demand and supply informs government decisions on taxation and subsidies.

Attitudes and Values

7. Examine why understanding elasticity is important for consumers, producers, and policymakers in making informed economic decisions.

UNIT 4

PRICE MECHANISM



Key Unit Competence:

Evaluate how market forces guide resource allocation and economic decisions in real-life markets.

Learning objectives

Knowledge and understanding

- Explain the meaning of the price mechanism and state its key assumptions.
 - Describe how the price mechanism operates in a free market.
 - Explain the role of the price mechanism in allocating resources efficiently through demand, supply, and price.
 - Identify reasons why governments interfere in the price mechanism.
 - Describe methods of government intervention in prices, including subsidies, taxes, price controls, and regulation.
- 1**
- Define price fluctuations, particularly for agricultural products.
 - Explain the causes of agricultural price fluctuations, such as weather changes, supply conditions, and demand variations.
 - Describe the effects of agricultural price fluctuations on farmers, consumers, and the economy.
 - Explain measures used to control price fluctuations, including buffer stocks, stabilization funds, and international commodity agreements.
 - Define price controls and identify different forms such as minimum prices, maximum prices.
 - Explain the positive and negative effects of price controls on markets and resource allocation.

Skills

2

- Analyze how market forces of demand and supply determine prices and guide the allocation of resources.
- Evaluate real-life market situations to explain how the price mechanism influences production and consumption decisions.
- Analyze the impact of government intervention on prices, market efficiency, and resource allocation.
- Interpret causes and effects of price fluctuations, especially in agricultural markets.
- Assess the effectiveness of price control measures in achieving economic and social objectives.
- Make informed economic judgments about when markets should be left free and when government intervention is necessary.
- Apply economic reasoning to explain real-life market outcomes such as shortages, surpluses, and price instability.

Attitudes and values

3

- Appreciate the role of market forces in guiding efficient use of scarce resources.
- Demonstrate a balanced attitude toward free markets and government intervention.
- Value price stability as a factor that supports economic welfare and planning.
- Develop concern for vulnerable groups, such as small farmers and low-income consumers, affected by price fluctuations.
- Show responsibility and fairness in evaluating economic policies like price controls.
- Recognize the importance of informed economic decision-making in promoting sustainable development.

Every day, millions of buyers and sellers make decisions about what to produce, what to buy, and how much to exchange without direct government control. Prices constantly rise and fall as demand and supply change, sending signals and incentives that guide both producers and consumers. For example, high prices encourage producers to supply more while making consumers more careful in their spending, while low prices do the opposite. Through this continuous interaction, markets are able to allocate resources, balance needs, and maintain order even in a complex economy. This automatic and self-regulating process is known as **the price mechanism**. However, when these market interactions fail to produce fair or stable outcomes, the government may step in to restore balance through measures known as **price controls**.

By studying this unit, you will understand how markets work, how resources are shared, and how order is created in the economy, as well as when and why government intervention becomes necessary.



Introductory activity

Carefully analyse the scenario below and answer the questions that follow:

One Saturday morning, students from a nearby school visit a local market to buy food for a school activity. They notice that the price of tomatoes is very high, while the price of bananas is very low. Some traders say tomatoes are scarce because the harvest season has not yet started, while bananas are plenty because many farmers harvested them at the same time.

1. Why do you think the price of tomatoes is higher than the price of bananas in the market?
2. What factors might cause prices of goods to increase or decrease in the market?
3. Should the government control prices, or should prices be determined by buyers and sellers in the market? Give reasons for your answer.
4. What might happen if there were no organizations to regulate the quality and pricing of goods and services in the country?

4.1 Introduction to Price mechanism

Learning Activity 4.1



Read the situation below carefully and answer the questions that follow.

Situation:

In a local market, people are buying and selling goods such as Irish potatoes and cassava. On some days, Irish potatoes are expensive while cassava is cheap. On other days, the prices change depending on how much is available and how many people want to buy them.

Questions

1. a) What usually happens when many people want to buy the same product?
b) What happens when a product is available in large quantities in the market?
c) How do sellers react when their goods are not being bought quickly?

2. True or False

- a) High prices encourage producers to produce more goods.
- b) Prices do not change in the market.
- c) When the supply of a good increases, its price may fall.

3. In your own words, explain how buyers and sellers can influence the price of goods in the market.

4. a) What are some advantages of allowing prices to be determined in the market?

b) What are some disadvantages of this system?

In any economy, buyers and sellers interact in markets to determine the value of goods and services. As people make choices based on their wants and limited resources, prices naturally adjust depending on changes in demand and supply. Under certain conditions, these price changes guide producers on what to produce and consumers on what to buy, helping to distribute resources where they are most needed. While this system helps promote efficiency and flexibility in economic activities, it can also have some shortcomings, which makes it important to understand how it works and its overall impact on the economy.

4.1.1 Meaning and assumption of Price mechanism

i. Meaning of price mechanism

The price mechanism, also known as the **price system**, is an automatic system in a free-market economy where prices are determined by the interaction of demand and supply without direct government control. Instead of a central authority deciding what to produce or who buys goods, changing prices act like an “invisible hand” that guides the decisions of buyers and sellers. Through these price changes, the market coordinates activities, ensuring that resources are produced and distributed efficiently. The price mechanism is explained through the demand and supply curves together. When demand or supply changes, these curves shift, and a new equilibrium is formed—this shows how the price mechanism adjusts in real life.

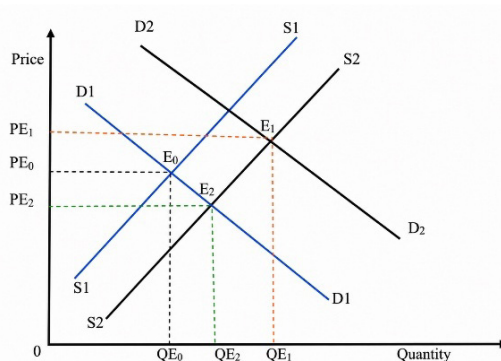


Figure 4.1: Interaction of Demand and Supply.

The diagram shows that price is determined by the interaction of demand and supply. At equilibrium (E_0), demand equals supply at price PE_0 and quantity QE_0 . When demand increases, a shortage pushes price up to a new equilibrium (E_1). When supply increases, a surplus forces price down to a new equilibrium (E_2). Thus, the price mechanism adjusts prices to restore balance in the market. <https://www.economicshelp.org/microessays/equilibrium/market-equilibrium/>

ii. Assumptions of price mechanism:

For the price mechanism to function effectively as an **“invisible hand”** that automatically balances demand and supply, economic theory relies on several foundational assumptions. In the real world, these assumptions describe a perfectly competitive free-market system:

- **Rational economic behavior:** The price mechanism assumes that people act in their own best interest. Consumers try to get goods at the lowest prices to gain maximum satisfaction, while producers aim to sell at higher prices to earn more profit. This self-interest drives decision-making in the market.
- **Absence of government intervention:** It is assumed that the government does not interfere in the market. There are no controls like price limits, subsidies, or regulations, so prices are free to change naturally based on demand and supply.
- **Perfect market information:** The market is assumed to be fully transparent, meaning both buyers and sellers have complete and accurate information. Consumers know all prices, and producers understand costs and competition, preventing unfair pricing.
- **Large number of buyers and sellers:** The market consists of many buyers and sellers, none of whom can control the price alone. Everyone is a **“price taker,”** meaning prices are determined by overall market forces, not individual decisions.
- **Homogeneous products:** Goods in the market are assumed to be identical in quality and nature. Since there is no difference between products, consumers choose based on price, which encourages strong competition among sellers.
- **Perfect mobility of factors of production:** Resources like labor and capital can easily move from one activity to another. If a product becomes more profitable, resources quickly shift toward its production, helping the market respond efficiently to price changes.
- **Freedom of entry and exit:** The price mechanism assumes that firms can easily enter a market when profits are high and leave when profits are low. This ensures competition increases when prices rise, helping to bring prices down and stabilize the market.
- **Private ownership of property:** It is assumed that individuals and firms have the right to own and control resources such as land, capital, and businesses. This ownership encourages them to use resources efficiently and respond to price signals in order to maximize profits, supporting the smooth functioning of the price mechanism.

4.1.2 Operation and role of price mechanism:

In every market, prices are constantly changing and behind these changes lies a powerful system that keeps the economy functioning smoothly. Understanding how the price mechanism works helps us see how simple price movements can guide decisions, solve problems of scarcity, and bring balance between buyers and sellers.

i. Operation of Price mechanism:

The operation of the price mechanism is how prices adjust automatically with changes in demand and supply, rising during shortages and falling during surpluses to restore equilibrium. Through signaling, incentives, and rationing, it guides producers and consumers and ensures efficient allocation of resources as explained below.

Through its key functions; **signaling, incentives, and rationing** the price mechanism quietly organizes economic activity and ensures resources are used where they are needed most as explained below:

- **Signaling function:** Prices show what is happening in the market by indicating whether goods are scarce or abundant. When prices rise, it means demand is high, encouraging producers to supply more; when prices fall, it shows oversupply, leading producers to reduce output. For example, as more people move to Gasabo and Kicukiro, rent increases, signaling developers to build more houses to meet the growing demand and balance the market.
- **Incentive function:** When prices are high, they motivate producers to improve production and earn more. For example, when international buyers pay higher prices for high-quality Rwandan coffee, farmers are encouraged to work harder, use better inputs, and carefully select the best coffee cherries so they can earn higher profits.
- **Rationing function:** Prices help distribute scarce goods by limiting access to those who can afford them. When a product is in short supply, its price rises, reducing demand and ensuring that only those willing and able to pay get it. For example, during the peak tourist season at Akagera National Park, the price of safari guides increases due to high demand and limited supply. This higher price rations the available guides, allocating them to tourists who can pay, while others choose cheaper options or visit during off-peak periods.

ii. Role/Functions of Price Mechanism.

The role of the price mechanism is seen in what it does in the economy, its key functions such as allocating resources, balancing demand and supply, guiding production, and distributing goods. It plays a vital role in a free-market system by coordinating decisions and activities automatically, ensuring efficient use of resources without central control. Through these functions, the price mechanism helps maintain order, efficiency, and stability in the economy, as explained below.

- **Determining What to Produce:** Prices guide producers on what goods to produce. Goods with high demand and high prices attract more resources, ensuring production matches what consumers want.
- **Determining When to Produce:** Producers choose to produce goods when demand and prices are high so they can earn more profit, adjusting their production timing accordingly.

- **Determining How to Produce:** Firms select the most efficient and lowest-cost methods of production. Inefficient producers with high costs cannot compete and may leave the market.
- **Determining Quantities to Produce:** When prices rise, producers increase output; when prices fall, they reduce production. This helps balance supply with demand.
- **Encouraging Innovation:** High prices and competition push firms to innovate by creating better products and improving production methods.
- **Facilitating Geographical Distribution of Resources:** Prices encourage the movement of goods from areas where they are in excess to areas where they are scarce, a process called arbitrage, helping stabilize markets.
- **Encouraging Specialization:** Firms focus on producing goods they can make efficiently, increasing productivity and overall supply.
- **Incentivizing Growth:** The price mechanism promotes economic growth by encouraging competition and allowing firms to enter or leave markets freely.
- **Optimal Resource Utilization:** Prices guide resources to their most profitable and productive uses, avoiding waste.
- **Encouraging Competition and Efficiency:** High prices attract more producers into the market, increasing competition and improving efficiency.
- **Distribution of National Income:** Income is distributed based on ownership of resources and ability to pay, as prices determine earnings.
- **Automatic Resource Allocation:** Resources are allocated automatically through demand and supply without the need for central control.
- **Investment Signals:** Prices indicate where profits are high, guiding businesses on where to invest their resources.

4.1.3 Advantages, disadvantages and government intervention in the price mechanism

The price mechanism plays an important role in the economy, but it is not perfect. While it provides many benefits, it also has some limitations that may lead to problems. Therefore, it is important to understand its advantages, limitations, and the control measures used to manage its weaknesses.

i. Advantages of price mechanism

The **price mechanism offers several important advantages** in a free-market economy by guiding decisions, promoting efficiency, and ensuring that resources are used where they are most needed.

as explained in the points below:

- **Efficient allocation of resources:** The price mechanism directs resources to where they are most needed, ensuring that goods in high demand receive more resources while less demanded goods receive fewer.
- **Automatic adjustment of demand and supply:** Prices rise and fall to balance demand and supply without the need for government control, helping the market stay stable.

- **Encourages competition:** High prices attract more firms into the market, increasing competition and leading to better quality goods and lower prices over time.
- **Promotes efficiency:** Firms are encouraged to reduce costs and use resources wisely in order to maximize profits and survive in the market.
- **Encourages innovation:** The desire to earn more profits motivates firms to develop new products and improve production methods.
- **Consumer sovereignty:** Consumers influence what is produced through their spending choices, ensuring that production is based on their preferences.
- **Flexibility:** The system quickly responds to changes in demand, supply, and consumer tastes, allowing the economy to adapt easily.
- **No need for central planning:** The price mechanism works automatically without heavy government control, reducing administrative costs and bureaucracy.
- **Promotes economic growth:** Through competition, efficiency, and investment, the price mechanism helps increase production and overall economic development.

ii. Disadvantages of price mechanism

The price mechanism also has several disadvantages, as it may lead to unequal outcomes, market failures, and situations where important social needs are not fully met as explained below:

- **Neglects the provision of public goods:** Markets fail to produce socially vital infrastructure like street lighting, flood defense dams, and national security because these items are non-excludable and non-rivalrous, meaning private firms cannot charge users a direct price to make a profit.
- **Causes extreme price instability in strategic sectors:** In industries with highly inelastic demand and supply, such as agriculture and fuel, tiny shifts in weather or global logistics trigger wild price rollercoasters that destabilize the entire economy.
- **Creates severe income inequality:** The price mechanism leads to unequal distribution of income because people who own more resources (like land or capital) earn higher incomes, while those with fewer resources earn less, increasing the gap between the rich and the poor.
- **Possibility of monopoly:** Strong and efficient firms may drive weaker ones out of the market. Over time, this can reduce competition and allow a few firms to dominate, leading to monopolies. Once a monopoly is formed, it destroys the free market by artificially restricting supply to drive up consumer prices.
- **Profit motive may harm consumers:** Since producers aim to maximize profits, they may sell low-quality or fake products or exploit consumers who lack full market information.
- **No control over harmful (demerit) goods:** Without government regulation, harmful or illegal products may enter the market because producers focus mainly on profit rather than safety.
- **Elimination of small firms:** Intense competition can push out small or new firms, preventing them from growing and leading to underuse of some resources.

- **Neglect of essential but unprofitable goods:** Private producers focus on profitable goods, so important services or products that do not generate profits may not be produced, even if society needs them.
- **Slow economic changes:** The price mechanism may not quickly bring about major changes or development in the economy because it depends only on gradual changes in demand and supply.
- **Failure to provide public goods:** Goods like street lights or roads may not be provided because they do not generate direct profit, even though they are important for society. <https://www.tutorchase.com/notes/aqa-a-level/economics/8-1-2-pros-and-cons-of-the-price-mechanism>

iii. Government intervention in the price mechanism

Although the price mechanism helps markets operate, it does not always produce fair or stable outcomes. Therefore, the government may intervene to correct these problems, protect both consumers and producers, and ensure the smooth functioning of the economy as explained below:

- **Imposing price controls (Ceilings and Floors):** The government sets legal price boundaries to stabilize unstable markets. It uses **price ceilings** (maximum prices) to keep survival essentials affordable for the poor, and **price floors** (minimum prices) to guarantee farmers earn a livable income.
- **Taxation:** Governments use taxes to reduce income inequalities created by the price mechanism. Wealthier people are taxed more, and the revenue can be used to support the poor, helping to reduce unfair distribution of income.
- **Charging indirect taxes on demerit goods:** Governments impose heavy excise duties on harmful items like tobacco and alcohol to artificially inflate their retail prices, using the price mechanism itself to discourage consumer demand.
- **Providing strategic subsidies to merit goods:** Financial grants are given to essential but underproduced sectors like healthcare, education, and solar energy to lower production costs, making them cheap and highly accessible to the general public.
- **Direct state provision of public goods:** Because the private sector completely ignores non-profitable infrastructure, the state uses taxpayer revenue to directly build and maintain public goods like national defense systems, public roads, and street lighting.
- **Introducing anti-monopoly regulations (Competition Laws):** Governments establish regulatory bodies to review corporate mergers, break up abusive monopolies, and fine businesses that engage in price-fixing cartels, forcing the marketplace to remain open and competitive.
- **Formation of Consumer Associations:** Consumers form groups to protect themselves from exploitation. These associations give consumers stronger bargaining power and provide useful information about prices and product quality.
- **Government Support:** The government may give subsidies or financial help to small or weak firms so they can grow and compete. This promotes competition and improves the quality and availability of goods.

- **Nationalisation:** The government may take control of certain private businesses to ensure that public interest is protected. This helps prevent private firms from focusing only on profit at the expense of society. <https://masomomsingi.com/price-mechanism-notes/>

Application activity 4.1



1. a) What is meant by price mechanism?
b) Explain the assumptions under which price mechanism operates.
2. a) Describe how price mechanism allocates resources.
b) Examine the role of price mechanism in allocation of resources.
3. a) Why does government interfere with the price system?
b) Discuss the various methods government can use to correct the failures of price mechanism.

4.2 Price fluctuations

Learning Activity 4.2



Read the scenario below carefully and answer the questions that follow.

Scenario:

In Rwanda, the price of bananas changes at different times of the year. Before the harvest season, bananas are scarce, so their prices rise. After the harvest, many farmers bring bananas to the market, making them plentiful and causing prices to fall. At times, unpredictable rainfall and pests reduce banana production, leading to sudden increases in prices. These changes make it difficult for farmers to plan and can affect their income.

Questions

- i. What pattern or trend in banana prices is described in the scenario?
- ii. Why do banana prices rise before harvest and fall after harvest?
- iii. What factors mentioned in the scenario cause banana prices to change?
- iv. How do these changes in prices affect banana farmers?
- v. How might these price changes affect consumers and the wider society?
- vi. What challenges do price changes create for farmers when planning their production?

In everyday markets, prices of goods and services change over time as economic conditions vary. These changes are more noticeable in agricultural products, where

seasons, weather, and harvest periods affect the quantity available. When goods are scarce, prices rise, and when they are plentiful, prices fall, creating instability that affects farmers, consumers, and planning decisions. In this lesson, you will examine how these changes occur in agricultural markets, their effects on different groups in society, and the role of the government in promoting stability. These continuous movements in prices are known as **price fluctuations**

4.2.1 Meaning and causes of agricultural products price fluctuations

i. Meaning of price fluctuations

Price fluctuation refers to the continuous rise and falls in the prices of goods and services over time due to changes in factors such as demand, supply, and other market conditions. It is especially common in agricultural markets, where factors such as seasons, weather, and harvest periods affect supply, causing prices to rise during scarcity and fall during abundance.

ii. Causes of price fluctuations of agricultural products

The extreme and frequent price fluctuations of agricultural products are a major challenge in economics. Unlike manufactured goods, which usually have stable and predictable production, agricultural prices often rise and fall sharply due to strong imbalances between supply and demand.

The main causes of price fluctuations in agricultural products include:

- **High dependence on weather and natural shocks:** Agricultural production relies heavily on natural conditions, which are often unpredictable. Events such as droughts, floods, pests, and diseases can suddenly reduce output. This causes shortages and sharp price increases, while good weather can create excess supply and lead to falling prices.
- **Long gestation periods/time lags in production:** Agricultural goods take time to produce, so supply cannot quickly respond to changes in market prices. This delay means that sudden increases in demand push prices up rapidly, while oversupply cannot be corrected immediately, causing instability.
- **Price inelastic demand for food:** Food is a basic necessity, so people must continue buying it even when prices rise. This causes prices to rise sharply during shortages. However, when supply increases, demand cannot expand much, leading to significant price drops.
- **Perishability of agricultural goods:** Many farm products spoil quickly and must be sold soon after harvest. This puts pressure on farmers to sell immediately, often at low prices when supply is high, causing sudden price declines.
- **The cobweb phenomenon (delayed responses by farmers):** Farmers base production decisions on current prices, which can lead to overproduction or underproduction in the future. This results in repeated cycles of surplus and shortage, causing continuous rises and falls in prices.
- **Small-scale production:** Many farmers produce on a small scale using traditional methods, making it difficult to plan and match market demand. This leads to irregular supply and unstable prices.

- **Market competition:** When many farmers grow the same crops, large amounts may flood the market at the same time. This oversupply increases competition and forces prices down.
- **Weak regulation:** Poor government control and lack of organized marketing systems make it harder to manage supply and stabilize prices, leading to frequent fluctuations.
- **Limited industrial demand:** Low demand for agricultural products from industries and poor systems for handling surplus mean excess produce is not absorbed, causing prices to drop when supply is high.

4.2.2 Effects and measures to control agricultural products price fluctuations

Price fluctuations in agricultural products are caused by various underlying factors and have wide-ranging effects on producers, consumers, and the entire economy. These fluctuations create significant economic challenges, making it important to understand not only their causes but also their effects and the measures used to control them.

i. Effects of price fluctuations in agricultural products

The causes of price fluctuations in agricultural products lead to several important consequences that affect farmers, consumers, and the overall economy. The following are the main effects of price fluctuations in agricultural products:

- Price fluctuations make planning difficult because farmers' incomes and government revenue become unpredictable.
- Agriculture is a major source of government revenue, especially in developing countries like Rwanda.
- Unstable prices can discourage farmers, leading some to abandon agriculture and causing employment fluctuations in the sector.
- Modernizing agriculture is harder since unreliable agricultural revenue limits investment in improvements.
- Fluctuating prices also affect foreign exchange earnings from exports, which can lead to Balance of Payment deficits when export prices fall. <https://masomomsingi.com/price-mechanism-notes/>

ii. How to control price fluctuations

Due to the negative effects of price fluctuations, governments and other stakeholders take various actions to stabilize prices and protect both producers and consumers. The following are the main measures used to control price fluctuations in agricultural products:

Measures to control price fluctuations in agricultural products

Governments use various methods to control fluctuations in agricultural product prices and protect both farmers and consumers. Traditional measures, such as setting minimum prices and maintaining crop reserves, help manage short-term price changes but do not address deeper market challenges. To overcome these issues, modern approaches combine broader strategies, including international trade agreements, import and export controls, improved processing and marketing, and support systems like insurance and farmer cooperatives. Overall, the focus is on strengthening the entire agricultural system to ensure long-term stability and resilience.

1. Direct Government Interventions.

Direct government interventions control agricultural price fluctuations by enabling the government to directly influence prices and supply. These measures help balance supply and demand, protect farmers' incomes, safeguard consumers, and reduce sudden and extreme changes in market prices. They include:

- **Price stabilization policies:** Governments set minimum prices (price floors) to protect farmers when prices fall too low, and maximum prices (price ceilings) to protect consumers when prices rise too high. This helps maintain a fair balance in the market.
- **Buffer stock schemes:** The government purchases excess produce during periods of high supply and stores it in reserves, then releases these stocks during shortages to increase supply and prevent sharp price increases. This policy assumes the government has adequate storage facilities, timely market information, and sufficient funds to buy and maintain stocks effectively. When well-managed, buffer stocks help stabilize prices, reduce extreme fluctuations, and ensure continuous availability of food, thereby protecting both farmers' incomes and consumers from sudden price shocks.
- **Export promotion and import control:** Governments encourage exports when there is surplus production to reduce oversupply in the domestic market. During shortages, imports can be increased to boost supply and stabilize prices.

2. Structural and Supply-Side Infrastructure

Structural and supply-side infrastructure measures help control price fluctuations by improving production, storage, and distribution systems. They ensure a steady supply of agricultural products by reducing losses, preventing oversupply, and easing movement of goods from surplus to shortage areas, which helps maintain more stable prices. They include;

- **Improved storage facilities:** The provision of modern storage systems such as warehouses, storage tower (silos), and cold rooms allows farmers to keep their produce for longer periods. This reduces the need to sell immediately after harvest, helping to avoid sudden price drops.
- **Agricultural diversification:** This involves encouraging farmers to grow different types of crops instead of relying on one. By doing so, farmers reduce risk because if the price of one crop falls, they can still earn income from other crops.

- **Processing and value addition:** Developing agro-processing industries (such as making juice, flour, or canned goods) helps absorb excess production, reduces waste, and creates steady demand, which supports stable prices.
- **Improved transport and infrastructure:** Efficient roads, transport systems, and market access reduce delays and allow goods to move easily from surplus areas to deficit areas, minimizing local shortages and excess supply.
- **Agricultural insurance:** Insurance schemes protect farmers from losses due to natural disasters like droughts or floods, helping them continue production and maintain stable supply in the long run.
- **Cooperative marketing:** Farmers can form cooperatives to sell their produce collectively, which strengthens their bargaining power, reduces exploitation by middlemen, and leads to more stable and fair prices.
- **Modernizing production and technology:** Moving away from weather-dependent production by investing in structural infrastructure such as irrigation networks, greenhouses, and alternative power grids, ensures a predictable and consistent flow of goods, minimizing the risk of supply-shock price spikes.

3. Market Information and Financial Planning.

These measures in controlling agricultural price fluctuations include tools that help farmers make better decisions and manage financial risks rather than directly controlling prices. These measures include;

- **Digital market information systems:** These systems use mobile phones (apps or SMS) to give farmers up-to-date information on market prices and demand. This helps farmers avoid being cheated by middlemen and make better decisions about what crops to grow, reducing chances of too many farmers producing the same crop at the same time.
- **Forward contracts and futures markets:** These are agreements where farmers and buyers decide on a fixed price before planting or harvesting. This protects both sides from sudden price changes, giving farmers guaranteed income and buyers price certainty.
- **A stabilisation fund policy:** This is a financial mechanism used by the government to reduce the impact of price fluctuations in agricultural markets by saving surplus earnings during periods of high prices and using those savings to support farmers when prices fall. In practice, when agricultural products fetch high prices, part of the extra revenue is set aside in the fund; during periods of low prices, the government uses this fund to provide subsidies or guarantee minimum payments to farmers. This helps stabilize farmers' incomes, encourages consistent production, and prevents sharp declines or disruptions in supply, ultimately contributing to more stable agricultural prices over time.

4. International / Multilateral Trade Measures

International / multilateral trade measures focus on cooperation between countries to manage agricultural markets beyond national borders. Their main purpose is to reduce extreme price fluctuations by coordinating production, trade policies, and market behavior at the global level. They include the International Commodity Agreement (ICA)

- **International commodity agreements (ICA):** These are formal agreements between countries that produce and trade specific agricultural products (such as coffee, cocoa, or sugar). These agreements help stabilize prices by setting production limits, export quotas, or price ranges. By controlling how much is supplied to the global market, they prevent oversupply (which causes prices to fall) and shortages (which cause prices to rise). As a result, they promote more predictable and stable incomes for farmers and more consistent prices for consumers worldwide. For example, International Coffee Agreement (ICA), International Tropical Timber Agreement (ITTA), International Cocoa Agreement (ICCO). Etc.

<https://tinyurl.com/yfum2mxa>,

https://kawa.ac.ug/kawa_economics/measures_to_stabilize_prices_in_agriculture.html

Application activity 4.2



Answer the following questions using your knowledge of price fluctuations.

1. a) What is meant by price fluctuation?
b) Why do prices of agricultural products fluctuate more than those of manufactured goods?
2. a) A prolonged drought in a farming region leads to a sharp fall in crop output. Explain how this situation is likely to affect:
 - i. the prices of agricultural products.
 - ii. farmers' incomes.
- b) The government introduces a buffer stock scheme to stabilize food prices. Explain:
 - i. how this measure works.
 - ii. how it helps to control price fluctuations of agricultural products.

4.3 Price controls

Learning Activity 4.3



Read the situation below carefully and answer the questions that follow.

Situation:

In a certain town, traders were free to set prices for essential goods such as food and medicine. During a shortage, prices rose sharply, making it difficult for many people to afford basic necessities. At other times, when there was a very large supply, prices fell so much that farmers and traders made very little profit. To address these problems, the government decided to set limits on how high and how low prices could go. It also supplied some goods through selected outlets and restricted the quantity each person could buy.

Questions

- i. Why did prices rise during the shortage, and why did they fall when supply was high?
- ii. What actions did the government take to influence prices in the market?
- iii. How do these actions help consumers during periods of high prices?
- iv. How do these actions help producers during periods of very low prices?
- v. Why do you think the government limited the quantity each person could buy?
- vi. What problems might arise from the government controlling prices in this way?

Governments sometimes intervene in markets when prices become a concern for society, especially during periods of economic instability or rising living costs. In such situations, authorities may take action to influence price levels in order to protect consumers, support producers, and promote equity. These interventions reflect efforts to balance market outcomes with social and economic objectives.

4.3.1 Meaning and forms of price controls

i. Meaning of price controls

While a free-market system relies entirely on the forces of demand and supply to establish the prices of goods and services, governments frequently intervene to alter these market outcomes through **price controls** or **price administration**. Price controls **are government-mandated legal limits placed on the prices that sellers can charge for goods and services in a market**. Governments typically implement these regulations to protect consumers from high costs or to guarantee a stable income for producers during economic shocks. Price mechanism is not allowed to influence prices in the market.

ii. Forms of price control

Price controls are implemented depending on whether the goal is to limit high prices or to support low prices in the market. These different approaches are referred to as forms of price control. The main forms of price control are outlined below.:

- Maximum price (price ceiling)
- Minimum price (price floor)

1. Maximum Price Legislation (Price Ceiling).

A maximum price, also known as a price ceiling, is the highest price legally fixed by the government, above which it is not permitted to buy or sell a good or service. It is usually set below the equilibrium price with the aim of making essential goods and services affordable to consumers and protecting them from excessively high prices. This policy is enforced particularly in situations where high demand or limited supply would otherwise lead to a significant rise in prices.

Reasons for government setting a maximum price (price ceiling):

Governments introduce a maximum price (price ceiling) to intervene in the free market when the equilibrium price of an essential item rises too high for the average citizen to afford. The primary objectives for setting a maximum price include;

- **To make essential goods affordable:** Governments want to ensure that basic needs like food, water, electricity, and medicine remain within reach for all citizens, especially low-income groups.
- **To protect consumers from high prices:** When market prices rise too much, a price ceiling helps prevent people from being overburdened by the high cost of living.
- **To prevent exploitation and price gouging (overcharging):** It stops sellers or powerful firms from charging extremely high prices during shortages or emergencies.
- **To control inflation:** By limiting how high prices can rise, the government helps reduce rapid increases in the overall cost of living.
- **To ensure fair distribution of scarce goods:** Lower prices encourage wider access so that more people can obtain important goods instead of only the rich.
- **To promote social stability:** When people can afford necessities, it reduces the chances of protests, unrest, or economic hardship in society.
- **To support housing affordability (rent control):** Governments may set maximum rents to help people afford housing in cities where demand is very high.

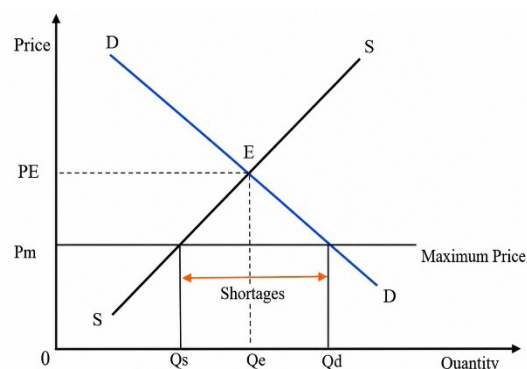


Figure 4.2 Maximum price curve

OPE is the equilibrium price and OQE is the equilibrium quantity at point E. When the government sets a maximum price below the equilibrium at OP_m, the quantity supplied falls from OQ_e to OQ_s, while the quantity demanded rises from OQ_e to OQ_d. This creates a shortage equal to the difference between OQ_d and OQ_s. The maximum price protects consumers from being overcharged by profit-seeking sellers. It also helps stabilize prices, control monopolies, and encourage consumption. Additionally, it improves population welfare and helps control inflation.

Effects of Maximum price legislation (price ceiling):

The imposition of a maximum price below the equilibrium level leads to several significant effects in the market, influencing consumer demand, producer supply, and the allocation of goods and services; while some of these effects improve affordability and protect consumers, others create shortages and inefficiencies as explained below.

Positive effects

The imposition of a maximum price can give rise to a number of beneficial effects, particularly in enhancing consumer welfare in the market, as discussed below.

- **Increases affordability of essential goods:** Price ceilings make goods cheaper than they would be in a free market. This enables consumers, especially low-income earners, to afford basic necessities such as food, fuel, and housing that might otherwise be too expensive.
- **Protects consumers from exploitation:** In situations where producers have monopoly power or where demand is very high, price ceilings prevent sellers from charging excessively high prices, thus protecting consumers from exploitation.
- **Improves access to basic needs:** By keeping prices low, more people are able to access essential goods and services. This promotes fairness and helps reduce inequality in access to necessities.
- **Promotes social equity:** Price ceilings help redistribute access to scarce goods by ensuring that even poorer members of society can purchase essential commodities at controlled prices.

Negative effects of the maximum price legislation:

Although governments set maximum prices with the intention of helping vulnerable consumers, placing a price below the market equilibrium disrupts the normal working of supply and demand. This interference removes the natural price signals that guide buyers and sellers, resulting in several serious negative effects in the market as explained below;

- **Chronic market shortages:** When the government fixes a low price, more people want to buy the good because it is cheaper, while producers lose motivation to supply it. This creates a shortage where demand is greater than supply, leading to empty shelves, long lines, and many customers failing to get the product.
- **Artificial hoarding:** Because people expect goods to run out quickly at low prices, they buy more than they actually need. This panic buying reduces the available stock even faster, making shortages more severe and widespread.
- **Emergence of black markets:** When goods become scarce at the official price, illegal markets develop where they are sold secretly at much higher prices. Sellers take advantage of the shortage, and desperate buyers end up paying more than they would have in a free market.
- **Decline in product quality and maintenance:** Since sellers cannot raise prices to cover costs, they reduce quality to save money. This may include using cheaper materials, offering smaller quantities, or neglecting maintenance, especially in housing under rent control.
- **Non-price rationing and corruption:** With price no longer controlling distribution, goods are allocated through other unfair means such as long queues, favoritism, or bribery. People with connections or extra money benefit, while others are left without access.

Application of Price Ceiling

A price ceiling is applied by the government in markets for essential goods and services where prices may become too high for consumers. It involves legally fixing a maximum price below the equilibrium level, which sellers are not allowed to exceed. This policy is commonly used in sectors such as food, fuel, and housing to improve affordability and access, particularly during periods of high demand, shortages, or rising prices. A common and important application of price ceilings in practice is **rent control** in the housing market.

Rent Control

Rent control is a specific application of a price ceiling in which the government sets a maximum rent that landlords can charge tenants. Its main objective is to make housing more affordable, especially for low-income earners in urban areas where rental prices tend to be high. As a result, more people are able to access accommodation at controlled rents. However, rent control can also lead to negative effects, as landlords may be discouraged from constructing new houses or maintaining existing ones due to reduced profitability. Over time, this can reduce the supply of rental housing and lead to shortages.

Complementary Measures to Price Ceilings

Although price ceilings such as rent control are intended to make essential goods and services more affordable, they often result in shortages and other market distortions. In order to address these challenges and improve the effectiveness of price control policies, governments introduce a range of complementary measures to regulate supply, ensure fair distribution, and maintain market stability.

- **Rationing** is one of the primary measures used to manage shortages created by price ceilings. Rationing refers to a system of limiting the quantity of a good or service that each consumer can obtain. By restricting individual consumption, rationing helps ensure that the available supply is shared more equitably among consumers, particularly when demand exceeds supply. In practice, goods may also be distributed through non-price mechanisms such as queues or on a **first-come, first-served basis**, although these methods may not always ensure equitable allocation. For example, as of 2026, Rwanda's water utility WASAC rationed water in Kigali through rotational supply schedules rather than letting prices rise to clear the shortage. Similarly, during a regional fuel shortage earlier that year, Rwandan petrol stations limited the litres each customer could buy instead of allowing pump prices to rise freely.
- Another important measure is the **provision of subsidies**, which are financial payments given by the government to producers or suppliers. Subsidies help to offset the losses that firms may face due to lower controlled prices, thereby encouraging them to continue producing and supplying goods. In the context of rent control, subsidies may be given to landlords or housing developers to support construction and maintenance of rental properties. For example, as of 2026, the Rwandan government spent about Rwf47.7 billion on fuel subsidies to keep diesel priced at Rwf2,927 per litre instead of the market price of roughly Rwf3,600, cushioning suppliers from rising global oil costs (The New Times, 2026, July 10).
- Governments may also engage in **direct provision of goods and services**, especially in sectors where private supply is insufficient. For example, the construction of

public housing helps to increase the supply of affordable accommodation, thereby reducing the pressure created by rent control policies. For example, the Rwandan government has directly built housing through model villages (imidugudu), such as the Rugerero Model Village in Rubavu District and the Karama Integrated Model Village in Kigali; two of over 130 such settlements constructed countrywide to provide housing for vulnerable citizens who could not otherwise access it through the private market.

- In addition, **production incentives** such as tax relief, grants, or low-interest loans may be offered to encourage producers to expand output. These incentives are particularly important in industries where profit margins are reduced due to price controls, as they help sustain long-term supply. For example, the Rwandan government forgave about Rwf291.6 billion in import taxes during the 2024/25 financial year to support manufacturers, particularly those in the Kigali Special Economic Zone (KT Press, 2026, July 21).
- To prevent artificial shortages, governments may enforce **anti-hoarding regulations**, which prohibit individuals or firms from stockpiling goods for speculative purposes. Such measures ensure that goods remain available in the market and are not deliberately withheld to sell at higher prices later. For example, Rwanda's Competition and Consumer Protection Act (Law n° 011/2026 of 26 February 2026) legally prohibits speculative hoarding and price fixing, and empowers the Rwanda Inspectorate, Competition and Consumer Protection Authority (RICA) to investigate and act against traders who withhold goods to manipulate prices.
- Finally, governments may allow or facilitate the importation of goods to supplement domestic supply. By increasing the availability of goods in the market, imports help to reduce shortages and stabilize prices. In Rwanda, imports play an important role in addressing shortages of goods such as fuel and certain food products, thereby helping to maintain market stability when domestic supply is insufficient.

2. Minimum price (Price floor)

A minimum price, also known as a price floor, is the lowest price that is legally fixed by the government above the equilibrium price, below which buying and selling of a good or service is not allowed. It is introduced to protect producers by ensuring they receive a fair income for their products. Examples include minimum wage set for workers and minimum prices fixed for agricultural products to protect farmers from exploitation by profit-oriented traders or middlemen.

Reasons for government setting a minimum price (price floor):

Governments set a minimum price (price floor) to legally prevent prices from dropping below a specific level, primarily to safeguard the financial survival of producers and workers. To be legally effective, this price floor must be placed above the natural market equilibrium price. The primary objectives for setting a minimum price include;

- **Protecting producer incomes (especially farmers):** A minimum price helps farmers avoid losses when market prices fall too low due to oversupply or good harvests.

It ensures they earn enough to cover their costs and continue farming, reducing the risk of poverty and business failure.

- **Safeguarding workers (minimum wage):** In the labor market, a minimum wage prevents employers from paying very low wages, especially when jobs are scarce. It helps workers earn enough to meet basic needs like food, housing, and healthcare.
- **Discouraging consumption of harmful goods:** Governments may set high minimum prices on goods like alcohol and cigarettes to make them less affordable. This reduces their consumption and helps protect public health.
- **Encouraging production of essential goods:** When prices for key goods like food crops fall too low, producers may stop making them. A minimum price encourages continued production, helping ensure food security and reducing reliance on imports.
- **Protecting small producers from unfair trade:** Small producers often face powerful buyers who can force prices down. A price floor sets a legal limit to prevent such unfair pricing, helping weaker producers get a fair deal. https://en.wikipedia.org/wiki/Price_controls

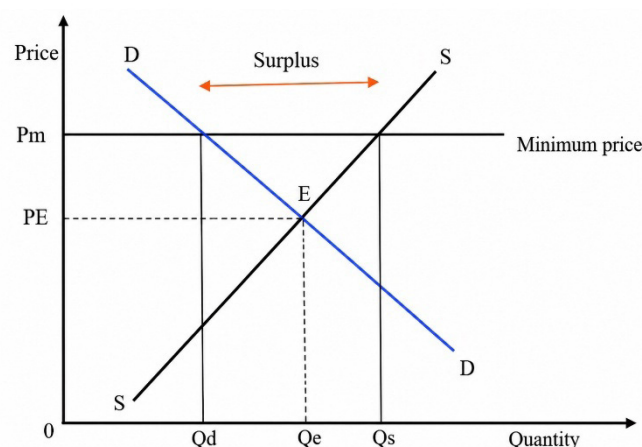


Figure 4.3 Minimum price legislation

OPE is the equilibrium price and OQe is the equilibrium quantity at point E. When the government sets a minimum price above equilibrium at OPm, the quantity supplied rises from OQe to OQs, while the quantity demanded falls from OQe to OQd. This creates excess supply, equal to the difference between OQs and OQd.

Effects of Minimum Price Legislation (Price Floor):

The imposition of a minimum price above the equilibrium level leads to several significant effects in the market, influencing consumer demand, producer supply, and the allocation of goods and services; while some of these effects improve producer welfare and income stability, others create surpluses and inefficiencies as explained below.

Positive Effects of Minimum Price:

The imposition of a minimum price can give rise to a number of beneficial effects, particularly in enhancing producer welfare in the market, as discussed below.

- **Higher Producer Income:** A minimum price ensures that producers receive a higher and more stable income for their goods or services, which improves their standard of living and financial security.
- **Encouragement of Production:** By guaranteeing higher returns, minimum prices encourage producers to increase output, thereby supporting long-term production, especially in essential sectors like agriculture.
- **Price Stability:** Minimum price policies help stabilize prices by preventing them from falling below a certain level, which protects producers from unpredictable market fluctuations.

Negative effects of the minimum price legislation:

Although minimum price legislation aims to protect producers, setting the price above equilibrium disrupts the normal functioning of demand and supply. This intervention distorts the price mechanism, leading to several adverse effects in the market, as discussed below.

- **Chronic market surpluses (excess supply):** When the government sets a higher minimum price, producers are encouraged to supply more goods to earn higher profits, while consumers buy less because prices are high. This leads to excess goods that remain unsold.
- **Financial burden on the government:** To maintain the minimum price, the government may have to buy the unsold surplus, which costs a lot of public money that could have been used for services like education and healthcare.
- **Heavy storage and waste costs:** After buying surplus goods, the government must store them, which is expensive. If the goods are perishable, such as food, many may spoil and go to waste.
- **Rise of black markets (illegal selling):** When goods cannot be sold at the high legal price, some producers may sell them secretly at lower prices just to avoid losses, leading to illegal markets.
- **Unintended unemployment (in labor markets):** If a minimum wage is set too high, firms may not afford to hire as many workers. This can lead to job cuts, reduced working hours, or fewer job opportunities.
- **Inefficiency and lack of innovation:** Since producers are guaranteed a high price, they may not try to reduce costs, improve quality, or innovate, which makes production less efficient. <https://shorturl.at/bgMb8>

Application of Minimum Price Legislation

Minimum price legislation is applied in markets where the government seeks to protect producers or workers from excessively low prices and incomes. It involves setting a legal minimum price above the equilibrium level, below which goods or services cannot be sold. This policy is commonly used in sectors where income stability is important, such as agriculture and labour markets, to promote economic security and encourage continued production.

A common application of minimum price legislation is in the form of **agricultural price supports** and **minimum wage laws**. In agriculture, the government may set a minimum price for crops to ensure farmers receive a stable income despite market fluctuations. Similarly, the minimum wage sets the lowest legal payment for labour, aimed at protecting workers from exploitation and ensuring a basic standard of living.

These applications help improve producer and worker welfare, although they may also lead to market imbalances such as surpluses or unemployment.

Complementary measures to minimum price legislation

When a minimum price is set above the equilibrium level, it often results in surplus and market inefficiencies. To address these challenges, governments introduce complementary measures to manage excess supply and improve the effectiveness of the policy, as discussed below.

- **Government purchase of surplus:** One important measure is the purchase of surplus output by the government. When a minimum price leads to excess supply, the government buys the unsold goods to support producers and prevent prices from falling. This helps maintain producer incomes and stabilizes the market. For example, in Rwanda, the government operates a National Strategic Grain Reserve, purchasing surplus maize and beans from farmers for storage, with capacity set to grow from 46,000 to 200,000 tonnes by 2029, helping absorb excess harvests and support farm gate prices.
- **Buffer stock schemes:** Closely related to surplus purchase is the use of buffer stock schemes, where the government stores excess goods bought during periods of surplus and releases them when there is a shortage. This helps stabilize both supply and prices over time, reducing extreme fluctuations in the market. For example, in Rwanda, the National Strategic Grain Reserve buys and stores surplus maize and beans during good harvests, and, when adequately stocked, can release grain to the market during droughts, food price spikes, or other shocks, acting as a "shock absorber" that helps stabilize prices and keep staples affordable during shortages.
- **Export promotion:** Governments may also encourage exportation of surplus goods to foreign markets. By selling excess production abroad, domestic surplus is reduced, helping to balance supply and demand while earning foreign exchange for the country. For example, in Rwanda, the National Agricultural Export Development Board (NAEB) supports exporters of surplus fruits and vegetables with storage and export infrastructure, helping agricultural export revenues reach a record \$1.1 billion in the 2025/26 financial year (AllAfrica, 2026, August 13).
- **Production controls:** To avoid persistent surpluses, the government may introduce production controls, such as limiting the quantity that producers can produce. This helps align output more closely with demand and prevents unnecessary wastage of resources.
- **Subsidies:** In some cases, the government may provide subsidies to producers to support their income without requiring excessively high minimum prices. This reduces the need for large surpluses while still ensuring that producers earn a reasonable return. For example, in Rwanda, the government subsidises fertiliser costs for farmers, with the subsidy budget reaching about Rwf64 billion for 2026/27, supporting farmer incomes without relying solely on high minimum prices (The New Times, 2026).
- **Storage and preservation facilities:** The government may invest in storage and preservation facilities, particularly for perishable goods. This ensures that surplus production is not wasted and can be used or sold in the future, thereby improving resource utilization. For example in Rwanda, the National Strategic Grain Reserve is expanding maize storage capacity from 46,000 to 200,000 tonnes by 2029, reducing post-harvest losses and preserving surplus grain for future use (AllAfrica, 2026).

4.3.2 Effects of price controls

Price controls have both positive and negative effects depending on how they are applied, as they can help protect consumers and producers while at the same time disrupting the normal functioning of the market and creating unintended challenges.

i. Positive Effects of price controls

Price controls can bring several benefits when they are well designed and properly enforced, as they help improve the welfare of citizens, protect vulnerable groups, and promote economic and social stability in a country. The following points explain the main positive effects of price controls.

- **Reduces Poverty and Enhances Welfare:** Price ceilings keep survival essentials (such as food, basic utilities, and medicine) affordable for low-income households, protecting them from extreme poverty.
- **Guarantees Income Stability for Farmers:** Price floors protect rural agricultural producers from sudden global price crashes, ensuring they earn enough to cover their production costs and remain financially solvent.
- **Protects low-skilled workers:** Minimum wage laws establish a legal baseline for earnings, ensuring that employers cannot exploit desperate job seekers with below-subsistence wages.
- **Controls inflationary panic:** During natural disasters, pandemics, or wars, maximum prices stop predatory sellers from inflating prices (price gouging) on scarce necessities like fuel or emergency supplies.
- **Protects national food security:** By guaranteeing a minimum profit for major staples like corn, wheat, or rice, price floors incentivize farmers to keep growing food, reducing a nation's reliance on risky foreign imports.
- **Reduces income inequality and wealth gaps:** Price ceilings on basic utilities and rent prevent wealth from concentrating solely in the hands of landlords and utility monopolies, keeping more disposable income in the pockets of the working class.
- **Encourages long-term agricultural investment:** When farmers are protected by predictable price floors, they gain the financial confidence to invest in long-term farm improvements, such as modern irrigation systems, high-quality seeds, and advanced tractors
- **Maintains social and political stability:** By capping prices on critical goods like fuel and bread during global economic shocks, governments prevent the rapid cost-of-living spikes that historically spark public strikes, protests, and civil unrest
- **Improves public health outcomes (Demerit Goods):** High minimum price floors placed on items like alcohol and tobacco reduce overall consumption, directly lowering the long-term rate of chronic illnesses and reducing the financial burden on public healthcare systems.
- **Enhances labor productivity:** Minimum wage laws ensure workers can afford nutritious food, stable housing, and healthcare; healthy and financially secure employees demonstrate higher motivation, lower absenteeism, and increased efficiency on the job.

ii. Negative Effects of price controls.

Price controls can also lead to several negative effects when they interfere with the normal functioning of the market, often creating shortages, inefficiencies, and unintended economic problems or secondary distortions. The following points explain the main disadvantages of price controls.

- **Creates chronic shortages or surpluses:** Price ceilings inevitably cause a permanent state of excess demand (shortages) where items disappear from store shelves. Price floors cause excess supply (surpluses) where unsold goods pile up and rot in warehouses.
- **Spurs parallel black markets:** When legal controls make a good scarce or too expensive, illegal underground economies instantly emerge. Under a ceiling, buyers secretly pay massive premiums under the counter to get scarce goods; under a floor, desperate sellers secretly discount their products below the legal limit just to make a sale.
- **Causes resource misallocation and waste:** With price floors, governments must spend massive amounts of taxpayer money to purchase and store unsold product surpluses. If these items are perishable agricultural goods, tons of food frequently rot before being consumed.
- **Causes rising unemployment:** When a minimum price floor is set too high in labor markets (minimum wage), it elevates production costs for businesses. To survive, firms often reduce their hiring or lay off low-skilled workers.
- **Deteriorates product quality:** Under maximum price limits, manufacturers can no longer raise prices to cover rising raw material costs. To protect their profit margins, they cut corners by reducing product sizes (shrinkflation), using cheaper ingredients, or neglecting property maintenance (in rent-controlled housing)
- **Destroys market-driven allocation (loss of allocative efficiency):** In a free market, high prices signal factories to produce more of a scarce item. Capping prices via a ceiling destroys this signal, ensuring that resources fail to flow to industries where they are needed most.
- **Imposes massive taxpayer opportunity costs:** When governments use public funds to buy up unsold crop surpluses caused by price floors, that tax revenue is diverted away from critical development areas like building schools, expanding electrical grids, or paving rural roads.
- **Encourages destructive "shrinkflation" and bribery:** Under strict price ceilings, businesses can no longer raise prices to cover rising costs; they instead reduce product sizes, lower ingredient quality, or resort to under-the-table favoritism and bribery to distribute scarce goods.
- **Drives long-term industrial exit:** If a price ceiling forces a factory to operate at a financial loss for too long, the owners will shut down operations entirely or relocate to another country, permanently damaging the domestic manufacturing sector.
- **Incentivizes the expansion of the shadow economy:** Long-term price controls cause citizens to lose faith in official retail systems, expanding a parallel underground economy where tax collection is impossible and consumer protection laws do not apply.

<https://www.economicshelp.org/blog/621/economics/price-controls-advantages-and-disadvantages>.

Application activity 4.3



1. a) Distinguish between maximum and minimum Price legislations.
b) Why is it important for the government to set a minimum price legislation?
2. Examine the impact of price controls on consumers, producers and the entire economy.

End of unit assessment



Knowledge

1. Give the meaning of price mechanism and explain how it determines the allocation of resources in a free market.
2. Explain the causes of price fluctuations in agricultural products and discuss their effects on farmers' incomes and planning.
3. a) List any five roles of the price mechanism.
b) State any four assumptions of the price mechanism.
c) State the advantages and disadvantages of price mechanism.

Skills

4. Illustrate how price controls (ceilings and floors) affect the market.
5. a) Discuss the effects of price fluctuations on producers and consumers.
b. How are buffer stocks helpful in correcting the problems caused by price fluctuations in agriculture products?
6. Analyze how the price mechanism allocates resources efficiently.

Attitudes

7. Explain why it is important for the government to intervene in the price mechanism.
8. Explain the importance of consumer associations in protecting buyers.

UNIT 5

CONSUMER BEHAVIOR



Key Unit Competence:

Analyse how rational consumers allocate limited income under budget constraints to maximize utility and reach equilibrium in consumption.

Learning objectives

Knowledge and understanding

1

- Define consumer and consumer behaviour,
- state the basic assumptions about consumer choices
- Explain the relationship between total utility and marginal utility using numerical examples and curves.
- State and explain the law of diminishing marginal utility, including its assumptions and limitations.
- Define the budget line and explain the concept of a budget constraint.
- Explain the effects of changes in income and prices on the budget line.
- Define an indifference curve and explain its assumptions and properties.
- Explain the concept of an indifference map.
- Define the marginal rate of substitution (MRS) and explain its relationship with indifference curves.
- Explain the conditions for consumer equilibrium using the budget line and indifference curve.
- Explain how consumers maximize utility when allocating limited income.
- Define and explain the income effect and substitution effect of a price change.
- Explain how price changes affect consumer choice.
- Define consumer surplus and producer surplus and explain how they are measured and interpreted in real markets.

Skills

2

- Analyze consumer behavior using utility, budget constraints, and indifference curve analysis.
- Calculate and interpret total and marginal utility from given data.
- Apply the law of diminishing marginal utility to explain consumer decision-making.
- Draw and interpret budget lines, showing the effects of income and price changes.
- Draw and interpret indifference curves and indifference maps.
- Analyze marginal rate of substitution (MRS) and explain its diminishing nature.
- Determine consumer equilibrium graphically using indifference curves and budget lines.
- Analyze income and substitution effects to explain changes in consumption patterns following price changes.
- Calculate and interpret consumer and producer surplus using numerical and graphical methods.
- Apply consumer behaviour analysis to real-life situations, such as budgeting and consumption choices.

Attitudes and values

3

- Appreciate the importance of rational decision-making in consumption under limited income.
- Demonstrate responsible use of limited resources through informed consumer choices.
- Value budgeting and planning as tools for maximizing satisfaction.
- Develop awareness of opportunity cost in consumption decisions.
- Show fairness and efficiency awareness when interpreting consumer and producer surplus.
- Recognize the relevance of consumer behaviour analysis in improving personal wellbeing and market efficiency.

If you had 1,000 FRW to spend today, how would you choose the goods that would give you the greatest satisfaction? This reflects how individuals make decisions about the use of their limited income to satisfy unlimited wants. In doing so, people aim to maximize the benefits they gain from consumption while making choices between different goods. These decisions involve understanding how satisfaction changes as more of a good is consumed, how income and prices influence purchasing power, and how consumers balance their preferences within their budget constraints. As a result, consumer behavior shapes demand in the market and influences how resources are allocated in the economy.



Introductory activity

Scenario:

During lunch break, three students go to a nearby shop with a limited amount of money. One student buys bread and milk because he is very hungry, another buys a soft drink because she is thirsty, while the third student compares prices before deciding what to buy so that the money can buy more items. Each student makes a different choice depending on their needs and the money they have.

Questions:

1. Why did the students choose different items to buy at the shop?
2. How did the amount of money they had influence their choices?

5.1 Introduction to consumer behaviour

Learning Activity 5.1



Analyse the scenario below and answer the questions that follow.

Jude had a fixed amount of money to spend on food and chose to buy slices of cake to satisfy his wants. At first, he enjoyed the cake and gained high satisfaction, but as he continued eating more, he began to feel discomfort.

- i. Why did Jude choose to spend his money on cake instead of other foods?
- ii. How did Jude feel after eating the first slices of cake compared to later ones?
- iii. Why do you think Jude continued buying more cake even after he started feeling discomfort?
- iv. What could Jude have done differently with his money to feel more satisfied?

People make many choices every day about what to buy and how to spend their money. These choices are influenced by factors such as income, prices, tastes, and advertising. Different people respond differently in the market, and these differences affect demand and business activities. Studying these patterns helps us better understand how consumers act in everyday life and in the economy.

5.1.1 Consumer Behaviour and Its Assumptions

i. Meaning of Consumer and consumer behaviour

A consumer is a person who buys goods or services for his or her own use to satisfy needs and wants. The consumer is the final user in the chain of distribution. Goods move from the producer through channels such as wholesalers, retailers, or agents before reaching the consumer.

Consumer behaviour refers to the way consumers choose, buy, use, and react to goods and services to satisfy their needs and wants. It studies the decisions consumers make when spending their income on different products.

ii. Assumptions of Consumer Behaviour

Consumer behaviour is based on several simplifying assumptions that help economists understand and explain how individuals make choices when faced with limited resources. These assumptions provide a foundation for analyzing how consumers aim to maximize satisfaction and make consistent and logical decisions in the market.

- **Rationality:** Consumers are rational, meaning they make decisions aimed at maximizing their satisfaction from available resources. For example, a student with 1,000 RWF for lunch must choose between buying a cake and juice, or a plate of rice and beans. A rational student chooses the option that gives the most satisfaction for the money available, rather than spending randomly.
- **Utility maximization:** Consumers aim to get the highest possible satisfaction (utility) from their limited income. For example, a student with 1,000 RWF chooses to spend it on food that gives the most satisfaction, rather than on items that give less value for the same amount of money.
- **Consistency of preferences:** Consumer choices are consistent over time; given the same situation, they will make the same decision. For example, if a student prefers chapati to bread when both cost the same, they will always choose chapati under the same conditions, rather than choosing differently each time.
- **Transitivity of preferences:** If a consumer prefers good A over B and B over C, then they will also prefer A over C. For example, if a student prefers rice to chapati, and chapati to bread, then the student will also prefer rice to bread.
- **Diminishing marginal utility:** As a consumer consumes more units of a good, the additional satisfaction gained from each extra unit decreases. For example, the first samosa a hungry student eats gives a lot of satisfaction, but each additional samosa gives less satisfaction than the one before.
- **Perfect information:** Consumers are assumed to have full knowledge about prices, quality, and availability of goods. For example, a student is assumed to already know the prices and quality of snacks sold by different vendors near school before deciding where to buy.
- **Given income and prices:** Consumer income and prices of goods are assumed to be fixed when making decision. For example, a student with 1,000 RWF for lunch assumes both the amount of money and the prices of food will not change while deciding what to buy.

5.2 The concept of utility

In our daily lives, we are constantly making choices—what to eat, what to buy, how to spend our time, and how to use our resources. Because resources such as money and time are limited while human wants are unlimited, individuals must make careful decisions. At the heart of these choices lies a key concept in economics known as utility, which helps explain how people allocate their scarce resources in ways that lead to the greatest possible satisfaction.

5.2.1 Meaning of utility

Utility refers to the satisfaction or pleasure a consumer derives from using a good or service. It helps explain why people choose certain goods and how they try to get the most satisfaction from limited resources. Traditionally, economists measure utility using a theoretical unit called **utils**, but modern economics often treats it as **ordinal**, meaning satisfaction is ranked rather than precisely measured.

Utility is a subjective measure of personal satisfaction rather than practical usefulness, meaning its value depends entirely on individual preferences and varies from person to person.

There are two ways to view utility:

- **Total Utility (TU):** This refers to the overall amount of satisfaction a consumer receives from consuming all units of a particular commodity. For example, if someone eats three apples and gains 20 utils from the first apple, 15 utils from the second, and 10 utils from the third, their total utility from consuming all three apples would be 45 utils.

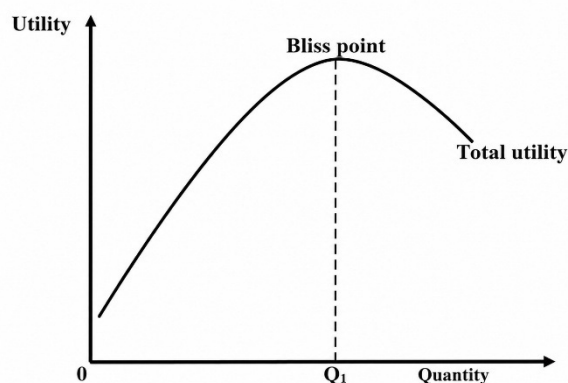


Figure 5. 1 The TU curve

Total utility first increases until quantity Q_1 then it starts diminishing. As more units of a commodity are consumed total utility increases because each new unit consumed adds to the total. This means that as one adds more units of a commodity, there is extra or additional satisfaction to him/her. This is called **Marginal utility (MU)**.

- **Marginal Utility (MU):** This refers to the additional satisfaction gained from consuming one more unit of the commodity. Using the same apples example, if the consumer eats the second apple, the marginal utility of the second apple is the difference between the total utility after consuming two apples and after consuming one apple: (35 utils - 20 utils = 15 utils).

$$MU = \frac{\Delta TU}{\Delta Q}$$

Where; ΔTU = change in total satisfaction (New total utility – old total utility) .

ΔQ = Change in the number of units consumed (New quantity – old quantity).

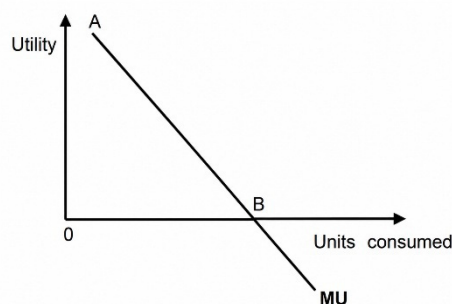


Figure 5. 2 the MU curve

From point A to point B MU is still positive. At point B MU is zero (0) and after that point, MU becomes negative (disutility)

5.2.2 The relationship between Total Utility (TU) and marginal Utility (MU).

The relationship between Total Utility (TU) and Marginal Utility (MU) shows how a consumer's total satisfaction changes as more units of a good are consumed. TU represents overall satisfaction, while MU measures the extra satisfaction from each additional unit. This relationship explains why satisfaction increases at first, reaches a maximum, and may eventually decline.

This relationship can be clearly illustrated in two main ways:

- Using a schedule (table): where TU and MU values are calculated for different units consumed.
- Using curves (graphs): where the TU and MU values are plotted to visually show how they move in relation to each other.

Table 5.1 A hypothetical utility schedule

Units Consumed (Q)	Total Utility (TU)	Marginal Utility (MU)
0	0	-
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2
8	24	-4

Observation from the schedule

Total Utility (TU) increases at a decreasing rate up to 6 units of consumption, where it reaches its maximum value of 30 units, this point is known as the **saturation or bliss point or point of satiety**. At the same time, Marginal Utility (MU) is positive as satisfaction increases, becomes zero when TU is at its maximum, and turns negative beyond this point, leading to **disutility** as additional consumption reduces overall satisfaction.

The first MU is a dash if no initial utility is given, but equals the first TU when consumption starts from zero. If the table counts by consecutive ones (0, 1, 2, 3, ...), the first unit's (MU) equals (TU). If the table skips numbers (1, 3, 5....), the first unit's (MU) must be a dash (-).

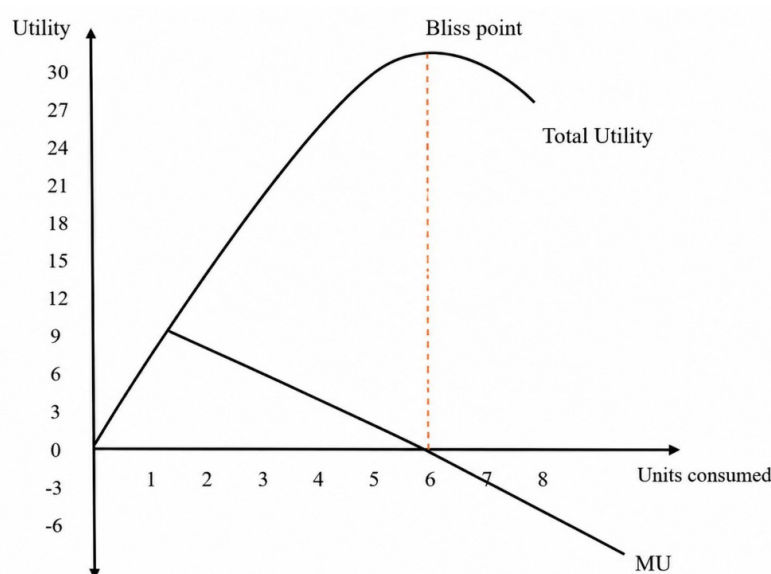


Fig 5.3 Relationship between TU and MU using curves

Observation

The graph above illustrates how Total Utility (TU) and Marginal Utility (MU) change as more units of a commodity are consumed.

The TU curve rises initially, showing that total satisfaction increases, but it does so at a decreasing rate because each additional unit adds less satisfaction than the previous one. TU reaches its highest point at about 6 units, known as **the bliss point or saturation point**, where the consumer achieves maximum satisfaction. Beyond this point, consuming more units leads to a decline in TU, indicating dissatisfaction.

The MU curve, on the other hand, slopes downward throughout, reflecting **the law of diminishing marginal utility**. It becomes zero at the 6th unit, which corresponds to the maximum TU, and turns negative thereafter, showing **disutility**, where additional consumption reduces overall satisfaction.

Total Utility (TU) depends on Marginal Utility (MU), meaning that any change in MU causes a change in TU. When MU is positive, TU increases; when MU is zero, TU is at

its maximum; and when MU is negative, TU decreases.

A consumer maximizes total satisfaction by stopping consumption the exact moment **Marginal Utility hits zero**. Eating anything beyond the satiety point turns a good into a **"bad,"** causing Total Utility to fall due to the onset of disutility.

Table 5. 2 Summary of the Relationship Between TU and MU (Table Form)

Marginal Utility (MU)	Total Utility (TU)	Explanation
MU is positive	TU is increasing	Each additional unit adds satisfaction, so total satisfaction rises
MU is decreasing	TU increases at a decreasing rate	Satisfaction still increases, but by smaller amounts
MU = zero	TU is maximum	Consumer reaches the saturation (bliss) point
MU is negative	TU is declining	Extra consumption causes disutility (loss of satisfaction)

Exercise 5.1

Use the table below to answer the questions that follow:

Quantity (Q)	1	3	5	7	9	11	13
Total Utility (TU)	50	80	100	110	110	106	100

- Calculate the MU at each unit of output consumed.
- Illustrate the information on graphs

5.2.3 The law of diminishing Utility

From the table and the graph, we can clearly see that as more units are consumed, the marginal utility keeps decreasing. This observation leads us to an important economic principle called the Law of Diminishing Marginal Utility." The law states that, **"as a consumer consumes more units of a commodity, the marginal utility derived from each additional unit decreases, other things remaining constant"**

<https://2012books.lardbucket.org/books/microeconomics-principles-v1.0/s10-01-the-concept-of-utility.html>

As a consumer continues to consume more units of a good, the additional satisfaction gained from each unit gradually decreases. For example, when very hungry, the first plate of food gives high satisfaction, the second gives less, and this continues with each extra plate. Eventually, a point is reached where no more satisfaction is gained, this is the point of satiety or bliss point. If consumption continues beyond this point, satisfaction becomes negative, leading to disutility, such as discomfort from overeating.

ii) Assumptions of the Law of Diminishing Marginal Utility

- **Rational consumer:** The consumer makes logical choices aimed at getting the highest possible satisfaction from what they consume.
- **Suitable unit size:** Goods are consumed in normal, reasonable quantities so that satisfaction changes can be clearly observed.
- **Constant tastes:** The consumer's preferences do not change during consumption, so satisfaction depends only on the quantity consumed.
- **Continuous consumption:** The good is consumed without long time gaps, ensuring that the reduction in satisfaction happens steadily with each extra unit.
- **Income remains constant:** The consumer's income does not change during consumption, so their level of satisfaction is not affected by changes in purchasing power.
- **Price remains unchanged:** The price of the commodity stays the same, ensuring that changes in satisfaction are due only to consumption, not price fluctuations.
- **Units are identical (homogeneous):** All units of the good consumed are the same in size, quality, and nature, so each unit can be compared fairly in terms of satisfaction.

iii) Limitations of the law of diminishing Marginal utility:

- **Assumption of Homogeneity:** The law assumes all units are identical, but in reality, units may differ in quality. If later units are better than earlier ones, they may give more satisfaction, contradicting the law.
- **Change in Consumer Preferences:** Consumer tastes and income can change over time. Such changes may increase satisfaction from additional units, instead of reducing it as the law predicts.
- **Continuity in Consumption:** The law applies only when goods are consumed continuously. If there are long gaps between consumption, satisfaction from extra units may increase rather than decrease.
- **Indivisibility of Commodities:** Durable goods like cars or houses are not consumed in small units, making it difficult to measure marginal utility and apply the law accurately.
- **Extraordinary Goods:** For goods like diamonds or alcohol, more consumption may increase satisfaction due to status or addiction, going against diminishing utility.
- **Increasing Need for Money:** As people earn more money, their desire for it may also grow, preventing marginal utility from reaching zero quickly.
- **Influence of Intoxicants:** Addictive substances can increase desire with more consumption, causing marginal utility to rise instead of fall.
- **Price Changes of Substitutes:** Changes in prices of substitute goods can affect satisfaction, making marginal utility behave differently from what the law suggests.
- **Applies mainly to short-run consumption:** Over a long period, consumption patterns and satisfaction may change.

iv) Relationship between MU and demand curves.

The Law of Diminishing Marginal Utility explains the downward slope of a demand curve because consumers place less value on each additional unit of a good, they buy.

Because marginal utility (extra satisfaction) decreases with each additional unit, a consumer's willingness to pay also falls. The first unit is valued most because it satisfies the strongest need, for example, when very thirsty, the first bottle gives 10 utils, so the consumer is willing to pay 500Frw. The second bottle gives less satisfaction (5 utils), so the willingness to pay drops to 400Frw, and the third, giving only 1 util, may only be bought at 300Frw. This shows that as quantity increases, the price a consumer is willing to pay decreases, giving rise to a downward-sloping demand curve. Thus, the demand curve reflects the portion of the MU curve above the x-axis, since prices cannot be negative even though marginal utility can become negative.

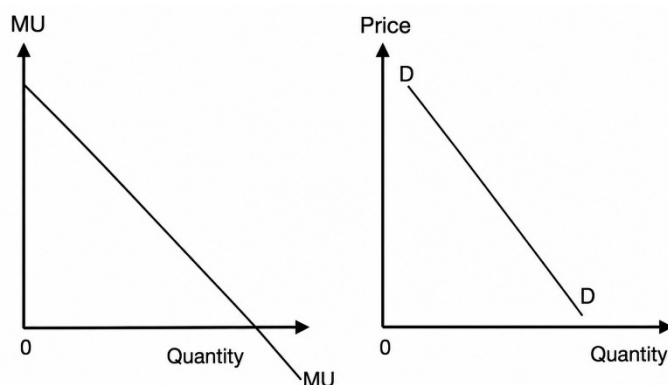


Fig 5.4 The relationship between MU and the demand curve.

The figure above shows that the marginal utility (MU) curve slopes downward as more units are consumed, indicating decreasing satisfaction. The demand curve also slopes downward, reflecting the price consumers are willing to pay. This shows that demand is derived from MU, when MU is high, price and willingness to pay are high, and as MU falls, price and willingness to pay decrease.

Application activity 5.2



1. a) What is meant by Utility
- b) State the law of diminishing marginal utility.
2. Calculate marginal utility and fill the table

Quantity (Q)	2	4	6	8	10	12	14
Total Utility (TU)	40	70	90	100	100	94	86
Marginal Utility							

3. The table below shows the total utility (TU) a consumer derives from consuming different quantities of a drink:

Quantity (Q)	0	1	2	3	4	5	6
Total Utility (TU)	0	20	35	45	50	50	45

- Identify the point of saturation in the table.
 - Explain what happens to total utility when consumption goes beyond the point of saturation.
 - Define disutility and indicate at which quantities it occurs in the table.
4. A customer eats slices of cake one after another and notices how much enjoyment each slice gives. How can the customer tell when eating another slice will start to reduce their satisfaction?

5.3 The Budget Constraint

Learning Activity 5.2



A student has a fixed amount of money to spend on snacks and juice and needs to decide how to divide it between the two.

Question: How can the student choose different combinations of snacks and juice so that all the money is spent, and which combinations are not possible with the budget?

While marginal utility explains how satisfaction is derived from the consumption of additional units of a commodity, it is important to recognize that in reality, consumers do not have unlimited resources to satisfy all their wants. Their choices are constrained by limited income and the prices of goods. This limitation gives rise to the concept of **the budget constraint**, which defines the boundaries within which consumers must make their decisions.

5.3.1 Budget Constraint and Budget Line

A budget constraint refers to the limit on the combinations of goods and services that a consumer can afford to purchase given their income and the prices of those goods. It shows that a consumer cannot buy everything they want, but must make choices based on what they can afford. It marks the boundary between what is affordable and what is unaffordable.

Within this limit, the consumer aims to achieve utility maximization, which means obtaining the highest possible satisfaction from their budget by allocating their spending in such a way that the marginal utility per unit of money spent is equal across all goods. The budget constraint is expressed using

a) the linear algebraic equation:

$$I = (P_x \times Q_x) + (P_y \times Q_y)$$

Where;

I: Total disposable income

P_x : Price of Good X

P_y : Price of Good Y

Q_x : Quantity purchased of Good X

Q_y : Quantity purchased of Good Y

For example: If a consumer has an income (I) of 20,000Frw, the price of oranges (P_x) is 400Frw per kilogram, and the price of Rice (P_y) is 200Frw, the equation is written as:

$$20,000 = 400Q_x + 200Q_y$$

b) Graphical Expression (The Budget Line)

When plotted on a graph, the budget constraint forms a downward-sloping straight line called a budget line: **A budget line** refers to a graphical representation of a budget constraint showing all the possible combinations of two goods that a consumer can afford to buy with a given income and given prices. From the above equation we can have a budget line as illustrated below:

- Maximum oranges (if all income is spent on oranges): $20,000 \div 400 = 50 \rightarrow$ point (50, 0)
- Maximum rice (if all income is spent on rice): $20,000 \div 200 = 100 \rightarrow$ point (0, 100)

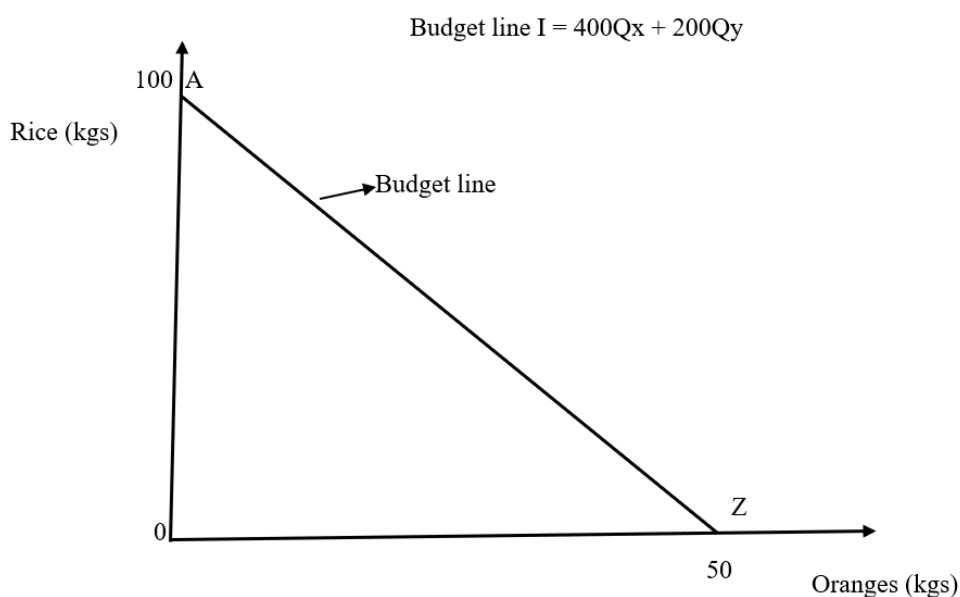


Fig 5.5 The budget line

The curve shows a budget line representing all the combinations of rice and oranges a consumer can afford with a fixed income. The intercepts indicate that the consumer can buy either 100 kg of rice (at point A) or 50 kg of oranges (at point Z) if all income is spent on one good. The downward slope of the line shows the trade-off between the two goods getting more oranges requires giving up some rice.

From the curve, the budget line runs from point **A (0, 100)** on the Y-axis (Rice) to point **Z (50, 0)** on the X-axis (Oranges). The slope of the budget line is calculated as:

$$\text{Slope} = \frac{\Delta X}{\Delta Y} = \frac{0-100}{50-0} = \frac{-100}{50} = -2$$

Slope = -2

The slope of -2 means that to gain 1 unit of oranges (Good X), the consumer must give up 2 units of rice (Good Y). It also reflects the price ratio.

$$-\frac{P_x}{P_y} = \frac{400}{200} = -2$$

So, the slope shows the opportunity cost of one good in terms of the other.

c) Assumptions of the Budget Constraint.

The budget line is based on several simplifying assumptions which help explain consumer behavior clearly."

- **Income is fixed:** The consumer has a given level of income (**I**) which does not change during decision-making, so choices are made within a constant budget.
- **Prices of goods are constant:** The prices of the goods remain unchanged, ensuring that any change in consumption is not due to price fluctuations.
- **Only two goods are considered:** The analysis is simplified by assuming the consumer allocates income between two goods, making it easier to illustrate trade-offs.
- **All income is spent:** The consumer is assumed to spend the entire income on the two goods, (no saving) meaning all points on the budget line represent full utilization of income.
- **Rational behavior:** The consumer is assumed to make logical decisions aimed at maximizing satisfaction (utility) within the given budget.
- **Perfect divisibility:** Goods are assumed to be perfectly divisible, meaning a consumer can buy fractional units (e.g., 2.5 litres of juice) if necessary to spend their last coin.

d) Affordable and Unaffordable Combinations on the Budget Line

The budget line shows the combinations of goods a consumer can afford and helps distinguish between what is affordable and unaffordable. Points on the line represent full use of income, points inside show underutilization of income, and points outside indicate combinations the consumer cannot afford.

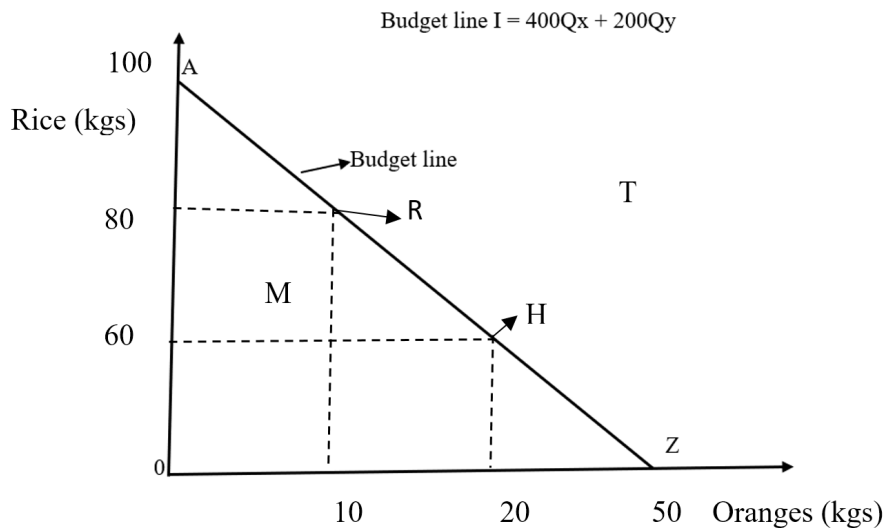


Fig 5.6 Affordable and Unaffordable Combinations on the Budget Line

The curve shows the budget line (A–Z). All points on it, such as **R** and **H**, represent combinations of rice and oranges that the consumer can afford by fully using their income. Points inside the budget line (**such as M**) are also **affordable** but show that the consumer is not using all their income. However, points above the budget line (**such as T**) are **unaffordable** because they require more income than the consumer has, even though they may be desirable.

5.3.2 Effects of income and price changes on the budget line

In the assumptions of a budget constraint, income and prices are assumed to remain constant. However, in reality, both income and prices can change, affecting what a consumer can afford. These changes cause the budget line to shift or rotate, thereby altering the set of possible consumption choices and influencing consumer decisions. This can be seen as below;

a) Effect of Change in Income on the Budget Line (Assuming prices remain constant)

When only income changes while prices remain constant, the budget line shifts parallel without changing its slope because relative prices remain the same. An increase in income causes the budget line to shift outward (to the right), meaning the consumer can afford more of both goods. In contrast, a decrease in income causes the budget line to shift inward (to the left), so the consumer can afford less of both goods. This shows that a change in income affects the consumer's purchasing power, but not the relative prices of goods.

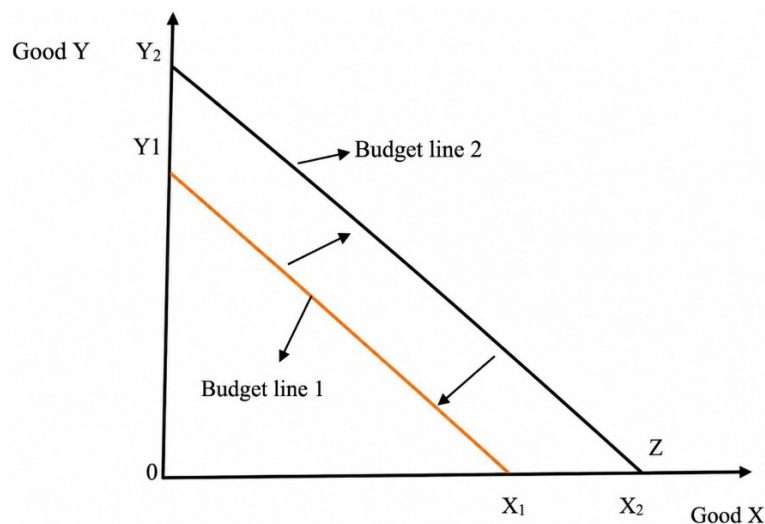


Fig. 5.7 Effect of Change in Income on the Budget Line

The diagram shows that a change in income shifts the budget line parallel without changing its slope. Budget line 1 represents a lower income, meaning the consumer can afford fewer units of both goods, while Budget line 2 shifts outward, indicating a higher income that allows the consumer to afford more of both goods (from X_1 to X_2 and Y_1 to Y_2).

b) Effect of changes in price on the budget line (Assuming income remain constant)

A change in price affects the budget line by causing it to rotate (pivot) around one intercept, while income is usually held constant. If both prices change in the same proportion, the budget line shifts parallel (outward if prices fall, inward if prices rise, as illustrated in the case of income changes). However, if the prices change by different proportions, the budget line rotates, reflecting a change in the relative prices of the two goods.

When the price of one good (for example, Good X) falls, the budget line rotates outward from the X-axis intercept, allowing the consumer to buy more of Good X while the Y-intercept remains unchanged. Conversely, when the price of Good X rises, the budget line rotates inward toward the X-axis, reducing the quantity of Good X that the consumer can afford, with the Y-intercept still staying the same.

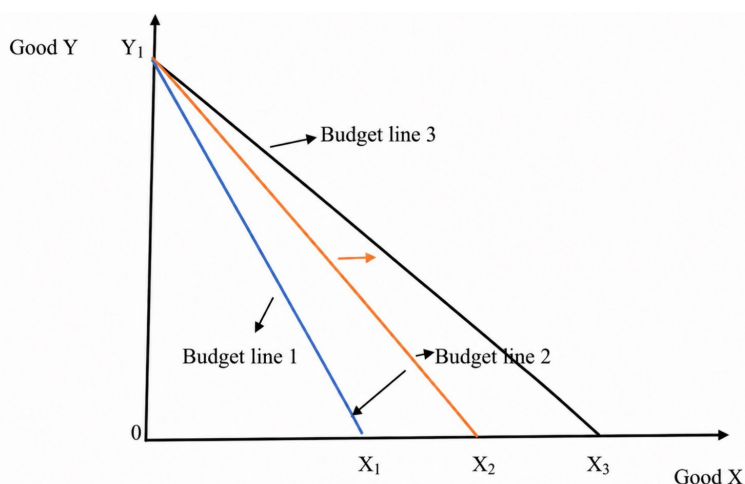


Fig 5.8 Effect of changes in price of one commodity on the budget line

The diagram shows that when the price of Good X changes, the budget line rotates around the fixed Y-intercept (Y_1). A higher price of X (Budget line 1) makes the line steeper and limits how much X can be bought (X_1), while a lower price of X (Budget lines 2 and 3) flattens the line and extends it outward to X_2 and X_3 , allowing more of X to be purchased. This illustrates that price changes affect the slope and purchasing power for that good, causing rotation rather than a parallel shift.

Application activity 5.3



Read carefully and answer the questions that follow.

1. a) Define a budget constraint.
b) State the assumptions underlying a budget constraint.
2. If commodity "K" is 50frw per unit, while commodity "L" is 20 units per unit, what would be the budget constraint for a consumer who has 8000frw. Illustrate the information on a curve.
3. Given that the price of meat is 2000frw per kilogram and that of sugar is 2500frw per kilogram. The income of the consumer is fixed at 50,000frw. Derive a budget line for this consumer.
4. Given that price of commodity "A" is 300frw per unit while that of commodity "B" costs 200frw per unit. Jane has a fixed income of 40,000frw, which she wants to spend on the two commodities.

Find how many units of each commodity he will purchase if;

- i. She spends 20,000frw on each commodity.
- ii. She spends 10,000frw on each commodity.
- iii. She spends all her income on commodity "A"
- iv. She spends all her income on commodity "B".
- v. Illustrate your findings on a curve.
- vi. What will happen to the amount of each commodity purchased if Jane's income doubled? Illustrate your findings

5.4 Indifference curves

Learning Activity 5.4



Observe the following table and answer the question

Combination	Tea	Fish
A	1	10
B	2	7
C	3	5
D	4	3
E	5	2

- Which combination (A to E) represents the highest consumption of tea and what is the corresponding amount of fish?
- Plot the information on a graph

Consumers' choices begin with the budget constraint, which shows the combinations of goods they can afford given their income and prices. However, affordability alone does not explain what a consumer prefers. This naturally leads to the concept of indifference curves, which represent combinations of goods that provide the same level of satisfaction.

5.4.1 Meaning and assumptions indifference curves

i. Meaning of indifference curves

An indifference curve is a graphical representation that shows all combinations of two goods that give a consumer the same level of satisfaction (utility). Along any point on the curve, the consumer is indifferent because each combination provides equal enjoyment or benefit.

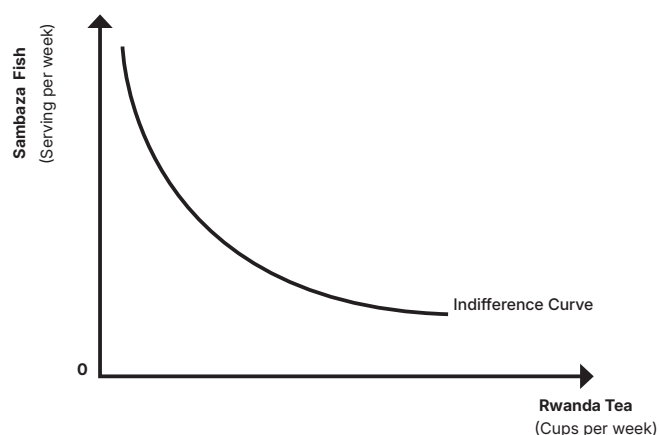


Fig 5.10 The indifference curve

ii. Assumptions of indifference curves.

Assumptions in indifference curve analysis are the basic rules that describe how a consumer thinks and makes decisions. They explain the behavior and reasoning behind consumer choices and guide how consumers evaluate options, forming the foundation for understanding their preferences.

- **Two goods with known prices:** The consumer chooses between two goods (A and B), and their prices are fixed, making analysis simple and easy to show on a graph.
- **Constant tastes, habits, and income:** Consumer preferences and income do not change, so any change in choice is due only to decision-making, not external factors.
- **Utility is ordinal:** The consumer can rank combinations of goods (more preferred vs less preferred) without measuring exact satisfaction.
- **Rationality (non-satiation):** The consumer is rational and prefers more goods to less, aiming to achieve the highest possible satisfaction.

- **Consistency in choice:** Preferences remain stable; if one combination is preferred once, it will be preferred again under the same conditions.
- **Transitivity:** Preferences follow a logical order (if A is preferred to B and B to C, then A is preferred to C).
- **Diminishing MRS:** As a consumer has more of one good, they are willing to give up less of the other, making indifference curves convex to the origin.

5.4.2 Properties of indifference curves and the indifference map:

Properties of indifference curves refer to the visible features or shape of the curves when drawn on a graph. They describe how the curves look, such as being downward sloping, convex to the origin, and never intersecting, which reflect the behavior implied by the underlying assumptions.

- **Same satisfaction:** All points on an indifference curve give equal utility, so the consumer is indifferent among them. This is because the loss of satisfaction from giving up some of one good is exactly offset by the gain from receiving more of the other, for example, a bundle with more of Good X and less of Good Y can lie on the same curve as one with less of Good X and more of Good Y.
- **Do not touch axes:** ICs usually do not touch the axes because consumers prefer having some of both goods rather than only one. If a curve touched the Y-axis, it would mean the consumer holds zero units of Good X yet still enjoys the same satisfaction as bundles containing both goods, which contradicts the assumption that more variety, not just more quantity, adds to satisfaction.
- **Higher IC = higher satisfaction:** Curves farther from the origin represent higher levels of utility due to greater quantities of goods.

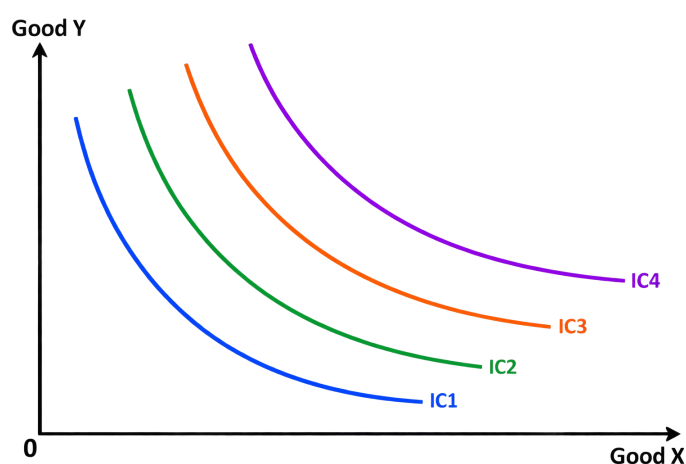


Fig 5.11 Indifference curves showing different levels of satisfaction

IC4 gives the highest satisfaction and IC1 the lowest, since each curve farther from the origin represents bundles with more of both goods.

- **Do not intersect:** ICs never cross, as each curve represents a unique level of satisfaction. If two curves crossed, the point of intersection would mean the same combination of goods gives two different levels of satisfaction at once, a contradiction, since a rational consumer would be indifferent among all points on one curve but not the other.

- **Downward sloping:** ICs slope downward because gaining more of one good requires giving up some of the other to maintain the same level of satisfaction. If a consumer could get more of both goods without giving anything up, they would move to a higher indifference curve rather than staying on the same one.
- **Convex to origin** ICs are convex due to diminishing marginal rate of substitution (MRS), meaning willingness to substitute decreases. For example, a consumer with little food will trade a lot of water for some food, but once they have plenty of food, they'll only give up a little water for more of it.
- **Many curves exist:** There are infinitely many ICs between any two curves, reflecting continuous levels of utility. Between any two curves, say IC1 and IC2, you could always draw another curve representing a satisfaction level in between, and another between those two, and so on without limit.
- **Higher curves shift right:** Increased satisfaction is shown by curves shifting outward (to the right). For example, Basing on the above figure, each curve farther to the right (IC1 → IC4) represents a higher level of satisfaction, since the consumer holds more of Good X (and Good Y) than on the curve before it.
- **Constant utility:** Utility remains unchanged along a single IC. This is because every point on the curve represents a different combination of the two goods that gives the consumer exactly the same level of satisfaction, so moving along the curve involves a trade-off, not a gain or loss in overall utility.
- **Indifference map:** A set of ICs forms a map that shows different levels of consumer satisfaction

The Indifference map

An indifference map is a collection of several indifference curves plotted on the same graph, each representing a different level of satisfaction (utility). It shows how a consumer's satisfaction changes with different combinations of two goods, with curves farther from the origin indicating higher levels of utility.

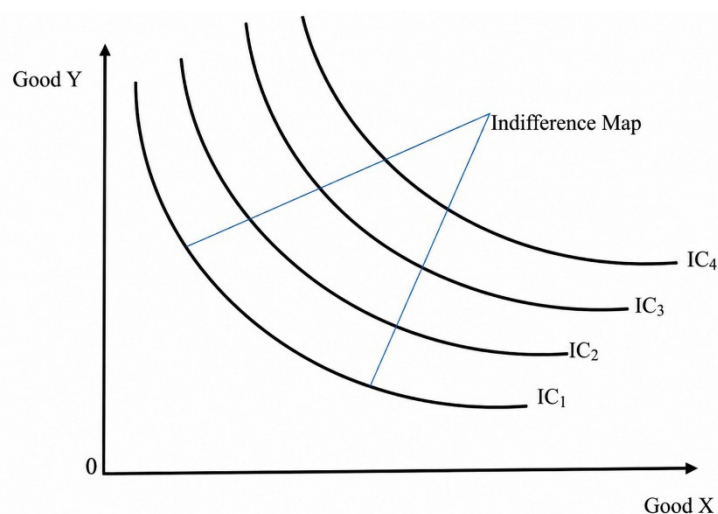


Fig 5.12 The indifference map

Any point along a single indifference curve (such as IC₁) shows combinations of Commodity X and Y that give equal satisfaction, while points on higher curves (such

as IC_3 and IC_4 are more preferred than those on lower curves (such as IC_1) because they provide higher utility.

Application activity 5.4



1. Why can two indifference curves never intersect?
2. What does a point along a single indifference curve represent?
3. If a consumer has three indifference curves, IC_1 , IC_2 , and IC_3 , explain which combinations of goods are most preferred and why.

5.5 Marginal rate of substitution

Learning Activity 5.5



A consumer enjoys chapati and beans. To get more chapati while remaining equally satisfied, the consumer is willing to give up some beans. For example, the consumer may give up 2 plates of beans to get 1 more chapati, without changing their level of satisfaction.

Questions

- i. If the consumer wants more chapati while staying equally satisfied, what must happen to the amount of beans?
- ii. When the consumer gains one more chapati, how many plates of beans must be given up?
- iii. Does the number of beans given up stay the same or change as more chapati is consumed? Why?
- iv. What does the number of beans the consumer is willing to give up for one more chapati tell us about their preferences?

Having seen that an indifference curve shows combinations that give equal satisfaction, the next question is: how much of one good is a consumer willing to give up for another? This leads to the concept of the Marginal Rate of Substitution (MRS). While the indifference curve shows the consumer's preferences, the MRS helps us measure the rate of trade-off between the two goods along the curve.

5.5.1 Marginal Rate of Substitution and Indifference Curves

The Marginal Rate of Substitution (MRS) is the rate at which a consumer is willing to give up one good in exchange for an additional unit of another good while maintaining the exact same level of total satisfaction (utility).

The MRS is the slope of the indifference curve at any point, showing the rate at which a consumer is willing to trade one good for another while maintaining the same satisfaction. Due to diminishing marginal utility, this rate decreases as more of one good is consumed, making the curve convex.

Formula:

$$MRS = -\frac{\Delta Y}{\Delta X}$$

Where:

- Δ = Change
- Commodity Y = the good given up
- Commodity X = the good gained

The MRS shows how much of Good Y a consumer is willing to give up to gain more of Good X while keeping satisfaction constant, and **the negative sign reflects that the indifference curve slopes downward.**

In terms of marginal utilities:

$$MRS_{xy} = -\frac{MU_y}{MU_x}$$

Where; MU_x = marginal utility of Good X

MU_y = marginal utility of Good Y

5.5.2 Relationship between MRS and indifference curves

The Marginal Rate of Substitution (MRS) measures the slope of the indifference curve at any given point. As a consumer moves along an indifference curve, the MRS shows how much of one good they are willing to give up to gain more of another while maintaining the same level of satisfaction. Due to diminishing marginal utility, the MRS decreases as more of one good is consumed, which explains why indifference curves are downward sloping and convex to the origin.

Example 1:

The Scenario

Sarah is choosing between bundles of two goods: cakes (Good X) and Guavas (Good Y). Because both bundles sit on the exact same indifference curve, Sarah is perfectly indifferent between them.

- Bundle A: 2 cakes (X) and 12 Guavas (Y)
- Bundle B: 3 cakes (X) and 8 Guavas (Y)

Question: Calculate Sarah's Marginal Rate of Substitution of cakes for Guavas (MRS_{xy}) when she moves from Bundle A to Bundle B. Interpret the meaning of the final value.

$$\Delta x = 3 - 2 = 1$$

$$\Delta y = 8 - 12 = -4$$

$$MRS_{xy} = -\frac{MU_y}{MU_x}$$

$$MRS_{xy} = -\frac{-4}{1}$$

MRS_{xy} = 4

This means Sarah is willing to give up 4 guavas to gain 1 additional cake while remaining on the same level of satisfaction. The value shows that because she has relatively few cakes and many guavas, she places a higher value on cakes compared to guavas.

Example 2: Fruits (X) and Juice (Y)

Choice	Fruits (X)	Juice (Y)	ΔFruits (X)	ΔJuice (Y)	MRS (ΔX/ΔY)
1	8	1	-	-	-
2	6	2	2	1	2
3	3	3	3	1	3
4	1	4	2	1	2

Application activity 5.5



1. A student has 4 cups of tea and 3 cups of coffee. To drink 1 more cup of coffee, she reduces tea consumption from 4 cups to 3 cups. Find MRS
2. Maria consumes fruits and juice. Initially, she has 5 fruits and 3 juices. She wants one additional unit of juice and is willing to give up 2 fruits to obtain it. Calculate Maria's Marginal Rate of Substitution (MRS) of juice for fruits and interpret its meaning.

<https://openstax.org/books/principles-economics-2e/pages/6-1-consumption-choices>

5.6 Consumer equilibrium

Learning Activity 5.6



A consumer in Karongi buys tea and fish with a limited amount of money and chooses a combination that gives them the most satisfaction. They reach equilibrium when changing the amount of tea or fish no longer increases their satisfaction.

1. What does it mean for the consumer to be in equilibrium?
2. What might make the consumer change their choice of tea and fish?

Having analyzed both consumer preferences (indifference curves) and financial limitations (the budget line), we can now combine them to determine how consumers make final purchasing decisions. Consumer equilibrium occurs when a consumer allocates their limited income in a way that maximizes total utility. Graphically and mathematically, this is the point where the consumer's willingness to trade between goods fits smoothly with the prices they face in the market.

5.6.1 Meaning and Conditions of Consumer Equilibrium

i. Meaning of consumer equilibrium

Consumer equilibrium is the point at which a consumer achieves the highest possible level of satisfaction (utility) given their limited income and the prices of goods. It represents the best combination of goods a consumer can afford, where their preferences are fully satisfied within their budget constraint.

ii. Conditions of consumer equilibrium

To fully understand consumer equilibrium, it is important to identify the **conditions that must be satisfied** for a consumer to achieve maximum satisfaction. These conditions explain how a consumer's choices are guided by both their preferences and budget, ensuring the most efficient use of limited resources. In this regard, consumer equilibrium is determined by two conditions: the **necessary condition**, where the Marginal Rate of Substitution (MRS) equals the price ratio, and the **sufficient condition**, which requires the indifference curve to be convex to the origin.

a) First-order (Necessary) Condition

Also known as the **Equi-Marginal Principle**, it states that a consumer reaches equilibrium when the Marginal Rate of Substitution (MRS) is equal to the ratio of prices of the two goods.

$$MRS_{xy} = \frac{P_x}{P_y} \text{ or } \frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

This condition ensures that the consumer's willingness to trade between goods matches market prices. It means that at equilibrium, the last franc spent on Good X provides the same level of satisfaction as the last franc spent on Good Y.

If $\frac{MU_x}{P_x} > \frac{MU_y}{P_y}$, a franc spent on Good X gives more satisfaction than one spent on Good Y. The consumer will therefore reduce spending on Good Y and increase spending on Good X. As more of Good X is consumed, its marginal utility (MU_x) decreases due to the law of diminishing marginal utility. At the same time, consuming less of Good Y causes its marginal utility (MU_y) to increase. This adjustment continues until the satisfaction gained per franc spent is equal for both goods. At that point, the consumer has no incentive to change their choice, as maximum satisfaction has been achieved given their income.

b) Second-order (Sufficient) Condition

The second-order condition requires **the indifference curve to be convex to the origin**, reflecting a diminishing Marginal Rate of Substitution (MRS). This shape ensures that when **the budget line is tangent to the indifference curve**, the point of tangency represents the maximum level of satisfaction, rather than just any point where the slopes are equal or a lower, unstable position.

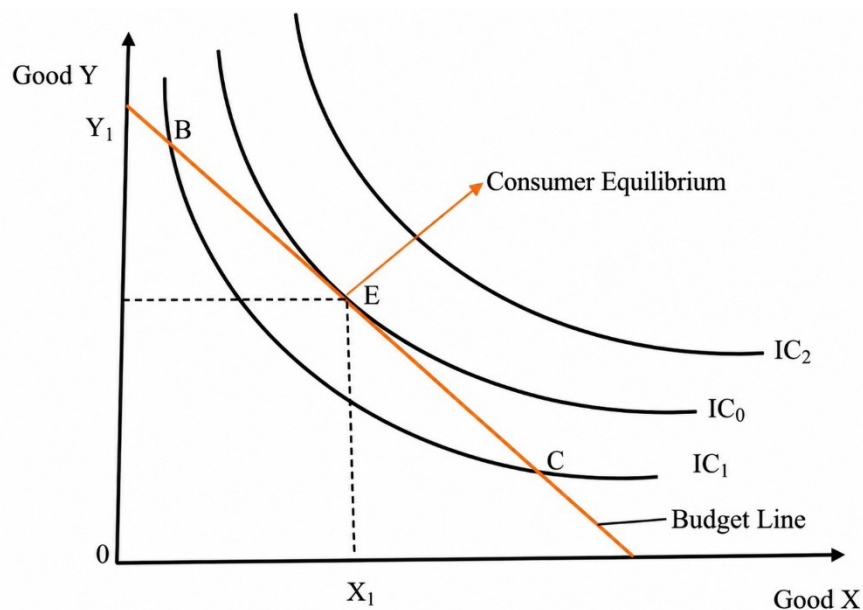


Fig 5. 13 Consumer equilibrium

Consumer equilibrium occurs at point E, where the budget line is tangent to indifference curve IC_0 , indicating that the consumer attains the maximum possible satisfaction given their income. Points such as B and C, which lie on a lower indifference curve (IC_1), are affordable but yield less utility. In contrast, IC_2 represents a higher level of satisfaction but is unattainable with the current budget. Therefore, point E represents the most optimal choice available to the consumer.

Application activity 5.6



1. A consumer purchases two commodities **X** and **Y**. The marginal utilities and prices are given below.

Commodity	Marginal Utility (MU)	Price (P)
X	40	5
Y	30	5

- i. Calculate the **marginal utility per franc (MU/P)** for each commodity.
- ii. Determine whether the consumer is in **equilibrium**.

2. Analyse the information in the table below and answer the questions that follow

Commodity	Marginal Utility (MU)	Price (P)
A	60	10frw
B	30	5frw

- i. Calculate the **marginal utility per franc (MU/P)** for each commodity.
- ii. Determine whether the consumer is in **equilibrium**.

5.7 Real income and substitution Effect

Learning Activity 5.7



Consumer A has 20,000 FRW as her fixed money income. She intends to spend it on sugar. The price of sugar is 1,000 FRW per kilogram.

1. How many kilograms of sugar will she purchased?
2. What will happen to the kilograms she can purchase if the price:
 - (i) Increases to 2,500 FRW per kg.
 - (ii) Reduces to 500 FRW per kilogram.

After understanding how consumer equilibrium is achieved, it is important to examine how a change in price affects this position. Such a change shifts the budget line and leads the consumer to adjust their choice, a process explained by the substitution effect and the real income effect, which show how the consumer moves from one equilibrium to another.

5.7.1 Meaning of income and substitution effects

The income effect and substitution effect explain how a consumer's choice changes when the price of a good change.

The real income effect: This refers to the change in consumption that occurs because a change in price affects a consumer's purchasing power (real income). Real income is the amount of goods and services a consumer can buy with their income. When the price of a good falls, real income rises because the consumer can afford more goods, leading to increased consumption. When the price rises, real income falls, reducing the consumer's ability to purchase goods.

Substitution Effect: The substitution effect refers to the change in consumption that occurs when a change in price makes one good relatively cheaper or more expensive than another. As a result, the consumer substitutes the relatively cheaper good for the more expensive one, while holding the level of satisfaction (utility) constant. Exemple, a decrease in the price of Close up from 1,500 FRW to 700 FRW per unit makes it cheap compared to Colgate whose price has remained constant at 1,500 FRW. The effect will be that some consumers will shift their demand from Colgate, which is expensive to Close up which has become cheap. If that is the case, quantity demanded of Close up shall increase while that of Colgate shall reduce. <https://openstax.org/books/principleseconomics2e/pages/62consumerchoiceandthebudgetconstraint>

5.7.2 Impact of Price Change on Consumer Choice

The impact of a price change on consumer choice explains how consumers adjust their buying decisions when the price of a good rises or falls.

1. When the Price Decreases

- Consumers tend to buy more of the product because it becomes more affordable.
- The fall in price increases the consumer's purchasing power (real income), allowing them to buy more goods with the same income.

- The good becomes relatively cheaper, so consumers may substitute it for other more expensive goods.
- Consumers adjust their spending to make better use of their budget.
- The range of affordable combinations of goods increases, shifting consumer choice outward. Example: If the price of juice falls, consumers may buy more juice and reduce consumption of substitutes like soda.

2. When the Price Increases

- Consumers tend to buy less of the product because it becomes more expensive.
- The rise in price reduces the consumer's purchasing power, limiting what they can afford.
- The good becomes relatively more expensive, so consumers may switch to cheaper alternatives (substitutes).
- Consumers adjust their spending by reducing consumption or reallocating income.
- The set of affordable combinations of goods decreases, restricting consumer choice. Example: If the price of fish rises, consumers may reduce fish consumption and buy cheaper alternatives like beans or meat.

Application activity 5.7



A consumer buys two brands of toothpaste: Brand X costs 1,500 FRW and brand Y costs 1,500 FRW. The price of brand X falls to 700 FRW while Brand Y remains the same.

Questions:

1. What will happen to the demand for brand X?
2. What will happen to the demand for brand Y?
3. Which concept explains this behavior?

5.8 Consumer and producer surplus.

Learning Activity 5.8



In a Karangazi market, farmers sell tomatoes while buyers with different willingness to pay come to purchase them. Some buyers pay less than what they were ready to pay, and some farmers sell at prices higher than their minimum expectations, so both sides benefit from the exchange.

Questions

- a) Explain how buyers benefit when they pay a price lower than what they were willing to pay.
- b) Describe how sellers benefit when they sell at a price higher than the minimum, they expected.

Having examined how consumers make choices and reach equilibrium, it is useful to consider the benefits gained from participating in the market. When consumers purchase goods at given prices, they may receive more satisfaction than what they actually pay for, while producers also gain from selling their goods. These gains are captured by the concepts of consumer surplus and producer surplus, which help explain the overall welfare created in a market.

5.8.1 Consumer surplus

i. Meaning of consumer surplus

Consumer surplus is the extra benefit or satisfaction a consumer gains when they pay a price lower than the maximum amount they are willing to pay for a good. It is the difference between the consumer's willingness to pay and the actual market price, representing the gain or saving enjoyed from the purchase. Suppose a consumer is willing to pay 1000Frw for a bottle of juice, but the market price is 600Frw. The consumer buys the juice at 600Frw and gains an extra benefit of 400Frw.

Consumer Surplus = $1000 - 600 = 400\text{Frw}$. So, the consumer enjoys a surplus of 400Frw, which represents the extra satisfaction or saving from the purchase.

ii. Measurement and interpretation of consumer's surplus.

The concept of consumer surplus helps to measure the benefits gained from market transactions. To better understand its significance, it is important to examine how it is measured and interpreted in real markets. This involves using simple calculations and graphical analysis to illustrate the gains enjoyed by consumers and how these contribute to overall economic welfare.

Consumer's surplus is expressed as:

- Basic formular: **Consumer surplus = Willingness to pay - Market Price**
- For multiple units: **Consumer surplus = $\Sigma(\text{Individual willingness to pay} - \text{price})$**
- In graphical form (from a demand curve):

$$\text{Consumer surplus} = \frac{1}{2} \times b \times h$$

- Where; Base (b) = quantity of goods.
- Height (h) = difference between maximum price consumers is willing to pay (intercept of demand curve) and actual market price (P).

Note: The $\frac{1}{2} \times \text{base} \times \text{height}$ formula only applies when demand is linear and forms a complete triangle. Where price controls, quotas, or a curved demand curve leave a trapezoid or irregular shape, a different method must be used instead.

Exemple 1

If a consumer goes to the market intending to buy sugar and is willing to pay 500 FRW per kilogram, it means they expect to receive satisfaction equal to that amount. However, if the market price is only 300 FRW, the consumer will still obtain the expected level of satisfaction but at a lower cost. As a result, the consumer gains a consumer surplus of **200 FRW (500 - 300)**, which represents the extra benefit enjoyed from paying less than what they were willing to pay.

By illustration, the consumer's surplus is the area below the demand curve and above the equilibrium price.

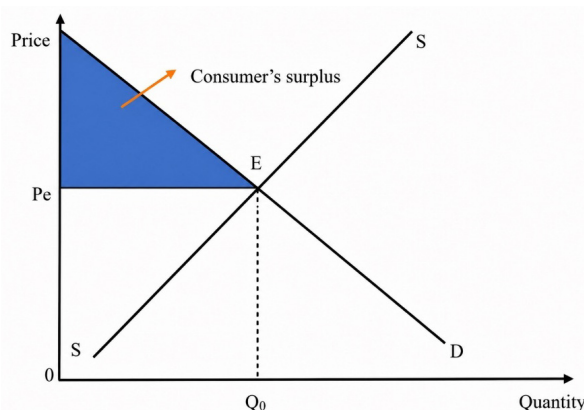


Fig 5.14 Consumer's surplus

The curve illustrates consumer surplus as the shaded triangular area above the equilibrium price (P_e) and below the downward sloping demand curve, up to the equilibrium quantity (Q_0). This shaded region represents the extra satisfaction or benefit consumers receive because they are willing to pay a higher price for the goods than the actual market price they pay at equilibrium.

Example 2:

Study the demand schedule below, where the market price is 100 FRW. Use the information provided to answer the questions that follow.

Price per unit. (FRW)	Quantity demanded
500	1
400	2
300	3
200	4
100	5

1. Calculate the consumer surplus
2. Draw a diagram to show consumer surplus

Answer

Step 1: Calculate Consumer Surplus

Consumer surplus is the difference between what consumers are willing to pay and what they actually pay.

From the table:

Quantity	Willingness to Pay (FRW)	Market Price (FRW)	Surplus per unit	Total Surplus
1	500	100	400	400
2	400	100	300	600
3	300	100	200	600
4	200	100	100	400
5	100	100	0	0

- Surplus per unit = Price willing to pay – Market price
- Total surplus = Surplus per unit × Quantity demanded
- This table shows how consumer surplus decreases for each additional unit consumed, because willingness to pay falls.

Total Consumer Surplus = 400 + 600 + 600 + 400 + 0 = 2000 FRW

Step 2: Diagram

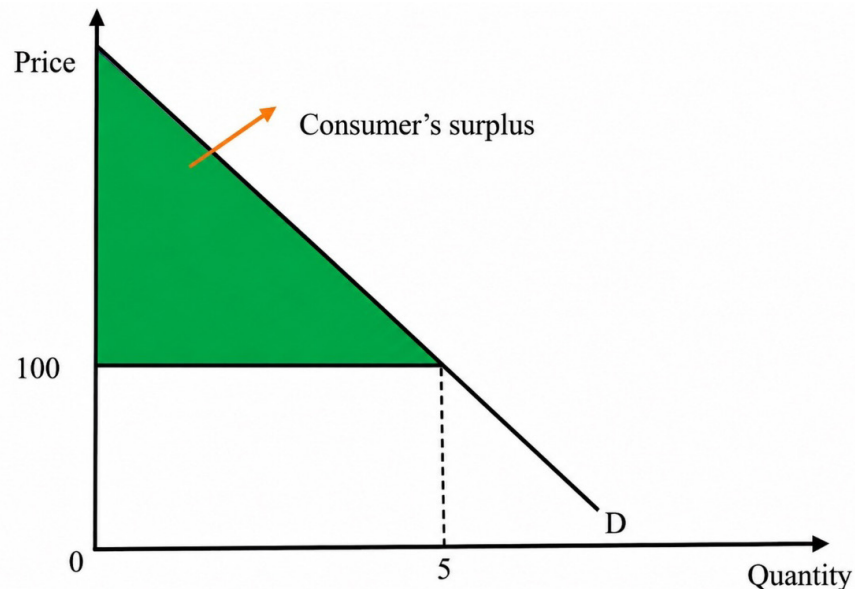


Fig 5.15 Consumer's surplus

When plotting consumer surplus on a curve, the graph ends at the last unit the consumer actually purchases, where the willingness to pay equals the market price, because at this point consumer surplus becomes zero. In this example, the stopping point is the 5th unit, where both willingness to pay and market price is 100 FRW, meaning the consumer gains no extra benefit from that unit, and any additional units are not purchased since their value would be lower than the market price.

Example 3:

A consumer is willing to pay 500 FRW for a good, but the market price is 100 FRW, and the quantity demanded is 5 units. Calculate the consumer surplus.

Base = 5 units

Height = 500-100 =400

$$\text{Consumer surplus} = \frac{1}{2} \times b \times h$$

$$\text{Consumer surplus} = \frac{1}{2} \times 5 \times 400$$

Consumer surplus= 1000

5.8.2 Producer surplus

i. Meaning of producer surplus:

Producer surplus refers to the difference between the price that the producer is ready and willing to accept for his or her commodity and what he or she actually receives. The producer usually has a particular price at which he is ready to offer his product for sale.

ii. Measurement and interpretation of producer's surplus.

The concept of producer surplus helps to measure the benefits gained by producers from market transactions. To better understand its importance, it is necessary to examine how it is measured and interpreted in real markets. This involves using simple calculations and graphical analysis to show the gains enjoyed by producers and how these contribute to overall economic welfare.

Producer surplus is expressed as:

- Basic formular: **Producer surplus = Market price - Supply price (Cost of production)**
- For multiple units: **Consumer surplus = $\Sigma(\text{Market price} - \text{Cost}) \times \text{Quantity supplied}$**
- In graphical form (from supply curve):
$$\text{surplus} = \frac{1}{2} \times b \times h$$
- Where; Base (b) = quantity supplied
- Height (h) = Market price – minimum price

Producer surplus is the extra benefit a producer gains by selling goods at a price higher than the cost of producing them.

The producer surplus is illustrated as shown below. It is the area below the equilibrium price and above the supply curve.

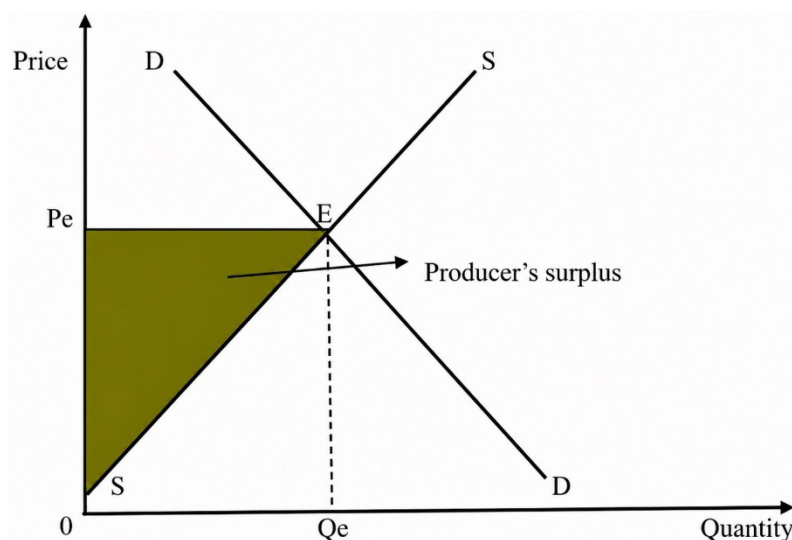


Fig 5.16 Producer's surplus

The diagram shows producer surplus as the shaded triangular area between the equilibrium price (P_e) and the supply curve, from zero up to the equilibrium quantity (Q_e). This shaded region represents the extra benefit producers receive because they sell goods at the market price, which is higher than the minimum price they are willing to accept for each unit.

Example 1

Study the demand schedule below, where the market price is 4000 FRW. Use the information provided to answer the questions that follow.

Price per unit. (FRW)	Quantity supplied.
2,000	1
2,500	2
3,000	3
3,500	4
4,000	5

1. Calculate the producer surplus
2. Draw a diagram to show producer surplus

Step 1: Calculate Producer Surplus

Producer surplus is the difference between the market price and the minimum price producers are willing to accept.

From the table:

Quantity	Minimum Price (FRW)	Acceptable	Market Price (FRW)	Surplus per unit	Total Surplus
1	2,000		4,000	2,000	2,000
2	2,500		4,000	1,500	3,000
3	3,000		4,000	1,000	3,000
4	3,500		4,000	500	2,000
5	4,000		4,000	0	0

Total Producer Surplus = 2,000 + 3,000 + 3,000 + 2,000 + 0 = 10,000 FRW

To find producer surplus:

- Surplus per unit = Market Price – Cost (supply price)
- Total Surplus = Surplus per unit × Quantity supplied
- In simple terms, surplus per unit shows the extra gain on each unit, while total surplus shows the total gain for all units supplied.
- Producer surplus decreases for each additional unit, because the cost of supplying each extra unit rises
- The last unit (5th unit) has zero surplus, since price = cost

Step 2: Diagram of Producer Surplus

The supply curve slopes upward, showing the minimum price producers are willing to accept. The producer surplus is the triangular area **below the market price** and **above the supply curve**, up to quantity 5.

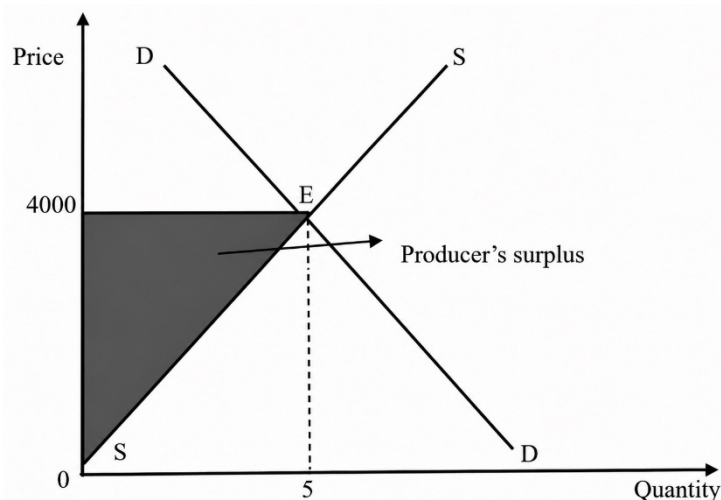


Fig 5.17 The produce's surplus

The curve illustrates producer surplus as the shaded triangular area below the market price (4000) and above the upward-sloping supply curve, up to the equilibrium quantity of 5 units. This shaded region represents the extra benefit producers receive because they are able to sell their goods at the equilibrium price, which is higher than the minimum price they are willing to accept for each unit.

Example 2

A producer is willing to supply a good at a minimum price of 2,000 FRW, while the market price is 4,000 FRW, and the quantity supplied is 5 units. Calculate the producer surplus

B = 5 units

Height = 4000 – 2000 = 2000

$$\text{Producer surplus} = \frac{1}{2} \times b \times h$$

$$\text{Producer surplus} = \frac{1}{2} \times 5 \times 2000$$

Producer surplus= 5,000

Application activity 5.8



Read and answer the questions below:

In a certain market, consumers are willing to pay 800 FRW for the first unit, 700 FRW for the second, 600 FRW for the third, 500 FRW for the fourth, and 400 FRW for the fifth unit. The market price is fixed at 400 FRW.

- i. Calculate the total consumer surplus.
 - ii. Illustrate the consumer surplus on a diagram.
3. In a certain market, producers are willing to sell units of a product at different minimum prices. The first unit is supplied at 200 FRW, the second at 250 FRW, the third at 300 FRW, the fourth at 350 FRW, and the fifth at 400 FRW. The market price is fixed at 400 FRW.

Questions:

- i. Calculate the total producer surplus.
- ii. Illustrate the producer surplus on a diagram.

End of unit assessment



Knowledge

1. Explain the difference between total utility and marginal utility using a real-life example.
2. Describe the law of diminishing marginal utility and explain why it causes the demand curve to slope downward.
3. When is a consumer in equilibrium?

Skills

4. A consumer has the following total utility: 20, 35, 45, 50, 50 48. Calculate the marginal utility and identify the point of saturation.
5. Given the prices $P_x = 1,000$ FRW, $P_y = 500$ FRW and income = 10,000 FRW, construct a budget line and determine two possible combinations of X and Y.
6. A consumer gets $MU_x = 40$, $P_x = 10$ and $MU_y = 20$, $P_y = 5$. Determine whether the consumer is in equilibrium and justify your answer.

Attitudes and Values

7. Explain how understanding utility can help a student make better spending decisions in daily life.
8. Discuss the dangers of overconsumption and how it relates to disutility.
9. Suggest ways a consumer can maximize satisfaction with limited income in real-life situations.

UNIT 6

PRODUCTION THEORY



Key Unit Competence:

Analyze how producers combine inputs and make efficient production decisions across different planning periods.

Learning objectives

Knowledge and understanding

- Explain the meaning, purpose, types, and levels of production.
 - Describe the factors of production (land, labour, capital, and entrepreneurship) and explain their roles in the production process.
 - Explain the concept of a production function and distinguish between short-run and long-run planning periods.
 - Define and differentiate between Total Product (TP), Average Product (AP), and Marginal Product (MP).
- 1**
- Explain the production function schedule and curves in the short run.
 - State and explain the law of diminishing returns (law of variable proportions) and its assumptions.
 - Explain the concept of returns to scale and distinguish between increasing, constant, and decreasing returns to scale.
 - State and explain the law of returns to scale and its assumptions.
 - Explain the meaning of isoquants and Isocosts and their relevance to production decisions.
 - Describe how producers make efficient production choices in the short run and the long run.

Skills

- 2**
- Classify production into levels and types in real-life situations.
 - Construct and interpret production schedules showing TP, AP, and MP.
 - Draw and analyze production curves (TP, AP, and MP) and explain their shapes.

- Apply the law of diminishing returns to real or simulated production scenarios.
- Analyze production decisions under short-run constraints where some factors are fixed.
- Compare production outcomes under different returns to scale in the long run.
- Draw and interpret isoquant maps and isocost lines.
- Analyze how changes in factor combinations affect output and production efficiency.
- Use diagrams, tables, and equations to evaluate efficient input combinations.
- Differentiate producer decisions across short-run and long-run planning periods.
- Solve basic economic problems related to input use, productivity, and efficiency.

Attitudes and values

3

- Appreciate the importance of efficient use of scarce resources.
- Demonstrate positive attitudes toward planning and decision-making in production.
- Recognize the role of entrepreneurship and innovation in economic development.
- Exhibit willingness to analyze production activities critically and objectively.
- Show awareness of the need for sustainable and responsible production practices.
- Demonstrate interest in applying economic principles to real-life business and production situations.
- Respect productive factors, especially human labour and environmental resources.

This unit develops a clear understanding of how production decisions are made by linking together key ideas in a logical sequence, starting with how and why producers engage in productive activities and how these activities vary across different areas of the economy. It then focuses on the resources used in production and shows how their nature and availability influence costs and output, leading to an explanation of how producers combine these resources and how changes in their use affect the level of output. The unit also highlights how production decisions differ over time, explaining how output responds when some or all resources are adjusted, and concludes by illustrating how producers make efficient and cost-effective choices when combining resources to achieve their desired level of output in real-life situations.



Introductory activity

Analyse the scenario below and answer the questions that follow.

A woman named Aline wants to start a small business making fruit juice to sell in her town. She plans to buy fruits from farmers, use a machine to make juice, and sell it to customers. To do this, she needs land for her workplace, workers to help her, money to buy machines, and good ideas to organize the business. At first, Aline can only use a small space, a few workers, and one machine. Later, she hopes to grow her business by using better machines and more workers so that she can produce more juice.

Questions:

- What is Aline trying to do?
- Why is she making juice?
- Who is a producer in this story?
- Is Aline making juice for her own use or for selling?
- What has been done to the fruits before they are sold as juice?
- What things does Aline need to make juice?
- What happens if Aline uses more workers or a better machine?
- Can she change everything at once, or does it take time?
- What will happen if Aline uses more machines and workers in the future?

6.1 Introduction to the Production theory

Learning Activity 6.1



Analyse the scenario below and answer the questions that follow;

In many communities, people depend on goods and services that pass through several stages before reaching consumers. For example, bread is a common food item used every day. Its making begins with people who work directly with nature by growing maize. After harvesting, other people change the maize into new forms by grinding it into flour and later baking it into bread. Once the bread is produced, traders sell it in shops so that consumers can buy it. In addition, other people support this process by transporting the products, handling payments, and organizing sales. Although these people are involved in different activities, all of them contribute in one way or another to making bread available and useful to consumers in the community.

Questions

- What do you think is the act of making bread?
- Are all the people involved in the scenario doing the same kind of work? Explain.

- c) Who works directly with nature, and what do they produce?
- d) Who changes maize into new products, and how do these products differ from the original maize?
- e) Who helps to ensure that bread reaches consumers without changing it physically?
- f) Can bread reach consumers if one of these activities is missing? Explain your answer.
- g) How many stages are involved in the bread production as described in the scenario above?
- h) Explain how the different activities involved in the bread production depend on one another.

This lesson introduces production theory, which examines how firms and producers combine various resources to produce goods and services. It focuses on the relationship between inputs and output, helping to explain how production decisions are made in order to achieve efficiency and maximize output. Understanding production theory provides a foundation for analyzing how businesses operate and how resources are utilized in real-life economic activities.

6.1.1 Meaning of production and purpose of production

Production: Production is the process of creating goods or services by combining factors of production such as land, labour, capital, and entrepreneurship. It involves adding value to resources in order to satisfy human needs and contribute to society's welfare. Production is not limited to manufacturing alone; it includes activities such as extraction, processing, storage, transportation, distribution, and direct services like teaching and medical care. For example, turning raw milk into yogurt or teaching students in a school are both forms of production because they create utility. The production process is considered complete only when the final goods or services reach the consumer and provide satisfaction.

Purpose of Production: Generally, the purpose of production is to satisfy human wants and improve economic wellbeing. According to **Adam Smith**, production exists mainly for consumption, meaning goods and services are produced to meet people's needs through exchange for example, a bakery producing bread so that people can buy and consume it. **Keynes**, on the other hand, emphasized that production aims to create wealth, increase income and employment, and improve overall economic wellbeing for instance, the same bakery creates jobs, generates income for workers and the owner, and contributes to economic growth. Together, these views show that production serves both to meet human needs and to promote economic growth and welfare.

6.1.2: Types of production and levels of production.

i. Types of production

Production can be carried out in different ways depending on the purpose for which goods and services are produced. Some production activities are meant to meet the basic needs of the producer and family, while others are carried out mainly for sale in the market. These different ways of producing help us understand how people produce for survival, income, and profits and thus forming the types of production which include the following;

a) Direct production. Direct production is the production of goods and services mainly for one's own consumption and is also called **subsistence production**. It is usually carried out by individuals with the help of family members, friends, or relatives using simple tools. Production is generally low and goods may be of poor quality because skills and technology are limited. Since most of the output is consumed at home, trade and competition are minimal. An example of direct production is a rural family that grows beans, maize, and sweet potatoes on their small farm mainly for their own consumption. The family members work together using simple tools such as hoes and machetes. Most of the food produced is eaten at home, and only the small surplus, if any, is taken to the local market. This type of farming is common in many rural areas where, households practice subsistence agriculture

b) Indirect production: Indirect (**commercial**) production refers to production carried out mainly for exchange in the market with the aim of earning profit. Goods are produced not for the producer's own use but for sale, and the income earned is used to buy other goods and services the producer cannot produce personally. For example, a smartphone factory produces phones for sale, a factory producing bread to sell to consumers and the profits are used to pay workers, buy raw materials, and expand production. This type of production is usually done on a large scale, uses modern machinery and skilled labour, and relies on wage employment, thereby creating jobs. It emphasizes mass production, high productivity, and often encourages research and innovation to remain competitive, all of which contribute to economic growth and improved livelihoods.

ii. Levels of production

Production does not take place in a single step. Before goods and services reach consumers, they pass through different stages of economic activity, with each stage adding value to the product. Some activities involve obtaining resources directly from nature, others involve changing these resources into usable goods, while others provide services that help distribute goods to consumers. These stages are connected and work together to ensure that goods and services are made available to satisfy human needs and support economic development as explained in the levels below;

a) Primary level of production

Primary production is the first stage of the economic production process, involving the direct extraction of raw materials from nature in their unprocessed form. It includes activities such as farming, fishing, forestry, mining, hunting, and quarrying, all of which focus on obtaining natural resources that serve as the foundational inputs for manufacturing and other economic activities. Classified as **extractive industries**, primary production plays a vital role by supplying essential raw materials required for secondary production (manufacturing) and tertiary sectors (services). Without this stage, subsequent production processes could not operate, making primary production indispensable to the functioning and sustainability of the entire economy.

b) Secondary level of production

Secondary production is the stage of production that focuses on converting raw materials and intermediate goods obtained from primary production into semi-finished or finished products that are ready for use or sale. At this stage, natural resources such as crops, minerals, or animal products are processed, refined, assembled, or manufactured to add value and utility. For instance, sugarcane is processed into sugar, maize is ground into flour, and flour is baked into bread. In many cases, primary products are first transformed into semi-finished goods such as timber into planks or iron ore into steel which are then transferred to other manufacturing units for further processing into final consumer or capital goods. Secondary production includes a wide range of activities such as manufacturing, construction, processing, oil refining, leather tanning, and textile production. This stage is crucial to economic development because it creates employment, increases the value of raw materials, supports industrial growth, and supplies goods needed by consumers and other sectors of the economy.

c) Tertiary level of production

Tertiary production focuses on the provision of services rather than the production of physical goods, aiming to enhance the value, accessibility, and usefulness of goods after they have been produced. This stage plays a vital role in facilitating the distribution, exchange, and consumption of final products by linking producers to consumers. Tertiary production includes **commercial services** related to trade and business, such as retailing and wholesaling, banking and insurance, transportation and logistics, and communication services, which support the movement and exchange of goods. It also involves **personal services** that directly meet individual and societal needs, including education, healthcare, legal services, entertainment, security, and personal care services like hairdressing and **professional services** which involve specialized knowledge and skills offered by trained experts. Examples include consultancy, engineering, accounting, architecture, and information technology services.

Application activity 6.1



1. Consider a small Rwandan community with an interconnected production and distribution system. Farmers cultivate maize, which is then processed into flour by a local miller, while shops and bakeries sell bread and other maize-based products in nearby markets.

Questions:

- i. Identify which activities in this community represent primary, secondary, and tertiary levels of production.
- ii. Explain how raw materials are transformed into finished goods in the secondary level of production. Give a local example.
- iii. How do tertiary services, like transport or banking, support the production and sale of goods in this community

2. Reflect on the different production activities that take place in your own household, family, or community. Think about who produces the goods or services, the purpose of the production, and whether the output is mainly for personal use or for sale. Use real and familiar examples from your everyday environment to support your answers, and explain your ideas clearly.

Questions:

- a) Identify one example of direct production in your household or neighborhood and explain why it is considered direct production.
- b) Identify one example of indirect production in your community and explain why it is considered indirect production.
- c) Compare the benefits and challenges of direct and indirect production for families and businesses in Rwanda.

6.2 Factors of production.

Factors of production refer to all productive inputs used in the production of goods and services and the major ones are capital, labor, land and entrepreneurship. The following are the main factors of production: **Land, Labor, Capital and Entrepreneurship.**

6.2.1 Land as a factor of production.

Learning Activity 6.2.1



Think about the activities you see around you such as farming, mining, building houses, or setting up a factory. Using your understanding and everyday experiences, carefully read each of the questions below and discuss your answers.

Questions:

- i. When farming, mining, building houses, or opening a factory, what must be available before any work can begin?
- ii. What things do we get directly from nature that are used to produce goods and services?
- iii. Can more land be created by humans in the same way machines or tools can be made? Why or why not?
- iv. How would production and human life be affected if there were no natural resources available?

Land as a factor of production focuses on the wide range of natural resources that support economic activities. In this section, attention is given to how these resources contribute to production, the way they are used, and their significance in enabling other factors of production to function effectively.

i. Meaning of Land

Land refers to all the free gifts of nature, which are found on earth, beneath or above. It includes soil, forests, mountains, oceans, rivers, minerals and air. The reward to land for its contribution in the production process is rent or rates.

ii. Characteristics of Land

- Land has a fixed supply: The total amount of land is limited and cannot be increased or reduced, regardless of changes in demand.
- Land is a free gift of nature: Land exists naturally and is not created through human effort or production.
- Land is permanent (non-perishable): Unlike other factors of production, land does not wear out or get used up, although its productivity may change.
- Land is a primary factor of production: All production activities begin with land, since natural resources are required to produce goods and services.
- Land is immobile: Land cannot be moved from one geographic location to another, nor can its use be easily changed.
- Land has inelastic supply: The supply of land does not respond to changes in demand; even if demand rises, the quantity of land remains the same.
- Land is differentiated: Land varies in quality, fertility, and productivity, which causes differences in value and price.

iii. Reward for Land:

The reward paid for the use of land is called **rent**. The amount of rent earned depends on factors such as the size and location of the property, available amenities, demand and supply conditions, tenants' ability to pay, the level of competition, and prevailing economic conditions.

Types of Rent

- Quasi rent:** Quasi rent is the payment made to a factor of production whose supply is fixed in the short run but can be increased in the long run. Example: A factory machine earns high income in the short run because few such machines exist, but as more machines are produced over time, this extra payment reduces.
- Commercial rent:** Commercial rent is the payment made for the use of durable assets such as houses, shops, or offices. Example: A shop owner pays monthly rent to use a building in a trading centre.
- Scarcity rent:** Scarcity rent arises because land is limited in supply. When demand for land increases, rent increases even though land supply remains fixed. Example: Rising demand for land in Kigali leads to higher land rents due to limited available land.
- Site rent:** Site rent is paid based on the location of land. Land in well-developed or busy areas earns higher rent than land in remote areas. Example: A plot in the city centre earns more rent than the same size of land in a rural village.

- e) **Economic rent:** Economic rent is the payment to a factor of production above what is necessary to keep it in its current use. Exemple: A farmer continues farming on land because he earns more income than he would in another activity.
- f) **Transfer earnings** Transfer earnings are the minimum amount a factor of production must be paid to keep it in its present use, rather than transferring it to its next best alternative. It represents the opportunity cost of a factor. Exemple: The income a piece of land must earn in its current use (for exemple, farming) to prevent it from being shifted to its next best alternative use (such as building rental houses) represents **transfer earnings**.

Transfer earnings = Actual Earnings - Economic rent.

Actual earning in this case means the total payment a factor of production actually receives for its services. While economic rent as seen above is economic rent is the extra payment earned over and above what is necessary to keep that factor in its current use.

Exemple: A piece of land earns 500,000Frw per year in its current use. The land could earn 300,000Frw per year in its next best alternative use. Calculate the transfer earnings.

Transfer earnings = Actual earnings – Economic rent

Transfer earnings = 500,000Frw – 200,000Frw = 300,000Frw

Transfer earnings = 300,000Frw

This shows that 300,000Frw is the minimum income required to keep the land in its current use.

iv. Role of land in production:

Land plays a crucial role in production by providing the natural resources and physical space needed for all economic activities to take place.

- **Source of raw materials:** Land provides natural resources such as minerals, forests, water, soil, and fisheries used in production.
- **Provides space for production:** It offers sites for agriculture, factories, businesses, roads, and housing.
- **Supports agricultural production:** Land is essential for growing crops and rearing livestock, which provide food and raw materials.
- **Basis of all economic activity:** All production begins on land, making it the foundation of economic activities.
- **Store of natural wealth:** Land contains valuable resources that contribute to national income and economic development.
- **Generates income:** Land earns rent for its owners when used in production.
- **Supports capital and labour:** Capital equipment and labour require land as a location to operate effectively.

Application activity 6.2.1



Clearly analyse the questions below and provide relevant answers for them.

1. a) Identify the examples of land in your community.
b) How is land important in everyday economic activities in your local area?
2. Given that a factor of production receives a transfer earning of 2,000,000Frw and its economic rent is twice its transfer earnings, calculate the factors actual earnings.
3. Given that a factor of production receives an economic rent of 3,000,000frw and its actual earning is 11,000,000frw, calculate the factors transfer earning.

6.2.2 Labor as a factor of production

Learning Activity 6.2.2



1. Imagine a farmer who owns land, seeds, and tools. The farmer leaves everything in the field but does no work for months.
 - a) Will crops be harvested? explain?
 - b) When land and raw materials are available, what next must be available to enable production go on?
 - c) What would happen to farming, factories, or services if no one was willing or able to work?
2. Do all people involved in production have the same capacity? Explain.
3. Do all people involved in production do the same type of work? How do their tasks differ?

Labour as a factor of production highlights the human effort used in producing goods and services. It covers its categories, key characteristics, and shows how productivity can be improved through specialization and the division of labour, emphasizing the essential role of workers in production.

i. Meaning and categories of Labour.

Labour is the total mental and physical effort people contribute to the production of goods and services. It may be **productive**, directly involved in production and rewarded with wages or salaries, or unproductive, which does not create goods or services and earns no wage. Labour is also classified by skill level into skilled labour, requiring high training and independence (such as doctors and teachers), semi-skilled labour, needing some training and supervision (such as drivers and tailors), and unskilled labour, which requires little or no training (such as cleaners and housemaids).

ii. Characteristics of labour

Unlike other factors of production, labour involves human effort and therefore has unique features that set it apart. The following points outline the main characteristics of labour.

- Labour is human: Labour consists of human effort used in production. It includes both physical works, such as farming or construction, and mental work, such as teaching or management.
- Labour is perishable: Labour cannot be stored for future use. If a worker does not work on a particular day, the labour for that day is lost permanently.
- Labour is heterogeneous: Labour is not uniform because workers differ in skills, experience, education, health, and efficiency. This leads to differences in productivity and wages.
- Labour has inseparable ownership: The services of labour cannot be separated from the worker. A person must be present to provide labour, unlike machines or tools.
- Labour is relatively immobile: Labour does not easily move from one place or occupation to another due to family ties, cultural barriers, cost of movement, and skill specialization.
- Labour is influenced by human feelings: Labour is affected by emotions, motivation, attitudes, and working conditions. These factors can influence productivity either positively or negatively.
- Labour requires training: Most labour needs education or training to improve skills and efficiency. Training increases productivity and the quality of output.
- Labour is subject to fatigue: Human beings get tired after working for long hours. Fatigue reduces efficiency and output, unlike machines which can operate continuously with maintenance.

iii. Specialisation and Division of Labour

Specialization refers to the situation where a worker, firm, or region concentrates on performing one specific task or producing one particular good or service in which they are most efficient. For example, A mechanic may specialise in motorcycle repairs, while another focuses only on car engines, or a country may specialize in producing coffee or tea instead of growing many crops.

Division of labour on the other hand, refers to the process of breaking down the production of a good or service into many separate tasks, with each worker performing only a small part of the whole job. This means several workers contribute different parts to produce a final good. For example, in house construction, one group of workers digs the foundation, others lay bricks, while others do roofing and painting. Thus, division of labour is about splitting work into different tasks, while specialization is about focusing on one task or activity to improve efficiency and productivity.

Forms of Specialisation

- a) **Specialisation by craft:** This is when groups of people focus on one type of skill or job that they are good at. For example, farmers produce food, carpenters make furniture, blacksmiths work with metal, and tailors sew clothes.

- b) Specialisation by process:** This happens when different workers focus on different stages of producing the same product. For example, in a printing industry, some workers type, others print, some bind books, others repair machines, and others package the final products.
- c) Regional specialisation:** This is when different regions concentrate on producing goods that they can make best because of their climate, resources, or skills, and then exchange them with other regions. Urban areas specialise in industrial activities such as manufacturing, transport, and trade, while rural areas focus on agriculture.
- d) International specialisation:** This occurs when countries focus on producing goods they can make at the lowest cost compared to other countries and then trade these goods internationally to benefit all countries involved.

Advantages of specialisation and division of labour.

Specialisation and division of labour offer several benefits in the production of goods and services. These advantages mainly relate to improved efficiency, higher output, and better use of resources as explained below.

- **Increase productivity and efficiency:** When workers focus on a specific task or activity, they become more skilled and work faster, leading to higher output.
- **Improve quality of goods and services:** Repeated practice helps workers gain experience, resulting in better-quality products.
- **Save time in production:** Time is saved because workers do not switch between tasks, and production becomes more organized.
- **Lower cost of production:** Increased efficiency reduces unit costs, making goods cheaper and more affordable.
- **Encourage use of machinery and technology:** Task specialization allows machines to be used effectively, improving productivity.
- **Promote trade and economic growth:** Individuals, firms, and regions depend on each other, encouraging exchange and development.

Disadvantages of specialisation and division of labour.

Despite their benefits, specialisation and division of labour also have some drawbacks. These disadvantages mainly affect workers, production flexibility, and job security as explained below:

- **Cause boredom and loss of job satisfaction:** Repetitive tasks can make work monotonous and reduce worker morale.
- **Lead to over-dependence:** Workers and producers rely heavily on others; if one part fails, production may stop.
- **Increase risk of unemployment:** If demand for a specialized product falls or machines replace workers, unemployment may occur.
- **Limit skills and flexibility:** Workers trained in one task may find it difficult to adapt to other jobs.
- **Reduce creativity and initiative:** Narrow tasks prevent workers from thinking broadly or developing new skills.

Application activity 6.2.2



1. a) What do you understand by the term Labour?
b) How does labour differ in capabilities?
2. If Rwanda concentrates on agriculture production, what are the likely benefits and dangers?
3. Describe the different forms of specialisation.

6.2.3 Capital as a factor of production.

Learning Activity 6.2.3



- i. When people want to produce goods faster or in larger quantities, what do they use besides land and their own effort?
- ii. In a factory or workshop, what helps workers make goods more efficiently and quickly?
- iii. Are tools, machines, buildings, and equipment used to directly satisfy wants? Explain
- iv. Are tools and machines freely provided by nature?
- v. Can more of these production items be created if people decide to make them?

Capital as a factor of production looks at the man-made resources used in producing goods and services. It highlights its categories, the idea of interest as a reward, its role in improving productivity, and how capital can increase over time through accumulation.

i. Meaning and categories of capital

In economics, capital refers to man-made resources used in the production of goods and services, such as machinery, tools, buildings, and equipment. It represents wealth that is not meant for immediate consumption but is used to produce further wealth, thereby increasing productive capacity. Creating capital requires sacrificing current consumption through investment, but this sacrifice is necessary for long-term economic growth and higher productivity.

Categories of capital:

- **Fixed Capital:** Fixed capital consists of durable man-made assets that are used repeatedly over a long period in the production process and are not used up at once. They help in production while remaining largely intact, though they may wear out over time. Examples: factory buildings, machinery, tools, trucks, tractors.
- **Working (Circulating) Capital:** Working capital refers to capital items that are used up entirely during one production process and must be replaced regularly to keep production going. These items support daily operations. Examples: raw materials, fuel, wages paid to workers, electricity, office stationery.

- **Physical (Real) Capital:** Physical capital includes tangible, man made goods that are directly used to produce other goods and services and increase productive efficiency. Examples: machines, equipment, factories, tools.
- **Human Capital:** Human capital refers to the skills, knowledge, training, experience, and health of workers that improve their ability to produce goods and services efficiently. Examples: skilled technicians, educated teachers, trained engineers, healthy workers.
- **Financial/liquid Capital:** Financial capital consists of money and financial assets that provide funds needed to carry out production and investment, though they do not directly produce goods. Examples: bank savings, loans, shares, stocks, bonds.
- **Social Capital:** Social capital refers to networks, trust, cooperation, and relationships among individuals or institutions that make economic activities easier and more efficient. Examples: trust between traders, business partnerships, community cooperation groups.
- **Private Capital:** Private capital is capital owned by individuals or private businesses, and any income earned from it belongs to the owner. Its main aim is to generate personal profit or income. Examples: a farmer's tractor, a shopkeeper's building, privately owned factory machines.
- **Social Overhead Capital (Public Capital):** Social overhead capital refers to capital owned and provided by the government to support economic and social activities rather than to earn profit. It benefits society as a whole. Examples: roads, schools, hospitals, water supply systems, bridges, national defence installations.

ii. Reward for capital:

The reward for capital is called **interest**, which is the payment made to the owner for allowing capital to be used in production. Interest compensates the lender for providing capital, and it may be paid as gross interest, which includes allowances for risk, inconvenience, and management, or as net (pure) interest, which represents the actual payment for the use of capital after these allowances are deducted.

Determinants of Interest

- **Inflation:** When inflation is high, interest rates rise to protect the value of money lent, but during deflation or low inflation, interest rates tend to fall.
- **Degree of Risk:** Loans with higher risk of nonrepayment attract higher interest rates, while lowrisk loans are charged lower interest.
- **Government Monetary Policy:** Restrictive monetary policy raises interest rates to reduce money supply, while expansionary policy lowers interest rates to encourage borrowing and spending.
- **Duration of Credit:** Longterm loans carry higher interest because of greater uncertainty over time, while shortterm loans have lower interest rates.
- **Amount of Credit (Loan):** Larger loans are usually charged higher interest due to increased risk and commitment, whereas smaller loans attract lower interest rates.
- **Demand and Supply of Credit** The rate of interest is determined by the interaction between the demand for capital by borrowers and the supply of capital by lenders

in the capital market. When demand and supply are equal, the market reaches equilibrium, and the interest rate at this point is known as **the natural rate of interest**, which represents the true price paid for borrowing capital.

iii. Role of capital in production.

Capital contributes to the production process by supporting labour, improving efficiency, and increasing output in economic activities as explained below:

- **Increases productive capacity:** Capital such as machines and tools helps produce more goods and services than manual effort alone.
- **Improves efficiency:** Modern capital reduces time, effort, and cost of production.
- **Supports specialization:** Capital equipment allows workers to specialize, increasing output and quality.
- **Enhances quality of output:** Use of advanced machinery leads to better and more uniform products.
- **Promotes technological progress:** Capital investment introduces new technology and innovation in production.
- **Facilitates large scale production:** Capital makes mass production possible, lowering average costs.
- **Contributes to economic growth:** By raising productivity, capital helps increase national income and living standards.

iv. Capital Accumulation:

Capital accumulation is the process by which a country increases its stock of capital assets over time through investment. It includes **gross capital accumulation**, which is the total spending on capital goods such as machinery and buildings, covering both the replacement of worn-out assets and the purchase of new ones and **net capital accumulation (net investment)** which is the real increase in capital stock after deducting depreciation, and it represents the actual growth in a country's productive capacity.

Determinants of a country's capital accumulation:

Capital accumulation is determined by several factors which include the following:

- **Level of income:** Higher income increases people's ability to save and invest, leading to greater capital accumulation, while low-income limits savings and investment.
- **Rate of saving and investment:** When a country saves and invests more, more funds are available to build machinery, infrastructure, and other capital goods
- **Political stability and security:** Peace and stability create confidence for investors, encouraging long-term investments that increase capital stock.
- **Government policies:** Policies such as tax incentives, subsidies, and investment support encourage saving and investment, boosting capital accumulation.
- **Development of financial institutions:** Strong banks and financial markets mobilise savings and channel them into productive investments, increasing capital formation.
- **Population structure:** A large working-age population saves and invests more, while a high number of dependants reduces savings and capital accumulation.

- **Capital inflows:** Foreign direct investment and external capital provide additional funds for building capital assets, increasing accumulation.
- **Level of entrepreneurship:** Skilled and innovative entrepreneurs create new businesses and invest in capital goods, expanding productive capacity.

Application activity 6.2.3



1. a) What do you understand by the term capita?
b) How does fixed capital differ from working capital?
2. Examine the role of capital in production
3. a) Explain what you understand by the term capital accumulation.
b) Account for the low levels of capital accumulation in Rwanda

6.2.4 Entrepreneurship as a factor of production

Learning Activity 6.2.3



1. In a certain community, many people complain that there is no nearby place to buy affordable bread. One day, Mugeni notices this problem, and after some time a small bakery appears in the area.

Questions:

- a) What kind of person do you think Mugeni is, to notice this problem and take action?
- b) Do you think people in the community were too poor to start a bread selling business, or could there have been another reason why no one had done it before?
- c) What do you think Mugeni needed or had to do in order to start the bread selling business?
- d) How is Mugeni different from other people who simply wait for opportunities to come their way?
- e) Why do you think some people, like Mugeni, create new ways of producing or selling goods instead of copying exactly what already exists?
- f) Suppose Mugeni faced challenges after starting the bakery. Do you think Mugeni would quit the business? Give reasons for your answer.

Entrepreneurship as a factor of production highlights the role of individuals who organize and manage resources to produce goods and services. It covers its meaning, key characteristics, and the functions entrepreneurs perform, while also examining the challenges and limitations that can affect the development of entrepreneurship.

i. Meaning and characteristics of Entrepreneurship:

Meaning of Entrepreneurship

Entrepreneurship is the ability and willingness to identify business opportunities, organize and combine the factors of production, take risks, and produce goods or services with the aim of making profit. It involves initiating, planning, and managing business activities under conditions of uncertainty. As a factor of production, entrepreneurship plays a coordinating role by bringing together land, labor, and capital for productive use.

Characteristics of Entrepreneurship.

The characteristics of entrepreneurship refer to the key qualities and features that describe how entrepreneurship operates and what distinguishes it from other factors of production. These include the following.

- **Risk taking:** Entrepreneurship involves bearing uncertainty in business activities. The entrepreneur commits resources with no guarantee of success and accepts the possibility of loss if the business fails.
- **Innovation:** Entrepreneurship is characterized by the introduction of new ideas, products, services, or production methods. This innovation helps businesses remain competitive and contributes to economic development.
- **Decision making:** An important characteristic of entrepreneurship is the ability to make key business decisions. These include decisions about what to produce, how to produce, pricing, and investment choices.
- **Organizational ability:** Entrepreneurship involves the ability to organize and coordinate land, labor, and capital efficiently. This ensures that resources are used productively to achieve business objectives.
- **Profit motive:** Entrepreneurship is driven by the desire to earn profit. Profit serves as a reward for the entrepreneur's effort, innovation, and risk-taking.
- **Planning and forecasting:** Entrepreneurs engage in planning and forecasting future market conditions. This helps them set goals, allocate resources, and prepare for potential challenges.

ii. Functions of Entrepreneurship.

The functions of entrepreneurship explain the important roles entrepreneurship plays in organizing production, managing businesses, and contributing to economic activity. These include the following.

- **Organizing factors of production:** Entrepreneurship brings together land, labor, and capital and combines them efficiently to produce goods and services.
- **Risk bearing:** Entrepreneurship involves bearing business risks that arise from market fluctuations, competition, changes in demand, or economic conditions.
- **Decision making and control:** Entrepreneurs make decisions regarding production levels, methods of production, and distribution of goods, and they control the overall direction of the business.

- **Innovation and improvement:** Entrepreneurship promotes innovation by introducing new products, improving existing ones, or adopting better production techniques.
- **Management and supervision:** Entrepreneurs manage business operations by supervising workers, coordinating activities, and ensuring that goals are achieved.
- **Job creation:** Entrepreneurship creates employment opportunities by establishing businesses that require labor, thereby reducing unemployment.
- **Economic growth and development:** Through increased production, investment, and innovation, entrepreneurship contributes to economic growth and national development.

iii. Limitations to Entrepreneurship Development.

The limitations to entrepreneurship development are the challenges and obstacles that hinder the growth and effectiveness of entrepreneurial activities in an economy. These include the following.

- **Lack of capital:** Many potential entrepreneurs fail to start or expand businesses due to inadequate financial resources and limited access to credit facilities.
- **Poor infrastructure:** Inadequate roads, unreliable electricity, poor water supply, and weak communication systems increase production costs and discourage entrepreneurship.
- **Limited skills and education:** A lack of entrepreneurial, managerial, and technical skills reduces the ability of individuals to successfully start and manage businesses.
- **High competition:** Intense competition from established local and foreign firms discourages new entrepreneurs and limits the growth of small businesses.
- **Government regulations and bureaucracy:** Complex licensing procedures, high taxes, corruption, and excessive regulations make it difficult for entrepreneurs to operate businesses profitably.
- **Market uncertainty:** Fluctuations in prices and demand create uncertainty, making it difficult for entrepreneurs to plan and maintain stable businesses.
- **Fear of risk and failure:** Some individuals are unwilling to engage in entrepreneurship due to fear of losing capital, income, or social status.

Application activity 6.2.4



1. a) What do you understand by the term entrepreneurship as a factor of production?
b) What the characteristics of a good entrepreneur?
1. a) Account for the low levels of entrepreneurship in Rwanda.
b) Suggest possible measures Rwanda should undertake to develop entrepreneurship among citizens.

Learning Activity 6.4



Analyse the situation below and answer the questions that follow.

Imagine a farmer using land, workers, and tools to grow maize.

- What things does the farmer use to produce maize?
- If the farmer uses more workers or better tools, what do you expect to happen to the amount of maize produced?
- If the farmer uses fewer workers or poor tools, how will output change?
- Does increasing all inputs always increase output by the same amount? Explain

Production explains how firms transform inputs into output, and the production function provides a clear way of expressing this relationship using tables, equations, and graphs. How this relationship operates depends on time, as firms face different conditions in the short run and the long run when adjusting inputs. In the long run, changes in all inputs reveal how output responds through returns to scale, highlighting efficiency and limitations in production. These ideas are brought together through isoquants and isocosts, which visually show how firms choose input combinations that achieve the desired level of output at the lowest possible cost.

6.3.1 Production Function, Planning Periods.

i. Meaning of Production Function

A production function shows the relationship between inputs (factors of production such as labour, capital, land, and entrepreneurship) and the maximum output that can be produced with a given level of technology during a specific period of time. It explains how changes in inputs affect the level of output, assuming efficient use of resources. The production function can be expressed in the following forms;

a) Tabular Form

The production function can be presented in a table showing different quantities of inputs and the corresponding levels of output produced.

Table 6.1 production function

Labour (L)	Output (Q)
0	0
1	5
2	11
3	18
4	24
5	29

The table shows that as labour increases, output increases, showing how labour contributes to production when capital is constant.

b) Equation (Mathematical) Form

It can be expressed as a mathematical relationship, for example: $Q = f(L, K)$; where Q represents output, L labour, and K capital. If capital is fixed, it can be simplified to: $Q = f(L)$. This means that Output is a function of labour, given a fixed level of capital and technology.

c) Graphical Form

The relationship between inputs and output can also be shown graphically, using curves or lines to illustrate how output changes as one or more inputs are varied.

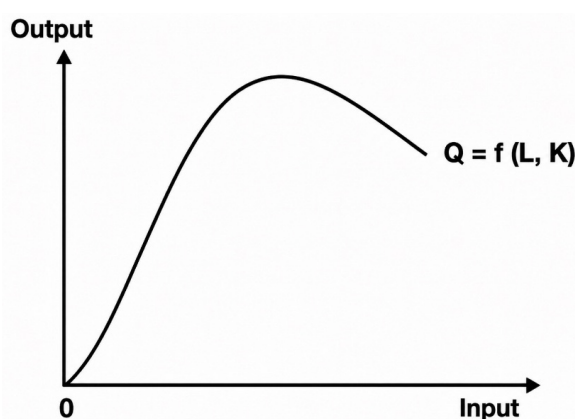


Fig 6.1 production function curve

The graph shows how total output changes as more input is used. At first, output increases rapidly as additional input is used efficiently. After reaching a maximum level, output begins to fall when too much input is applied, causing inefficiencies.

ii. Planning Periods.

Planning periods refer to the time a firm considers when making production decisions, reflecting how quickly it can adjust its factors of production. Since not all inputs can be changed at once, four main planning periods exist: the very short run, short run, long run, and very long run. In practice, the short run and long run are the two most commonly used in economic analysis, as they capture the key distinction between fixed and variable costs. The very short run and very long run are included for completeness but are used less often.

- a) **Very short-run (market) period:** The very short run, or market period, is so brief that all factors of production are fixed, and firms cannot change output at all, regardless of price changes. For example, a fishmonger at a local market has only the fish already caught for the day to sell, no matter how much prices rise.
- b) **Short-run period:** The short run is a planning period in which at least one factor of production is fixed while others can be changed. Firms cannot immediately alter inputs such as buildings or heavy machinery, but they can adjust variable inputs like labour and raw materials to change output. Production decisions in the short run are therefore limited by existing capacity. A local bakery cannot quickly change its building or ovens, but it can hire extra workers or buy more flour to increase production.

c) Long Run: The long run is a planning period in which all factors of production are variable. Firms have enough time to adjust every input, including plant size, machinery, and technology, allowing them to change their scale of production fully. Decisions in the long run focus on expansion, contraction, and overall efficiency rather than short-term limitations. For example, the same bakery we used in the short run, can now build a larger shop, buy more ovens, or open another branch to increase its production capacity.

d) Very Long Run: The very long run goes even further, allowing not just the scale of production to change, but the technology and methods of production themselves. For example, the bakery might eventually adopt entirely new baking technology or ingredients that change how bread is produced, not just how much.

6.3.2 Product of the Firm.

In economics, the product of the firm refers to the total output that a firm produces over a given period of time by combining and using its factors of production such as land, labour, capital, and entrepreneurship. In simple terms, it represents the goods or services generated through the firm's production process. In production theory, the concept of the firm's product is commonly analyzed through three related measures:

a) Total Product (TP) is the total quantity of output produced by a firm using a given combination of inputs during a specific period of time. It represents the overall result of employing all factors of production and is always measured in physical units such as kilograms, litres, or tons, not in monetary terms.

Total product is expressed mathematically as; $TP = AP \times \text{Labour}$

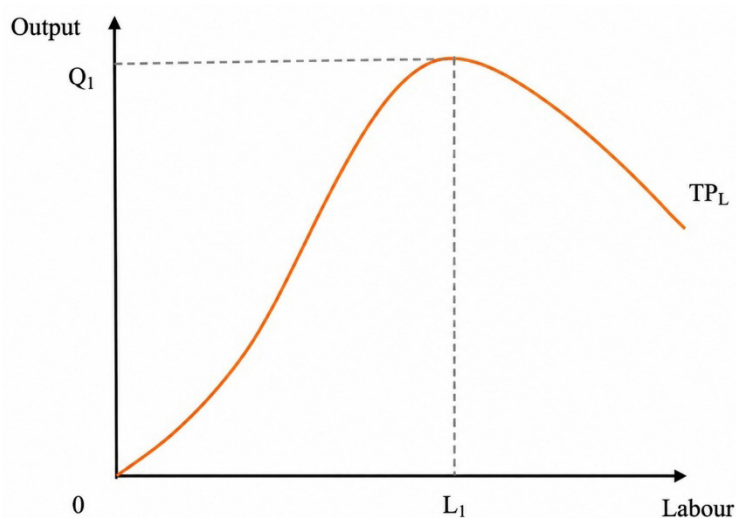
For example; If a juice firm employs 40 workers and each worker produces 20 cartons of juice, calculate the total output produced by the firm.

Solution;

TP = Average Product x Labour

$TP = 20 \text{ carton} \times 40 \text{ TP} = 800$

Total Product can be graphically be represented graphically as below



ATPC

Fig 6.1 Total product curve

b) Average Product (AP) is the output produced per unit of a particular variable input, usually labour, when other inputs are held constant. It shows how productive each unit of the variable factor is and is calculated by dividing total product by the number of units of the variable input used.

$$AP = \frac{\text{Total Product}}{\text{Units of a variable factor (Labour)}}$$

$$AP = \frac{TP}{L}$$

For example: Suppose a farm employs 4 workers and produces 80 bags of maize in a day.

$$AP = \frac{TP}{L}$$

$$AP = \frac{80}{4} = 20$$

AP = 20 bags of maize

The Average Product of labour is 20 bags, meaning each worker produces 20 bags of maize on average per day.

Average Product can be graphically be represented graphically as below;

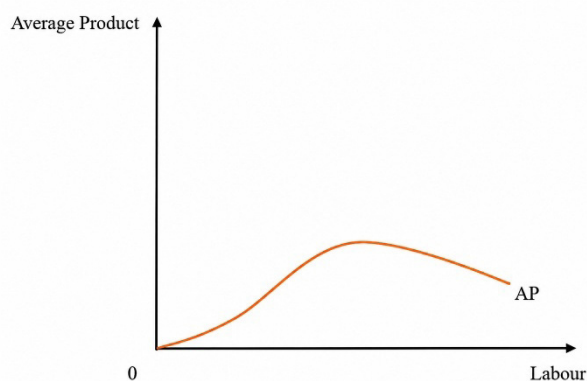


Fig 6.3 Average product curve

c) Marginal Product (MP) refers to the additional output obtained by employing one more unit of a variable factor, while all other inputs remain constant. It measures the change in total product resulting from a change in the quantity of the variable input.

$$MP = \frac{\text{Change in Total Product}}{\text{Change in the units of variable factor (Labour)}}$$

$$MP = \frac{\Delta TP}{\Delta L}$$

For example: Suppose a small bakery increases the number of workers it employs and observes a change in its daily output: when it has 2 workers, the bakery produces 40 loaves of bread, and after increasing the workforce to 3 workers, daily production rises to 55 loaves. Calculate its marginal product.

Solution:

$$MP = \frac{\Delta TP}{\Delta L}$$

$$MP = \frac{55 - 40}{3 - 2} = \frac{15}{1}$$

MP = 15 loaves

The Marginal Product of the 3rd worker is 15 loaves, meaning that hiring one additional worker increased output by 15 loaves.

Marginal Product can be graphically be represented graphically as below.

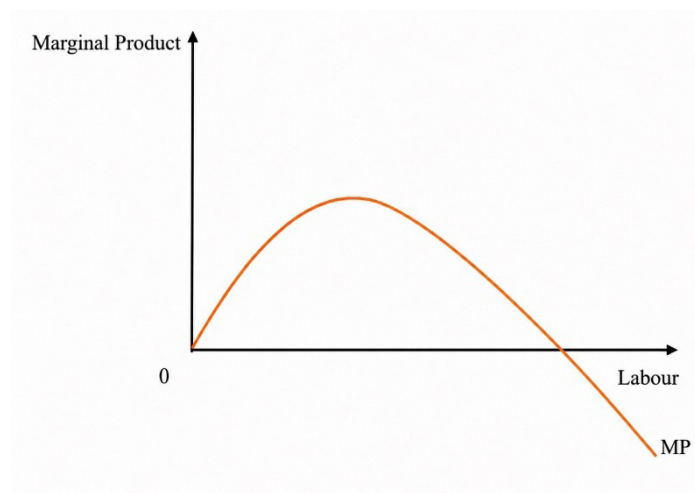


Fig 6.4 Marginal product curve

Exercise. 6.1

Answer the following questions:

1. A farmer employs workers to grow tomatoes on a fixed piece of land. When one worker is employed, total output is 10 bags of tomatoes. When two workers are employed, total output rises to 24 bags. With three workers, output becomes 36 bags, and with four workers, output increases to 44 bags.
 - i. Calculate average product at each level of employing labour.

ii. At which number of workers is the average product highest, and what does this suggest about labour efficiency?

iii. Plot the information on a graph.

2. Using the same tomatoes farm, suppose total output is 10 bags with one worker, 24 bags with two workers, 36 bags with three workers, 44 bags with four workers, and 48 bags with five workers.

i. Calculate marginal product at each level of employing labour.

ii. What trend do you observe in marginal product as more workers are employed, and what economic idea does this illustrate?

iii. Plot the information on a graph.

6.3.3 Production function in the short run.

In this lesson, attention is given to how a firm's output changes in the short run, when at least one factor of production is fixed. Using the production function schedule, the relationship between variable inputs and output is clearly illustrated to show how production responds as more of a variable factor is employed. This analysis leads to the law of diminishing returns (or variable proportions), which explains why continuous increases in a variable input eventually result in smaller additions to total output. Understanding these ideas helps students grasp short-run production efficiency and the limits firms face when expanding output.

i. Production function schedule in the short run

A short-run production function schedule is a table that shows how a firm's output changes as more units of a variable factor (such as labour) are employed while at least one factor remains fixed (such as land or machinery).

Table 6.2: A hypothetical production function schedule in the short run

Labour (L)	Total Product (TP)
0	0
1	5
2	12
3	18
4	22
5	24
6	24
7	22

The schedule above shows that as labour increases while capital remains fixed, total output initially rises quickly due to better use of the fixed factor, but the increase in output gradually slows down as the fixed input becomes overcrowded. This illustrates the law of diminishing returns, which applies in the short run when one or more factors of production are held constant.

The short-run production function schedule shows how total output varies with changes in labour and therefore provides a basis for measuring productivity. By using the total product figures in the schedule, firms can calculate **Average Product (AP)** and **Marginal Product (MP)** of labour and evaluate how efficiently labour is used in the short run.

ii. Relationship between TP, AP and MP.

The relationship between Total Product (TP), Average Product (AP), and Marginal Product (MP) explains how output changes as more units of a variable factor are employed, and it is shown using a product schedule and product curves to illustrate the stages of production and the operation of the law of diminishing returns.

Table 6.3: A hypothetical production function schedule showing the relationship between AP and MP in the short run

Labour (L)	Total Product (TP) (kg)	Average Product (AP) (kg)	Marginal Product (MP) (kg)
0	0	–	–
1	15	15.0	15
2	35	17.5	20
3	60	20.0	25
4	80	20.0	20
5	95	19.0	15
6	105	17.5	10
7	110	15.7	5
8	110	13.8	0
9	105	11.7	–5

From the table, it can be seen that,

- MP rises first and reaches its maximum earlier (at 3 units of labour).
- P reaches its maximum when AP = MP (at 4 units of labour).
- TP is maximum at 8 units of labour, where MP is close to zero.
- Beyond this point, MP becomes negative and TP starts to fall, showing diminishing returns

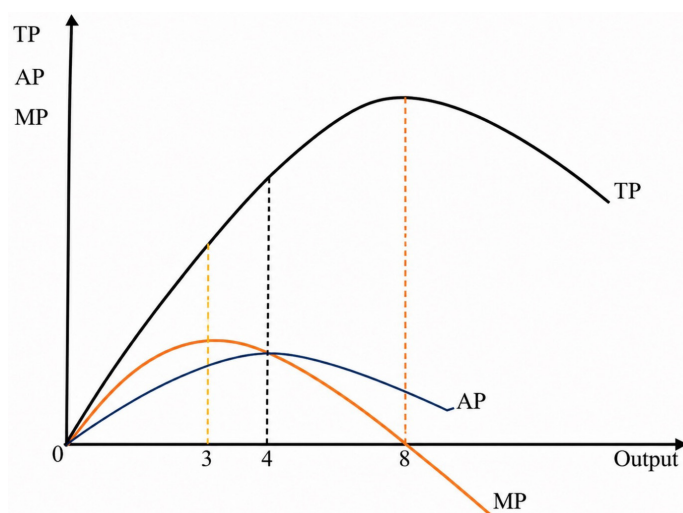


Fig 6.5 Relationship between TP, AP and MP curves

Using the curve shown, the relationship between Total Product (TP), Average Product (AP), and Marginal Product (MP) can be explained as follows:

- At the beginning of production, when output (or labour input) is low, TP increases at an increasing rate. This is because resources are being used more efficiently. During this stage, MP is rising and lies above AP, which causes AP also to rise. MP reaches its maximum point before AP, shown around output level 3, and this is where TP is rising most steeply.
- As production continues, MP begins to fall but remains positive, meaning TP is still increasing but now at a decreasing rate. AP continues to rise until it reaches its maximum, which occurs at output level 4, where MP intersects AP. This intersection is very important: when MP equals AP, AP is at its highest point.
- Beyond this point, MP falls below AP, causing AP to decline. TP continues to increase until MP becomes zero, shown at output level 8. At this point, TP is at its maximum. After this, MP becomes negative, which leads to a fall in TP, indicating inefficiency due to overuse of the variable factor.

iii. The stages of production in the short run

The Total Product (TP) curve initially rises at an increasing rate, then rises at a decreasing rate, reaches a maximum point, and eventually declines. This shape shows how variable factors such as labour interact with fixed factors at different stages of production, as illustrated by the curve. The movements of the Average Product (AP) and Marginal Product (MP) curves help explain the behavior of total output across these stages.

Stage 1: Region of Increasing Returns.

In this stage, TP increases at an increasing rate, as shown by the steepening TP curve. At the same time, both AP and MP rise, with MP lying above AP. This occurs because fixed factors are underutilized relative to the variable factor, so adding more labour improves specialization, coordination, and efficiency. As a result, output per worker continues to increase. Since AP is rising throughout this stage, it would be irrational for a firm to stop production here. For example, at a coffee washing station in Huye District, a single worker cannot sort, wash, and dry cherries at once, leaving the equipment underused. Adding more workers to specialize in each task keeps the machines running continuously, so output rises faster with each extra worker.

Stage 2: Region of Decreasing Returns

During this stage, TP continues to increase but at a decreasing rate, indicated by the flattening of the TP curve. MP begins to decline due to diminishing efficiency but remains positive, and it eventually intersects AP at the point where AP reaches its maximum. Beyond this point, MP falls below AP, causing AP to decline as well. This situation arises because the variable factor becomes excessive relative to the fixed factor, leading to congestion and less effective use of resources. This stage represents the most realistic and economically optimal level of production for a price-taking firm. For example, as more and more workers are added to the same coffee washing station, they start getting in each other's way around the machines. Each new worker still helps produce more coffee, but adds less than the last one, since there isn't enough equipment for everyone to work efficiently.

Stage 3: Region of Negative Returns

In the final stage, MP becomes negative and TP begins to decline, as shown by the downward-sloping portion of the TP curve. Excessive use of the variable factor leads to serious inefficiencies such as overcrowding and poor coordination, which reduce total output. AP continues to fall, confirming a general decline in productivity. Firms should avoid operating in this stage and instead adjust their use of fixed factors or scale of production. For example, if too many workers crowd around the same coffee washing station, they get in each other's way and start spilling cherries or breaking equipment, so adding one more worker actually lowers total coffee output instead of raising it.

Overall, the curve demonstrates that increasing the use of variable inputs does not lead to a continuous increase in output. Once TP, AP, and MP begin to decline, further expansion becomes unprofitable. Firms must therefore carefully determine the optimal quantity of variable factors to employ, as **the law of diminishing returns** sets in when variable inputs are increased relative to fixed factors.

Exercise: 6.2

Use the table below to complete the columns for AP & MP and illustrate your information on a graph. Assume land and capital are fixed, and only labour varies.

Labour (units)	Total Product (TP)	Average Product (AP)	Marginal Product (MP)
0	0		
1	8		
2	18		
3	30		
4	40		
5	48		
6	54		
7	56		
8	56		
9	54		

iv. The law of diminishing marginal returns.

The law of diminishing returns, also known as **the law of variable proportions**, states that **given a short-run situation with two factors of production, one variable and one fixed factor, as more and more units of a variable input are applied to the fixed factor in a given state of technology, the marginal product of the variable factor will eventually diminish**. This law is reflected in the behavior of TP, AP, and MP, particularly when MP begins to fall.

This means that after a certain point, each additional unit of the variable input contributes less to output than the previous one, holding other factors constant. For example, in agriculture, when more workers, fertilizer, or seeds are applied to a fixed piece of land, output initially rises but later increases at a decreasing rate, such as when too many laborers are added to cultivate the same hectare of land. Similarly, in manufacturing, adding more workers to a factory with fixed machinery may eventually reduce efficiency due to congestion. In the service sector, overcrowding in offices or classrooms can lower productivity, and in short-run business operations, increasing labour without expanding fixed capital leads to diminishing efficiency.

Assumptions of the Law of diminishing returns.

The law operates under the following assumptions:

- **At least one factor is fixed:** The law works only in the short run, when one factor such as land or machinery cannot be changed.
- **Variable units are homogeneous:** All units of the variable factor (e.g. labour) are of the same quality and efficiency, so output changes are not due to skill differences.
- **Technology is constant:** Methods of production do not change; otherwise, output changes would reflect technological improvement.
- **Variable factor is added in equal units:** Inputs are increased one unit at a time, making it easier to observe changes in output.
- **Factors are imperfect substitutes:** Variable and fixed factors cannot fully replace each other, causing inefficiency when one is overused.
- **Output is measurable:** Production must be quantifiable so changes in TP, AP, and MP can be clearly observed.

The law can be illustrated as below:

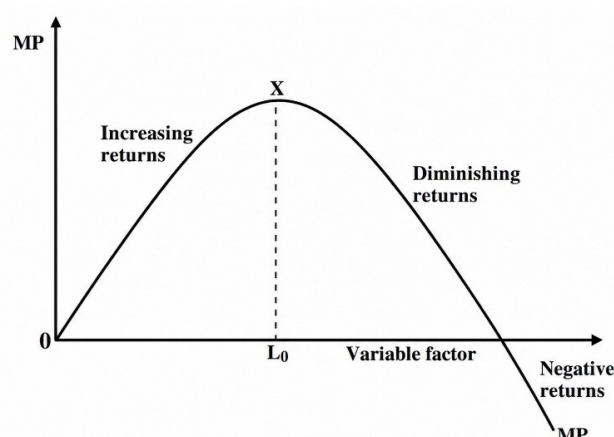


Fig 6.6 The law of diminishing returns curve

The diagram illustrates the Marginal Product (MP) curve. At first, MP rises, showing increasing returns as additional units of the variable factor are used more efficiently. MP reaches its maximum at point X, indicating the highest productivity. Beyond this point, MP falls, showing diminishing returns, and eventually becomes negative, meaning additional units of the input reduce total output due to inefficiency and congestion.

Importance of the Law

The law is significant for both theory and practice:

- It helps firms determine the optimal level of input use.
- It explains the shape of TP, AP, and MP curves and the stages of production.
- It guides firms in avoiding inefficiency and excessive costs.
- It forms a foundation for cost theory and helps explain rising marginal costs.
- It aids in production planning and short-run decision-making.

Limitations of the Law

Despite its usefulness, the law has some limitations:

- It applies only in the short run, not the long run where all factors are variable.
- It assumes constant technology, which may not hold in real-world production.
- The law does not operate if fixed factors are indivisible.
- It may not apply where specialization continuously improves efficiency for long periods.
- The law is less relevant in industries where automation and technological change dominate

The law of diminishing returns explains why increasing a variable input does not lead to unlimited increases in output. As shown by TP, AP, and MP curves, productivity eventually declines when variable factors are increased relative to fixed factors. Understanding this law enables firms to operate efficiently and avoid unproductive stages of production.

6.3.4 Production function in the long run (Returns to scale):

In the long run, producers can change the scale of production by varying all factors of production, including labour, land, and capital. Unlike the short run where at least one factor is fixed, the long run allows firms **enough time** to adjust plant size, machinery, and land use alongside labour. As a result, changes in output are caused by proportionate changes in all inputs, not just one. This relationship between the proportionate change in inputs and the resulting change in output is known as **returns to scale**. Therefore, the long-run production function explains how output responds when a firm expands or contracts its entire combination of inputs, reflecting the full flexibility available to producers over a longer time period.

i. Meaning and types of Returns to scale:

Returns to scale describes how total output responds when all inputs are increased together, and it can be explained using marginal returns, which show the extra output gained from each additional unit of scale of production. Marginal returns are calculated by dividing the change in total returns by the change in scale of production

$$\text{MR} = \frac{\Delta \text{Total Returns}}{\Delta \text{Scale of production}}$$

When marginal returns are increasing, each additional unit of scale adds more output than the previous one, indicating increasing returns to scale; when marginal returns are constant, each extra unit adds the same amount to output, showing constant returns to scale; and when marginal returns fall, each additional unit contributes less output than before, indicating decreasing returns to scale, often due to inefficiencies or overcrowding of resources.

Table 4: Returns to Scale Schedule

Scale of Production	Land Units	Labour Units	Total Returns	Marginal Returns
1	1	2	8	–
2	2	4	20	12 – Increasing returns
3	3	6	35	15
4	4	8	51	16 – Constant returns
5	5	10	67	16
6	6	12	83	16
7	7	14	94	11 – Decreasing returns
8	8	16	102	8
9	9	18	108	6

This table shows how output changes as land and labour are increased together. Total returns increase at every stage, but marginal returns change. The first marginal return is shown as a dash because there is no previous level of output to compare with, so the change in total returns cannot be worked out. From scale 2 to 3, marginal returns increase, **showing increasing returns to scale**. From scale 4 to 6, marginal returns remain the same, indicating **constant returns to scale**. After scale 6, marginal returns begin to fall, showing **decreasing returns to scale** due to inefficiency and difficulty in managing large-scale production.

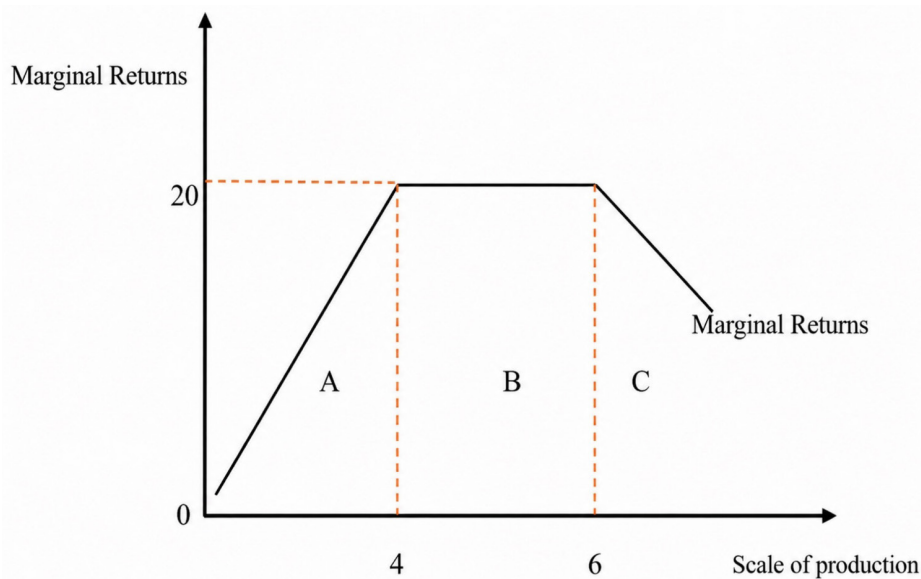


Fig. 6.7 Marginal returns curve

This curve shows how marginal returns change as the scale of production increases.

- **In Region A**, the ratio of output to inputs is increasing, which means that each additional unit of scale of production produces more output than the previous one. This happens because resources are used more efficiently through better organization and specialization. This situation is called **increasing returns to scale**, and it means that *when inputs are doubled, output more than doubles*. For example, if Bralirwa doubles both its workers and machines, output could more than double, since workers can now specialize in brewing, bottling, or packaging rather than each handling every task alone.
- **In Region B**, the ratio of output to inputs remains the same, meaning that each extra unit of scale adds an equal amount of output. This shows **constant returns to scale**, where efficiency is stable. Under constant returns to scale, *when inputs are doubled, output also doubles*. For example, once Bralirwa's brewery is already well organized, doubling its workers and machines simply doubles its output too, since there is no more room for extra efficiency gains.
- **In Region C**, the ratio of output to inputs is falling, meaning that each additional unit of scale produces less output than the previous one. This is known as **decreasing returns to scale** and usually occurs because of overcrowding, poor coordination, or management difficulties. Under decreasing returns to scale, *when inputs are doubled, output increases by less than double*, meaning output grows at a decreasing rate. For example, if Bralirwa keeps doubling its workers and machines beyond its ideal brewery size, output grows by less than double, since managers struggle to coordinate so many workers and machines effectively, leading to overcrowding and communication problems on the production floor.

Overall, the curve illustrates **the law of returns to scale**, where marginal returns first increase, then remain constant, and finally decrease as production expands.

ii. The law of returns to scale:

The law of returns to scale, states **that when all factors of production are increased in the same proportion, output will increase but the increase may be at increasing rate or constant rate or decreasing rate depending on the scale of production.**

Assumptions of the Law of Returns to Scale.

The law of returns to scale is based on the following key assumptions:

- Production takes place in the long run. i.e. Sufficient time is available to adjust all factors of production.
- All factors of production are variable i.e., Labour, land, and capital can all be increased or decreased in the long run.
- Technology remains constant i.e. There is no change in production techniques during the analysis.
- Factors of production are divisible and homogeneous i.e. Inputs can be increased in the same proportion and are of uniform quality.
- Inputs are changed in the same proportion i.e. All factors are varied simultaneously and proportionately.
- Managerial efficiency is assumed constant i.e. Efficiency does not improve or deteriorate unless scale effects arise.
- These assumptions ensure that changes in output are due solely to changes in the scale of inputs, allowing the effects of returns to scale to be clearly observed.

Ref: <https://study.com/learn/lesson/returns-to-scale-economics.html>

Application activity 6.3



1. A firm produces goods using labor and capital. The table below shows different combinations of inputs and the resulting output:

Capital (K)	Labor (L)	Output (Q)
2	2	4
4	4	10
8	8	20

Questions:

- a) Identify whether the firm is experiencing constant, increasing, or decreasing returns to scale. Explain your answer using the data in the table.
- b) If the firm wants to double all inputs from 4 units of capital and 4 units of labor, predict the expected output based on the type of returns to scale you identified.

6.4 Isoquants and Isocosts

Learning Activity 6.4



Combination	Units of capital	Units of labor	Total output
A	10	4	100
B	8	9	100
C	6	14	100
D	4	19	100

From the above table, answer the following question:

1. What factors of production are used by the firm?
2. Explain the relationship between capital and labor.
3. How many units produced after combining capital and labor?

6.4.1 Isoquants.

i. Meaning of Isoquants

An isoquant is a curve that shows all the different combinations of two inputs (usually labour and capital) that produce the same level of output. An isoquant is also known as an equal product curve, production indifference curve, or iso-product curve. Each point on an isoquant represents a technically efficient input combination, meaning the firm can substitute one input for another without changing total output. Isoquants slope downward from left to right, reflecting the trade-off between inputs, and higher isoquants represent higher levels of output.

ii. Isoquant schedule.

An isoquant schedule is a table that shows a list of different combinations of two inputs that can be used to produce the same level of output. It helps producers analyze how one input can be substituted for another without changing total production and serves as the basis for drawing an isoquant curve to aid in efficient production decisions.

Table 6.4: A hypothetical isoquant schedule.

Point	Labour (L)	Capital (K)	Output
A	4	10	100
B	9	8	100
C	14	6	100
D	19	4	100

Each combination shows a different mix of labour and capital that produces the same level of output (100 units). As labour increases from A to D, capital decreases, illustrating the downward sloping isoquant and the substitutability of labour and capital as shown below;

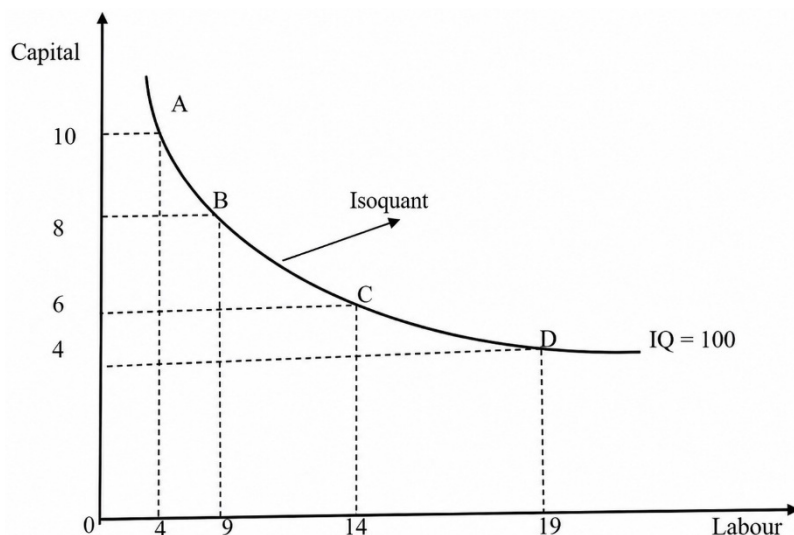


Fig. 6.8: An Isoquant curve

The curve shows all the different combinations of labour and capital that can be used to produce the same level of output (IQ = 100). It slopes downward from left to right, showing that labour and capital can substitute for each other, as more labour is used, less capital is required to maintain the same output, and vice versa. The curve's shape reflects the **diminishing marginal rate of technical substitution (MRTS)**, which is the rate at which capital can be replaced by labour while keeping output constant, meaning each additional unit of labour replaces progressively smaller amounts of capital.

iii. Properties of the Isoquants.

Isoquants share several key characteristics with indifference curves:

- An isoquant represents all combinations of labour and capital that yield the same level of output.
- Isoquants are downward sloping, showing that labour and capital can be substituted for one another.
- Isoquants are convex to the origin, reflecting the diminishing marginal rate of technical substitution (MRTS).
- Higher isoquants indicate higher levels of output, while lower isoquants indicate lower levels of output.
- Isoquants do not intersect, because one combination of inputs cannot produce two different output levels.
- An isoquant does not touch the axes, since production requires positive amounts of both labour and capital.

- More than one isoquant curve together forms **an isoquant map**, which shows different output levels achievable with various input combinations. (a set of isoquants that shows the maximum attainable output from any given combination inputs.)

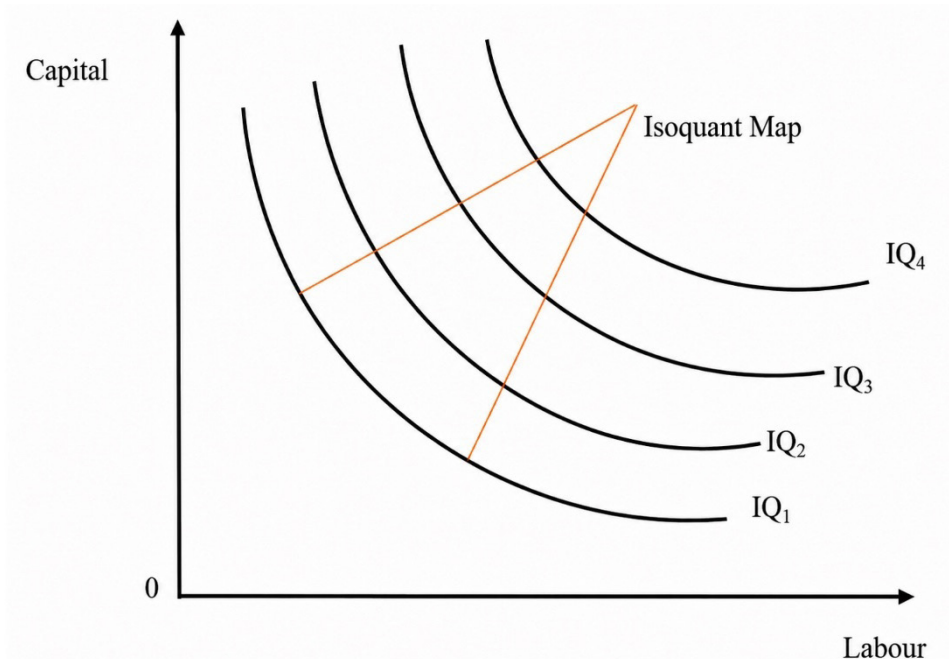


Figure 6.9 An isoquant map

Isoquants IQ_1 – IQ_4 form an isoquant map that is downward-sloping and convex to the origin, showing combinations of labour and capital that produce equal output, with IQ_4 representing a higher level of output than IQ_3 , IQ_2 , and IQ_1 .

6.4.2 Isocosts.

i. Meaning of Isocost

An isocost is a line that shows all combinations of labour and capital that a firm can purchase at the same total cost, given the prices of labour and capital. It represents a firm's budget constraint in production, with every point on the isocost indicating the same level of expenditure on inputs. Isocosts are straight lines because factor prices remain the same whatever the outlay of the firm on the two factors.

ii. Isocost schedule

This is a table that shows a list of the different combinations of two inputs (such as labour and capital) that a producer can purchase with the same total cost, given the prices of the inputs and the producer's budget.

Table 6.5: A hypothetical Isocost schedule.

Labor Price= 20,000 FRW	Capital Price= 10,000 FRW
10	0
8	4
6	8
4	12
2	16
0	20

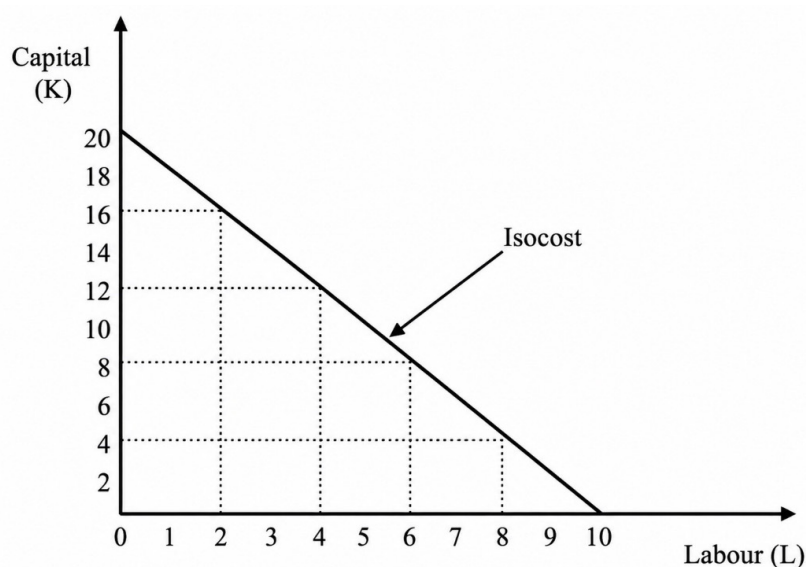


Fig 6.9: Isocost line

The curve is an isocost line showing all combinations of labour and capital that cost the same amount. It slopes downward, indicating that using more labour requires less capital to keep total cost constant. The line shows a maximum of 20 units of capital when labour is zero and a maximum of 10 units of labour when capital is zero, with every point between representing an equally affordable mix of labour and capital.

iii. Properties of Isocosts

- An isocost curve shows all combinations of labour and capital that can be purchased for a given total cost.
- It slopes downward from left to right, reflecting the trade-off between labour and capital when total cost is fixed.
- Isocost curves are straight lines because input prices are assumed to be constant.
- The slope of an isocost curve equals the ratio of input prices (wage rate divided by rental cost of capital).
- Higher isocost curves represent higher total cost, while lower ones represent lower total cost.

- Isocost curves do not intersect, since each curve represents a different level of total cost.

6.4.3. Producer equilibrium.

Producer equilibrium occurs at the point where **an isoquant is tangent to an isocost line**. At this point, the producer achieves a given level of output at the lowest possible cost or maximizes output for a given cost. The tangency shows that the marginal rate of technical substitution (MRTS) between labour and capital equals the ratio of their prices. Any other combination on a different point would either increase cost for the same output or reduce output for the same cost, so the tangency point represents the most efficient and optimal combination of inputs for the producer.

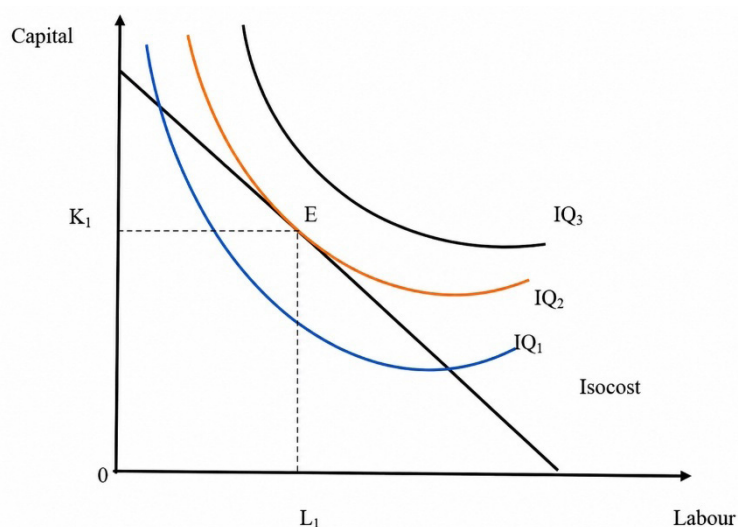


Fig 6.10: Producer equilibrium

The curve above illustrates a firm's cost-minimizing choice of inputs: the curved isoquants (IQ_1 , IQ_2 , IQ_3) show different levels of output produced using combinations of labour and capital, with higher isoquants representing higher output, while the straight downward-sloping isocost line shows all input combinations that cost the same; the point E, where the isocost is tangent to IQ_2 , indicates the optimal mix of labour (L_1) and capital (K_1) that produces that level of output at the lowest possible cost, where the marginal rate of technical substitution equals the ratio of factor prices. <https://www.economicshelp.org/blog/glossary/isoquant-and-isocosts>

"Producer Equilibrium Using Isoquants and Isocosts"

Worked example:

Assume;

- Output level: **Q = 100 units** (all combinations lie on the same isoquant)
- Wage rate (price of labor): **w = 10**
- Rental rate of capital: **r = 20**
- Total cost: **TC = 200**

Table 6.5: Isoquant schedule (constant output Q = 100)

Combination	Labour (L)	Capital (K)
A	2	11.5
B	4	9
C	6	7
D	8	6
E	10	5.5

- a) state
- the isoquant and Isocost equation
 - the slope of the isocost
- a) Compare the MRTS at each isoquant combination with the factor price ratio.
- b) Identify the combination where: $MRTS = w/r$.
- c) Explain why this combination represents producer equilibrium.

Solution

- a) i) Isoquant and Isocost equations

Isoquant: Represents all combinations that produce the same output

Q=100

Isocost

$$TC = wL + rK$$

$$200 = 10L + 20K$$

Simplified:

$$L + 2K = 20$$

ii. Slope of isocost

$$\text{Slope} = -\frac{w}{r} = -\frac{10}{20} = -0.5$$

b) MRTS at each isoquant combination

$$MRTS = \frac{\Delta K}{\Delta L}$$

Table 6.6. calculating MRTS

Combination	Labour (L)	Capital (K)	Change in Capital (ΔK)	Change in Labour (ΔL)	MRTS _{L, K} ($\frac{\Delta K}{\Delta L}$)
A	2	11.5	-	-	-
B	4	9	9 - 11 = -2.5	4 - 2 = 2	$ \frac{-2.5}{2} = 1.25$
C	6	7	7.5 - 9 = -2	6 - 4 = 2	$ \frac{-2}{2} = 1$
D	8	6	6 - 7.5 = -1	8 - 6 = 2	$ \frac{-1}{2} = 0.5$
E	10	5.5	5.25 - 6 = -0.5	10 - 8 = 2	$ \frac{-0.5}{2} = 0.25$

From the table, as labour increases from A to E, the MRTS falls steadily from 1.25 to 0.25, indicating that each additional unit of labour replaces a decreasing amount of capital. Initially, when labour is scarce, one extra unit of labour can replace 1.25 units of capital, but as labour becomes more abundant, its effectiveness declines, so at later stages it replaces only 0.5 and eventually 0.25 units of capital, demonstrating the Law of Diminishing Marginal Rate of Technical Substitution.

Factor price ratio:

Comparison of MRTS with factor price ratio ($w/r = 0.5$)

A → B: MRTS = **1.25** > 0.5

B → C: MRTS = **1** > 0.5

C → D: MRTS = **0.5** = 0.5

D → E: MRTS = **0.25** < 0.5

c) Combination where MRTS = w/r

This occurs at Combination D (L = 8, K = 6)

d) Why this represents producer equilibrium

At combination D, MRTS = w/r , meaning the rate at which labour can substitute for capital is exactly equal to their price ratio. This implies that the firm is using the least-cost combination of inputs to produce the given output. Any change in the combination would either increase cost or reduce efficiency. Therefore, combination D represents the point where the isoquant is tangent to the isocost line, achieving producer equilibrium.

- Using given values of labour, determine the corresponding values of capital to derive different factor combinations to form an isocost schedule.

Using the isocost equation:

$$200 = 10L + 20K \rightarrow K = \frac{200 - 10L}{20}$$

Table 6. 7: Deriving Capital (K) from Given Labour (L)

Combination	Labour (L)	Capital (K) Calculation	Capital (K)
A	2	$K = \frac{200-20}{20} = \frac{180}{20}$	9
B	4	$K = \frac{200-40}{20} = \frac{160}{20}$	8
C	6	$K = \frac{200-60}{20} = \frac{140}{20}$	7
D	8	$K = \frac{200-80}{20} = \frac{120}{20}$	6
E	10	$K = \frac{200-100}{20} = \frac{100}{20}$	5

Application activity 6.3



1. Use the Isoquant information (Output = 300 units)

Point	Labour (L)	Capital (K)
A	4	12
B	8	8
C	13	5
D	19	3

- i. Plot the isoquant for output = 300 units using the data above.
 - ii. Explain why the isoquant slopes downward and is convex to the origin.
2. Cost information.
- a) Price of labour (W) = 20,000 FRW per unit.
 - b) Price of capital (R) = 10,000 FRW per unit.
 - c) Total cost (TC) = 200,000 FRW.
- i. Calculate and show at least three combinations of labour and capital that the firm can afford with this total cost.
 - ii. Plot the isocost curve on a graph

3. Based on the schedule in question 1 and 2,
 - i. Identify the point where the isoquant is tangent to the isocost line.
 - ii. State:
 - a) The optimal quantity of labour and capital used.
 - b) Why this point represents producer equilibrium.
4. Explain what would happen if the firm:
 1. Chose a point on the same isoquant but above the isocost.
 2. Chose a point on the same isocost but on a lower isoquant.

End of unit assessment



A. Knowledge and understanding

1. Define total product (TP), average product (AP), and marginal product (MP).
2. State the law of diminishing returns and explain its assumptions.
3. What are isoquants and isocosts, and what do they represent in production?
4. Identify the factors of production that may be used by a farmer in growing maize.
5. Explain the difference between returns to a variable factor and returns to scale.

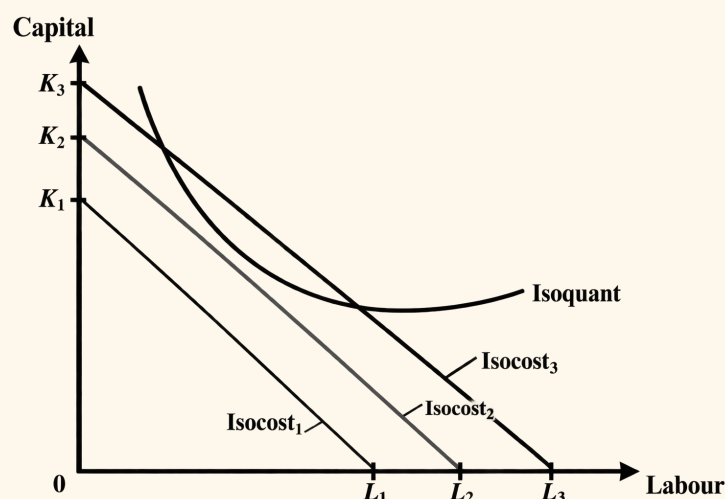
Skills

6. a) Calculate AP and MP given the following data:

Labour (L)	Total Product (TP)
1	10
2	28
3	48
4	64
5	72
6	72
7	68

- b) Draw a TP, AP, and MP curve from given data and interpret the stages of production.

7. Use the figure below to answer the questions that follow.



- On the diagram, mark the point of producer equilibrium and label it E. Explain why.
- State the approximate combination of labour and capital used at the equilibrium point
- Explain what is meant by tangency between the isoquant and an isocost line.
- Explain why any other point on the same isoquant but on a higher isocost is not optimal for the firm.
- What would happen to the firm's cost if it moved away from point E along the isoquant?

Attitudes and Values

- Discuss why a farmer should not keep adding unlimited fertilizer or labor when producing maize.
- Explain why it is important for firms to choose the least-cost combination of inputs when producing goods.

UNIT 7

FIRM BEHAVIOR



Key Unit Competence:

Analyze how firms make production decisions by balancing costs, revenue, and profits to achieve growth and efficiency.

Learning objectives

Knowledge and understanding

- Define a firm, plant, and industry, and explain the main objectives of a firm.
- Explain the long-term decisions of a firm and identify factors influencing such decisions.
- Explain the meaning of firm location and analyze the factors influencing the location of firms and industries.
- Describe localization of firms, including its meaning, causes, merits, and demerits.
- Explain how small-scale firms survive alongside large-scale firms.
- Define cost of production and distinguish between short-run and long-run costs.
- Explain the meaning of total fixed cost (TFC), total variable cost (TVC), and total cost (TC), and describe their short-run cost curves.
- Explain the variation of costs in the short run, including AFC, AVC, ATC (AC), and MC, and describe their shapes.
- Explain the relationships among short-run cost curves using calculations and diagrams.
- Explain how the long-run average cost (LAC) curve is derived.
- Define scale of production and explain the growth of firms.
- Explain the meaning and types of mergers, and discuss their reasons, advantages, disadvantages, and limiting factors.

1

- Define and explain economies of scale (internal and external) and diseconomies of scale (internal and external).
- Define revenue and explain different forms of revenue.
- Explain how to calculate total revenue (TR), average revenue (AR), and marginal revenue (MR) and describe their relationships.
- Explain revenue positions under perfect and imperfect competition.
- Define profit, identify its forms, and explain the role of profit in production.
- Explain profit maximization under perfect competition and identify the determinants of profits.

Skills

2

- Analyze firm behavior by examining how costs, revenues, and profits influence production decisions.
- Calculate and interpret short-run costs (TFC, TVC, TC, AFC, AVC, ATC, and MC) from given data.
- Draw and interpret cost curves and explain their relationships.
- Analyze long-run production decisions, including scale of production and growth of firms.
- Evaluate the impact of firm location and localization on efficiency and costs.
- Assess mergers and economies of scale in terms of efficiency, growth, and competitiveness.
- Calculate revenues (TR, AR, MR) and analyze their relationships under different market conditions.
- Determine profit-maximizing output levels under perfect competition using cost and revenue analysis.
- Apply cost, revenue, and profit analysis to real-life firm decisions, such as expansion, pricing, and production planning.

Attitudes and values

3

- Appreciate the importance of efficiency and cost control in business decision making.
- Develop a positive attitude toward entrepreneurship and firm growth.
- Value careful planning and informed decision making in production and investment choices.
- Demonstrate fairness and responsibility when evaluating firm behavior, mergers, and market competition.
- Recognize the role of profits in encouraging innovation, production, and economic development.
- Show awareness of the social and economic impacts of firm growth and market power.

Why do you think many similar businesses (like shops, banks, or factories) are often found located close to each other in the same area instead of being spread far apart? How might this benefit both the firms and the customers?

This unit on firm behavior aims to help students understand how firms operate and make decisions in the production of goods and services. It explores the meaning of a firm, plant, and industry, as well as the objectives of firms such as profit maximization, sales growth, and long-run survival. The unit also examines factors that influence firms' long-term decisions, location, and the localization of industries. Key concepts in this unit include firm, plant, industry, profit maximization, location of firms, localization, and industrial development.



Introductory activity

Read the scenario below and answer the questions that follow:

Jesca runs a small juice business in her town. She buys fruits, uses workers and machines to produce juice, and sells it to customers to earn income. She has located her business near a busy road, close to fruit sellers and other food businesses. Each day, she monitors sales, costs, and how the business is performing before deciding whether to expand or keep the business at the same level.

Questions

- i. Why do you think the owner makes and sells juice instead of giving it away for free?
- ii. Based on the scenario, what do we call a business that combines these things to produce and sell juice?
- iii. Why might the owner have chosen to locate the shop near a busy road and close to fruit sellers and other food businesses?
- iv. How does being close to similar or related businesses affect the day-to-day operations of the juice business?
- v. Why do you think the owner closely observes how the business performs each day before deciding to expand or remain the same size?
- vi. From the scenario, what can you conclude about the relationship between what the business puts into production and what it gets in return?

7.1 Introduction to the firm behaviour

Learning Activity 7.1



Carefully read the scenario below and answer the questions that follow:

In a local town, there is a business called Sunrise Bread. The business makes bread and sells it to people in the community. All the bread is made in one baking place using ovens and workers. In the same town, there are other businesses that also make bread and other bakery products. The owner of Sunrise Bread makes daily decisions about what to bake, how much to bake, and how to attract more customers so that the business can continue operating.

Guiding Questions

1. What do you think Sunrise Bread is in terms of business activity?
2. What is the purpose of the baking place where the bread is made?
3. How is Sunrise Bread related to the other bakeries in the town?
4. Why does the owner decide what and how much bread to bake each day?
5. What do you think the owner wants to achieve by running this business?

This lesson introduces firm behaviour, focusing on how firms make decisions regarding production, pricing, and the use of resources. It examines the objectives of firms and how they respond to different market conditions in order to achieve their goals. Understanding firm behaviour provides a basis for analyzing how businesses operate and interact within the market economy.

7.1.1: Meaning of a firm, plant and industry.

- i. A firm:** refers to a single business unit or enterprise under one ownership, management and control. It is an independently administered business organization that makes decision on how to use resources to produce goods and services in order to make profit. For example, Inyange Industries is a firm in Rwanda that produces milk, juice, and other dairy products to earn profit. Firms vary in size i.e., small firms (sole traders or partnerships) and large firms (public corporations) and in ownership i.e., private businesses, joint stock companies, and government-owned firms.
- ii. A plant:** A plant is the physical production unit or establishment, a factory, workshop, or site, where actual production takes place. A firm may own and operate one or several plants. Example: The Inyange juice-processing factory in Kigali is a plant, a physical site where fruits are processed and packaged into juice.
- iii. An industry:** An industry is a group of firms or organizations that produce the same or closely related goods or provide similar services and therefore compete in the same market. Firms within an industry use comparable production methods and serve similar customers; for example, all firms involved in processing and selling milk and milk products such as Inyange Industries and other dairy processors form the dairy industry in Rwanda, while tea factories and exporters across the country together make up the tea industry, and hotels, tour operators, and lodges collectively form the tourism industry.

Table 7.1: The differences between a firm, a plant, and an industry with examples:

Aspect	Firm	Plant	Industry
Meaning	A firm is a business organization that owns and controls one or more production units.	A plant is a single physical location where production takes place.	An industry is a group of firms producing similar or related goods and services.
Scope	Broader than a plant but narrower than an industry.	Narrowest unit of production.	Broadest concept covering many firms.
Ownership	Owned and managed by an individual or group.	Owned by a firm.	Consists of many firms (different owners).
Function	Makes decisions about production, pricing, and management.	Carries out the actual production process.	Represents the whole sector of production.
Number	A firm may have one or many plants.	A plant is usually one unit.	An industry contains many firms and plants.
Exemple	A company like a bakery business operating in Kirehe.	One bakery production site where bread is baked.	The entire baking industry or food processing industry in Rwanda.

iv. Objectives of the firms:

The objective of a firm is the main goal or aim that the business wants to achieve. It shows what the firm is trying to do and guides its decisions and actions. Objective helps the firm plan how to use its resources, organize activities, and measure success. It gives direction to employees and managers, so everyone works toward the same purpose. The objectives of the firms are the followings:

- **Profit maximization:** The firm aims to earn the highest possible profit by increasing revenue and reducing costs. Exemple: A bakery increases bread production and reduces wastage to earn more profit.
- **Sales maximization:** The firm focuses on increasing the volume of sales, even if profit is lower in the short run. Exemple: A shop lowers prices to attract more customers.
- **Growth and expansion:** The firm aims to increase its size by opening new branches or entering new markets. Exemple: A maize milling business expands to other districts.
- **Survival:** The firm aims to remain in operation, especially during tough competition or low profits. Exemple: A small shop keeps operating despite earning little profit.
- **Market share maximization:** The firm aims to control a larger portion of the market compared to competitors. Exemple: A company uses advertising to attract more customers than rivals.

- **Customer satisfaction:** The firm aims to provide quality goods and services to meet customer needs and build loyalty. Example: A restaurant improves food quality and service.
- **Innovation:** The firm introduces new products or improves existing ones to remain competitive. Example: A company introduces better packaging for its products.
- **Cost minimization:** The firm aims to reduce production costs while maintaining quality. Example: Using machines to reduce labor costs.
- **Social responsibility:** The firm considers the welfare of society and the environment. Example: A factory reduces pollution and creates jobs.
- **Employee welfare:** The firm aims to improve workers' conditions and motivation. Example: Providing good wages, bonuses, and safe working conditions.

7.1.2 Long term decisions of the firm.

i. Meaning of long-term decisions of a firm:

A longterm decision of a firm is a decision that affects the business for many years. It usually involves spending a lot of money and cannot be changed easily. These decisions include buying machines, expanding the business, choosing a location, using new technology, or introducing new products. Longterm decisions are important because they determine the future growth, profits, and survival of the firm.

ii. Factors Influencing LongTerm Decisions of a Firm

1. **Objective of the firm:** A firm makes long-term decisions based on what it wants to achieve. If the goal is to make more profit, it may focus on reducing costs or charging higher prices. If the goal is to sell more, it may expand production or lower prices to attract many customers.
2. **Level of competition:** The amount of competition affects how a firm plans. In markets with many competitors, firms must be careful and improve efficiency. A monopolistic firm, with little or no competition, can make different decisions such as setting higher prices or investing more freely.
3. **Cost of production:** Firms consider how much it costs to produce goods. High production costs may limit expansion, while low costs allow firms to grow and compete better. Costs also influence how firms set prices.
4. **Government policies on business:** Government actions such as taxes and subsidies affect firm decisions. High taxes may force firms to raise prices or reduce costs, while subsidies encourage firms to expand and invest more.
5. **Location of the firm:** The place where a firm operates influences its decisions. Urban firms may expand due to better markets and services, while rural firms may remain small because of limited customers and resources.
6. **Economic conditions:** General economic conditions like inflation, unemployment, and economic growth affect firm decisions. Stable conditions encourage long-term investment, while unstable conditions make firms cautious.

- 7. Business expectations:** What a firm expects about future profits influences its plans. If a firm expects good returns, it is more likely to invest and expand. If expectations are poor, the firm may delay or avoid expansion.
- 8. Corporate Social Responsibility (CSR) and Reputation:** Firms make long-term decisions that protect the environment and support the community to build a good reputation and gain public trust, which helps them survive and grow.
- 9. Diversification and Risk Management:** Firms invest in different products or markets to reduce risk, so that poor performance in one area does not cause major losses.

Application activity 7.1



Read the situation carefully and apply what you have learned to identify and explain key business concepts. Use the information provided to answer the questions that follow clearly and logically.

In Musanze District, a business called Green Valley Dairy Ltd produces and sells milk, yogurt, and cheese to shops and supermarkets. The business owns two production sites: one where milk is collected and pasteurized, and another where yogurt and cheese are processed and packed. In the same district, there are other businesses such as Nyagatare Dairy Cooperative and Masaka Creamery, which also process and sell dairy products.

Green Valley Dairy Ltd has recently invested in better machines to reduce wastage, improve product quality, and attract more customers. The owner plans to open another branch in a neighboring district and has started advertising to increase sales. At the same time, the company ensures fair wages for workers and follows environmental guidelines to reduce pollution.

Questions:

1. Identify Green Valley Dairy Ltd as a firm, plant, or industry. Give a reason for your answer.
2. Identify the plants mentioned in the situation and explain their role.
3. Name the industry to which Green Valley Dairy Ltd and the other dairy businesses belong.
4. From the situation, identify three objectives of Green Valley Dairy Ltd and explain each briefly.
5. Match each action below to the correct objective of the firm:
 - a) Investing in better machines
 - b) Advertising products
 - c) Opening a new branch
 - d) Paying fair wages and reducing pollution
6. Explain how achieving these objectives can help the firm grow and remain competitive.

7.2 Location of firms.

Learning Activity 7.2



Carefully read through these questions and share your views with others.

- In your community, where do you usually see shops, markets, salons, or small workshops situated?
- Why do you think these businesses are built in those places and not in deserted or hidden areas?
- What things in those places help the businesses operate well?

Firms play a very important role in economic development because they produce goods and provide services to people. When starting a business, an entrepreneur must carefully choose the most suitable place where the firm will operate. Different businesses are found in different places depending on their needs and the conditions of the area. These choices are influenced by several economic, social, and physical factors. Understanding these factors helps students to appreciate why firms are found in particular areas and not others.

7.2.1 Meaning and factors influencing location of firms

i. Meaning of location of firms.

The location of a firm is the place where a business is set up and operates from. It is where goods are made or where services are provided to people. Choosing a good location is very important for a firm because it affects how easily the firm gets customers, raw materials, workers, and services like roads, water, and electricity. A firm that chooses the right location is more likely to succeed.

Table 7.2: Key Location Concepts

Concept	Explanation
Footloose Industries	These are businesses that can work from almost anywhere because they only need skilled people and good internet. They do not depend on natural resources. Example: A software company can move to another town and still operate easily.
Industrial Clusters	Industrial clusters are groups of related businesses and supporting institutions located in the same area and working together in connected economic activities. Example: A coffee-processing cluster may include coffee farmers, processing factories, transport services, packaging companies, banks, and export agencies operating in the same region.
Industrial Inertia	This is when a business stays in the same place even if it is no longer the best location, because moving is expensive. Example: Some older factories in Rwanda may remain in their locations even if better sites exist elsewhere.

Weight-Losing Industries	Industries that lose weight during production, so they locate near raw materials to reduce transport costs. Example: Coffee processing plants near coffee farms in Rwanda.
Weight-Gaining Industries	Industries that gain weight during production, so they locate near the market (customers) to reduce transport costs. Example: Soft drink bottling companies locate near urban centres where people buy the products.

ii. Factors influencing location.

The location of a firm is influenced by various factors, which may be **geographical**, **economic**, or **political**. These factors affect production costs, efficiency, and the overall success of a firm as explained below.

- **Raw Materials:** Industries that use heavy or bulky raw materials usually locate near their sources to reduce transport costs, such as CIMERWA cement factory in Rusizi, which is close to limestone deposits. These are known as **weight-losing industries**. In contrast, firms like electronics shops and assembly businesses rely on light, imported raw materials and can locate more flexibly, making them **footloose industries**.
- **Power Supply:** A reliable power supply is essential for firms that use machines and equipment. Without steady electricity or energy, production can be interrupted, leading to losses. Therefore, firms prefer areas with stable and sufficient power supply. For example, many industries are located near areas with stable electricity supply, such as Kigali City Special Economic Zone (KSEZ).
- **Transport and Accessibility:** Good transport networks make it easier for firms to receive raw materials and deliver finished goods to markets. Roads, railways, and airports reduce travel time and costs. Firms usually locate in places that are easy to access. For example, manufacturing firms in the Kigali Industrial Park benefit from good road networks and proximity to Kigali International Airport, which supports import and export activities.
- **Market Proximity:** Firms that sell perishable or bulky goods prefer to be close to customers. Being near markets reduces spoilage, lowers delivery costs, and allows goods to reach consumers quickly. This helps firms increase sales and customer satisfaction.
- **Water Supply:** Some industries require large amounts of water for processing and cleaning. These firms must locate near reliable water sources to ensure smooth production since the lack of water can slow down or stop operations. For example, BRALIRWA in Rubavu District and agro-processing firms like Inyange Industries in Masaka (Kicukiro District) are located in areas with reliable water supply, which is essential for their production processes.
- **Government Policies:** Government decisions such as tax incentives, industrial zones, and regulations influence where firms choose to locate. Supportive policies attract firms, while strict rules or high taxes may discourage them. Governments use these policies to guide industrial development for example, the Kigali Special Economic Zone (KSEZ).

- **Site Requirements (Land Availability):** Firms need enough suitable land for buildings, machinery, storage, and expansion. If land is too expensive or unavailable, firms may be unable to locate in that area. As a result, many firms choose places with ample and affordable land. For example, industrial zones such as Kigali Special Economic Zone (Gahanga) and Bugesera Industrial Area provide spacious and flat land.
- **Skill Base in the Area:** The availability of skilled and unskilled workers affects firm location. Firms prefer areas where they can easily find workers with the right skills. This helps improve productivity and reduces training costs.
- **Security:** Firms prefer to operate in safe and stable environments. Insecure areas increase the risk of theft, damage, and business disruption. Good security gives firms confidence to invest and operate successfully.
- **Existing Industrial Structure:** Firms often choose to locate near other similar businesses so they can benefit from shared skilled workers, new ideas, and technology. Being close to other firms makes it easier to learn, hire employees, and grow faster.

Application activity 7.2



Analyse the case study below and answer the questions that follow.

The Kigali Special Economic Zone (KSEZ) in Gahanga was established by the Rwandan government to attract industries and support economic development. Many firms locate there because of reliable infrastructure, security, available land, good transport, and government incentives that reduce production costs. However, the concentration of industries has also caused challenges such as traffic congestion, strain on infrastructure, and environmental pollution. Why do you think many manufacturing and processing firms choose to locate in KSEZ?

- Which facilities mentioned in the case study make KSEZ attractive to firms?
- How does concentration of firms in KSEZ help to reduce production and operating costs?
- State two advantages of having many firms concentrated in one area like KSEZ.
- What problems or dangers can result from concentrating many firms in one place?

7.3. Localization of firms.

Learning Activity 7.3



Answer the following questions

- Why is security important when deciding where to establish a firm?
- How do commercial institutions like banks and advertising companies influence a firm's location choice?

In the modern economy, firms are interdependent, and the waste products of one firm can serve as raw materials for another. Because of this relationship, many entrepreneurs choose to locate their firms close to those that supply raw materials or provide markets for their output. This proximity reduces transport costs and lowers expenses associated with waste disposal. As a result, many firms tend to concentrate in particular areas due to these advantages, leading to the localisation of firms.

7.3.1 Meaning and causes of localization of firms

i. Meaning of localization of firms

Localization of firms is the process where firms, especially those in the same or related industries, concentrate in a particular area in order to benefit from advantages such as shared resources, reduced costs, availability of skilled labour, and stronger business linkages.

Localization of firms occurs partly because of backward and forward linkages, where firms locate close to their suppliers to reduce transport costs and ensure a steady supply of raw materials (backward linkages), and near customers or processing industries that use their products to make selling easier and reduce distribution costs (forward linkages), thereby improving efficiency and strengthening production relationships.

Localized firms do not necessarily have to produce the same goods. Localisation mainly emphasizes the closeness of firms in a particular area, where what matters most is proximity and the benefits firms gain from each other. However, it tends to work best when firms are either similar or economically connected, rather than completely unrelated, since this allows them to share resources, labour, and services more effectively. For example, the Kigali Special Economic Zone (KSEZ) illustrates this well. It hosts garment, agro-processing, and light industrial firms that make different products but benefit from shared infrastructure, a common labour pool, and shared logistics, reducing costs for all firms involved despite their differing outputs.

ii. Causes of Localization of Firms.

The causes of localization of firms explain the factors that influence businesses to concentrate in particular areas rather than being evenly spread out. These causes explain why firms choose specific locations where they can reduce costs, improve efficiency, and benefit from shared resources, markets, infrastructure, and close interaction with other firms. These may include the following:

- **Availability of raw materials:** Firms often locate near sources of raw materials to reduce transport costs. In Rwanda, tea factories in Nyamagabe, Rusizi, and Nyamasheke are located close to tea plantations so that fresh tea leaves are processed quickly and cheaply.
- **Availability of labour:** Areas with abundant skilled or semi-skilled labour attract firms because labour is easier to obtain. For example, many manufacturing firms locate in Kigali due to the availability of trained labour.
- **Good transport and infrastructure:** Reliable roads, electricity, water, and communication services encourage localization. The Kigali Special Economic Zone (SEZ) attracts many firms because it provides good roads, power supply, and industrial services.

- **Closeness to the market:** Firms producing bulky or perishable goods prefer to locate near markets. Many construction-material firms in Kigali and Gikondo are close to urban markets where demand is high.
- **Interdependence among firms:** Firms may localize because the output or waste of one firm is used by another. In Kigali SEZ, manufacturing firms, packaging companies, and transport firms operate close to one another to reduce costs.
- **Government policies:** Government initiatives such as industrial parks and incentives encourage firms to locate in specific areas. Examples include Kigali SEZ, Rwamagana Industrial Park, and Muhanga Industrial Area.
- **Economies of scale and shared services:** When firms locate together, they share infrastructure and services, reducing production costs. This is seen among metal and steel firms in Kigali and Rwamagana.

7.3.2 Merits and demerits of localization of firms.

i. Merits of localization of firms.

Localization of firms brings several advantages not only to the localized firms themselves, but also to the surrounding community and the entire nation. By locating close to one another, firms reduce costs, increase efficiency, and improve production, while communities gain employment and services, and the nation benefits from economic growth and improved use of resources. These may include:

- **Lowers production costs through easy access to inputs and services:** When firms locate close to one another, they can easily access raw materials, skilled labour, transport, power supply, and business services. Shared facilities such as roads, warehouses, and repair services reduce individual firm expenses, leading to lower production costs.
- **Ensures quick supply of raw materials and disposal of waste:** Localization allows firms to obtain raw materials quickly from nearby suppliers, reducing delays and losses. At the same time, waste products can be easily disposed of or reused by neighboring firms, improving efficiency and environmental management.
- **Promotes specialization and division of labour:** Firms concentrate on specific stages of production while relying on nearby firms for other stages. This specialization improves worker skills, increases output quality, and boosts overall productivity within the industrial area.
- **Improves reliability of supply chains:** Short distances between suppliers, manufacturers, and distributors reduce transport risks and delays. This makes production more predictable and stable, helping firms meet market demand consistently.
- **Enhances firm survival through mutual support:** Localised firms support each other by sharing information, markets, and services. In times of difficulty, firms benefit from nearby suppliers, buyers, and financial service providers, increasing their chances of survival.
- **Generates steady employment opportunities:** Localised industries create many direct and indirect jobs in factories, services, transport, and trade. This reduces unemployment and provides stable incomes for local communities.

- **Stimulates growth of towns and urban centres:** As industries concentrate in one area, demand for housing, trade, and services increases. This encourages the growth of towns into major urban and industrial centres.
- **Attracts social services such as schools and hospitals:** Rising population and income levels in industrial areas encourage government and private investment in social services, improving education, health, and overall living standards.
- **Encourages development of small and medium enterprises (SMEs):** Large localised firms create opportunities for smaller businesses such as suppliers, food vendors, transport operators, and maintenance services, strengthening local entrepreneurship.
- **Raises land values and local government revenues:** Industrial concentration increases demand for land and property, raising land values. Local authorities also benefit from higher tax revenues, enabling better public service provision.
- **Builds reputation of the area and its products:** Localized industries give an area a strong industrial identity. Over time, products from the area gain recognition for quality, attracting more buyers and investors.
- **Expands industrial output and productivity:** By concentrating resources, skills, and infrastructure, localization increases industrial production and efficiency, contributing to economic growth.
- **Increases export earnings and foreign exchange:** Localised industries that produce export goods generate foreign income, strengthening the country's balance of payments.
- **Encourages efficient use of national resources:** Resources such as labour, capital, and infrastructure are used more effectively when firms operate close together, reducing wastage and duplication.
- **Supports technology transfer and skill development:** Firms learn from each other through worker mobility, training programs, and shared technologies, improving the national skill base.
- **Promotes overall economic stability and growth:** Localization attracts foreign investors, encourages knowledge sharing, and creates **backward linkages** (suppliers of raw materials) and **forward linkages** (processors, distributors, exporters). These linkages strengthen the entire economy, making it more resilient and dynamic.

ii. Demerits of localization of firms.

Localiation of firms, while beneficial in many ways, also has disadvantages that affect the firms themselves, the surrounding community, and the entire nation. Concentrating many firms in one area can lead to problems such as intense competition, congestion, environmental pollution, pressure on social services, and economic vulnerability if the dominant industry declines. These challenges may reduce efficiency and harm economic and social well-being if not well managed. These may include the following:

- **Increased competition among firms:** Many firms producing similar or related goods in one area may face stiff competition, reducing profits and causing weaker firms to close.

- **Overdependence on one industry:** Firms and areas become vulnerable if the dominant industry fails, leading to business closures and economic decline.
- **Rising cost of land and labour:** High demand for land and workers in localized areas increases rents and wages, raising production costs for firms.
- **Congestion and overcrowding:** Concentration of firms and people causes traffic congestion, housing shortages, and pressure on infrastructure.
- **Environmental pollution:** Industrial concentration increases air, water, and land pollution, affecting health and reducing the quality of the local environment.
- **Strain on social services:** Rapid population growth in localized areas puts pressure on schools, hospitals, water, and sanitation systems.
- **Uneven regional development:** Localization may lead to overdevelopment of some areas while other regions remain underdeveloped.
- **Risk of mass unemployment:** If the dominant industry collapses or relocates, many workers lose jobs at the same time, affecting the wider economy.
- **High cost of living in the local area:** Influx of workers raises prices of housing and basic services, making life expensive for residents.
- **Encourages internal migration:** People move to localized areas, causing labour shortages in rural or less developed regions.
- **Limits diversification of the economy:** Too much focus on one industrial cluster can slow the development of other sectors nationally.
- **National economic instability:** Heavy reliance on a few industrial centres makes the economy vulnerable to market changes, global price fluctuations, or disasters.

7.3.3 Survival of small- scale firms alongside large scale firms.

A smallscale firm operates with few workers, low capital, and limited output, while a largescale firm operates on a wide scale using many workers, heavy capital, and advanced machinery to produce large quantities of goods or services. Although largescale firms enjoy advantages such as economies of scale and wider markets, smallscale firms continue to survive because expansion is not always possible or desirable. Various factors enable smallscale firms to remain competitive and operate successfully alongside largescale firms. They include;

- **Ease of management:** Small-scale firms are easier to establish and manage because they involve few workers and simple procedures. Owners can make decisions quickly without complex administrative structures, making small businesses attractive to entrepreneurs.
- **Market size limitations:** In areas where demand is limited, small firms are better suited to serve small markets. They provide specialized or customized products that large firms may ignore because such markets are not profitable at large scale.
- **Government policies and support:** Governments often support small firms through subsidies, tax incentives, and grants due to their role in employment creation and efficient use of local resources. This support enhances their ability to survive alongside large firms.

- **Fear of diseconomies of scale:** Some firms avoid expansion because growth may lead to inefficiencies such as poor management, weak supervision, and communication problems, especially in service industries.
- **Capital limitations:** Many small firms lack access to sufficient capital to expand operations. Limited financing from banks or investors forces them to remain small even when opportunities for growth exist.
- **Provision of personal services:** Businesses that require personal interaction, such as hair salons or tailoring firms, remain small to maintain service quality and customer satisfaction, which large firms may find difficult to achieve.
- **Fear of amalgamation:** Some entrepreneurs prefer to remain small to avoid mergers or takeovers, allowing them to operate independently and retain ownership control.
- **Entrepreneurial preference:** Some owners deliberately choose to remain small because they value independence, flexibility, and lifestyle satisfaction more than business expansion.
- **Use of by-products:** Small firms may rely on by-products from larger industries as inputs, enabling them to survive without competing directly with large firms.
- **Tax avoidance and regulation:** Remaining small may allow some businesses to avoid heavy tax obligations and strict legal requirements, reducing operating costs.
- **Supplementary income motive:** Some people operate small businesses as a secondary source of income rather than for full profit maximization, enabling them to remain small and sustainable.
- **Geographical separation:** When firms operate in remote or underserved areas, distance from large competitors allows small firms to thrive without intense competition.

Application activity 7.3



Carefully read the scenario below and answer the questions that follow.

In Kigali, several small tailoring shops continue to operate successfully even though large clothing manufacturers have opened nearby. The owners of the small shops make quick decisions, maintain personal relationships with customers, and sometimes repair or tailor clothes for the larger firms. Some customers prefer the personalized service and convenience offered by these small businesses.

Questions:

1. Identify three reasons from the scenario that explain why the small tailoring shops continue to exist alongside the large clothing manufacturers.
2. Explain how personal service and customer relationships help small firms survive in a competitive market.
3. Suggest two ways the small shops could improve their chances of longterm survival.

7.4: Costs of the firm:

Learning Activity 7.4



Carefully read through the questions below and provide your views based on real life experiences.

- When you go to school each day, what things do you need to spend money on?
- If you start a small business-like selling fruits, what items must you buy first before you can sell anything?
- Which of these expenses are clearly linked to the product you are making or selling?
- Are there expenses that support the activity but are not directly seen in the final product? Name them.
- Why do you think some expenses increase when you produce more goods, while others remain the same?
- Can you give examples of expenses you pay at home that support many activities at once?

Every firm, whether small or large, exists to produce goods or services by using resources such as labour, raw materials, and machinery, which are not freely available but come at a price. In the process of production, firms must sacrifice something of value, and these sacrifices are known as costs. Understanding costs is essential because they influence key decisions about production, pricing, efficiency, and profitability, and help explain how firms operate and compete in the market.

7.4.1 Meaning and types of costs.

i. Meaning of costs:

Costs are the total expenses a person, household, or firm incurs in carrying out an activity or producing goods and services. They include what is spent or given up, such as money, time, and resources like labour, raw materials, and capital. For example, cooking involves spending on food and fuel as well as time, while in business costs include wages, rent, electricity, transport, and raw materials. Costs represent payments to factors of production and may be explicit (actual cash payments) or implicit (opportunity costs). Understanding costs helps firms operate efficiently and make sound production and profit decisions.

ii. Types of costs.

Costs are divided into two:

- Implicit costs:** Implicit costs are costs that do not involve direct cash payments by a firm and therefore are not recorded in its accounting books or included in accounting profit. They are often called **hidden or social costs** because they are difficult to measure in monetary terms but represent real sacrifices made by the firm or society, such as opportunity cost, time wasted through inefficiency, noise pollution, and the value of the owner's unpaid labour.

b) Explicit costs: Explicit costs are costs that involve actual and direct payments made by a firm for the use of inputs in the production process. These costs represent the real expenditure of the firm and can be easily identified, measured, and recorded in the firm's financial accounts. They are included when calculating the firm's profits and cover payments for items such as raw materials, labour wages, transport, and energy. Explicit costs include both fixed costs, which remain constant regardless of output, and variable costs, which change with the level of production.

7.4.2 Short run costs of the firm

Short run is a period in which the firm cannot change its plant, equipment and the scale of operations. To meet increased demand, it can only raise output by hiring more labour and raw materials or asking the existing labour force to work overtime. The short-run total costs are divided into:

a) Total Fixed Costs (TFC).

b) Total Variable costs (TVC).

a) Fixed Costs. (FC): These are costs that do not change as output changes. They include payment for rent, interest on borrowed money, insurance charges, wages and salaries of permanent staff and maintenance expenditure. They are also called **overhead costs, supplementary costs and indirect costs**. The total fixed cost curve is therefore horizontal and parallel to the X-axis.

Total Fixed Costs = Total Costs – Total Variable Costs.

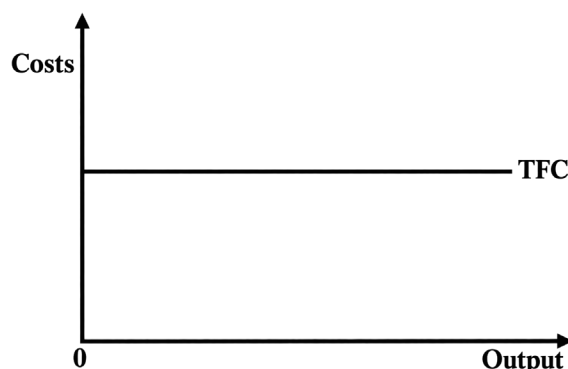


Fig 7.1: Total fixed cost curve

b) Variable costs (VC)

Also known as direct costs, prime costs or operating costs, Variable costs are costs that change directly with the level of output, meaning they increase when output increases and decrease when output falls; at zero output, variable costs are also zero. Examples include expenses on labour and raw materials, taxes, power and advertising which rise as more units are produced. Because of this, the total variable cost (TVC) curve begins at the origin. The TVC curve has an inverted-S shape due to changes in productive efficiency over the course of production: in the early stages, better use of fixed factors with additional variable inputs leads to increased output at a diminishing cost, causing TVC to rise at a decreasing rate and appear concave; however, as production expands further, congestion of variable inputs with fixed factors reduces efficiency, making each additional unit more costly, so TVC rises at an increasing rate and becomes convex, resulting in the overall inverted-S shape.

$$\text{Total Variable Costs} = \text{Total Costs} - \text{Total Fixed Costs}$$

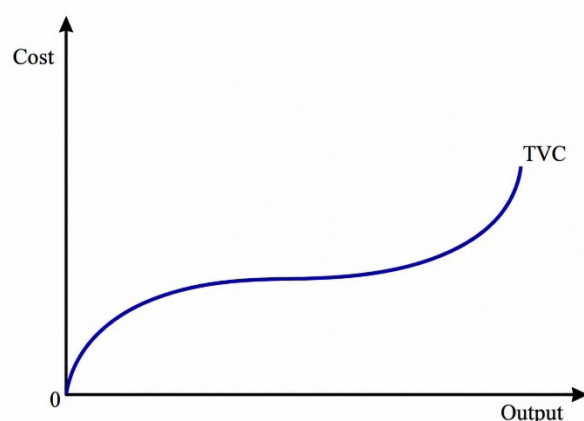


Fig 7.2: Total Variable Cost curve

c) The total cost (TC)

This is the total sum of the total fixed costs and the total variable costs in the business.

$$\text{TC} = \text{TFC} + \text{TVC}$$

When output is equal to zero, total cost is equal to fixed costs since variable costs will be zero. When production begins to increase, total costs will continue to rise as variable costs increase since variable costs must increase as output expands.

7.4.3 Calculating and illustrating TFC, TVC and TC.

Calculations and illustrations of TC, TFC, and TVC are used to show the relationship between them through schedules and curves as below.

Table 7.3 A hypothetical Total Cost (TC), Variable Cost (VC) and Fixed Cost (FC) Schedule

Output (Units)	Fixed Cost (FC)	Variable Cost (VC)	Total Cost (TC)
0	100	0	100
1	100	40	140
2	100	70	170
3	100	95	195
4	100	130	230
5	100	180	280
6	100	245	345

The table shows how a firm's Total Cost (**TC**) changes as output increases, based on its Fixed Cost (**FC**) and Variable Cost (**VC**). The fixed cost remains constant (100) at all levels of output because it does not depend on production, while variable cost increases as more units are produced since more inputs are required. Total cost rises with output because it is the sum of fixed and variable costs (**TC = FC + VC**), and the increasing rate of TC reflects the rise in variable costs as production expands.

From the cost schedule, we can derive cost curves that graphically represent the behaviour of fixed cost, variable cost, and total cost as output increases. These curves make it easier to analyse how a firm's costs respond to changes in production.

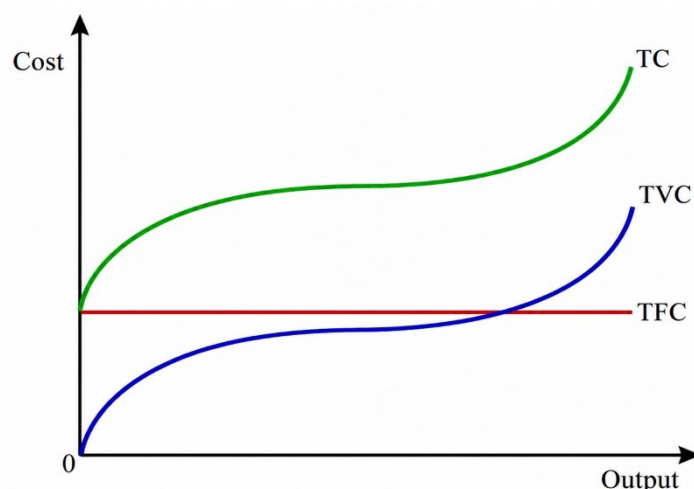


Fig 7.3. Total cost, Variable and fixed cost curves

The total cost curve cuts the vertical axis at a point above the origin and rises continuously from left to right. This is because even when no output is produced the firm has to incur fixed costs. The TVC curve starts from the origin (0) because when output is zero, the variable costs are also zero. They increase as output increases.

The relationship between total cost (TC), total variable cost (TVC), and total fixed cost (TFC) shows that TFC remains constant at all levels of output, represented by a straight line parallel to the output axis. In contrast, both TC and TVC are upward sloping, indicating that they increase as output rises. Additionally, TVC and TC move in parallel and never intersect, since total cost is always equal to the sum of total variable cost and total fixed cost.

Exercise: 7.1

Use the information in the table below to answer the questions that follow.

Output (Bottles of Juice) (Q)	Total Cost (TC) (FRW)
0	200,000
50	280,000
100	340,000
150	410,000
200	500,000

1. Calculate the Total Fixed Cost (TFC) and Total Variable Cost (TVC) for each level of output.
2. Illustrate the information on graphs.

7.4.4: Variation of costs in the short run.

Learning Activity 7.4.4



Analyse the information in the table below and answer the questions that follow:

Loaves baked per day	Rent & oven cost (francs)	Ingredient cost (francs)	Total cost (francs)
0	50,000	0	50,000
10	50,000	10,000	60,000
20	50,000	20,000	70,000
30	50,000	30,000	80,000

Questions:

- When no bread is baked, why does the baker still have to pay some cost?
- How does the ingredient cost change as more loaves are baked? Why?
- If the baker produces 10 loaves, how much does one loaf cost on average?
- If the baker produces 30 loaves, how much is the cost on rent per loaf?
- As production increases from 10 to 20 loaves, does the cost per loaf rise or fall? Why?
- How much extra cost is added when production increases from 10 to 20 loaves?

In short-run analysis, average costs are more important than total costs because the cost of producing each unit of output is not the same, yet all units must be sold at the same price. For this reason, a firm needs to know the cost per unit of output, known as the average cost, in order to make effective production and pricing decisions. In the short run, a firm's average costs consist of **average fixed cost (AFC)**, **average variable cost (AVC)**, **average total cost (ATC)**, and **marginal cost (MC)**, all of which help explain how costs behave as output changes.

a) The Average Fixed Cost (AFC)

This is the total fixed cost at each level of output divided by number of units produced. That is:

$$AFC = \frac{TFC}{Q}$$

$$AFC = AC - AVC$$

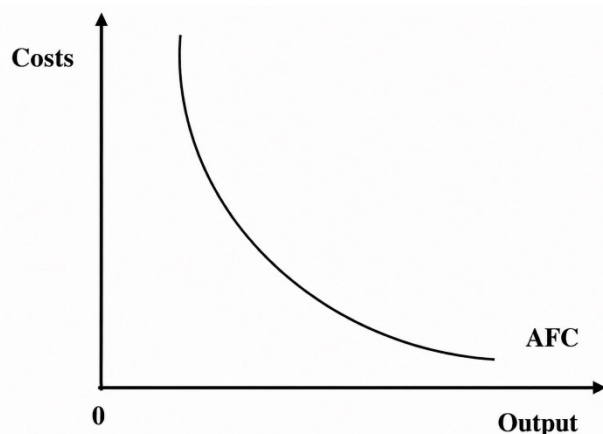


Fig. 7.4 Average Fixed Cost (AFC) curve.

Average fixed cost (AFC) falls as output increases. This is because fixed costs stay the same, no matter how much is produced. When a fixed cost is divided among more units of output, the cost per unit becomes smaller. As production increases, the fixed cost is shared across many units, so average fixed cost keeps decreasing. This is why the AFC curve slopes downward. However, the AFC curve never touches the output (quantity) axis. If it did, average fixed cost would be zero, which would mean fixed costs are zero. Since fixed costs always exist, this cannot happen.

b) The Average Variable Costs (AVC)

Average variable cost (AVC) refers to the cost of producing one unit of output using only variable costs, such as raw materials, labour, and fuel, which change with the level of production; it is calculated by dividing total variable cost by the quantity of output and shows how variable costs per unit behave as output changes.

$$AVC = \frac{TVC}{Q}$$

$$AVC = AC - AFC$$

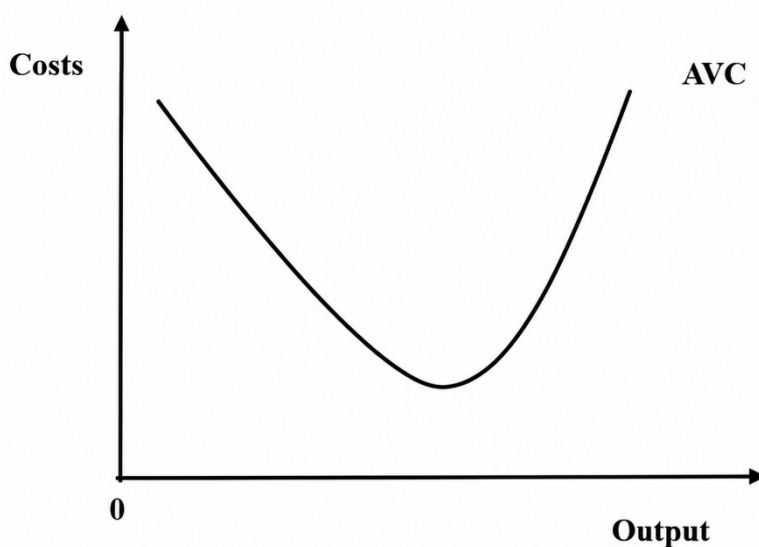


Fig 7.5 The Average Variable Cost

The AVC curve is U-shaped because of the law of diminishing marginal returns in the short run. At first, when output increases, more units of the variable factor (such as labour) are combined with fixed factors (such as land or machines), leading to better efficiency and lower average variable cost, so the AVC falls. However, after a certain level of output, the fixed factors become overused, causing congestion and inefficiency, which means each additional unit of output costs more to produce. As a result, average variable cost begins to rise, giving the AVC curve its U- shape.

c) The Average Total Costs (ATC/AC)

Average total cost (ATC) is the cost of producing one unit of output when both fixed costs and variable costs are included. It is calculated by dividing total cost by the level of output, showing how much, on average, it costs a firm to produce each unit of output.

$$ATC = \frac{TC}{Q} = \frac{TFC + TVC}{Q}$$

$$ATC = \frac{TC}{Q}$$

$$ATC = AFC + AVC$$

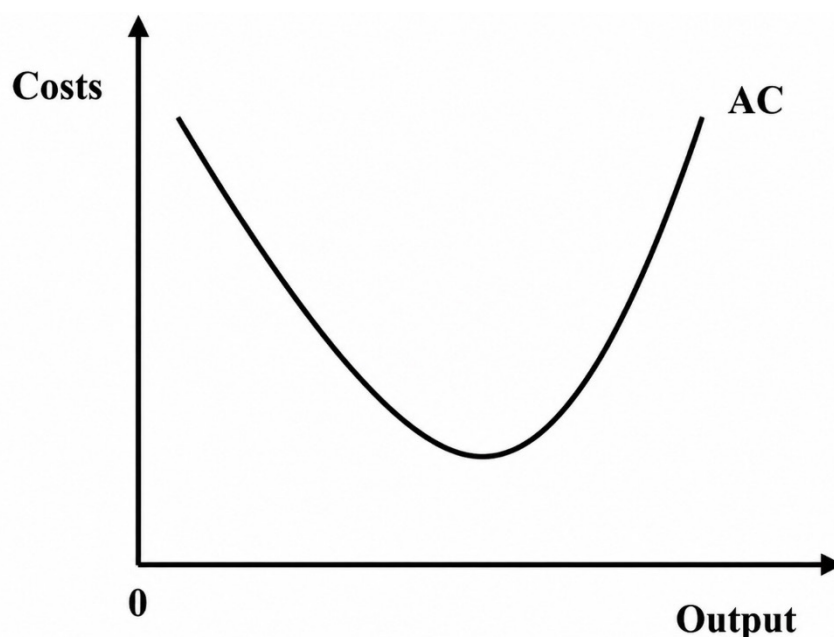


Fig: 7.6 The Average Total Cost Curve

d) Marginal Cost (MC)

Marginal cost (MC) is an important idea used to decide how much a firm should produce. It refers to the extra cost incurred when a firm produces one additional unit of output. In other words, marginal cost shows how much total cost increases when output is increased by one unit. Firms use marginal cost to determine the most suitable level of production, because comparing marginal cost with revenue helps them decide whether producing more units is worthwhile.

$$MC = \frac{\Delta TC}{\Delta Q}$$

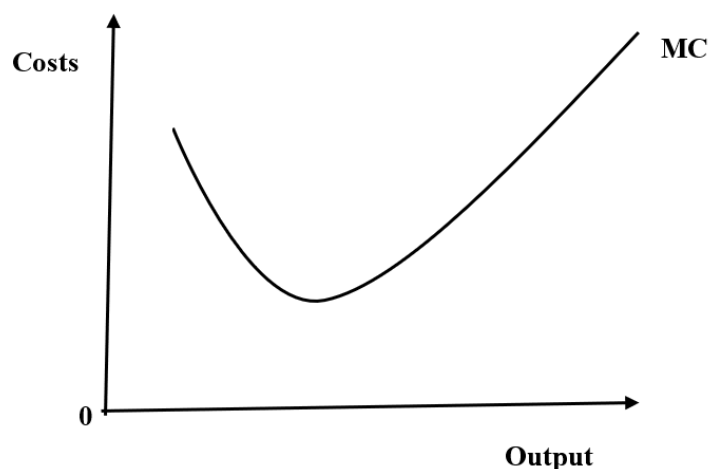


Fig. 7.7 Marginal Cost curve

7.4.5 Relationship between short run costs curves.

The relationship between short-run cost curves is shown by how AFC, AVC, ATC, and MC change with output in a unit cost schedule. In the short run, fixed and variable costs cause AFC to continuously fall, AVC and ATC to form U-shaped curves, and MC to first fall and then rise. When these costs are illustrated using curves, they clearly show how different cost measures are interconnected and how they guide a firm's short-run production decisions.

a) Unit cost schedule

Table 7.4: Hypothetical Unit Cost Schedule

Output (Units)	TFC	TVC	TC	AFC	AVC	ATC	MC
0	50	0	50	–	–	–	–
1	50	20	70	50.0	20.0	70.0	20
2	50	35	85	25.0	17.5	42.5	15
3	50	45	95	16.7	15.0	31.7	10
4	50	60	110	12.5	15.0	27.5	15
5	50	85	135	10.0	17.0	27.0	25
6	50	115	165	8.3	19.2	27.5	30
7	50	150	200	7.1	21.4	28.6	35
8	50	190	240	6.3	23.8	30.0	40

In the hypothetical unit cost schedule, total fixed cost (TFC) remains constant at all levels of output, while total variable cost (TVC) first rises slowly and then more rapidly as output increases, reflecting increasing efficiency followed by diminishing marginal returns; as a result, average fixed cost (AFC) continuously falls because the same fixed cost is spread over more units, average variable cost (AVC) first falls and then rises giving it a Ushape, average total cost (ATC) follows a similar Ushape because it is the sum of AFC and AVC, and marginal cost (MC) first falls then rises as additional units become more expensive to produce.

At zero output, AC (or ATC), AFC, AVC, and MC are shown as dashes because average and marginal costs cannot be calculated when no units are produced: averages require division by output, which would involve dividing by zero, and marginal cost cannot be defined since no additional unit is being produced at zero output.

b) Unit cost curves

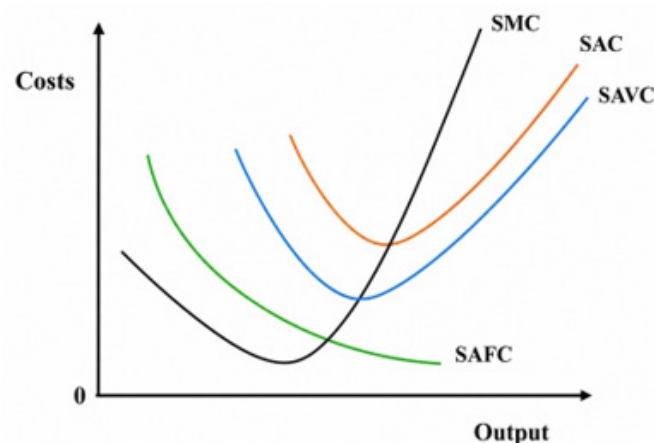


Fig. 7.8 Short run unit cost curves

From the above curve, it can be observed that;

- The short run Average Fixed Cost (**SAFC**) curve continually declines as output increases because fixed costs are spread over more units of production. It is asymptotic to both axes, meaning it approaches but never touches either the cost axis or the output axis. AFC can never become zero since fixed costs always exist, and it cannot start at zero output because average costs cannot be calculated when output is zero.
- The short run Average Variable Cost (**SAVC**) curve initially declines due to better utilization of variable factors, leading to increased efficiency, but after some level of output it rises as diminishing marginal returns set in.
- The Average Total Cost (**ATC/AC**) curve also first falls as declining AFC and AVC reduce average cost, then rises when the increase in AVC becomes stronger than the fall in AFC.
- The short run Marginal Cost (**SMC**) curve intersects the ATC curve at its minimum level, showing that when marginal cost equals average total cost, ATC is at its lowest.

The relationship between MC and AC (ATC)

There is a direct relationship between the ATC and the MC

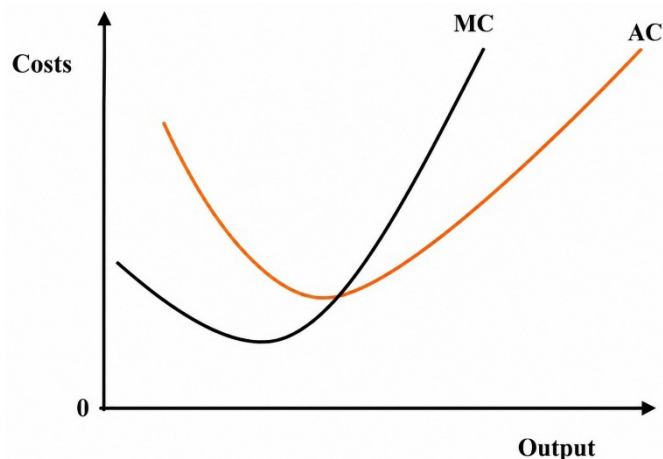


Fig. 7.9 Relationship between MC and AC curves

From the above curves, it can be observed that;

- Marginal cost (MC) and average total cost (AC or ATC) are closely related and both are **U-shaped** due to **the law of variable proportions**. When AC is falling, MC is less than AC, because the cost of producing an extra unit is lower than the average cost of all units, pulling AC downward; this happens since changes in MC affect only one unit, while AC spreads the effect over all units.
- When AC reaches its minimum, MC equals AC, and MC cuts the AC curve from below at this point, which represents the firm's **optimum level** of output, also called optimum **capacity of the firm**, where average cost is lowest and resources are used most efficiently. Producing below this level shows **underutilization** of resources (excess capacity), while producing beyond the optimum leads to inefficient use of fixed factors (overcapacity) and rising costs.
- When AC is rising, MC is greater than AC, pushing it upward. Therefore, a rational firm operates at the output level where $MC = AC$, since this ensures production at optimum capacity and minimum average cost.

7.4.6 The long run cost curves

In the long run, all factors of production are variable, meaning a firm can change its use of labour, capital, and other inputs, and firms are free to enter or leave an industry. As a result, a firm can choose any level of output by adjusting inputs and bearing the full cost of production. In this period, the law of diminishing marginal returns does not apply because no factor is fixed. Instead, long-run costs are influenced by returns to scale, where costs fall due to economies of scale when output expands efficiently, or rise due to diseconomies of scale when expansion becomes inefficient.

a) Long run Average cost curves

The long-run average cost (LAC) curve shows the lowest possible average cost at which a firm can produce different levels of output when it is free to choose among all possible plant sizes. It is derived from a series of short-run average cost (SAC) curves, each representing a plant of a particular size suitable for a specific range of output. In the long run, the firm moves from one plant to another to produce output at the lowest average cost and will operate each plant only up to the level where its SAC is falling. Therefore, the LAC represents the most efficient combinations of output and cost achievable by selecting the best short-run production option at each output level.

b) Deriving the long run average cost curve (LAC)

In the long run, a firm can increase output by changing all inputs, including those that were fixed in the short run, which allows it to expand the scale of production. Each scale of production is represented by a short-run average cost curve (SAC), with each SAC corresponding to a particular plant size that is efficient over a certain range of output (optimum level). Since a plant size is fixed only in the short run, the firm can choose among many possible plants in the long run and will select the one that produces a given level of output at the **lowest average cost**.

Graphically, this choice gives rise to the long-run average cost (LAC) curve, which is drawn as a smooth curve tangent to the minimum points of the various SAC curves, making it an **envelope or planning curve**. As the firm expands its scale, minimum average cost initially falls due to **economies of scale**, but this decline does not continue indefinitely; beyond a certain size, **diseconomies of scale** set in, causing average costs to rise. As a result, the LAC curve is **U-shaped**, reflecting falling costs at low levels of output, a minimum efficient scale, and rising costs at higher levels of production as shown in the illustration below.

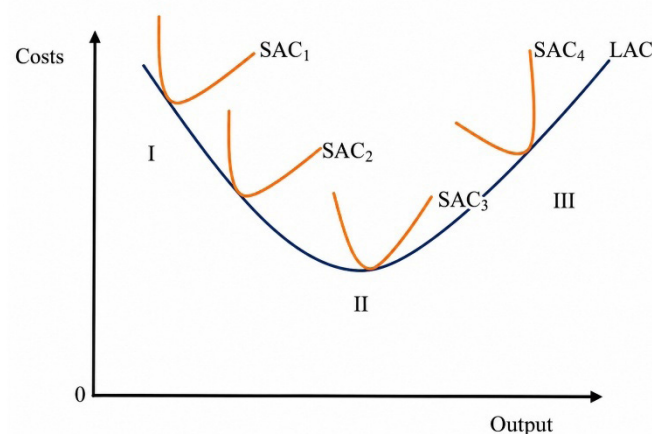


Fig 7. 10 Long run average cost curve

The illustration of the longrun average cost (LAC) curve shows how a firm's average cost changes as it expands its scale of production over time. The LAC curve, which is flatter and Ushaped, is derived from several shortrun average cost (SAC) curves and therefore serves as the firm's planning curve, indicating the most efficient plant size for each output level. The minimum point of the LAC, such as at **SAC₃** in the diagram, represents the firm's optimum size or optimum capacity, where it produces the highest level of output at the lowest average cost; up to this level, it is logical for the firm to operate using the plant represented by **SAC₃**.

Before this point, the firm is too small and can reduce average costs by expanding production due to economies of scale (*increasing returns to scale*). Beyond the optimum point, operating at a larger plant size such as **SAC₄** for higher outputs leads to higher costs because the firm becomes too large and faces diseconomies of scale (*diminishing returns to scale*). Thus, the LAC curve can be divided into three parts: falling average costs (I), constant average costs (II), and rising average costs (III), reflecting **increasing, constant, and decreasing returns to scale** respectively.

Application activity 7.4



Read through carefully and answer these questions

- i. What do you understand the term cost as used in production?
- ii. State two main differences between the short run and the long run costs?
- iii. What is the relationship between the long run Average Cost curve and the Short Run Average Cost curves.
- iv. Why are all factors of production variable in the long run?
- v. What is meant by the optimum plant size of a firm?
- vi. Explain why the law of diminishing marginal returns does not apply in the long run.
- vii. Analyse why a firm will operate each plant only up to the level where its Short Run Average Cost curve is falling in the long run.
- viii. Explain why the cost curves are U-shaped both in the short run and long run.
- ix. Analyse the significance of the minimum point of the long run Average Cost curve to a firm's production decisions.
- x. Distinguish between increasing returns to scale, constant returns to scale, and decreasing returns to scale, as shown by different sections of the Long run Average Cost curve.
- xi. Explain why the long run Average Cost curve is also referred to as a planning curve.

7.5. Scale of production.

Learning Activity 7.5



Analyse the scenario below and answer the questions that follow:

A small farmer in Kirehe grows 10 tomato plants in his backyard to sell to neighbors. As demand for his tomatoes increases, he expands production by planting 50 tomato plants in a larger garden.

- a) What difference is shown in the farmer's level of production when he increases from 10 tomato plants to 50 tomato plants?
- b) Why did the farmer need to move from his backyard to a bigger garden as demand for tomatoes increased?
- c) How does producing tomatoes on a larger scale help the farmer meet the needs of more customers compared to producing on a small scale?

7.5.1 Meaning of Scale of production and growth of firms.

Every firm aims to achieve its set objectives, one of which is growth and market expansion. Firms seek to increase their size and serve more markets both within and outside the country in order to improve sales and competitiveness. While some firms can expand independently, others lack enough resources and therefore join with other firms so that together they can grow, compete effectively, and withstand competition from larger firms in the market. A firm's aim to grow and compete in wider markets requires expanding output, which directly involves increasing its scale of production.

i. Meaning of scale of production

Scale of production refers to the size or level of output a firm produces by varying all its factors of production such as labour, capital, and land in the long run. It explains how production expands or contracts and how this affects costs over time, with average costs falling initially due to economies of scale, reaching a minimum at the optimum scale, and rising beyond this point because of diseconomies of scale like inefficiency and management challenges.

In economics, scale of production is explained through returns to scale: **increasing returns to scale**, where output rises more than proportionately as inputs increase; **constant returns to scale**, where output grows in the same proportion as inputs; and **decreasing returns to scale**, where output increases less than proportionately, causing higher average costs. Firms adjust their scale in practice by expanding or reducing all inputs for example, by investing in new machinery, hiring more labour, adopting new technology, merging with other firms, or downsizing operations, to move toward the optimum scale of production where average costs are lowest. Choosing the right scale of production helps a firm maximize profit, reduce costs, and achieve sustainable growth.

ii. Growth of firms

Growth of a firm refers to the steady increase in its productive capacity, shown by a sustained rise in the real output of goods and services over time. Firms grow in order to achieve their goals and objectives such as increasing sales, maximizing profits, and expanding market share, and this growth can occur either through **internal expansion** (*natural growth*), where the firm grows on its own, or through **external growth**, such as *mergers* with other firms.

a) Natural growth:

Natural growth refers to the expansion of a firm that occurs through an increase in output, which raises profits and gradually leads to a larger size of operation. This type of growth depends on a firm retaining part of its profits to buy new assets, improve technology, and expand production, and as the firm's resources increase over time, it gains stronger financial capacity and collateral that allows it to borrow funds to finance further expansion projects.

b) External growth.

External growth of firms occurs when a firm expands by combining with or taking over other firms rather than growing on its own. This happens through **mergers or acquisitions**, where two or more firms come together to form a larger business. External growth is usually faster than natural growth because it uses resources and assets from other firms.

7.5.2 Mergers.

i. Meaning of mergers.

A merger is an agreement in which two or more firms combine to form a single company, sharing ownership, management, and operations. After a merger, the firms involved cease to exist as separate entities and operate together as one new legal organization, usually to increase size, reduce competition, or achieve economies of scale. In contrast, an acquisition occurs when one firm (usually larger) takes over another firm (usually smaller), with the acquiring firm retaining control and identity while the acquired firm loses its independence.

ii. Types of mergers.

Mergers are of the following types:

- **A horizontal merger:** This occurs when firms operating in **the same industry and producing similar products** combine in order to reduce competition, increase market share, and enjoy economies of scale. These mergers usually involve former competitors and aim at creating a larger and more efficient firm. Exemple, if Bralirwa merged with another local beverage-producing company, since both would be operating at the same stage of production in the same industry, allowing the combined firm to strengthen its market position and reduce production costs.
- **A vertical merger** occurs when firms operating at **different stages of the same industry's supply chain** combine in order to reduce costs, improve efficiency, and secure supplies or markets. It may be **backward** (with suppliers) or **forward** (with distributors), to reduce costs and secure supply or markets. By integrating suppliers, manufacturers, or distributors, firms can achieve better coordination and economies of scale. Exemple, if Inyange Industries merged with a milk-collecting cooperative or a packaging company, allowing it to control raw material supply or distribution more effectively and reduce production and operating costs.
- **A conglomerate merger** occurs when two or more firms engaged in **unrelated business activities** combine, even though they operate in different industries or markets. Such mergers are usually aimed at diversifying activities, spreading risk, and creating synergy that can improve performance and shareholder value. For exemple, if Bank of Kigali merges with or acquires a construction or manufacturing company, where the businesses are completely different but combine to strengthen financial capacity and market stability.
- **A congeneric/concentric (product-extension) merger:** This occurs when firms operating in the **same or closely related market** combine in order to expand their range of products and reach more consumers, using similar technology, production processes, or marketing channels. Such mergers allow firms to add new product lines, increase market share, and benefit from shared expertise. For exemple, in the Rwandan context, if Inyange Industries were to merge with a local juice-processing company, the combined firm would offer dairy products alongside fruit juices, enabling it to serve a wider market using related production and distribution systems.

iii. Reasons why firms merge/amalgamate.

- **Cooperation (Synergy):** Companies merge to achieve synergy, which means that the combined firm is stronger and more valuable than the separate firms working alone. By cooperating, firms can share skills, technology, ideas, and resources, leading to better performance and higher profits than before.
- **Diversification:** Some companies merge or acquire others to diversify their activities and reduce risk. By operating in different industries or producing complementary products, firms reduce their dependence on one market and protect themselves from losses if one sector performs poorly.
- **Survival:** Companies may merge in order to survive difficult economic conditions, such as financial crises or intense competition. Small or struggling firms often join stronger firms to avoid failure, closure, or bankruptcy.
- **Economies of Scale:** Mergers help companies enjoy economies of scale, where producing on a larger scale reduces the average cost of production. By sharing factories, machines, suppliers, research facilities, and distribution networks, merged firms become more efficient and increase profits.
- **Elimination of competition:** Firms merge to reduce or eliminate competition between them. By joining together, rival firms no longer compete for customers, reduce spending on advertising, gain a larger market share, and may offer goods at lower prices.
- **Operating economies:** Companies may merge to avoid duplication of services such as accounting, marketing, and purchasing. This improves efficiency and allows better management, especially when small inefficient firms combine with larger and better-managed firms.
- **Growth purposes:** Mergers help firms grow faster and more easily than through internal expansion. Instead of slowly building new facilities, firms can expand quickly by acquiring existing businesses, saving time, costs, and risks.
- **Utilisation of tax shields:** Some companies merge to benefit from tax advantages. A loss-making firm can merge with a profit-making firm so that previous losses can be offset against future profits, reducing the total tax paid.
- **Increase in value:** One reason for merging is to increase the overall value of the business. The value of the merged firm is often greater than the total value of the firms operating separately because of better efficiency, stronger market power, and higher profits.
- **Better financial planning:** Merged firms are able to plan and use finances more effectively. Profits from one part of the business can support another part, improve cash flow and make long-term investment and expansion easier.
- **Economic Necessity;** Sometimes mergers happen due to economic or government pressure, especially to rescue sick firms. Governments may encourage or force mergers so as to protect jobs, improve efficiency, and ensure better use of national resources.

iv. Advantages, disadvantages and limitations of mergers.

a) Advantages of mergers.

- Mergers offer several benefits that explain why firms choose to combine their operations into a single business. By pooling resources, skills, and markets, merged firms can operate more efficiently, grow faster, and compete more effectively. The main advantages of mergers include:
- **Economies of scale:** One major reason why companies merge is to achieve economies of scale. By combining operations and resources, firms are able to produce on a larger scale, which reduces the average cost of production. Fixed costs such as machinery, buildings, and administration are spread over more output, making production cheaper and more efficient.
- **Increased market share:** Companies merge in order to increase their market share. A larger firm controls a greater portion of the market, attracts more customers, and strengthens its position against competitors. Increased market share often leads to higher sales, stronger influence over prices, and improved brand recognition.
- **Access to more resources:** Mergers enable firms to combine resources such as labour, capital, technology, machinery, and managerial skills. This improves production efficiency and allows firms to invest in better equipment, new technology, and improved marketing strategies, thereby enhancing competitiveness.
- **Faster growth:** Another reason for mergers is to achieve faster growth. Internal growth may be slow and costly, whereas mergers allow firms to expand quickly by acquiring existing capacity, skilled workers, and established markets. This helps firms respond quickly to market opportunities.
- **Diversification of products and services:** Firms merge to diversify their product lines or services. By offering a wider range of products, firms reduce their dependence on a single product or market. Diversification lowers business risks and helps firms remain stable when demand for one product declines.
- **Improved financial strength:** Mergers strengthen the financial position of firms by combining assets, profits, and capital. With stronger finances, firms can invest more in production, research, and advertising, and they also find it easier to obtain loans from financial institutions.
- **Reduction of competition:** Companies may merge to reduce or eliminate competition, especially in industries with many rival firms. Reduced competition allows firms to stabilize prices, increase profits, reduce advertising costs, and focus on improving efficiency and product quality.
- **Better utilization of resources:** Through mergers, firms can use resources more efficiently by avoiding duplication of facilities and services. Shared use of equipment, labour, and administrative departments reduces waste, lowers costs, and increases productivity.

- **Enhanced brand recognition:** A merger often leads to a stronger brand name by combining customer bases and reputations. A well-known brand increases consumer trust, improves marketing effectiveness, encourages customer loyalty, and may allow firms to charge better prices.
- **Ability to enter new markets:** Finally, mergers help firms enter new geographical or product markets that would be difficult to reach individually. Accessing new markets increases sales, reduces reliance on a single location, and promotes long-term growth and stability.

In conclusion, companies merge for many reasons, including to reduce costs, grow faster, strengthen market position, and improve efficiency. When well-planned and managed, mergers can greatly enhance a firm's performance and long-term success.

b) Disadvantages of mergers.

Although mergers bring many benefits, they also have drawbacks that may affect the merged firms, employees, consumers, and the wider economy. When firms combine, challenges related to management, competition, and costs can arise. The main disadvantages of mergers include:

- **Job losses:** One major disadvantage of mergers is loss of employment. When firms merge, duplicate roles such as accountants, drivers, or managers may arise, forcing the firm to lay off workers to cut costs. This leads to unemployment, low worker morale, and may damage the firm's image in society.
- **High costs of merging:** Mergers involve high initial costs such as legal fees, consultancy fees, restructuring expenses, and administrative costs. These costs may reduce short-term profits and, in some cases, outweigh the expected benefits of the merger.
- **Management problems:** Mergers may create management conflicts due to different leadership styles, policies, and organizational structures. Poor coordination and disagreements among managers can slow decision-making and reduce efficiency in the merged firm.
- **Loss of control:** After a merger, original owners may experience a loss of control over decision-making. Business decisions must be shared among partners or shareholders, which may cause dissatisfaction, delays, and conflicts over company goals.
- **Cultural differences:** Different firms often have different work cultures, and merging them can be difficult. Differences in working styles, rules, and values may cause resistance among workers, reduce productivity, and lower employee satisfaction.
- **Customer dissatisfaction:** A merger may lead to changes in product quality, pricing, or services, which may dissatisfy loyal customers. Standardization of products after a merger can result in loss of customers and reduced sales in the short run.
- **Risk of monopoly:** Large mergers may lead to monopoly power, reducing competition in the market. This can attract government regulation, harm consumers through higher prices, and damage the firm's public image.

- **Integration problems:** Merging systems, operations, and departments can be complex and challenging. Poor integration of accounting systems, supply chains, or production processes may cause inefficiency, confusion among workers, and increased operational costs.
- **Loss of identity:** Small or unique firms may lose their brand identity after merging with larger firms. Customers and employees may feel disconnected from the new firm, leading to reduced loyalty and weaker brand recognition.
- **Uncertain outcomes:** Mergers are often risky, and success is not guaranteed. Some mergers fail to achieve expected benefits due to conflicts, poor planning, or market changes, making mergers less predictable than internal growth.

In conclusion, although mergers can promote growth and efficiency, they also pose several disadvantages such as job losses, management challenges, and uncertainty. Firms should therefore carefully evaluate potential risks before deciding to merge.

c) Limitations of merging of mergers

Although mergers help firms to grow and become more competitive, not all firms are able or willing to merge. Several economic, financial, managerial, and market factors may limit or prevent firms from entering into merger agreements.

- **High Cost of Merging:** One important factor limiting mergers is the high cost involved. Mergers require legal services, consultants, restructuring expenses, and administrative costs, which many small firms may fail to afford, making mergers financially difficult.
- **Government Regulations:** Government laws and regulations may restrict mergers, especially those likely to create monopolies or reduce competition. Authorities may block mergers to protect consumers from high prices and ensure fair competition in the market.
- **Cultural Differences:** Differences in organizational culture, management style, and company values can limit mergers. Employees may find it hard to adapt to new working systems, leading to resistance, conflicts, and slow integration.
- **Loss of Control:** Fear of losing decision-making power discourages some firms from merging. Owners of small firms may be unwilling to share or give up control of their businesses, especially when merging with larger firms.
- **Employee Resistance:** Employees may resist mergers due to fear of job losses or role changes. Low morale, uncertainty, and lack of trust can reduce productivity and make the merger process difficult to implement successfully.
- **Financial Problems:** If one firm has debts or poor financial performance, stronger firms may avoid merging to prevent inheriting financial risks. Weak financial health reduces trust between merging partners.
- **Integration Challenges:** Merging firms face challenges in combining systems, processes, and operations. Poor planning during integration may lead to inefficiency, confusion, and increased operational costs, discouraging mergers.

- **Customer Reactions:** Possible negative reactions from customers may limit mergers. Changes in products, prices, or services after a merger can lead to customer dissatisfaction and reduced sales, making firms hesitant to merge.
- **Differences in Goals:** Firms may have different objectives, such as growth, profit maximization, or expansion, which may conflict. These differences in goals can limit cooperation and reduce the success of a merger.
- **Market Conditions:** Unfavourable market conditions, such as low demand, high competition, or economic instability, can make mergers risky. Firms may delay or avoid merging until market conditions improve.

In conclusion, although mergers can offer many benefits, factors such as high costs, regulations, financial risks, cultural differences, and market conditions may limit firms from merging. Firms must therefore carefully evaluate these limitations before deciding to merge.

7.5.3 Economies and diseconomies of scale.

As firms operate over time, they often aim to grow by increasing production, employing more workers, and expanding their markets in order to raise sales and profits. In the early stages of growth, expanding production can make a firm more efficient and reduce the cost of producing each unit, but as the firm grows larger, managing workers, resources, and operations can become more difficult and costly. These cost advantages that arise from growth are known as **economies of scale**, while the rising costs that may occur when a firm becomes too large are called **diseconomies of scale**.

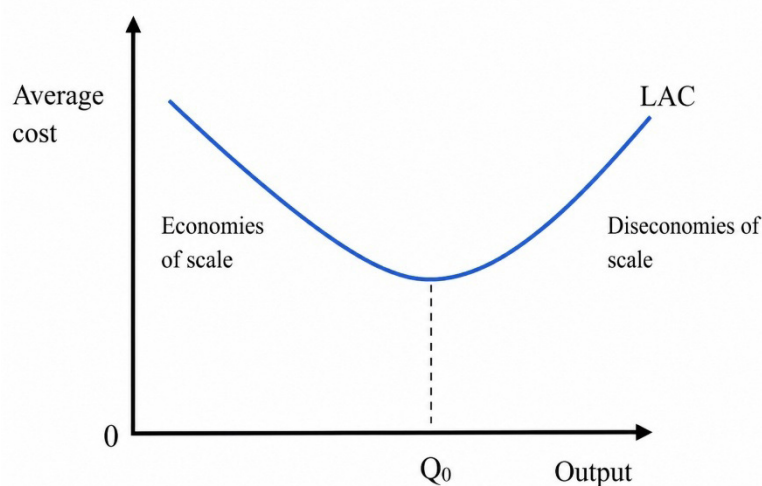


Fig. 7.11: Economies and diseconomies of scale curve

i. Economies of scale

Economies of scale are cost advantages gained when a firm expands production, causing the average cost per unit to fall. This happens mainly because some costs, such as management and marketing, are fixed and do not change with output; as production increases, these fixed costs are spread over more units. Since average cost equals total cost divided by quantity of output, producing more units lowers the average cost per unit. Economies of scale can be categorized as **internal or external economies of scale**.

a) Internal economies.

Internal economies of scale refer to the cost advantages enjoyed by a single firm when it increases its scale of production in the long run. As a firm grows, it becomes better organized, produces more efficiently, and spreads its costs over a larger output, leading to lower average costs. These benefits arise from the firm's own expansion and are not shared with other firms in the industry. Internal economies of scale improve long-run productive efficiency, allowing the firm to charge lower prices, earn higher profits, and gain a competitive advantage; they are shown by a falling long-run average total cost (LRAC) curve. It consists of the following:

- **Technical economies:** Large-scale firms enjoy technical economies because they can use modern and efficient machinery, which reduces the need for labour and lowers production costs. They also use raw materials more efficiently, including recycling waste to produce by-products, leading to lower unit costs.
- **Financial economies:** Large firms find it easier and cheaper to raise finance because banks and other financial institutions view them as less risky. As a result, they can access loans and credit at lower interest rates than small firms.
- **Marketing economies:** Large-scale firms benefit from marketing economies by buying raw materials in bulk at discounted prices and spreading advertising and distribution costs over large output. This allows them to promote products more widely and sell at lower average cost.
- **Managerial economies:** As firms grow, they can employ specialist and skilled managers and divide work among departments. Better organization, specialization, and improved communication increase efficiency and reduce errors, lowering average costs.
- **Risk-bearing economies:** Large firms can spread risks over many products and markets. Losses in one area can be offset by profits in another, making the business more stable and secure than small firms.
- **Network economies of scale:** Large firms connect with more users, suppliers, or customers, making their services more valuable. As the network grows, costs per user fall while efficiency, communication, and market reach increase, allowing firms to operate more effectively and gain greater benefits from expansion.
- **Research and development economies:** Large firms can afford to invest heavily in research and development, allowing them to create new products, improve quality, and adopt better technology—advantages usually unavailable to small firms.
- **Social and welfare economies:** As firms expand, they can provide better worker benefits such as housing, medical care, and recreation. These benefits improve worker morale and productivity, reducing costs per unit.
- **Storage economies:** Large firms enjoy storage economies because they can store goods in bulk, which reduces storage cost per unit. Efficient storage also prevents shortages and supports large-scale production efficiently.

b) External economies

These are advantages/benefits enjoyed by a firm as a result of the expansion or growth of the industry. The external economies of scale are shared with other firms operating within the industry. These benefits include:

- **Labor economies** occur as skilled workers are attracted to the industry, providing firms with cheaper access to expertise.
- **Economies of co-operation** arise when firms collaborate to establish common services like research centers and trade fairs.
- Expansion leads to the **development of commercial facilities** such as banks, insurance, transport, and advertising, helping smaller firms survive.
- **Social and economic infrastructure** like roads and communication networks improve, reducing production costs for all firms.
- **Economies of information** happen as research findings and innovations are shared, increasing efficiency and lowering costs.
- **Economies of concentration and specialization** allow firms to focus on specific processes, benefiting both individual firms and the industry as a whole.

ii. Diseconomies of scale.

Diseconomies of scale are increases in the cost per unit of output that occur when a firm grows too large and becomes less efficient. When a firm expands beyond its optimal size, it becomes harder to manage, leading to higher costs and reduced efficiency. They are also divided into internal and external diseconomies.

a) Internal diseconomies.

These are the disadvantages or problems faced by a firm which has increased its scale of production. These problems are only experienced by a single firm and they are not shared by other firms in the industry. They include:

- **Managerial diseconomies:** When a firm expands it faces the problem of supervision, coordination, control and communication of decisions, bureaucracy increases, time wastage and rigidity also increases which results into higher costs per unit of output.
- **Technical diseconomies:** A large firm realizes disadvantages of specialization and incurs high costs of maintaining complicated machines and buying raw materials hence higher average costs of production.
- **Financial diseconomies:** A large firm may face financial disadvantages especially if funds are not secured in time which delays production and limits expansion of the firm.
- **Risk bearing diseconomies:** Due to expansion a firm faces more disadvantages in case of uninsurable risks like a war since it loses on a very large scale.
- **Marketing diseconomies:** A large firm faces the problem of scarcity of raw materials, changes in tastes and preferences of consumers, declining demand for its products etc. which may result into higher costs and more losses.

b) External diseconomies

These are disadvantages realized by a firm from outside due to expansion of the industry. These problems are shared by the firms operating within the industry. They include:

- Labor diseconomies: As an industry expands it becomes more difficult to secure labor with the appropriate skills resulting into higher wages for labor and higher costs of production.
- Marketing diseconomies: As a result of expansion of an industry, competition for markets increases resulting into higher marketing costs and lower profits.
- High demand for raw materials: The expansion of an industry results into higher demand for raw materials hence higher costs of production.
- Scarcity of land: Expansion of an industry creates scarcity of land for expansion for individual firms which increase rent, congestion and transport costs hence higher average costs of production.
- High factor prices: When an industry expands, firms will be challenged by the shortage of infrastructure facilities and social amenities. This will mean that the cost of transport and communication will be very high hence leading to high cost of production.

Application activity 7.5



Analyse the scenario below and answer the questions that follow:

Two Rwandan manufacturing firms, one producing bottled water and the other producing soft drinks, are considering merging to expand their market share and compete more effectively with multinational beverage companies.

- a) Identify three internal economies of scale that the merged company might enjoy and explain how each reduces production costs.
- b) Explain two potential internal diseconomies of scale the merged company could face as a result of the merger.
- c) Describe two external economies that could benefit the firms because the beverage industry in Rwanda is growing.
- d) Give two examples of external diseconomies that might increase costs for the merged firm as the industry expands.

7.6 Revenues and profits of a firm.

Learning Activity 7.6



Analyse the scenario carefully and respond to the questions that follow.

A business lady sells juice at school during break time. In one day, she sells 20 bottles at 500 RWF each. To prepare the juice, she spent 6,000 RWF on fruits, sugar, and cups. After selling all the juice, she compares the money she earned with the money she spent.

Tasks:

Based on the scenario,

- i. How much money did she receive in total from sales?
- ii. What do we call the total money received from selling goods?
- iii. How much money did she spend on making the juice?
- iv. After subtracting the expenses from the money earned, is there any money left? How much money?
- v. What do we call the extra money left after expenses are paid?
- vi. What would happen if her expenses were higher than the money, she earned?
- vii. Why is it important for a seller to earn more than they spend?

Firms aim to maximize profits, sales, revenue, and their public image by minimizing costs and increasing output and sales. During production, various costs are incurred, and after selling goods to consumers, firms calculate profit as the difference between total sales and total costs. The profit earned can then be used to expand the business or invest in other ventures.

7.6.1 Meaning and forms of revenues.

i. Meaning of revenues

Revenue refers to the total money a firm earns from selling its goods or services over a given period, before any costs are deducted. For example, if a firm sells 300 units at 50 FRW each, its total revenue is 15,000 FRW per day.

ii. Forms of revenues.

In economics, revenue is classified into **three main forms**:

1. Total revenue (TR)

This is the total sum of money value received from the sales of various quantities of output of product produced during a given period of time at a certain price level. It can be calculated as the selling price of the firm's product times the quantity sold in a given period of time.

Total revenue = price × quantity, or letting TR be the total revenue function:

$$TR(Q) = P(Q) \times Q$$

Where Q is the quantity of output sold, and P(Q) is the inverse demand function (the demand function solved out for price in terms of quantity demanded).

Or simply: **Total revenue = Total product × price**

$$TR = Q \times P$$

TR curves differ depending on whether the firm is a **price maker** like a monopolist or a **price taker** like under perfect competition. Where a firm has price setting powers (e.g. monopoly) and faces a downward sloping demand curve, total revenue will initially rise but subsequently fall.

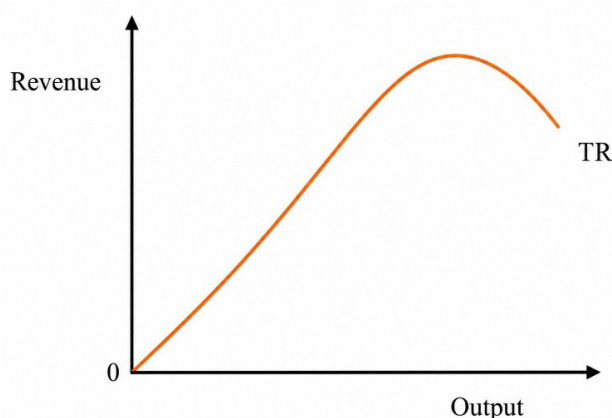


Fig 7.12 Total Revenue curve of a price maker

When the firm is a price taker like under perfect competition, it will face a straight total revenue curve. Because the average and marginal revenue will always increase by the same amount. This gives a straight TR curve.

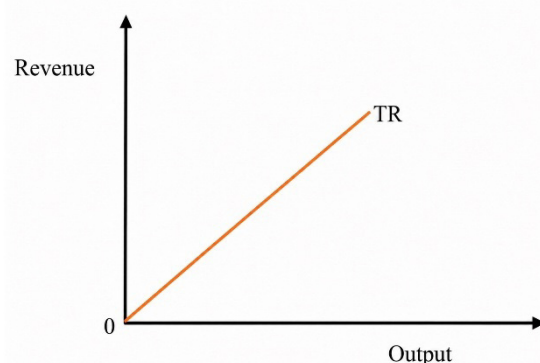


Fig. 7.13 Total Revenue curve of a price taker

2. Average revenue (AR)

This refers to the revenue per unit of output sold. It is obtained by dividing total revenue by output.

$$\text{Average Revenue} = \frac{\text{Total Revenue}}{\text{Total Output}}$$

$$\text{Average Revenue} = \frac{\text{TR}}{Q}$$

Given that TR will consist of price multiplied by the quantity ($P \times Q$) then:

$$AR = \frac{PQ}{Q} = P$$

This implies that the average revenue equals the price. Any curve relating average revenue to output is the same as the demand curve which relates price to output.

For example, if firm A sells its output worth 5,000kgs in a month and earns total revenue of 7,250,000RWF, the AR will be:

$$AR = \frac{7,250,000}{5000} = 1450 \text{ Frw}$$

AR curves differ depending on whether the firm is a price maker like a monopolist or a price taker like under perfect competition. Under the price takers (perfect competition) have a constant price and so their average revenue curve is a straight horizontal line. Price makers (monopolist and monopolistic competition) have a downward sloping AR curve.

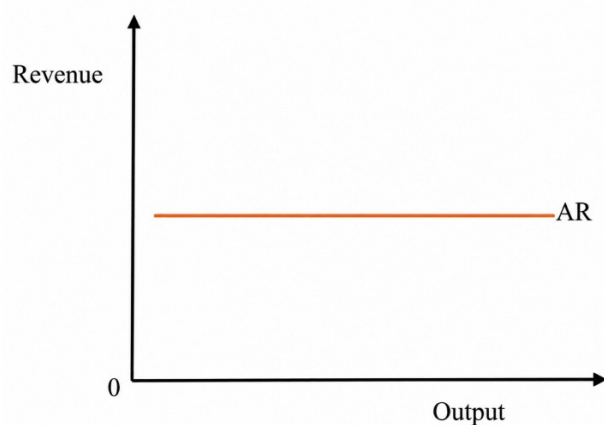


Fig 7.14 AR curve of a perfectly competitive firm.

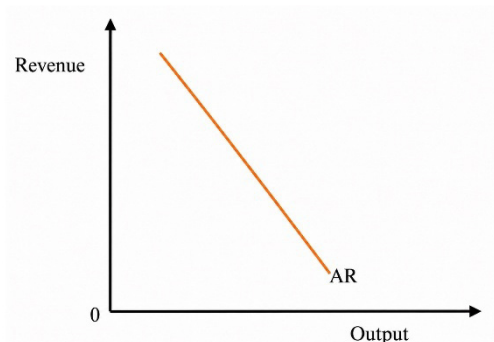


Fig 7.15 The AR of a price maker e.g. monopoly firms.

3. Marginal revenue (MR)

This refers to the additional revenue from an extra unit of output sold. Or it refers to a measure of the instantaneous rate of change of total revenue with respect to output Q. It is obtained by dividing the change in total revenue by the change in total product.

$$\text{Marginal Revenue} = \frac{\text{Change in Total Revenue}}{\text{Change in Total Output}}$$

$$\text{Or } MR = \frac{\Delta TR}{\Delta Q}$$

a) Exemple, if a firm sells 600 units of books for 5000RWF each, then if the firm later increases the quantity sold to 800 units of books at 4000RWF each, the MR will be?

$$MR = \frac{\Delta TR}{\Delta Q}$$

Initial total revenue: $P \times Q = 5000 \times 600 = 3,000,000\text{Frw}$

New total revenue: $P \times Q = 4000 \times 800 = 3,200,000\text{Frw}$

$$MR = \frac{3,200,000 - 3,000,000}{800 - 600}$$

Thus, $MR = 1000\text{Frw}$

The marginal revenue (MR) curve is influenced by the same factors as the demand curve, such as changes in income, prices of substitutes and complements, and population changes, which can cause it to shift or change position. However, its behaviour differs between market structures: in perfect competition, firms are price takers, so the price remains constant regardless of output and marginal revenue equals price ($MR = P$), making the MR curve horizontal and identical to the average revenue curve. In imperfect competition, firms have some market power, so the MR curve slopes downward, reflecting decreasing revenue from additional units sold.

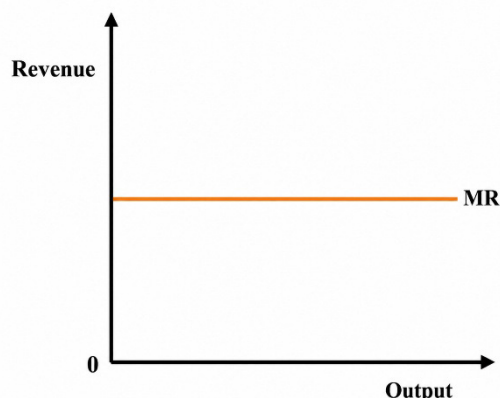


Fig 7.16 The MR curve of a price taker.

In imperfect competition, firms are large enough to influence market prices, so when they increase output they must lower the price of all units sold, which makes marginal revenue less than price ($MR < P$). As a result, the extra revenue from selling one more unit is reduced because the firm gains revenue from the additional unit but loses revenue on all previous units due to the price reduction. This makes marginal revenue

lower than average revenue and causes the MR curve to be downward sloping and lie below the demand (AR) curve, reflecting the firm's power to set prices in the market.

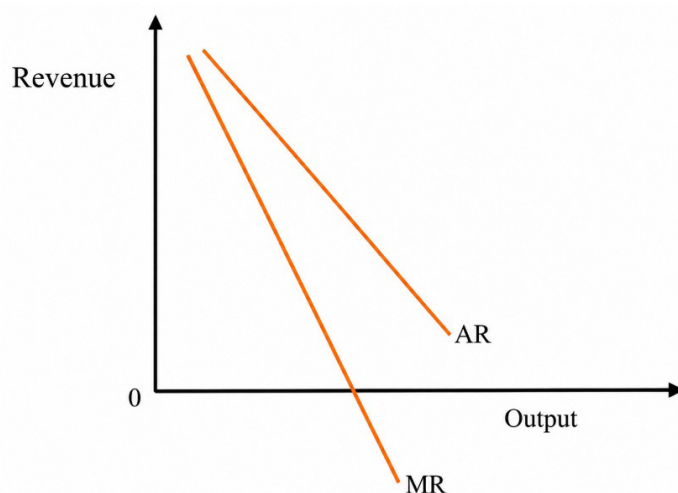


Fig 7.17 The MR curve of a price maker.

7.6.2 Calculating Revenues of a firm:

Calculating revenue is important in Economics because it helps firms understand how much income they earn from selling their output at different levels of production. For example:

Table 7.5: A hypothetical revenue schedule 1

Output (Units)	Price (RWF)	Total Revenue (TR) (RWF)	Average Revenue (AR)	Marginal Revenue (MR)
0	–	0	–	–
1	100	100	100	100
2	90	180	90	80
3	80	240	80	60
4	70	280	70	40
5	60	300	60	20

Table 7.6: Hypothetical Revenue Schedule 2:

Output (Units)	Price (RWF)	Total Revenue (TR) (RWF)	Average Revenue (AR) (RWF)	Marginal Revenue (MR) (RWF)
0	50	0	–	–
1	50	50	50	50
2	50	100	50	50
3	50	150	50	50
4	50	200	50	50
5	50	250	50	50

7.6.3 Relationship between TR, AR and MR.

The relationship between Total Revenue (TR), Average Revenue (AR), and Marginal Revenue (MR) depends on the market structure in which a firm operates. In perfectly competitive markets, firms are price takers, so AR and MR are equal to the constant market price and TR increases proportionately with output. In imperfect markets (such as monopoly or monopolistic competition), firms face a downward-sloping demand curve, causing AR to decline as output rises, MR to fall faster than AR, and TR to increase at a decreasing rate and eventually decline.

Assuming a hypothetical revenue schedule below, we can observe the relationship between TR, AR and MR under perfect markets with various units sold.

Table 7.7 A hypothetical Revenue schedule under perfect markets.

Units Sold (Q)	Price (P)	Total Revenue (TR = P × Q)	Average Revenue (AR = TR ÷ Q)	Marginal Revenue (MR = ΔTR ÷ ΔQ)
0	10	0	–	–
1	10	10	10	10
2	10	20	10	10
3	10	30	10	10
4	10	40	10	10
5	10	50	10	10

Under perfect competition, the relationship between Total Revenue (TR), Average Revenue (AR), and Marginal Revenue (MR) is simple and direct because firms are price takers. Total Revenue increases proportionately with output since it is calculated as price multiplied by quantity (**TR = P × Q**), and with price remaining constant, TR rises in a straight line from the origin as output increases.

Average Revenue, which represents revenue per unit sold (**AR = TR ÷ Q**), is equal to the constant market price at all levels of output, resulting in a horizontal AR curve. Marginal Revenue, defined as the additional revenue gained from selling one more unit (**MR = ΔTR ÷ ΔQ**), is also equal to the market price because each extra unit sold adds exactly the same amount to total revenue. As a result, under perfect competition, AR and MR are constant, equal to price, and coincide as horizontal lines on the graph, while TR increases linearly with output.

In a perfectly competitive market: Total Revenue increases linearly with output, while Average Revenue and Marginal Revenue remain constant and equal to the market price. This fundamental relationship explains why firms in perfect competition base output decisions on MR = MC (Marginal Revenue equals Marginal Cost).

Table 7.6. Summary of Perfect Competition Revenue Relationships

Concept	Relationship
TR	Increases at a constant rate
AR	Constant and equal to price
MR	Constant and equal to AR and price
AR vs MR	AR = MR = Price

Under perfect competition market, this relationship between Average and Marginal Revenue can be illustrated in a graph as shown in the figure below.

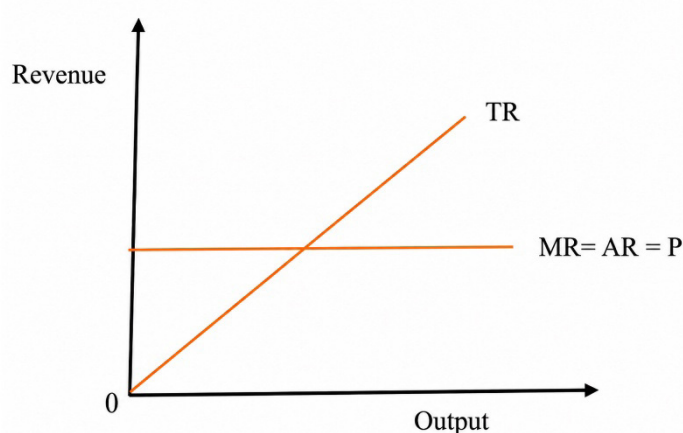


Fig. 7.18 Relationship between TR, AR and MR curves under perfect markets.

Assuming a hypothetical revenue schedule below, we can observe the relationship between TR, AR and MR under imperfect markets with various units sold.

Table 7.9 A hypothetical Revenue schedule under imperfect markets.

Units Sold (Q)	Price (AR)	Total Revenue	Marginal Revenue (MR)
0	–	0	–
1	20	20	20
2	18	36	16
3	16	48	12
4	14	56	8
5	12	60	4
6	10	60	0
7	8	56	–4

In an imperfectly competitive market, the relationship between Total Revenue (TR), Average Revenue (AR), and Marginal Revenue (MR) reflects the firm's need to lower price to sell more output. Total Revenue initially rises as sales increase, but it does so at a diminishing rate, reaches a maximum when marginal revenue becomes zero, and then falls as further price reductions reduce overall revenue. Average Revenue, which represents the price per unit, declines as output expands because the firm must charge a lower price to increase sales; consequently, the AR curve slopes downward and corresponds to the firm's demand curve. Marginal Revenue falls more rapidly than AR since each price cut applies to all units sold, not just the additional one, and for this reason MR may eventually become zero or even negative.

In an imperfectly competitive market therefore, selling additional units requires lowering the price, which causes average revenue to fall and marginal revenue to decline faster than average revenue. As a result, total revenue increases at first but eventually decreases, highlighting the fundamental difference between perfect and imperfect markets.

Table 7.10 Summary of imperfect Competition Revenue Relationships

Revenue Concept	Relationship in an Imperfect Market
Total Revenue (TR)	Increases as output rises, reaches a maximum, then decreases
Average Revenue (AR)	Downward sloping as price falls when output increases
Marginal Revenue (MR)	Downward sloping and always lies below the AR curve
At Maximum TR	Marginal Revenue (MR) equals zero

Under imperfect competition market, this relationship between Average and Marginal Revenue can be illustrated in a graph as shown in the figure below.

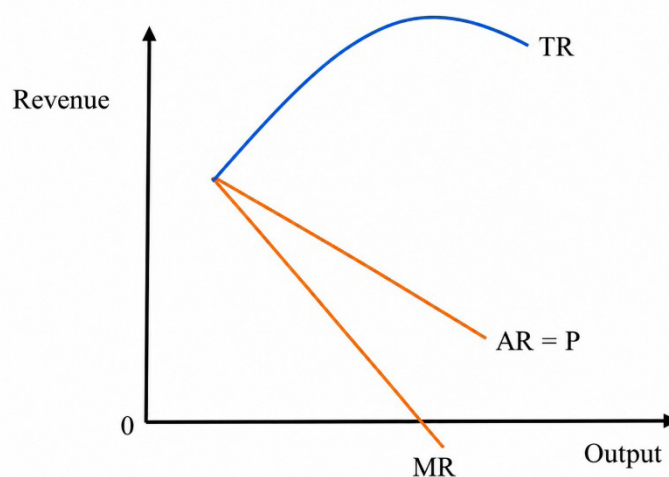


Fig. 7.19 Relationship between TR, AR and MR curves under imperfect markets

In imperfect markets (such as monopoly or monopolistic competition), the firm faces a downward-sloping demand curve, which shapes the behavior of Total Revenue (TR), Average Revenue (AR), and Marginal Revenue (MR). The AR curve, which represents price per unit, slopes downward because the firm must lower price to sell additional units. The MR curve also slopes downward but lies below the AR curve, since reducing price to sell one more unit applies to all units sold, causing marginal revenue to fall faster. Total Revenue (TR) initially increases as output expands, reaches a maximum at the level of output where MR equals zero, and then declines as further price reductions reduce overall revenue. These graphs together show that firms in imperfect markets are **price makers**, unlike price takers in perfect competition.

7.6.4: The revenue position of a perfect and imperfect competitive firms.

Under perfect competition, firms sell every unit at a fixed market price. Because this price does not change as output increases, both Average Revenue and Marginal Revenue remain constant and equal to the price. As a result, AR and MR coincide as the same horizontal line, showing that each additional unit sold adds exactly the same amount to revenue. Under perfect competition market, the relationship between Average and Marginal Revenue can be illustrated in a graph as shown in the figure below.

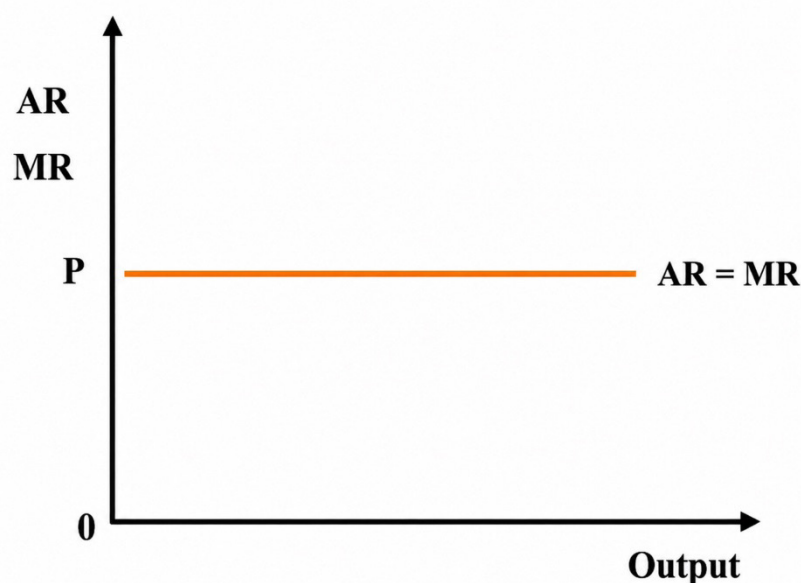


Fig 7.20 The revenue position of a perfect competitive firm.

Under imperfect markets, firms face a downward-sloping demand curve, so selling more output requires reducing price. As a result, Average Revenue (AR) falls as output increases. Marginal Revenue (MR) also declines but at a faster rate than AR because a lower price has to be charged on all units sold, not just the additional one. Consequently, MR always lies below AR, reflecting the firm's price-making power in imperfect competition. The relationship can be illustrated as below;

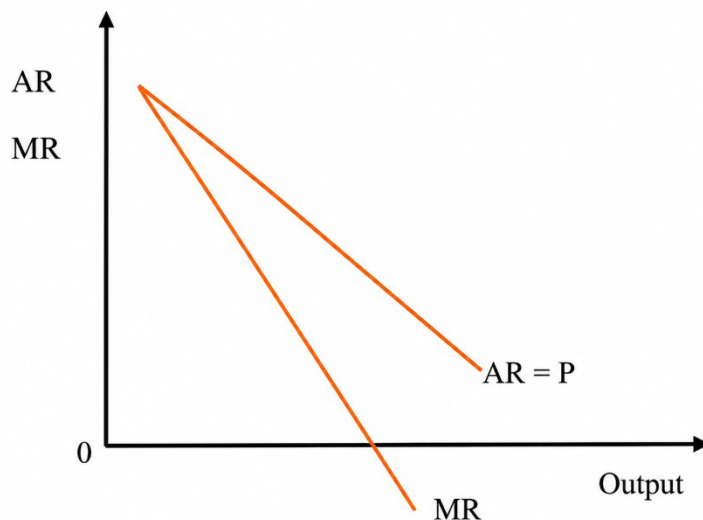


Fig 7.21 The revenue position of an imperfect competitive firm

Application activity 7.6



1. Firm A sells notebooks at different prices depending on how many are sold per day.

Quantity Sold (Q)	Price per Notebook (RWF)
100	500
200	450
300	400
400	350

a) Calculate

- i. Total Revenue (TR) for each level of output.
- ii. Average Revenue (AR) at each output level.
- iii. Marginal Revenue (MR) between each increase in output.

b) Answer the following questions:

- i. At which quantity does total revenue increase the most?
- ii. What happens to marginal revenue as output increases?
- iii. Is this firm acting as a price taker or a price maker? Explain your answer.
- iv. Based on your calculations, should the firm continue increasing output? Explain your answer.

7.7 Profits of a firm.

Learning Activity 7.7



Analyse the scenario below and answer the questions that follow.

A person opens a small juice stand near a busy road. Each day, the person sells juice and collects 20,000 francs. To run the stand, the person spends 12,000 francs on fruits, sugar, cups, and transport. At the end of the day, some money remains after all the expenses have been paid.

- i. How might the juice seller feel about the amount that remains at the end of the day?
- ii. What could the seller do with the money that remains?
- iii. What would happen to the remaining money if the seller sold more juice?
- iv. What if the costs of fruits increased but sales stayed the same, what would happen?
- v. Which seems more important for increasing the remaining money: selling more, reducing costs, or both?
- vi. How could the remaining money help the juice seller improve the business?
- vii. What might happen if, day after day, there was no money left after paying expenses?

Every firm operates with the aim of sustaining itself and achieving growth in a competitive economic environment. To do this, a firm must carefully compare the revenue it earns from selling its goods or services with the costs it incurs in producing them. The difference between these two outcomes determines whether the firm is performing well or poorly. Profits are significant to a firm, and thus, are central to business decision-making, survival, and expansion.

7.7.1 Meaning and forms of profits.

Profit refers to **the earnings to an entrepreneur as a factor of production**. It is the difference between total revenue received by the firm and the total costs incurred by the same firm. It is the residue which remains after paying all other factors of production. A firm realises profits when its average revenue is greater than average cost.

In economics, profits earned by a firm can be grouped into different types based on how they arise and how costs are considered. These types of profits help explain a firm's performance and guide decisions on production, expansion, or closure. The main types of profits include the following:

a) Normal Profit.

This is the minimum return a firm needs to remain in business. It is earned when total revenue is just equal to total cost, including both explicit and implicit costs. Normal profit is treated as part of the cost of production. It is also commonly referred to as **zero economic profit**, because the firm earns just enough to cover all its costs, including opportunity costs, with no extra gain.

b) Supernormal (Abnormal) Profit.

This occurs when a firm's total revenue exceeds its total cost. It represents earnings above the minimum required to keep the firm in operation and often arises due to efficiency, innovation, or favorable market conditions, especially in the short run. Abnormal (or supernormal) profits are also called **economic profits** because they represent earnings above all economic costs, including opportunity costs, showing a real net gain in resource use efficiency beyond the minimum required to keep the firm operating.

c) Subnormal Profit (Loss)

This arises when total revenue is less than total cost. In this case, the firm earns less than the normal level required to stay in business and may be forced to reduce output or exit the industry if the situation continues. It is also called **loss (or economic loss)**, because the firm's total revenue is insufficient to cover all its economic costs, including normal profit.

Profits are measured by comparing a firm's total revenue with its total costs at different stages of production. In practice, firms first calculate **gross profit** by deducting the cost of goods sold from sales revenue, and then determine **pure or net profit** by subtracting all other operating and incidental expenses. Profits arise in real situations when a firm sells its output above production costs due to factors such as efficient management, high demand, reduced costs, innovation, or favorable market conditions; in some cases, unexpected external changes may also generate sudden gains (**windfall profits**).

Exercise 7.3

1. A firm earns total revenue of 500,000 FRW and incurs total costs of 500,000 FRW.
 - a) Identify the type of profit earned.
 - b) Explain what this implies about the firm's future in the industry.
2. A textile company reports the following:
 - Total revenue = 800,000 FRW
 - Explicit costs = 600,000 FRW
 - Implicit costs = 150,000 FRW
 - i. Calculate the profit and classify it
 - ii. Advise whether the firm should expand production.

7.7.2 Profit maximization of the firm.

Firms operate with the aim of achieving the most favourable results from their production decisions while working within limited resources and constantly changing market conditions. As output increases, both revenue and costs change, but not always at the same pace, meaning higher production does not automatically lead to better outcomes. Because of this, firms must carefully decide how much to produce rather than simply producing as much as possible.

Profit maximization therefore involves choosing the level of output at which the difference between revenue and cost is greatest, based on a careful analysis of revenue–cost relationships in both the short run and the long run, a principle that guides firms in both perfect and imperfect markets, even though the way revenue is determined differs across these market structures.

a) Profit maximization under perfect competition.

Under perfect competition, firms operate in a market with many sellers, identical products, and prices determined by market forces rather than by individual firms. Since firms are price takers, they base their production decisions on the given market price. Profit maximization is achieved at the output level where marginal cost equals marginal revenue (**MC = MR**), with marginal revenue equal to the market price (**MR = AR**) as illustrated below;

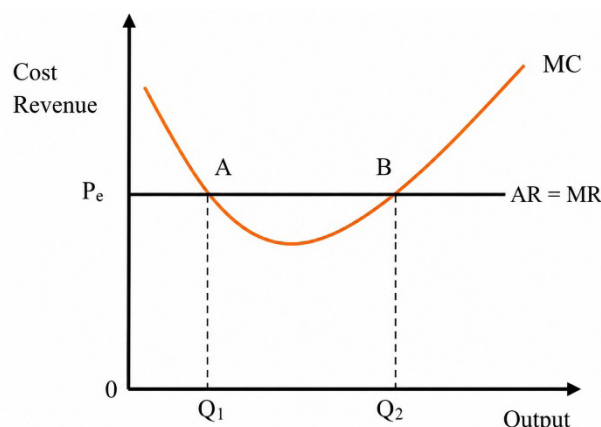


Fig. 7.22 Profit maximization under perfect competition.

The curve above shows profit maximization under perfect competition, where the firm faces a perfectly elastic demand curve such that Average Revenue (AR) equals Marginal Revenue (MR) at the market price P_e . The firm maximizes profit at the output level Q_2 , where the Marginal Cost (MC) curve intersects the $MR (= AR)$ curve from below. Although MC also equals MR at Q_1 , profit is not maximized there because MC is falling; only at Q_2 does the necessary condition for profit maximization hold.

b) Profit maximization under imperfect markets.

Under imperfect markets, firms have some control over price because their products are differentiated or competition is limited. To sell more output, a firm must lower its price, which causes marginal revenue to fall more quickly than price. Profit maximization is therefore achieved at the level of output where marginal cost equals marginal revenue, with marginal revenue lying below the average revenue (price) curve.

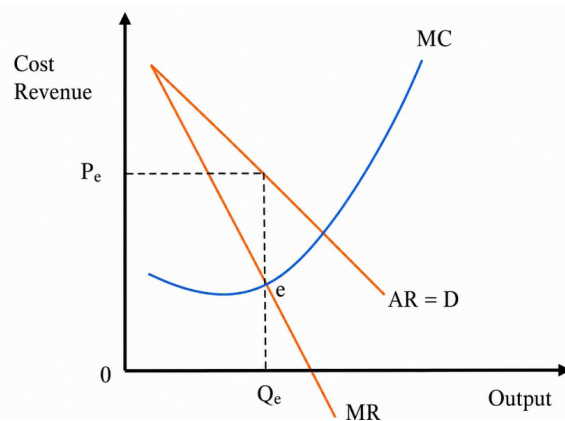


Fig 7.23 Profit maximization under imperfect markets.

7.7.3 Determinants and the role of profits in production

i. Determinants of profits.

Profit is a key objective of most business firms and an important indicator of business success. However, the level of profit earned by a firm does not arise by chance; it is influenced by a number of economic and business factors which affect a firm's revenue and costs, and therefore its ability to earn profits. They include the following;

- **Nature of competition:** Less competition allows firms to set higher prices and enjoy greater market control, leading to higher profits, while intense competition forces price reductions and lowers profit margins.
- **Risk and uncertainty:** Firms that undertake high risks may earn higher profits as compensation, but unfavorable outcomes due to uncertainty can reduce or eliminate profits.
- **Price level:** Higher selling prices increase total revenue and profits if costs remain unchanged, whereas lower prices reduce revenue and profits.
- **Cost of production:** Lower costs of inputs such as labour, raw materials, and power increase the difference between revenue and costs, raising profits, while high costs reduce profits.
- **Size of the market:** A large market provides more customers and higher sales volumes, leading to higher profits; a small market limits sales and profit potential.
- **Level of taxes:** High taxes on profits reduce the net earnings of firms, while low taxes allow firms to retain more profits for growth and expansion.
- **Production efficiency:** Efficient use of resources minimizes wastage and reduces costs, thereby increasing profits, while inefficiency raises costs and lowers profits.
- **Technology:** Modern technology improves productivity, reduces unit costs, and enhances output quality, leading to higher profits.
- **Quality and innovation:** High-quality products and innovation attract more customers, increase demand, and enable firms to earn higher profits.
- **Level of demand:** High demand increases sales and revenue, resulting in higher profits, while low demand reduces sales and profits.

Profits are determined by factors that affect a firm's costs and revenues, such as competition, demand, prices, production efficiency, technology, market size, taxation, and risk. Firms that operate efficiently, control costs, respond to market demand, and adapt to favorable business conditions are more likely to earn higher profits, while unfavorable conditions lead to low profits or losses.

ii. The role of profits in production.

In every economy, production involves the use of scarce resources such as land, labour, capital, and entrepreneurship. For producers to combine these resources and supply goods and services, they need a strong motivation. Profit is one of the most important incentives that drives production. It acts as a reward for entrepreneurs, guides decision-making, and influences how resources are allocated in the economy. Profits encourage efficiency, innovation, expansion, and economic growth thus, are essential for the survival and progress of production units as explained below.

- **Allocation of resources:** Profits guide resources to their most productive uses by encouraging producers to invest in profitable activities, ensuring efficient resource allocation.
- **Reward for entrepreneurship and risk:** Profits reward entrepreneurs for taking risks and dealing with uncertainty, motivating them to produce goods and services.
- **Source of finance for expansion:** Profits, especially retained earnings, provide funds for reinvestment, expansion, and improvement of production capacity.
- **Encouragement of innovation:** The prospect of higher profits motivates firms to innovate, adopt new technology, and improve production methods.
- **Promotion of efficiency:** Profit motive encourages firms to minimize costs, reduce waste, and use resources efficiently, shifting resources from unprofitable to profitable uses.
- **Improvement of economic welfare:** Higher profits indicate efficient production, increased output, and better availability of goods and services, raising overall economic welfare.
- **Reward for coordination:** Profits serve as a bonus to entrepreneurs for effectively combining and managing factors of production.
- **Increase in government revenue:** Profits raise taxable income, enabling governments to collect more taxes to fund public services and development.

Application activity 7.7



1. a) State the condition under which a firm earns profit.
b) Distinguish between:
 - i. Normal profit.
 - ii. Supernormal profit.
 - iii. Subnormal profit.
2. A firm has total revenue of 200,000 RWF and total cost of 150,000 RWF.
 - i. Calculate the profit.
 - ii. Identify the type of profit earned.
b) In your opinion, how can profits contribute to economic development?
3. a) If marginal cost is greater than marginal revenue, what should the firm do? Explain.
b) Compare profit maximization under perfect competition and imperfect markets.

End of unit assessment



Remembering and understanding questions

1. State three reasons why costs vary in the short run.
2. Define total revenue (TR), average revenue (AR), and marginal revenue (MR).
3. a) Define mergers and explain two advantages and two disadvantages of mergers.
b) Identify three factors that limit mergers.
4. a) Explain any three factors that influence the location of a firm.
b) Describe two merits and two demerits of localization of firms.
5. Explain the relationship between total revenue (TR), average revenue (AR), and marginal revenue (MR) using curves.
6. Explain the meaning of scale of production and describe two types of growth of firms.

Skills questions.

7. A firm produces the following output with the costs shown below:

Output (Units)	TFC (Frw)	TVC (Frw)
0	2,000	0
1	2,000	1,000
2	2,000	1,800
3	2,000	2,400

- Calculate total cost (TC), average fixed cost (AFC), average variable cost (AVC), and average total cost (ATC) for each output level.
 - Explain the relationship between marginal cost (MC) and average total cost (ATC) using the calculated results.
 - Compare the revenue position of a perfect competitive firm and an imperfect competitive firm.
8. Using a suitable diagram, distinguish between economies of scale and diseconomies of scale, and explain how each affects a firm's cost of production.
9. Explain how costs, revenues, and profits would influence a firm's decisions to relocate, adopt new technology, and expand output.
10. Examine the role and determinants of profits in guiding the firm's decisions.

Values and attitudes questions.

11. In what ways can a firm's profit making activities support community development in the area where it operates?

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